MATH 370

Lecture Notes · Fall 2026

Cody Carroll

University of San Francisco

Contents

Glossary of Abbreviations and Notation **3**

1 Probability, Sets, and Counting **6**

1.1 Probability, Sets, and Subsets . . . 6
1.2 Fundamental Results of Probability . . . 7
1.3 Set Operations and Disjointification . . . 10
1.4 De Morgan's Laws and Monotone Sequences of Events . . . 12
1.5 Definitions and Axioms of Probability . . . 15
1.6 Counting Methods . . . 19

2 Conditional Probability and Independence **25**

2.1 Conditional Probability and Bayes' Rule . . . 25
2.2 Independent Events . . . 30

3 Random Variables and Their Distributions **34**

3.1 Random Variables and Distribution Functions . . . 34
3.2 Discrete Random Variables: Worked Examples . . . 37
3.3 Continuous Random Variables . . . 41

4 Expectation and Variance **44**

4.1 Expectation of a Random Variable . . . 44
4.2 Functions of a Random Variable and Variance . . . 46
4.3 Moment Generating Functions and Probability Inequalities . . . 55
4.4 Median, Quantiles, and the Mode of a Random Variable . . . 60

5 Named Discrete Distributions **66**

5.1 Bernoulli and Binomial Distributions . . . 66
5.2 The Geometric Distribution . . . 71
5.3 The Poisson Distribution . . . 78

6 Named Continuous Distributions **82**

6.1 The Normal Distribution . . . 82
6.2 Uniform and Exponential Distributions . . . 88
6.3 Gamma and Chi-Squared Distributions . . . 93

7 Joint Distributions **98**

7.1 Joint Densities: Discrete and Continuous . . . 98
7.2 Joint Density Examples and Marginal Densities . . . 107
7.3 Marginal and Conditional Densities . . . 111
7.4 Conditional Expectation and Conditional Variance . . . 118
7.5 Conditional Expectation: Worked Examples . . . 122

8 Covariance and Correlation **127**

8.1 Joint Expectations and the Joint m.g.f. . . . 127
8.2 Covariance: Definition and Bounds . . . 130
8.3 Covariance Computations and Correlation . . . 133
8.4 Correlation: Worked Examples . . . 136

9 Independence and Sums of Random Variables **140**

9.1 Independence of Random Variables . . . 140
9.2 Expectation and Variance of Sums . . . 143
9.3 Sums of Normal Random Variables . . . 148
9.4 Sums of Independent Random Variables . . . 151
9.5 Linear Combinations of Normals . . . 156

10 Convergence and Limit Theorems **158**

10.1 The Central Limit Theorem . . . 158
10.2 Convergence in Distribution . . . 167
10.3 Proof of the CLT . . . 168
10.4 Convergence in Probability . . . 172
10.5 The Weak Law of Large Numbers . . . 174
10.6 The Sample Variance . . . 176

11 Transformations of Random Variables **177**

11.1 Squared Normals and the Chi-Squared Distribution . . . 177
11.2 Transformations of a Random Variable . . . 179
11.3 Min and Max Distributions; Transformations of Two Variables . . . 182

Glossary of Abbreviations and Notation

Everything below is used consistently throughout these notes. Where a convention could reasonably go either way, the choice made here is stated explicitly.

Abbreviations

RV	random variable
c.d.f.	cumulative distribution function, $F_X(x) = P(X \le x)$
p.d.f.	probability density function (continuous) or probability mass function (discrete); these notes use "p.d.f." for both
m.g.f.	moment generating function, $M_X(t) = \mathbb{E}[e^{tX}]$
i.i.d.	independent and identically distributed
o.w.	otherwise (used in piecewise definitions)
s.t.	such that
iff	if and only if
WLOG	without loss of generality
WLLN	Weak Law of Large Numbers
CLT	Central Limit Theorem
LHS / RHS	left-hand side / right-hand side

Sets and events

s	a sample point: one outcome of a random experiment
S	the sample space, the set of all possible s
A, B, E	events (subsets of S); B is also used for a Borel subset of $\mathbb{R}$
A^c	complement of A
$A \cup B$, $A \cap B$	union, intersection
$\varnothing$	the empty set
$\mathbb{R}$	the real line $(-\infty, \infty)$
$\mathbb{1}_{\{E\}}$	indicator of the event E: equals 1 if E occurs, 0 otherwise. The subscript is always the event, so a condition appears as the event it defines, e.g. $\mathbb{1}_{\{X \le x\}}$

Random variables and their distributions

Capital letters X, Y, Z denote random variables; the matching lowercase letters x, y, z denote particular values they may take. Subscripts naming a random variable are therefore always capital: f_X, F_X, M_X, μ_X, σ_X.

$P(A)$	probability of the event A
$P(A \mid B)$	probability of A given B
$F_X(x)$	c.d.f. of X

$f_X(x)$	p.d.f. of X
$M_X(t)$	m.g.f. of X
$\mathbb{E}[X]$, μ_X	expected value (mean) of X
$\mathrm{Var}[X]$, σ_X^2	variance of X; σ_X is the standard deviation
$\mathbb{E}[X^k]$	the kth moment of X
x_p	the pth quantile of X, i.e. the value with $F_X(x_p) = p$
p, q	a success probability and its complement, $q = 1 - p$
$X \sim D$	"X is distributed as D"
$\Phi(x)$	c.d.f. of the standard normal $Z \sim N(0,1)$

Pairs of random variables

Joint objects carry a comma in the subscript ($f_{X,Y}$); conditional objects carry a bar ($f_{Y|X=x}$).

$F_{X,Y}(x,y)$	joint c.d.f. of X and Y
$f_{X,Y}(x,y)$	joint p.d.f. of X and Y
$M_{X,Y}(t_1,t_2)$	joint m.g.f. of X and Y
$f_{Y\|X=x}(y)$	conditional p.d.f. of Y given $X = x$
$\mathbb{E}[Y \mid X = x]$	conditional expectation of Y given $X = x$
$\mathrm{Var}[Y \mid X = x]$	conditional variance of Y given $X = x$
$\mathrm{Cov}(X,Y)$	covariance of X and Y
$\rho(X,Y)$	correlation of X and Y

On brackets vs. parentheses: the one-argument operators take square brackets ($\mathbb{E}[\,\cdot\,]$, $\mathrm{Var}[\,\cdot\,]$), while the two-argument ones take parentheses ($\mathrm{Cov}(\cdot,\cdot)$, $\rho(\cdot,\cdot)$), as does $P(\cdot)$.

Samples and limits

$X_1, \ldots, X_n$	a random sample
$\bar{X}_n$	the sample mean $\frac{1}{n}\sum_{i=1}^n X_i$ (the subscript is dropped when n is clear)
S_n	the sample sum $\sum_{i=1}^n X_i$
S^2	the sample variance $\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$ (capital: it is itself a RV)
S_0^2	the known-mean sample variance $\frac{1}{n}\sum_{i=1}^n (X_i - \mu_X)^2$, used in Sections 10.6 and 11.1
$\hat{F}_n(x)$	the empirical c.d.f.
$Y_{(1)}$, $Y_{(n)}$	the minimum and maximum of $X_1, \ldots, X_n$
$Y_n \xrightarrow{p} Y$	converges in probability
$Y_n \xrightarrow{d} Y$	converges in distribution

Named distributions

Bernoulli(p)	single trial, 1 with probability p and 0 otherwise
Bin(n, p)	binomial: number of successes in n independent trials
Geo(p)	geometric: number of trials until the first success
Pois(λ)	Poisson: counts of rare events at rate λ
$\Gamma(\alpha, \beta)$	gamma, with shape α and rate β
Exp(λ)	exponential with rate λ, so $\mathbb{E}[X] = 1/\lambda$; the $\Gamma(\alpha = 1,\ \beta = \lambda)$ special case
$N(\mu_X, \sigma_X^2)$	normal with mean μ_X and variance σ_X^2; $Z \sim N(0, 1)$ is standard normal
χ_r^2	chi-squared with r degrees of freedom; the $\Gamma(\alpha = r/2,\ \beta = 1/2)$ special case
$U(\alpha, \beta)$	uniform on the interval (α, β)

Multi-letter distribution names are set upright (Bin, Pois, Exp) to distinguish them from variables; the conventional single-symbol families (N, U, Γ, χ^2) keep their usual italic form.

1 Probability, Sets, and Counting

1.1 Probability, Sets, and Subsets

Probability Sets and Subsets

Most of statistics is interested in the result of some experiment that has a random outcome.

Example 1.1: The outcome of the roll of a die.

Definitions

1. Let "s" be the result of a random experiment, called a "sample point".
2. Let S be the set of all possible values of "s".
3. Let subsets of S be denoted by capital letters A, B, C, D, etc.

An event A "occurs" when $s \in A$ ($\in$ = "is contained in").
In other words, when the outcome of the random experiment is one of the sample points in A.

An event A does not occur when $s \notin A$ ($\notin$ = "is not contained in")

S is also an event, but one that is certain to occur (since it contains all possible values of "s").

The event $\varnothing$ is defined to be the impossible event, where no $s \in S$ occurred.

4. $A \subseteq B$ means that event A is completely contained in event B, or every $s \in A$ is also an $s \in B$.

Example 1.2: Consider a box with three blue coins, and three red coins. Draw four coins from the box one by one, and record the color.
Let r = red, b = blue.

a) List S

$$S = \{rrrb,\ rrbr,\ rbrr,\ brrr,\ rrbb,\ rbrb,\ rbbr,\ bbrr,\ brrb,\ brbr,\ bbbr,\ bbrb,\ brbb,\ rbbb\}$$

For 14 distinct values of "s". (With only three coins of each color in the box, the outcomes $rrrr$ and $bbbb$ are impossible, so we get $2^4 - 2 = 14$ sample points rather than 16.)

Note: Unlike the die rolls and coin flips ahead, these 14 outcomes are *not* equally likely. Drawing sequentially without replacement, $P(rrrb) = \frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} \cdot 1 = \frac{1}{20}$, while $P(rrbb) = \frac{3}{6} \cdot \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{1}{10}$. Listing a sample space says nothing about the probabilities of its points; keep this in mind when we meet the classical definition of probability in Section 1.5, which *assumes* equally likely outcomes.

b) Let A = event of at least 2 blue coins. List all $s \in A$.

$$A = \{rrbb,\ rbrb,\ rbbr,\ bbrr,\ brrb,\ brbr,\ bbbr,\ bbrb,\ brbb,\ rbbb\}$$

c) Let B = event of exactly two red coins. List all $s \in B$.

$$B = \{rrbb,\ rbrb,\ rbbr,\ bbrr,\ brrb,\ brbr\}$$

d) Is $A \subseteq B$ or $B \subseteq A$? Explain.
Since if $s \in B$, that same $s \in A$, when B occurred A certainly occurred, and $B \subseteq A$.

1.2 Fundamental Results of Probability

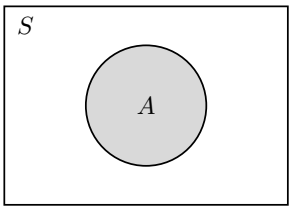
Fundamental Results

Let $A, A_1, A_2, \ldots, A_n$ be events.

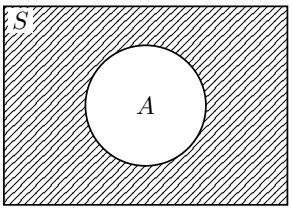
1. The complement of A, denoted A^c, is

$$A^c = \{s \in S : s \notin A\} \quad \text{(the set of all } s \in S \text{ such that } s \notin A\text{)}$$

A^c is where none of the events in A occurred.



A shaded



A^c hatched

2. The union of events A_1, A_2, denoted $A_1 \cup A_2$, is

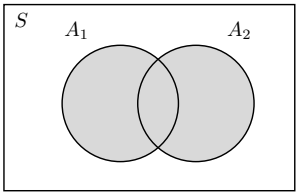
$$A_1 \cup A_2 = \{s \in S : s \in A_j \text{ for at least one } j = 1, 2\}$$

i.e. the event where at least one of the events A_1, A_2 occurred.

This can be extended to n events:

$$\bigcup_{j=1}^{n} A_j = \{s \in S : s \in A_j \text{ for at least one } j = 1, \ldots, n\}$$

In a Venn Diagram:



$A_1 \cup A_2$ shaded

3. The intersection of two events A_1, A_2, denoted $A_1 \cap A_2$, is:

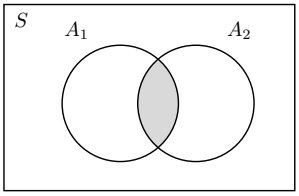
$$A_1 \cap A_2 = \{s \in S : s \in A_i \text{ for all } i = 1, 2\}$$

i.e. the event where both events A_1, A_2 occurred.

This can also be extended:

$$\bigcap_{j=1}^{n} A_j = \{s \in S : s \in A_i \text{ for all } i = 1, \ldots, n\}$$

In a Venn Diagram:

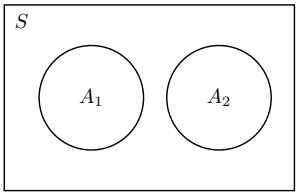


$A_1 \cap A_2$ shaded

4. Two events A_1 and A_2 are disjoint if $A_1 \cap A_2 = \varnothing$, and events $A_1, \ldots, A_n$ are mutually disjoint if

$$A_i \cap A_j = \varnothing \quad \forall\, i \neq j \qquad (\forall = \text{“for all”})$$

Venn Diagram:

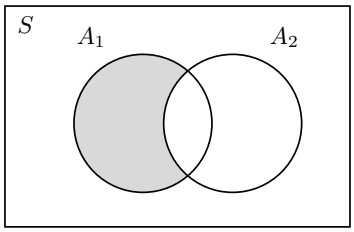


$A_1 \cap A_2 = \emptyset$

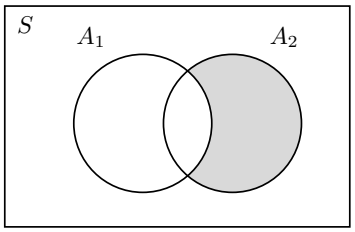
5. The <u>difference</u> between events A_1, A_2, denoted $A_1 \setminus A_2$, is

$$A_1 \setminus A_2 = \{s \in S : s \in A_1,\ s \notin A_2\} = \text{event where } A_1 \text{ occurred, but } A_2 \text{ did not.}$$

Notice that $A_1 \setminus A_2 = A_1 \cap A_2^c$.



$A_1 \cap A_2^c = A_1 \setminus A_2$



$A_2 \cap A_1^c = A_2 \setminus A_1$

Example 1.3: Roll a fair six-sided die, so $S = \{1, 2, 3, 4, 5, 6\}$ with each outcome equally likely. Let $A = \{1, 2, 3, 4\}$ ("at most a 4") and $B = \{2, 4, 6\}$ ("even").

a) List each of the following as a set:

$$A^c = \{5, 6\}, \qquad A \cup B = \{1, 2, 3, 4, 6\}, \qquad A \cap B = \{2, 4\},$$
$$A \setminus B = \{1, 3\}, \qquad B \setminus A = \{6\}.$$

b) Since all six outcomes are equally likely, each has probability $\frac{1}{6}$, so a probability is just (number of outcomes)/6:

$$P(A) = \tfrac{4}{6}, \quad P(B) = \tfrac{3}{6}, \quad P(A \cap B) = \tfrac{2}{6}, \quad P(A \cup B) = \tfrac{5}{6}.$$

c) Check that $A \setminus B = A \cap B^c$. Here $B^c = \{1, 3, 5\}$, so $A \cap B^c = \{1, 2, 3, 4\} \cap \{1, 3, 5\} = \{1, 3\}$, which is exactly $A \setminus B$. ✓

d) Notice that

$$P(A) + P(B) = \tfrac{4}{6} + \tfrac{3}{6} = \tfrac{7}{6} > 1,$$

which cannot be a probability. The outcomes 2 and 4 have been counted twice, once in A and once in B. Subtracting the overlap once fixes it:

$$P(A) + P(B) - P(A \cap B) = \tfrac{4}{6} + \tfrac{3}{6} - \tfrac{2}{6} = \tfrac{5}{6} = P(A \cup B). \checkmark$$

This is the inclusion–exclusion rule we prove in Section 1.5.

1.3 Set Operations and Disjointification

Fundamental Rules

Example 1.4: Let a sample point be the outcome of three coin flips, where the coin has a head on one side, and a tail on the other.

a) List all $s \in S$

$$S = \{HHH,\ HTH,\ HHT,\ THH,\ TTH,\ THT,\ HTT,\ TTT\}$$

b) Let A = event of at least one H, B = event of at least two tails.
List all sample points in:

i) $A \cup B$

$$A \cup B = \{HHH,\ HTH,\ HHT,\ THH,\ TTH,\ THT,\ HTT,\ TTT\}$$

ii) $A \cap B$

$$A \cap B = \{TTH,\ THT,\ HTT\}$$

iii) $A \cap B^c$

$$A \cap B^c = \{HHH,\ HTH,\ HHT,\ THH\}$$

iv) $A^c \cap B$

$$A^c \cap B = \{TTT\}$$

From the previous rules we have the following propositions:

Proposition 1.1:

i) $S^c = \varnothing, \quad \varnothing^c = S, \quad (A^c)^c = A$

$S^c = \{s \notin S\} = \varnothing, \quad \varnothing^c = \{s \notin \varnothing\} = \{s \in S\} = S$

$$(A^c)^c = \{s \in S : s \notin A^c\} = \{s \in S : s \in A\} = A$$

ii) $S \cup A = S, \quad \varnothing \cup A = A, \quad A \cup A^c = S, \quad A \cup A = A$

iii) $S \cap A = A, \quad \varnothing \cap A = \varnothing, \quad A \cap A^c = \varnothing, \quad A \cap A = A$

Proposition 1.2:

i) $A_1 \cup (A_2 \cup A_3) = (A_1 \cup A_2) \cup A_3$

$$A_1 \cap (A_2 \cap A_3) = (A_1 \cap A_2) \cap A_3$$

ii) $A_1 \cup A_2 = A_2 \cup A_1, \quad A_1 \cap A_2 = A_2 \cap A_1$

iii) $A \cap \left(\bigcup_j A_j\right) = \bigcup_j (A \cap A_j), \quad A \cup \left(\bigcap_j A_j\right) = \bigcap_j (A \cup A_j)$

Part (iii) of the proposition above:

Proof sketch: $A \cap \left(\bigcup_j A_j\right) \Rightarrow s \in A$ and $s \in A_j$ for at least one j, say $j = 1$. Then, if $s \in A$ and $s \in A_1$, $s \in (A \cap A_1)$, i.e. $s \in (A \cap A_j)$ for at least one $j = 1, \ldots, n$.
Thus, $A \cap \left(\bigcup_j A_j\right) = \bigcup_j (A \cap A_j)$.

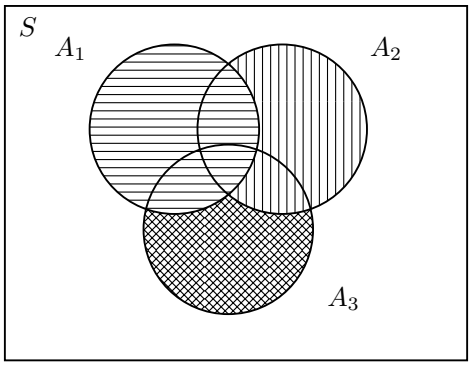
$\bigcup_j (A \cap A_j) \Rightarrow s \in (A \cap A_j)$ for at least one $j = 1, \ldots, n$, say $j = 1$. Then, if $s \in (A \cap A_1)$, $s \in A$ and $s \in A_1$, thus $s \in A \cap \left(\bigcup_j A_j\right)$, and

$$\bigcup_j (A \cap A_j) = A \cap \left(\bigcup_j A_j\right)$$

Proposition 1.3:

$$\bigcup_j A_j = A_1 \cup (A_1^c \cap A_2) \cup (A_1^c \cap A_2^c \cap A_3) \cup \cdots$$

Via Venn Diagram:



$\equiv A_1$ (horizontal lines pattern)

$\equiv A_1^c \cap A_2$ (vertical lines pattern)

$\equiv A_1^c \cap A_2^c \cap A_3$ (cross-hatch pattern)

Proof Sketch: Let $n = 2$. Then, the equation we wish to show is:

$$\underbrace{A_1 \cup A_2}_{\text{LHS}} = \underbrace{A_1 \cup (A_1^c \cap A_2)}_{\text{RHS}}$$

To show this, it is sufficient to show that when $s \in$ RHS, $s \in$ LHS, and vice versa.

Say $s \in$ RHS. Then,

- If $s \in A_1$, $s \in$ LHS (trivially).
- If $s \in (A_1^c \cap A_2)$, then $s \in A_2$ and $s \notin A_1$, so $s \in$ LHS.

Thus, if $s \in$ RHS, then $s \in$ LHS.

Now, say $s \in$ LHS:

- If $s \in A_1$, $s \in$ RHS trivially.
- If $s \in A_2$, then $s \in (A_1 \cap A_2)$ or $s \in (A_1^c \cap A_2)$.
  - If $s \in (A_1 \cap A_2)$, then $s \in$ RHS since $s \in A_1$.
  - If $s \in (A_1^c \cap A_2)$, then $s \in$ RHS.

Thus, if $s \in$ LHS, then $s \in$ RHS. □

The proof can be extended from here.

Example 1.5: Let $S = \{1, 2, \ldots, 10\}$ with all outcomes equally likely, and let

$$A_1 = \{1, 2, 3, 4, 5\}, \qquad A_2 = \{4, 5, 6, 7\}, \qquad A_3 = \{7, 8, 9\}.$$

a) Why can we not just add the probabilities?

$$P(A_1) + P(A_2) + P(A_3) = \tfrac{5}{10} + \tfrac{4}{10} + \tfrac{3}{10} = \tfrac{12}{10} > 1$$

The outcomes $4, 5$ lie in both A_1 and A_2, and 7 lies in both A_2 and A_3, so they are counted twice.

b) Disjointify. Apply the proposition term by term:

$$\begin{aligned} A_1 &= \{1, 2, 3, 4, 5\} \\ A_1^c \cap A_2 &= \{6, 7, 8, 9, 10\} \cap \{4, 5, 6, 7\} = \{6, 7\} \\ A_1^c \cap A_2^c \cap A_3 &= \{8, 9, 10\} \cap \{7, 8, 9\} = \{8, 9\} \end{aligned}$$

Each piece keeps only what is *new*: everything already counted is stripped out by the complements.

c) Check that the three pieces are pairwise disjoint and that their union is $A_1 \cup A_2 \cup A_3 = \{1, 2, \ldots, 9\}$. ✓

d) Now the probabilities may simply be added, because the pieces are disjoint:

$$P(A_1 \cup A_2 \cup A_3) = \tfrac{5}{10} + \tfrac{2}{10} + \tfrac{2}{10} = \tfrac{9}{10}$$

which agrees with counting $|A_1 \cup A_2 \cup A_3| = 9$ directly. ✓

Note: This is why disjointification matters: axiom iii) of probability only lets us add probabilities over *disjoint* events. Rewriting a union as a disjoint union is the standard way to make that axiom apply.

1.4 De Morgan's Laws and Monotone Sequences of Events

Proposition 1.4 *(De Morgan's Law)***:**

$$\left(\bigcup_j A_j\right)^c = \bigcap_j A_j^c, \qquad \left(\bigcap_j A_j\right)^c = \bigcup_j A_j^c$$

Proof when $n = 2$: $\underbrace{(A_1 \cup A_2)^c}_{\text{LHS}} = \underbrace{A_1^c \cap A_2^c}_{\text{RHS}}$

Say $s \in$ LHS. Then $s \notin A_1$, $s \notin A_2 \Rightarrow s \in A_1^c$ and $s \in A_2^c$

$$\Rightarrow s \in (A_1^c \cap A_2^c) \Rightarrow s \in \text{RHS}$$

Say $s \in$ RHS. Then, $s \in A_1^c$ and $s \in A_2^c \Rightarrow s \notin A_1$ and $s \notin A_2$

$$\Rightarrow s \in (A_1 \cup A_2)^c \Rightarrow s \in \text{LHS} \qquad \square$$

Example 1.6: The nine starter Pokémon of the original games form a two-attribute sample space: each is a pair (line, stage), where the line is Grass, Fire, or Water and the stage is basic, middle, or final. Written out by name,

	basic	middle	final
Grass	Bulbasaur	Ivysaur	Venusaur
Fire	Charmander	Charmeleon	Charizard
Water	Squirtle	Wartortle	Blastoise

One of the nine is drawn at random. Let A = "not from the Grass line" (Fire or Water), and B = "a middle-stage evolution."

a) List all $s \in S$, $s \in A$, $s \in B$

$$S = \{\text{all nine starters above}\}$$
$$A = \{\text{Charmander, Charmeleon, Charizard, Squirtle, Wartortle, Blastoise}\}$$
$$B = \{\text{Ivysaur, Charmeleon, Wartortle}\}$$

b) List all events in $A^c \cap B^c$, and $(A \cap B)^c$

$$A^c \cap B^c = \{\text{Bulbasaur, Venusaur}\} \quad \text{"Grass line, and not middle stage"}$$
$$A \cap B = \{\text{Charmeleon, Wartortle}\}$$
$$(A \cap B)^c = S \setminus (A \cap B) = \{\text{the other seven starters}\} = A^c \cup B^c$$
$$= \text{"The opposite of } A \text{ and } B\text{"}$$

The last equality is De Morgan's law, verified here by hand.

c) List all events in $A \cap B^c$, $B \cap A^c$

$$A \cap B^c = \{\text{Charmander, Charizard, Squirtle, Blastoise}\} = \text{"In } A \text{ but not } B\text{"}$$
$$A^c \cap B = \{\text{Ivysaur}\} = \text{"In } B \text{ only"}$$

Monotone Sequence of Events

A sequence of events $\{A_n\}$, $n \geq 1$, is called monotone if

i) $A_1 \subseteq A_2 \subseteq \cdots$ (monotone increasing)

ii) $A_1 \supseteq A_2 \supseteq \cdots$ (monotone decreasing)

If $\{A_n\}$ is monotone increasing,
$\bigcup_j A_j$ is called the limit of the sequence.

If $\{A_n\}$ is monotone decreasing,
$\bigcap_j A_j$ is called the sequence's limit.

These limits matter because probability is *continuous* along monotone sequences:

$$P\left(\lim_n A_n\right) = \lim_n P(A_n)$$

We will not use this result directly, but it is what makes limit statements about the c.d.f. rigorous, such as $F_X(\infty) = 1$ in Section 3.1.

Note: $A_1 \subseteq A_2$ means either $A_1 = A_2$ or $A_1 \subset A_2$ (A_1 is strictly smaller than A_2).

1.5 Definitions and Axioms of Probability

Basics of Probability

The probability of an event is a number within $[0, 1]$ which represents the likelihood of an event occurring.

There are two main definitions of probability:

1) Classical Definition: Where we have $S = \{s_1, s_2, \ldots, s_n\}$ is a finite space, where each outcome $\{s_i\}$, $i = 1, \ldots, n$, is equally likely. Then, if event A has m values of s_i,

$$P(A) = \frac{m}{n} \qquad (P(A) = \text{probability of event } A).$$

2) Frequentist Definition: Let an experiment be repeated N times. Let $M(A) = \#$ of times A occurred out of N. Then,

$$P(A) = \lim_{N\to\infty} \frac{M(A)}{N}$$

Note: When the classical definition and frequentist definition both apply, they are consistent with each other.

Axioms of Probability: Every probability function must satisfy three axioms:

i) $P(A) \geq 0$ for all events A

ii) $P(S) = 1$

iii) If events A_i are pairwise disjoint ($A_i \cap A_j = \varnothing,\ i \neq j$),

$$P\left(\bigcup_j A_j\right) = \sum_j P(A_j)$$

Note: The classical "definition" above is really a *consequence* of these axioms. Read literally it looks circular, since it seems to define P by helping itself to the values of P on the singletons. It does not have to. Assume only that the outcomes are equally likely, i.e.

$$P(\{s_1\}) = P(\{s_2\}) = \cdots = P(\{s_n\}),$$

an equality between unknown numbers rather than a stipulation of their value. Axioms ii) and iii) then pin the value down:

$$1 = P(S) = P\left(\bigcup_{i=1}^{n}\{s_i\}\right) = \sum_{i=1}^{n} P(\{s_i\}) = n\,P(\{s_1\}), \qquad \text{so} \qquad P(\{s_i\}) = \tfrac{1}{n},$$

and additivity gives $P(A) = m/n$ whenever A contains m outcomes. The $1/n$ is *derived*; the only assumption is the symmetry. The axioms say what a probability *is*; the two definitions above are recipes for choosing one (a symmetry assumption) and for interpreting one (a long-run frequency).

Axiom iii) is why the disjointification result of Section 1.3 matters: to compute the probability of an arbitrary union, we first rewrite it as a union of *disjoint* pieces, and then axiom iii) lets us add their probabilities.

Proposition 1.5 (consequences of the axioms): Everything else we know about probability follows from i)–iii). In particular:

iv) $P(\varnothing) = 0$

v) $P(A^c) = 1 - P(A)$

vi) $A_1 \subseteq A_2 \implies P(A_1) \leq P(A_2), \quad P(A_2 \setminus A_1) = P(A_2) - P(A_1)$

vii) $0 \leq P(A) \leq 1$

Proof of v): A and A^c are disjoint with $A \cup A^c = S$, so axioms ii) and iii) give $1 = P(S) = P(A) + P(A^c)$, and rearranging gives v). Taking $A = S$ in v) gives iv), and vii) follows from i) and v). The remaining proofs are similar.

Proposition 1.6:

i) For any two events A_1, A_2,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

ii) For any three events A_1, A_2, A_3,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) = {} & P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) \\ & - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3) \end{aligned}$$

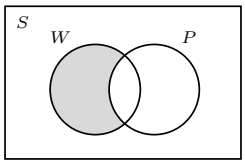
iii) For any events $A_1, A_2, \ldots$

$$P\left(\bigcup_i A_i\right) \leq \sum P(A_i), \quad i = 1, \ldots, \infty \text{ or } i = 1, \ldots, n.$$

Example 1.7: Choose one of the original 151 Pokémon uniformly at random. Among them, 32 are Water type and 33 are Poison type, and exactly 2 (Tentacool and Tentacruel) are both, since a Pokémon can carry two types at once. Let W = Water type, P = Poison type, so

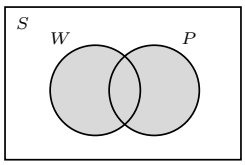
$$P(W) = \tfrac{32}{151}, \qquad P(P) = \tfrac{33}{151}, \qquad P(W \cap P) = \tfrac{2}{151}.$$

a) Find the probability of drawing a Water but not Poison type.



$$P(W \cap P^c) = P(W) - P(W \cap P) = \frac{32}{151} - \frac{2}{151} = \frac{30}{151} = 0.199$$

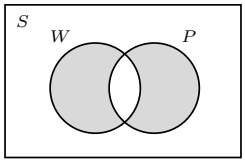
b) Find the probability of drawing a Water type, a Poison type, or both.



$$P(W \cup P) = P(W) + P(P) - P(W \cap P) = \frac{32}{151} + \frac{33}{151} - \frac{2}{151} = \frac{63}{151} = 0.417$$

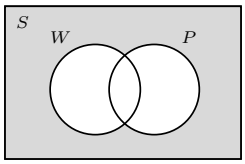
Without the subtraction, Tentacool and Tentacruel would each be counted twice.

c) Find the probability of drawing a Water type, or a Poison type, but not both.



$$\begin{aligned} P((P \cap W^c) \cup (W \cap P^c)) &= P(P \cap W^c) + P(W \cap P^c) \\ &= P(W \cup P) - P(W \cap P) = \frac{63}{151} - \frac{2}{151} = \frac{61}{151} = 0.404 \end{aligned}$$

d) Find the probability of drawing neither a Water nor a Poison type.



$$P(W^c \cap P^c) = P((W \cup P)^c) = 1 - P(W \cup P) = 1 - \tfrac{63}{151} = \tfrac{88}{151} = 0.583$$

Part d) is De Morgan's law followed by the complement rule.



Figure 1.3: *Kinkaku-ji (left) and Ginkaku-ji (right), Kyoto.* [1]

Example 1.8: Of 200 surveyed visitors to Kyoto, 120 went to Kinkaku-ji (金閣寺), 90 went to Ginkaku-ji (銀閣寺), and 50 went to both (Fig. 1.3). The two names are a matched pair: Kinkaku-ji is the Temple of the Golden Pavilion, whose top two floors are covered in gold leaf, and Ginkaku-ji is the Temple of the Silver Pavilion (which was never actually coated in silver). Pick a visitor at random and let K and G be the events that they went to each site, so $P(K) = 0.60$, $P(G) = 0.45$, and $P(K \cap G) = 0.25$.

a) At least one of the two, by inclusion–exclusion:

$$P(K \cup G) = 0.60 + 0.45 - 0.25 = 0.80.$$

b) Neither, by De Morgan's law and the complement rule:

$$P(K^c \cap G^c) = 1 - P(K \cup G) = 0.20.$$

c) Exactly one: the union minus the overlap,

$$P(K \cup G) - P(K \cap G) = 0.80 - 0.25 = 0.55.$$

Counting heads gives the same answers: $120 + 90 - 50 = 160$ visitors went to at least one site, 40 to neither, and $160 - 50 = 110$ to exactly one.

1.6 Counting Methods

It is often of particular interest to determine how many ways some outcomes could occur.

Example 1.9: If a salsa dancer has a performance wardrobe of 4 shirts, three pairs of pants, and two pairs of shoes, how many unique outfits do they have?

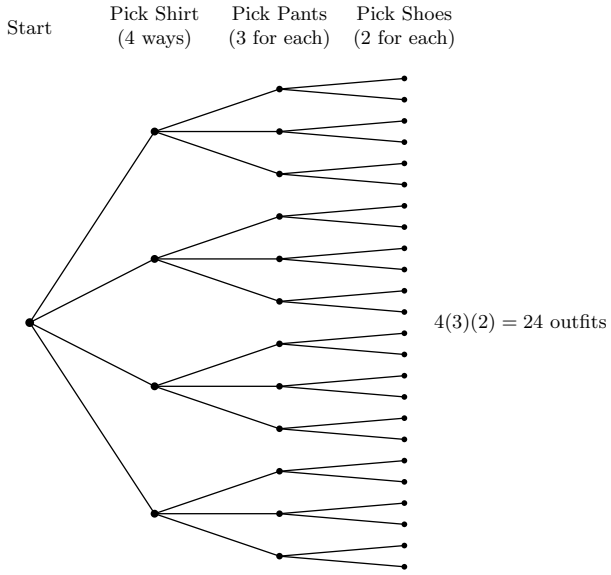
We could of course list out all the possible options, but this takes a lot of time and is often unnecessary.

Instead, we have theorems/results we may use to solve these problems.

Theorem 1.1 (Fundamental Rule of Counting): Suppose we have k tasks to carry out, and each task has $n_1, n_2, \ldots, n_k$ corresponding ways to carry them out.
The # of different ways to complete all k tasks is:

$$(n_1)(n_2)(n_3)\cdots(n_k)$$

Consider the outfit problem. We have



Now, sometimes the order of events matters, and sometimes it does not. This leads to:

Definition 1.1: A permutation of k objects from n assumes the order of the k objects matters.

Definition 1.2: A combination of k objects from n assumes the order of the k objects does not matter.

Example 1.10: Picking 3 people (out of 10) for 1st, 2nd, 3rd prize implies order matters.
Picking 3 people out of 10 to form a committee implies order does not matter.

Corollary 1.1:

i) The number of ordered arrangements of k objects from n is: (Permutations)

a) If repetitions are allowed, $n^k = n \cdot n \cdot n \cdots$

b) If no repetitions are allowed:

$$P_{n,k} = \#\text{ of ways to order } k \text{ objects from } n = n(n-1)(n-2)\cdots(n-k+1)$$

If $k = n$, $\quad P_{n,n} = n(n-1)(n-2)\cdots(3)(2)(1) \equiv n!$

ii) The number of unordered arrangements of k objects from n is: (Combinations)

$$\binom{n}{k} = \frac{P_{n,k}}{k!} = \frac{n!}{k!\,(n-k)!}$$

Note: $0! \equiv 1$

Example 1.11: A sushi chef has 20 kinds of fish on ice and builds an 8-piece omakase (a chef's-choice sushi course), served in a deliberate order with no kind repeated.

a) The order is part of the meal, so a sequence is a permutation: $P_{20,8} = 20 \cdot 19 \cdots 13 = 5{,}079{,}110{,}400$.

b) Ignoring order, each set of 8 kinds accounts for 8! of those sequences, so $\binom{20}{8} = P_{20,8}/8! = 125{,}970$.

c) If the chef must open with tuna and close with tamago (sweet egg omelet), the two ends are fixed and the six middle pieces are an ordered choice from the 18 kinds left: $P_{18,6} = 18 \cdot 17 \cdots 13 = 13{,}366{,}080$.

Example 1.12: Flip five coins and record the sequence, so by the fundamental rule of counting there are $2^5 = 32$ equally likely outcomes. How many have exactly k heads?
Choosing *which* of the five flips come up heads is an unordered choice, so the count is $\binom{5}{k}$:

k	0	1	2	3	4	5	total
$\binom{5}{k}$	1	5	10	10	5	1	32

Note: The row is symmetric because $\binom{5}{k} = \binom{5}{5-k}$: choosing which flips are heads is the same as choosing which are tails. The entries sum to 2^5, since every sequence has *some* number of heads.

Example 1.13: Suppose your Pokémon storage box holds 8 Grass types, 7 Fire types, and 10 Water types.
A team of 4 is to be entered in a tournament. How many different teams are possible if:

i) The team is all Fire types:

$$\binom{7}{4} = \frac{7!}{3!\,4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 7 \cdot 5 = 35$$

ii) The team has two Fire, one Grass, and one Water:

$$\binom{7}{2}\binom{8}{1}\binom{10}{1} = \frac{7!}{5!\,2!} \cdot \frac{8!}{7!\,1!} \cdot \frac{10!}{9!\,1!} = 21(8)(10) = 1680$$

Proposition 1.7: Consider n objects, where there are k groups, where each of the k groups has m_i objects,
s.t. $m_1 + m_2 + \cdots + m_k = n$.
The # of ways to order by group (not within group) is:

$$\frac{n!}{m_1!\, m_2! \cdots m_k!}$$

Example 1.14: A band has recorded 10 distinct songs and wants to split them across three releases: two EPs of 3 songs each, and one of 4. How many ways can this be done?

$$\frac{10!}{3!\,3!\,4!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{(3 \cdot 2 \cdot 1)(3 \cdot 2 \cdot 1)} = \frac{151200}{36} = 4200$$

Note: If the running order *within* each EP mattered, it would be 10!. Note also that this count treats the releases as *labeled* (a first EP of 3, a second EP of 3, and one of 4). If the two 3-song EPs are interchangeable, divide by 2! to get 2100.

Example 1.15: A music festival has 20 acts on its shortlist, where 13 are bands and 7 are solo DJs. Suppose all acts are equally likely to be booked, and 5 acts will fill the main stage.

a) How many ways are there to book 5 acts?

$$\binom{20}{5} = 15504$$

b) What is the probability the stage is all DJs?

$$\frac{\binom{7}{5}\binom{13}{0}}{\binom{20}{5}} = \frac{21}{15504} = 0.0013544$$

c) What is the probability of booking at least one band?
Let A_i = event where i bands are booked, $i = 0, \ldots, 5$.

$$\begin{aligned} P(A_1 \text{ or } A_2 \text{ or } A_3 \text{ or } A_4 \text{ or } A_5) &= 1 - P(A_0) = 1 - \binom{7}{5} \Big/ \binom{20}{5} \\ &= 1 - 0.0013544 \end{aligned}$$

Exercises

Worked solutions follow on the next page; try each exercise first. The course problem bank has many more problems on every topic.



Figure 1.5: *Sandhill cranes over the Sangre de Cristo Mountains.* [2]

Exercise 1.1. Three migrating cranes (Fig. 1.5) arrive over a valley with 10 fields in a row, numbered 1 through 10, and each lands in one of fields 2 through 10, equally likely and independently of the others (two cranes may share a field). What is the probability that the set of fields they land in is 3 consecutive fields?

Exercise 1.2. A *straight* in poker is a 5-card hand where the 5 cards have consecutive ranks. The possibilities are A2345, 23456, …, 9TJQK, TJQKA. ("T" represents a 10, and the Ace can be either low or high but not both.)

(a) What is the probability that a randomly selected 5-card poker hand from a well-shuffled standard deck is a straight?

(b) What is the probability that it is a straight but *not* a straight flush? (A straight flush is a straight where all the cards have the same suit.)



Figure 1.6: *The Shibuya scramble crossing.* [3]

Exercise 1.3 (A Tokyo itinerary)**.** A traveler has four days in Tokyo and a list of seven neighborhoods (Shibuya, Shinjuku, Asakusa, Akihabara, Harajuku, Ginza, Ueno; Fig. 1.6), one neighborhood per day with no repeats.

(a) How many itineraries are possible?

(b) How many if Akihabara must be day 1 and Asakusa must appear on one of the later days?

(c) A ramen stamp card lists 10 shops and the traveler will try 3 of them. How many different sets of shops are possible?

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 1.1. Each crane picks one of 9 fields, so there are $9^3 = 729$ equally likely outcomes. The set of fields landed in is 3 consecutive fields only if all three cranes land in different fields forming a run: the runs are $\{2,3,4\}, \{3,4,5\}, \ldots, \{8,9,10\}$, seven of them, and each can be assigned to the three cranes in $3! = 6$ orders. So $P = \dfrac{7 \cdot 6}{729} = \dfrac{42}{729} = 0.0576$.

Exercise 1.2. (a) There are 10 possible rank runs (A2345 through TJQKA), and each of the 5 cards can be any of 4 suits: $10 \cdot 4^5 = 10{,}240$ straights out of $\binom{52}{5} = 2{,}598{,}960$ hands, so $P = 0.00394$. **(b)** Straight flushes: 10 runs $\times$ 4 suits $= 40$. So $P(\text{straight, not a straight flush}) = \dfrac{10{,}240 - 40}{2{,}598{,}960} = 0.00392$.

Exercise 1.3. (a) Ordered choice of 4 of 7: $P_{7,4} = 7 \cdot 6 \cdot 5 \cdot 4 = 840$. **(b)** Day 1 is fixed as Akihabara, Asakusa takes one of the 3 remaining days, and the other two days are filled in order from the 5 neighborhoods left: $1 \cdot 3 \cdot (5 \cdot 4) = 60$. **(c)** Which shops, not in what order: $\binom{10}{3} = 120$.

2 Conditional Probability and Independence

2.1 Conditional Probability and Bayes' Rule

Conditional Probability

One extremely common type of probability to calculate is a conditional probability, which calculates the probability of event A under the assumption event B has occurred.

Example 2.1: The probability that a team wins, given that they were behind at halftime.

Example 2.2: The probability that an incoming email is spam, given that its subject line contains the word "free."

Definition 2.1: The conditional probability of event A, given event B ($P(B) > 0$) is denoted $P(A \mid B)$, and

$$P(A \mid B) = P(A \cap B) \,/\, P(B)$$

Conditional probabilities are essentially subsetting S, and calculating probabilities based on that subset. They can be used to find:

Theorem 2.1 (Multiplicative Theorem): For any n events $A_1, \ldots, A_n$

$$P\left(\bigcap_{j=1}^{n} A_j\right) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$$

Ex: $n = 3$, $\quad P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 \cap A_2)$

Example 2.3: Cards are drawn from a shuffled 52-card deck.

a) What is the probability the first two cards drawn are both aces?
Let A_j be the event that the jth card is an ace. The second factor is conditional: once an ace is gone, 3 aces remain among 51 cards.

$$P(A_1 \cap A_2) = P(A_1)\, P(A_2 \mid A_1) = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{52 \cdot 51} = \frac{1}{221}$$

b) A blackjack is a two-card hand of one ace and one ten-valued card, $T = \{K, Q, J, 10\}$, so there are 4 aces and 16 tens. What is the probability of being dealt one?
The ace can come first or second, and those two orders are disjoint, so the $-P(X \cap Y)$ term of the addition rule is 0 and we simply add:

$$\begin{aligned} P(\text{blackjack}) &= P(A_1 \cap T_2) + P(T_1 \cap A_2) = P(A_1)\, P(T_2 \mid A_1) + P(T_1)\, P(A_2 \mid T_1) \\ &= \frac{4}{52} \cdot \frac{16}{51} + \frac{16}{52} \cdot \frac{4}{51} = \frac{32}{663} \approx 0.0483 \end{aligned}$$

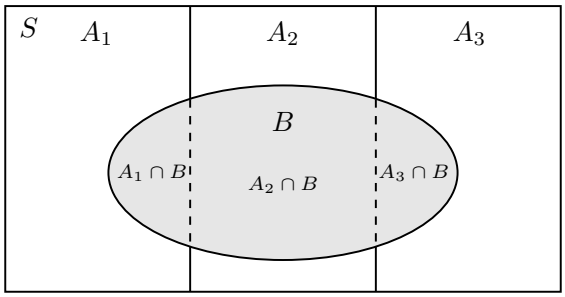
about one hand in 21.

Definition 2.2: $\{A_1, \ldots, A_n\}$ partition S if $A_i \cap A_j = \varnothing$, $i \neq j$ (pairwise disjoint), and $\bigcup_{j=1}^{n} A_j = S$.

Theorem 2.2 (Total Probability Theorem): Let $A_1, \ldots, A_n$ partition S. For any event B (assuming $P(A_j) > 0 \; \forall \, j$):

$$P(B) = \sum_j P(A_j)\, P(B \mid A_j)$$

Example 2.4: Say $n = 3$.



Notice if $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} \;\Rightarrow\; P(A \cap B) = P(B)\, P(A \mid B)$

$$\begin{aligned} P(B) &= P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) \\ &= P(A_1)\, P(B \mid A_1) + P(A_2)\, P(B \mid A_2) + P(A_3)\, P(B \mid A_3) = \sum_i P(A_i)\, P(B \mid A_i) \end{aligned}$$

Theorem 2.3 (Bayes' Theorem): This essentially combines conditional probability with the Total Probability Theorem.
Let $A_1, \ldots, A_n$ partition S, then, for any $j = 1, 2, \ldots$

$$P(A_j \mid B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(A_j)\, P(B \mid A_j)}{\sum_i P(A_i)\, P(B \mid A_i)}$$

Example 2.5: A disco label presses its records at three plants. Plant 1 presses 50% of the records, Plant 2 presses 30%, and Plant 3 presses 20%. Of the records pressed at Plant 1, 4% are warped; at Plant 2, 2% are warped; at Plant 3, 5% are warped.

Let A_j = event a record comes from Plant j (note A_1, A_2, A_3 partition S), and B = event a record is warped.

a) Find the probability that a randomly chosen record is warped.

By the Total Probability Theorem:

$$\begin{aligned} P(B) &= P(A_1)\, P(B \mid A_1) + P(A_2)\, P(B \mid A_2) + P(A_3)\, P(B \mid A_3) \\ &= (0.50)(0.04) + (0.30)(0.02) + (0.20)(0.05) \\ &= 0.020 + 0.006 + 0.010 = 0.036 \end{aligned}$$

b) Given that a record is warped, find the probability it came from Plant 3.

By Bayes' Theorem:

$$P(A_3 \mid B) = \frac{P(A_3)\,P(B \mid A_3)}{P(B)} = \frac{(0.20)(0.05)}{0.036} = \frac{0.010}{0.036} = 0.278$$

c) Which plant is the most likely source of a warped record?

$$P(A_1 \mid B) = \frac{0.020}{0.036} = 0.556, \qquad P(A_2 \mid B) = \frac{0.006}{0.036} = 0.167, \qquad P(A_3 \mid B) = 0.278$$

Plant 1 is the most likely source. Even though Plant 3 has the highest warp rate, Plant 1 presses so many more records that it accounts for over half of all the warped ones. The base rates $P(A_j)$ matter.

Note: The three posterior probabilities sum to 1, as they must, since A_1, A_2, A_3 still partition S after conditioning on B.

Example 2.6: In casino craps, the shooter rolls two dice. The pass-line bet wins immediately on a 7 or 11, loses immediately on 2, 3, or 12, and otherwise the number rolled becomes the *point*: the shooter then rolls repeatedly, winning if the point comes up again before a 7.

Partition on the first roll. Immediate outcomes: $P(7 \text{ or } 11) = \frac{8}{36}$ and $P(2, 3, \text{ or } 12) = \frac{4}{36}$. For a point p, only rolls of p or 7 matter, so $P(\text{make the point} \mid \text{point } p)$ is the chance p appears before 7:

$$P(\text{win} \mid 4) = \tfrac{3}{9}, \quad P(\text{win} \mid 5) = \tfrac{4}{10}, \quad P(\text{win} \mid 6) = \tfrac{5}{11},$$

with 10, 9, 8 matching 4, 5, 6 by symmetry. By the total probability theorem,

$$P(\text{win}) = \frac{8}{36} + 2\left(\frac{3}{36}\cdot\frac{3}{9} + \frac{4}{36}\cdot\frac{4}{10} + \frac{5}{36}\cdot\frac{5}{11}\right) = \frac{244}{495} = 0.4929.$$

Just under half. The casino pays even money, so its edge is $1 - 2(0.4929) = 1.41\%$ per dollar bet: among the best odds on the casino floor, and still a loss. Later, with the central limit theorem, we will see what that 1.41% does to a long night of play.

Exercises



Figure 2.1: *Egyptian scarab amulets.* [4]

Exercise 2.1. The scarab amulets (Fig. 2.1) in a museum's Egyptian collection came from three excavation sites: Saqqara, Abydos, and Amarna. The proportions of amulets from Saqqara, Abydos, and Amarna are 0.5, 0.3, and 0.2, respectively, and amulets from each site turn out to be modern forgeries with probabilities 0.01, 0.02, and 0.03, respectively. Given that an amulet is a forgery, what is the probability that it came from Saqqara?

Exercise 2.2. Fred is answering a multiple-choice problem on an exam, and has to choose one of n options (exactly one of which is correct). Let K be the event that he knows the answer, and R be the event that he gets the problem right (either through knowledge or through luck). If he knows the right answer he will definitely get the problem right, but if he does not know then he will guess completely randomly. Let $P(K) = p$.

(a) If Fred gets the problem right, what is the chance he actually knew the answer? That is, find $P(K \mid R)$ in terms of p and n.

(b) Show that $P(K \mid R) \geq p$, and explain why this makes sense intuitively. Are there particular values of n, p that would make $P(K \mid R) = p$?

Exercise 2.3 (Cyclades ferries)**.** The Cyclades are the Greek islands scattered across the Aegean southeast of Athens, and the only way between most of them is by boat. Of the departures between two of the islands, 70% are on the fast catamaran and 30% on the slow ferry. High wind cancels the catamaran 10% of the time and the slow ferry 2% of the time.

(a) What fraction of all departures are cancelled?

(b) Your departure was cancelled. What is the probability it was the catamaran?

(c) Your departure ran. What is the probability it was the slow ferry?

Exercise 2.4 (Karl the Fog)**.** On a summer morning in San Francisco, Ocean Beach is fogged in 60% of the time. When the morning is foggy, the sun breaks through by noon 30% of the time; when the morning is clear, it stays clear through noon 90% of the time.

(a) Find the probability that noon is clear.

(b) Noon turned out clear. Find the probability the morning was foggy.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 2.1. Let F = forgery. By total probability, $P(F) = 0.5(0.01) + 0.3(0.02) + 0.2(0.03) = 0.005 + 0.006 + 0.006 = 0.017$. Bayes gives $P(\text{Saqqara} \mid F) = \dfrac{0.5(0.01)}{0.017} = \dfrac{0.005}{0.017} = 0.294$: Saqqara supplies half the amulets but under a third of the forgeries.

Exercise 2.2. **(a)** $P(R) = P(K) + P(K^c) \cdot \frac{1}{n} = p + \frac{1-p}{n}$, so $P(K \mid R) = \dfrac{p}{p + (1-p)/n} = \dfrac{np}{np + 1 - p}$. **(b)** $P(K \mid R) \geq p$ because $np + 1 - p \leq np + 1 - p + \ldots$; concretely $\dfrac{np}{np + 1 - p} \geq p \iff np \geq np^2 + p - p^2 \iff p(1-p)(n-1) \geq 0$, which holds. A correct answer is evidence of knowledge, so the posterior cannot fall below the prior. Equality holds when $p = 0$, $p = 1$, or $n = 1$ (one option, so guessing always works and a right answer carries no information).

Exercise 2.3. Let C = catamaran, F = slow ferry, X = cancelled. **(a)** $P(X) = 0.7(0.10) + 0.3(0.02) = 0.076$. **(b)** $P(C \mid X) = 0.070/0.076 = 0.921$. **(c)** $P(F \mid X^c) = \dfrac{0.3(0.98)}{0.924} = 0.318$.

Exercise 2.4. Let F = foggy morning, C = clear at noon. **(a)** $P(C) = 0.6(0.3) + 0.4(0.9) = 0.54$. **(b)** $P(F \mid C) = 0.18/0.54 = \frac{1}{3}$.

2.2 Independent Events

Example 2.7: Suppose that 35% of families own a dog, 30% of families own a cat, and 22% of families who own a dog also own a cat.

a) Find the probability a family owns both a dog and a cat.

$$P(C \cap D) = P(D)\,P(C \mid D) = 0.35(0.22) = 0.077$$

b) If a family owns a cat, find the probability they own a dog.

$$P(D \mid C) = P(D \cap C) \,/\, P(C) = 0.077 \,/\, (0.30) = 0.2567$$

c) Find the probability someone owns a dog.
Since $P(C \mid D) = P(C \cap D)/P(D)$, we have

$$P(D) = P(C \cap D)/P(C \mid D) = 0.077/0.22 = 0.35$$

d) If a family does not own a dog, find the probability they do not own a cat.

$$\begin{aligned} P(C^c \mid D^c) &= P(C^c \cap D^c) \,/\, P(D^c) = P((C \cup D)^c) \,/\, (1 - P(D)) \\ &= \frac{1 - P(C \cup D)}{1 - P(D)} = \frac{1 - [P(D) + P(C) - P(D \cap C)]}{1 - P(D)} \\ &= \frac{1 - 0.35 - 0.30 + 0.077}{1 - 0.35} = 0.6569 \end{aligned}$$

Independent Events

When the presence (or absence) of one event does not affect the probability of another event, the events are said to be independent of one another. More specifically,

Definition 2.3: Events A_1 and A_2 are <u>independent</u> iff (assume $P(A_1) > 0$, $P(A_2) > 0$)

i) $P(A_1 \mid A_2) = P(A_1)$

ii) $P(A_1 \cap A_2) = P(A_1)\,P(A_2)$

Notice that i) $\Rightarrow$ ii), since $P(A_1 \cap A_2) = P(A_2)\,P(A_1 \mid A_2)$.
If either i) or ii) does not hold, events are said to be <u>dependent</u>.

Warning: Do not confuse <u>independent</u> with <u>disjoint</u>; the two are nearly opposites. If A_1, A_2 are disjoint (and both have positive probability), then knowing A_1 occurred tells you A_2 *cannot* have occurred, i.e. $P(A_2 \mid A_1) = 0 \neq P(A_2)$. So disjoint events with positive probability are always *dependent*.
Example 2.8: Roll a die. Let $A_1 = \{1, 2\}$ and $A_2 = \{3, 4\}$. These are disjoint, but $P(A_1 \cap A_2) = 0 \neq P(A_1)P(A_2) = \frac{1}{9}$, so they are dependent. Independence is about *information*, not about non-overlap.

This can be extended in the following manner:

Events $A_1, A_2, \ldots, A_n$ are mutually independent if for all k out of n events $(2 \le k \le n)$, and integers $j_1, \ldots, j_k$ $(1 \le j_1 < \cdots < j_k \le n)$

$$P\left(\bigcap_{i=1}^{k} A_{j_i}\right) = \prod_{i=1}^{k} P(A_{j_i})$$

A very common instance of independence we assume holds true in statistics is: take a random sample of size N, and assume subjects are independent.

Example 2.9: Pairwise independence does not imply mutual independence. Flip two fair coins. Let A_1 = first flip is H, A_2 = second flip is H, and A_3 = the two flips match. Each pair is independent, since $P(A_i \cap A_j) = \frac{1}{4} = P(A_i)\,P(A_j)$ for every pair $i \neq j$. But

$$P(A_1 \cap A_2 \cap A_3) = P(\text{HH}) = \tfrac{1}{4} \neq \tfrac{1}{8} = P(A_1)\,P(A_2)\,P(A_3)$$

so the three events are not mutually independent: any two of them together determine the third.

Theorem 2.4: Results of independent $A_1, \ldots, A_n$

i) If A_1, A_2 are independent, so are A_1, A_2^c; A_1^c, A_2; A_1^c, A_2^c

ii) More generally, if $A_1, \ldots, A_n$ are independent, then so are $A_1', \ldots, A_n'$, where A_i' can denote either A_i or A_i^c.

Proof of i): $P(A_1 \cap A_2^c) = P(A_1) - P(A_1 \cap A_2) = P(A_1) - P(A_1)\,P(A_2)$

$$= P(A_1)(1 - P(A_2)) = P(A_1)\,P(A_2^c) \quad \blacksquare$$

Similarly for $P(A_1^c \cap A_2)$.

$$\begin{aligned} P(A_1^c \cap A_2^c) &= P((A_1 \cup A_2)^c) = 1 - P(A_1 \cup A_2) \\ &= 1 - [P(A_1) + P(A_2) - P(A_1 \cap A_2)] \\ &= 1 - P(A_1) - P(A_2) + P(A_1)\,P(A_2) = (1 - P(A_1))(1 - P(A_2)) \\ &= P(A_1^c)\,P(A_2^c) \quad \blacksquare \end{aligned}$$

Example 2.10: A conservation group runs two reintroduction projects: a dragonfly release at a restored marsh has an 8% chance of establishing a breeding population, and a sea otter translocation has a 10% chance. Further, assume that the two projects succeed independently.

a) Find the probability that both projects succeed.
Let D = the dragonfly project succeeds, O = the otter project succeeds.

$$P(D \cap I) = P(D)\,P(O) = 0.08(0.10) = 0.008$$

b) Find the probability that neither project succeeds.

$$P(D^c \cap O^c) = P(D^c)\,P(O^c) = (1 - 0.08)(1 - 0.10) = 0.828$$

c) Find the probability exactly one project succeeds.

$$\begin{aligned} P((D \cap O^c) \cup (D^c \cap I)) &= P(D \cap O^c) + P(D^c \cap I) = P(D)\,P(O^c) + P(D^c)\,P(O) \\ &= 0.08(0.90) + 0.92(0.10) = 0.164 \end{aligned}$$

Exercises

Exercise 2.5. Alice and Bob both live in San Francisco. They are complete strangers. Every Saturday, they each decide separately what to do. If the weather is sunny, Alice goes to Dolores Park with 40% probability and Bob goes with 30% probability. If the weather is not sunny, Alice goes with 5% probability and Bob goes with 2% probability. On a particular Saturday, let A be the event that Alice goes to Dolores Park, B the event that Bob goes, and S the event that the weather is sunny.

(a) Are A and B conditionally independent given S? Given S^c? Write down the relevant equations and check numerically.

(b) Give an intuitive explanation for why A and B are NOT independent, even though Alice and Bob are strangers who decide separately. Do you expect $P(A \cap B)$ to be greater than or less than $P(A)P(B)$? Why?

(c) If $P(S) = 0.5$, what are the chances that Alice goes to Dolores Park, that Bob goes, and that both go? That is, compute $P(A)$, $P(B)$, and $P(A \cap B)$. Check that these are consistent with your answer to part (b).

Exercise 2.6 (A day in Rio)**.** A visitor to Rio de Janeiro plans to see Christ the Redeemer, the hilltop statue above the city, where fog hides the view on 25% of days, and to ride the cable car up Sugarloaf, the granite peak on the bay, which is closed for maintenance on 10% of days. Treat the two as independent.

(a) Probability both go to plan.

(b) Probability at least one is spoiled.

(c) Probability exactly one is spoiled.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 2.5. (a) Given S: $P(A \cap B \mid S) = 0.4(0.3) = 0.12 = P(A \mid S)P(B \mid S)$, so yes; given S^c: $0.05(0.02) = 0.001$, so yes. They decide separately once the weather is known. **(b)** Not independent unconditionally, because both react to the same weather: if Alice went, it was probably sunny, which makes Bob more likely to have gone too. So $P(A \cap B) > P(A)P(B)$. **(c)** $P(A) = 0.5(0.4) + 0.5(0.05) = 0.225$, $P(B) = 0.5(0.3) + 0.5(0.02) = 0.16$, and $P(A \cap B) = 0.5(0.12) + 0.5(0.001) = 0.0605$, which exceeds $0.225(0.16) = 0.036$, as predicted.

Exercise 2.6. Let A = fogged view (0.25), B = cable car closed (0.10), independent. **(a)** $0.75(0.90) = 0.675$. **(b)** $1 - 0.675 = 0.325$. **(c)** $0.25(0.90) + 0.75(0.10) = 0.30$.

3 Random Variables and Their Distributions

3.1 Random Variables and Distribution Functions

Random Variables

Instead of listing all possible values in a sample space, we often use random variables and their distributions to describe the sample space.

Definition 3.1: A random variable (RV) is a function defined on a sample space, which takes on values in the real line $\mathbb{R} = (-\infty, \infty)$. Denoted by capital letters X, Y, Z, etc.

$X(s)$ = RV at sample point s.
$X(S)$ = range of all possible values of X.
$X(s)$ is considered an unknown, random value, while a lowercase x ($x \in \mathbb{R}$) denotes a realized value of $X(s)$.

Example 3.1: If $X(s)$ = # of Pikmin (the tiny plant-like helpers you lead around in Nintendo's Pikmin games), out of a squad of 10, that make it back to the Onion, their ship-like home base, by sundown, $X(S) = \{0, 1, 2, \ldots, 10\}$ (all possible values)

If $X(s)$ = the price of a vintage disco 12-inch (a single pressed on a full-size record) at a record fair, $X(S) \in [0,$ (the most anyone has ever paid)$]$

There are two main types of RV's:

1. **Discrete RV's:** Where $X(S)$ takes on countably many values (typically have gaps).
2. **Continuous RV's:** Where $X(S)$ takes on all values in a non-degenerate interval in $\mathbb{R}$ (could hypothetically take on any value within an interval).

Distribution of RV

For both discrete and continuous RVs, recall $X \in B$ stands for $\{s \in S : X(s) \in B\}$. Then,

$$P(X \in B) = P(\{s \in S : X(s) \in B\}) \qquad B \subseteq \mathbb{R}$$

Definition 3.2: A distribution function of X is a set function P_X, s.t. $P(X \in B) = P_X(B)$.

Definition 3.3: The probability distribution function, or cumulative distribution function (c.d.f.) is:

$$F_X(x) = P(X \leq x) = P(\{s \in S : X(s) \leq x\})$$

Proposition 3.1: Properties of the c.d.f. include:

i) $0 \le F_X(x) \le 1 \quad x \in \mathbb{R}$

ii) F_X is non-decreasing: if $x_1 < x_2$, $F_X(x_1) \le F_X(x_2)$

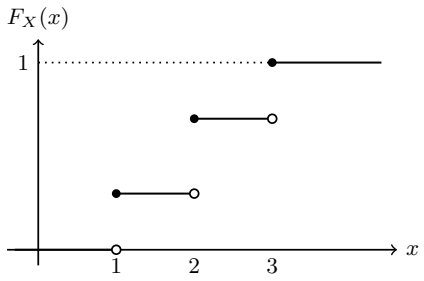
iii) F_X is right-continuous; as $n \to \infty$, $x_n \downarrow x$

$$\Rightarrow \; F_X(x_n) \to F_X(x)$$

iv) $F_X(\infty) = 1$, $F_X(-\infty) = 0$ where

$$F_X(\infty) = \lim_{n\to\infty} F_X(x_n) \; (x_n \uparrow \infty), \qquad F_X(-\infty) = \lim_{n\to\infty} F_X(x_n) \; (x_n \downarrow -\infty)$$

For a discrete RV, the c.d.f. is a step function:



The filled dot at the top of each jump is right-continuity, property iii): the jump at x belongs to $F_X(x)$, since "$\le$" includes x itself.

But, discrete RVs and continuous RVs have different forms of the probability density function (p.d.f.)

Definition 3.4: p.d.f. for discrete RVs:
Let X be a RV taking on values x_i, $i \ge 1$.

with corresponding probabilities $P(X = x_i)$. The p.d.f. of X is

$$f_X(x) = \begin{cases} P(X = x_i) & \text{if } x = x_i, \; i \ge 1 \\ 0 & \text{o.w.} \end{cases}$$

Note: Most other books call this discrete version the *probability mass function* (p.m.f.), reserving "density" for continuous RVs. As stated in the glossary, these notes use "p.d.f." for both; expect to see "p.m.f." in your later courses.

With properties:

i) $f_X(x) \ge 0 \quad \forall\, x \in \mathbb{R}$

ii) For any event $B \subseteq \mathbb{R}$, $P(X \in B) = \sum_{x_i \in B} f_X(x_i)$

iii) $F_X(x) = \sum_{x_i \le x} f_X(x_i)$, $x \in \mathbb{R}$, and $\sum_{x_i \in \mathbb{R}} f_X(x_i) = 1$

iv) If $x_i < x_{i+1}$, $i \geq 1$,

$$f_X(x_{i+1}) = F_X(x_{i+1}) - F_X(x_i), \qquad f_X(x_1) = F_X(x_1)$$

Example 3.2: Flip a fair coin twice and let X be the number of heads, so $f_X(0) = \frac{1}{4}$, $f_X(1) = \frac{1}{2}$, $f_X(2) = \frac{1}{4}$.

a) Write down the c.d.f.

Accumulate the p.d.f. from the left, using $F_X(x) = \sum_{x_i \leq x} f_X(x_i)$:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

b) Check the four properties.

i) Every value listed is between 0 and 1. ✓

ii) The values $0, \frac{1}{4}, \frac{3}{4}, 1$ only increase as x increases. ✓

iii) At $x = 1$ the function jumps from $\frac{1}{4}$ to $\frac{3}{4}$, and $F_X(1) = \frac{3}{4}$, the value *after* the jump. Approaching 1 from the right gives $\frac{3}{4}$, so F_X is right-continuous. ✓

iv) $F_X(x) = 0$ for all $x < 0$ and $F_X(x) = 1$ for all $x \geq 2$. ✓

c) Recover the p.d.f. from the c.d.f., using property iv) above:

$$f_X(1) = F_X(1) - F_X(0) = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}\checkmark$$

The p.d.f. is the size of the jump; the c.d.f. is the running total. Either one determines the other.

d) Compare $P(X \leq 1)$ with $P(X < 1)$.

$$P(X \leq 1) = F_X(1) = \tfrac{3}{4}, \qquad P(X < 1) = F_X(0) = \tfrac{1}{4}$$

Note: These differ by exactly $f_X(1) = \frac{1}{2}$, the size of the jump at 1. For a *discrete* RV the distinction between $<$ and $\leq$ matters; for a continuous RV it never will, since there are no jumps. Right-continuity, property iii), is precisely the convention that assigns each jump to the point at its right.

3.2 Discrete Random Variables: Worked Examples

Example 3.3: In *Settlers of Catan* every turn starts with a roll of two dice, and the sum decides which hexes produce resources. Let X be the sum of two fair dice, so $X(S) = \{2, 3, \ldots, 12\}$.

a) Of the 36 equally likely ordered pairs, $6 - |x - 7|$ have sum x, so

$$f_X(x) = P(X = x) = \frac{6 - |x - 7|}{36}, \qquad x = 2, \ldots, 12.$$

x	2	3	4	5	6	7	8	9	10	11	12
$f_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The number tokens on the board print the numerators as little pips: the 6 and 8 tokens carry five pips and the 2 and 12 carry one, so a settlement on a five-pip hex produces five times as often as one on a one-pip hex.

b) A 7 moves the robber instead of producing anything: $P(X = 7) = \frac{6}{36} = \frac{1}{6}$, so about one turn in six is a robber turn, the most likely single outcome on the board.

c) A settlement touching a 6 and an 8 produces on $P(X = 6) + P(X = 8) = \frac{10}{36}$ of rolls, better than one in four.

d) From the c.d.f., $F_X(6) = P(X \le 6) = \frac{1+2+3+4+5}{36} = \frac{15}{36}$, so $P(X \ge 7) = 1 - \frac{15}{36} = \frac{21}{36}$: the high half of the board produces slightly more often than the low half, because 7 sits on the high side of 6.



Figure 3.2: *Ozzy and Olive in boxes.* [5]

Example 3.4: Let X be the number of minutes, to the nearest 5, that a cat stays in a freshly delivered cardboard box (Fig. 3.2), and suppose X has c.d.f.:

x	5	10	15	20	25	30
$F_X(x)$	$\frac{1}{20}c$	$\frac{5}{20}c$	$\frac{8}{20}c$	$\frac{12}{20}c$	$\frac{14}{20}c$	$\frac{15}{20}c$

where $c > 0$, $c \in \mathbb{R}$.

a) What is the chance the cat stays in the box for each possible time, 5, 10, …, 30 minutes? That is, find the p.d.f. of X.
Using $f_X(x_{i+1}) = F_X(x_{i+1}) - F_X(x_i)$:

x	5	10	15	20	25	30
$f_X(x)$	$\frac{1}{20}c$	$\frac{5}{20}c - \frac{1}{20}c = \frac{4}{20}c$	$\frac{8}{20}c - \frac{5}{20}c = \frac{3}{20}c$	$\frac{12}{20}c - \frac{8}{20}c = \frac{4}{20}c$	$\frac{14}{20}c - \frac{12}{20}c = \frac{2}{20}c$	$\frac{15}{20}c - \frac{14}{20}c = \frac{1}{20}c$

b) Find the value of c s.t. f_X is a valid p.d.f.
$\sum_i f_X(x_i) = 1$. Thus

$$\left(\tfrac{1}{20}c + \tfrac{4}{20}c + \tfrac{3}{20}c + \tfrac{4}{20}c + \tfrac{2}{20}c + \tfrac{1}{20}c\right) = \tfrac{15}{20}c$$

$$c = \frac{20}{15} = \frac{4}{3}.$$

Note: Could also use $F_X(30) = 1 \;\Rightarrow\; \frac{15}{20}c = 1 \;\Rightarrow\; c = \frac{20}{15}$.

c) What is the chance the cat stays in the box longer than 6 minutes? That is, find the probability that $X > 6$

$$P(X > 6) = P(X \geq 10) = 1 - P(X \leq 6) = 1 - P(X = 5)$$
$$= 1 - \tfrac{1}{20}c = 1 - \tfrac{1}{20}\left(\tfrac{4}{3}\right) = 1 - \tfrac{4}{60} = 1 - \tfrac{1}{15} = \tfrac{14}{15}$$

Example 3.5: A peregrine falcon hunts by stooping: folding its wings and diving at a bird from above at over 300 km/h, and most stoops miss. Suppose its stoops are independent, and X is the stoop on which it finally catches prey. Suppose X has p.d.f.

$$f_X(x) = c\left(\frac{9}{10}\right)^{x-1}, \quad x = 1, 2, \ldots \quad \text{(and 0 o.w.)}$$

This is called the *geometric distribution*; we study it formally in Section 5.2.

i) Determine c.
Again, $\sum_{x_i} f_X(x_i) = 1$, so, recalling $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$:

$$\sum_{x=1}^{\infty} c\left(\frac{9}{10}\right)^{x-1} = c\sum_{x=1}^{\infty}\left(\frac{9}{10}\right)^{x-1} = c\left[1 + \left(\frac{9}{10}\right) + \left(\frac{9}{10}\right)^2 + \cdots\right] = c\left(\frac{(9/10)^0}{1 - 9/10}\right)$$
$$= c \cdot \frac{1}{1/10} = 10c \;\Rightarrow\; c = \tfrac{1}{10}$$

ii) What is the chance the falcon needs 20 or more stoops to catch prey? That is, calculate $P(X \geq 20)$.

$$P(X \geq 20) = c \sum_{x=20}^{\infty} \left(\frac{9}{10}\right)^{x-1} = c\left[\left(\frac{9}{10}\right)^{19} + \left(\frac{9}{10}\right)^{20} + \cdots\right] = c \sum_{n=19}^{\infty} \left(\frac{9}{10}\right)^{n}$$
$$= c\left[\frac{(9/10)^{19}}{1 - 9/10}\right] = \frac{1}{10}\left[\frac{(9/10)^{19}}{1/10}\right] = \left(\frac{9}{10}\right)^{19} \approx 0.135$$

iii) What is the chance the falcon has caught prey within its first x stoops? That is, find $F_X(x)$, the c.d.f.

$$F_X(x) = P(X \leq x) = \sum_{t=1}^{x} c\left(\frac{9}{10}\right)^{t-1} = 1 - P(X > x) = 1 - \sum_{t=x+1}^{\infty} c\left(\frac{9}{10}\right)^{t-1}$$
$$= 1 - c \sum_{n=x}^{\infty} \left(\frac{9}{10}\right)^{n} = 1 - \frac{1}{10}\left(\frac{(9/10)^{x}}{1 - 9/10}\right) = 1 - \left(\frac{9}{10}\right)^{x}$$

Exercises

Exercise 3.1. Dancers are arriving at a salsa social one at a time. While waiting for enough partners to start, they entertain themselves by comparing their birthdays. Let X be the number of people needed to obtain a birthday match: that is, before person X arrives no two people have the same birthday, but when person X arrives there is a match. Assume there are 365 equally likely birthdays and that different guests' birthdays are independent.

(a) What is the support of X? (Think about the smallest and largest possible values.)

(b) What is the chance the first birthday match comes when the 2nd, 3rd, or 4th dancer arrives? That is, compute $P(X = 2)$, $P(X = 3)$, and $P(X = 4)$. Then find a general formula for $P(X = k)$.



Figure 3.3: *Torii gates on the path at Fushimi Inari, Kyoto.* [6]

Exercise 3.2 (Photographing the torii at Fushimi Inari)**.** The number X of torii gates (Fig. 3.3) a visitor stops to photograph on a walk up the Senbon Torii path at Fushimi Inari, the Kyoto shrine whose paths are lined with thousands of vermilion gates, has p.m.f. $f(1) = 0.1$, $f(2) = 0.2$, $f(3) = 0.4$, $f(4) = 0.2$, $f(5) = 0.1$.

(a) Tabulate the c.d.f. of X.

(b) What is the chance a visitor photographs between 2 and 4 gates? That is, find $P(2 \leq X \leq 4)$.

(c) How many gates does a visitor photograph on average, and how much does that number vary from visitor to visitor? That is, find the mean, variance, and standard deviation of X.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 3.1. (a) A match needs at least two people, and by the pigeonhole principle it must happen by the 366th: the support is $\{2, 3, \ldots, 366\}$. **(b)** $P(X = 2) = \frac{1}{365}$. $P(X = 3) = \frac{364}{365} \cdot \frac{2}{365}$ (second person misses, third hits one of two taken days). $P(X = 4) = \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{3}{365}$. In general the first $k-1$ people are all distinct and the kth matches one of their $k-1$ days:

$$P(X = k) = \left(\prod_{j=1}^{k-2} \frac{365 - j}{365} \right) \frac{k-1}{365}, \qquad k = 2, \ldots, 366.$$

Exercise 3.2. (a) $F(1) = 0.1$, $F(2) = 0.3$, $F(3) = 0.7$, $F(4) = 0.9$, $F(5) = 1$. **(b)** $P(2 \le X \le 4) = F(4) - F(1) = 0.8$. **(c)** $\mathbb{E}(X) = 0.1 + 0.4 + 1.2 + 0.8 + 0.5 = 3$; $\mathbb{E}(X^2) = 0.1 + 0.8 + 3.6 + 3.2 + 2.5 = 10.2$; $\mathrm{Var}(X) = 10.2 - 9 = 1.2$; $\mathrm{SD}(X) = 1.095$.

3.3 Continuous Random Variables

Continuous RV

For a continuous RV, the possible values fill out an entire interval, so we cannot list them and sum over them as in the discrete case. Instead we turn to integration:

Definition 3.5: Let X be a continuous RV. The probability density function (p.d.f.) of X is any f_X s.t.

i) $f_X(x) \ge 0 \quad \forall\, x \in \mathbb{R}$

ii) $P(X \in B) = \int_B f_X(x)\, dx, \quad B \subseteq \mathbb{R}$ (probabilities are areas under curves)

iii) $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$

Note: For a continuous RV, $P(X = x) = \int_x^x f_X(t)\, dt = 0$ at every single point x: probability accumulates only over intervals. One consequence is that

$$P(a < X < b) = P(a \le X \le b),$$

so strict vs. non-strict inequalities make no difference for continuous RVs. They do matter for discrete RVs, where single points carry positive probability.

Further, additional properties are:

I) $F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)\, dt \quad \forall\, x \in \mathbb{R}$

II) $\frac{d}{dx} F_X(x) = f_X(x) \quad \forall\, x \in \mathbb{R}$ s.t. $f_X(x)$ is continuous.

Example 3.6: Let X be the number of hours a cat spends asleep in a sunbeam on a given afternoon, a continuous RV s.t.

$$f_X(x) = \begin{cases} c(4x - 2x^2) & 0 \le x \le 2 \\ 0 & \text{o.w.} \end{cases}$$

a) Determine the value of c.

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x)\,dx = \int_0^2 c(4x - 2x^2)\,dx &= c\left[\frac{4x^2}{2} - \frac{2}{3}x^3\right]_0^2 \\ &= c\left(\frac{4(4)}{2} - \frac{2}{3}(8)\right) = c\left(8 - \frac{16}{3}\right) = c\left(\frac{24}{3} - \frac{16}{3}\right) = c\left(\frac{8}{3}\right) = 1 \\ \Rightarrow\ c &= \frac{3}{8} \end{aligned}$$

b) What is the chance the cat sleeps in the sunbeam for at most x hours? That is, find $F_X(x)$

$$\begin{aligned} F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt &= \int_0^x c(4t - 2t^2)\,dt \\ &= c\left[\frac{4t^2}{2} - \frac{2}{3}t^3\right]_0^x = \frac{3}{8}\left(2x^2 - \frac{2}{3}x^3\right) \\ &= \frac{3}{4}x^2 - \frac{1}{4}x^3 = \frac{3x^2 - x^3}{4} \end{aligned}$$

Note: $F_X(2) = (3(4) - 8)/4 = (12 - 8)/4 = 4/4 = 1$

c) What is the chance the cat sleeps in the sunbeam for between half an hour and an hour? That is, find $P(1/2 < X < 1)$

$$\begin{aligned} P(1/2 < X < 1) = \int_{1/2}^{1} f_X(x)\,dx &= F_X(1) - F_X(1/2) \\ &= \left(\frac{3 - 1}{4}\right) - \left(\frac{3(1/4) - 1/8}{4}\right) \\ &= \frac{2}{4} - \frac{6/8 - 1/8}{4} = \frac{2}{4} - \frac{5/8}{4} = \frac{2}{4} - \frac{5}{32} = \frac{16}{32} - \frac{5}{32} = \frac{11}{32} \end{aligned}$$

Conditional probability (Section 2.1) combines directly with the c.d.f.:

Example 3.7: Suppose X is a continuous RV with c.d.f.

$$F_X(x) = \begin{cases} 0 & x \le -1 \\ \frac{1}{4}(3x - x^3 + 2) & -1 < x < 1 \\ 1 & x \ge 1 \end{cases}$$

a) Find $P(X > 1/2 \mid X > 0)$

$$\begin{aligned} P(X > 1/2 \mid X > 0) &= \frac{P(X > 1/2 \cap X > 0)}{P(X > 0)} = \frac{P(X > 1/2)}{P(X > 0)} \\ &= \frac{1 - P(X \leq 1/2)}{1 - P(X \leq 0)} = \frac{1 - F_X(1/2)}{1 - F_X(0)} \\ &= \frac{1 - \frac{1}{4}\left(\frac{3}{2} - \frac{1}{8} + 2\right)}{1 - \frac{1}{4}(2)} = \frac{1 - \frac{1}{4}\left(\frac{27}{8}\right)}{\frac{1}{2}} \\ &= \frac{1 - \frac{27}{32}}{\frac{1}{2}} = 0.3125 \end{aligned}$$

4 Expectation and Variance

4.1 Expectation of a Random Variable

Expectation and Variance of a RV

For a random variable X, we often want to try and quantify its behavior. For example,

i) What is the center of the distribution of X?

ii) What is a typical value of X?

iii) How much does X vary from the typical value?

If we have the range of X, and its p.d.f., these are relatively easy to find.

Expectation of X (Discrete)

Let X be a discrete RV with probabilities $f(x_i)$, $i = 1, \ldots, n$. The average, expected value, or mean of X is:

$$\mu_X = EX = \mathbb{E}[X] = \sum_{i=1}^{n} x_i f(x_i)$$

If $i = 1, 2, \ldots$ (countably infinite) values, then $\mathbb{E}[X] = \sum_{i=1}^{\infty} x_i f(x_i)$,

so long as $\sum_i |x_i| f(x_i) < \infty$.

Expectation of X (Continuous)

If X is a continuous RV with p.d.f. $f(x)$, the

$$\mu_X = \mathbb{E}[X] = EX = \int_{-\infty}^{\infty} x f(x)\, dx$$

The expected value can be considered as "the point where the distribution is balanced". Thus, while it is not always the literal center, it is often used as a measure of center.

Note: The integral is written over the whole real line, but $f(x) = 0$ off the support, so it really runs over the support of X only. Pay attention to the support: the limits of integration come from the p.d.f., not from habit.

Why this formula?

Let X be the age of a student picked uniformly at random from a class of 18, with p.d.f.

x	19	20	21	22	23
$f(x)$	$\frac{4}{18}$	$\frac{6}{18}$	$\frac{6}{18}$	$\frac{1}{18}$	$\frac{1}{18}$

The mean age is the ordinary average of the 18 ages. Regrouping that sum by age turns it into the expectation formula:

$$\frac{\overbrace{19+\cdots+19}^{4\times}+\overbrace{20+\cdots+20}^{6\times}+\overbrace{21+\cdots+21}^{6\times}+22+23}{18} = 19\left(\tfrac{4}{18}\right)+20\left(\tfrac{6}{18}\right)+21\left(\tfrac{6}{18}\right)+22\left(\tfrac{1}{18}\right)+23\left(\tfrac{1}{18}\right)$$
$$= \sum_{\text{all ages}} (\text{age})\cdot P(X=\text{age}) = \mathbb{E}[X].$$

Each age is weighted by the fraction of the class that has it, and that fraction is exactly $f(x)$. The expectation formula is an average in which the p.d.f. supplies the weights.

Example 4.1: Suppose a casino game has you spend \$1 per game, and you have the following probabilities of winning the following amounts:

Winnings	\$0	\$1	\$5	\$20	\$100	x_i
Prob	0.70	0.24	0.05	0.009	0.001	$f(x_i)$

Let X = *net* winnings for one game, i.e. the payout minus the \$1 spent to play. So X takes the values $-1, 0, 4, 19, 99$ with the probabilities above.

a) Verify that this is a valid p.d.f.

All probabilities ≥ 0, and $\sum f(x_i) = 1$. ✓

b) Find the expected value of take home winnings for this game.

$$\mathbb{E}[X] = \sum_{i=1}^{5} x_i f(x_i) = (-1)(0.70)+0(0.24)+4(0.05)+19(0.009)+99(0.001)$$
$$= -\$0.23$$

Note: Notice $\mathbb{E}[X] < 0$: the player loses 23 cents per play in the long run, the house edge that keeps the casino open; Section 10.5 shows why even a small negative mean guarantees the house's profit.

Example 4.2: Let X be the length, in minutes, of the instrumental break on a randomly chosen disco 12-inch single (a full-size record carrying one extended track), with p.d.f.

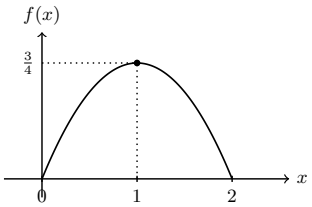
$$f(x) = \begin{cases} \dfrac{3}{54}x^2 - \dfrac{1}{54} & 2 \leq x \leq 4 \\ 0 & \text{o.w.} \end{cases}$$

How long, on average, is the instrumental break on a disco 12-inch, in minutes? That is, find $\mathbb{E}[X]$.

$$\mathbb{E}[X] = \int_2^4 x\left(\tfrac{3}{54}x^2 - \tfrac{1}{54}\right)dx = \tfrac{3}{54}\cdot\tfrac{1}{4}x^4 - \tfrac{1}{54}\cdot\tfrac{1}{2}x^2\Big|_2^4$$
$$= \tfrac{1}{54}\left(\left[\tfrac{3}{4}\cdot 4^4 - \tfrac{1}{2}\cdot 4^2\right] - \left[\tfrac{3}{4}\cdot 2^4 - \tfrac{1}{2}\cdot 2^2\right]\right) = \tfrac{1}{54}(3\cdot 64 - 8 - 12 + 2)$$
$$= \tfrac{1}{54}(192 - 20 + 2) = \tfrac{1}{54}(174) = 3.222$$

Example 4.3: Let X be a continuous RV with p.d.f.

$$f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & 0 \le x \le 2 \\ 0 & \text{o.w.} \end{cases}$$



Find $\mathbb{E}[X]$.

i) By symmetry. f is a parabola with vertex at $x = 1$, so the density is a mirror image about 1 and the balance point has to be there: $\mathbb{E}[X] = 1$.

ii) By integration, over the support $[0, 2]$:

$$\mathbb{E}[X] = \int_0^2 x \cdot \tfrac{3}{8}(4x - 2x^2)\,dx = \int_0^2 \left(\tfrac{3}{2}x^2 - \tfrac{3}{4}x^3\right) dx = \left[\tfrac{1}{2}x^3 - \tfrac{3}{16}x^4\right]_0^2$$
$$= \tfrac{1}{2}(2)^3 - \tfrac{3}{16}(2)^4 - 0 = 4 - 3 = 1. \qquad \checkmark$$

Proposition 4.1 (Properties of expectation): Let X be a RV and let c, d be constants (deterministic, not random).

i) *Expectation is linear:* $\mathbb{E}[c + dX] = c + d\,\mathbb{E}[X]$. In particular $\mathbb{E}[cX] = c\,\mathbb{E}[X]$.

ii) *Constants are their own expectation:* $\mathbb{E}[c] = c$.

iii) *Monotonicity:* if $X \ge c$, then $\mathbb{E}[X] \ge \mathbb{E}[c] = c$.

Why i) holds is answered by the next result, which is used so often that it has its own name.

4.2 Functions of a Random Variable and Variance

Theorem 4.1 (Law of the Unconscious Statistician (LOTUS)): Let X be a RV with p.d.f. f_X, let $g : \mathbb{R} \to \mathbb{R}$ (possibly nonlinear), and let $Y = g(X)$. Then

(Discrete X):

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_{\text{all } x} g(x)\, f_X(x)$$

(Continuous X):

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\,dx$$

Note: This is a result, not a definition. By definition $\mathbb{E}[Y] = \sum_y y\, f_Y(y)$, which would require first finding the p.d.f. of Y. LOTUS says that step can be skipped: weight each value $g(x)$ by the probability of the x that produced it. In the discrete case the proof is one line, since the values x with $g(x) = y$ are exactly the ones whose probability makes up $f_Y(y)$:

$$\sum_y y\, f_Y(y) = \sum_y y \sum_{x:\, g(x)=y} f_X(x) = \sum_{\text{all } x} g(x)\, f_X(x).$$

The name pokes fun at statisticians who use the formula as if it were the definition, without noticing it needs a proof.

Specifically, the kth moment of X is: $\mathbb{E}[X^k]$, $k \geq 1$ (i.e., $Y = g(X) = X^k$).

Why is expectation linear?

Take $g(x) = ax + b$ in LOTUS. In the continuous case (the discrete case replaces each integral by a sum),

$$\begin{aligned}\mathbb{E}[aX + b] = \mathbb{E}[g(X)] &= \int_{-\infty}^{\infty} (ax + b)\, f_X(x)\, dx = \int_{-\infty}^{\infty} ax\, f_X(x)\, dx + \int_{-\infty}^{\infty} b\, f_X(x)\, dx \\ &= a \underbrace{\int_{-\infty}^{\infty} x\, f_X(x)\, dx}_{=\mathbb{E}[X]} + b \underbrace{\int_{-\infty}^{\infty} f_X(x)\, dx}_{=1} = a\mathbb{E}[X] + b.\end{aligned}$$

The integral splits because integration is linear, the first piece is the definition of $\mathbb{E}[X]$, and the second is the total probability. Taking $a = 0$ gives $\mathbb{E}[b] = b$, which is property ii) above.

Example 4.1, cont'd: Find the 2nd moment of X.

$$\begin{aligned}\mathbb{E}[X^2] &= \sum x_i^2 f(x_i) = (-1)^2(0.70) + 0^2(0.24) + 4^2(0.05) + 19^2(0.009) + 99^2(0.001) \\ &= 14.55\end{aligned}$$

Example 4.2, cont'd: Find $\mathbb{E}[X^2 + c]$ for a constant c.

$\mathbb{E}[X^2 + c] = \mathbb{E}[X^2] + c$

$$\begin{aligned}\mathbb{E}[X^2] &= \tfrac{1}{54}\int_2^4 x^2(3x^2 - 1)\, dx = \tfrac{1}{54}\int_2^4 (3x^4 - x^2)\, dx = \tfrac{1}{54}\left(\tfrac{3}{5}x^5 - \tfrac{x^3}{3}\right)\Big|_2^4 \\ &= \tfrac{1}{54}\left[\left(\tfrac{3}{5}4^5 - \tfrac{1}{3}4^3\right) - \left(\tfrac{3}{5}2^5 - \tfrac{1}{3}2^3\right)\right] = 10.677\end{aligned}$$

$\mathbb{E}[X^2 + c] = 10.677 + c$

Definition 4.1: The variance of a RV X is denoted $\text{Var}[X]$, and defined as:

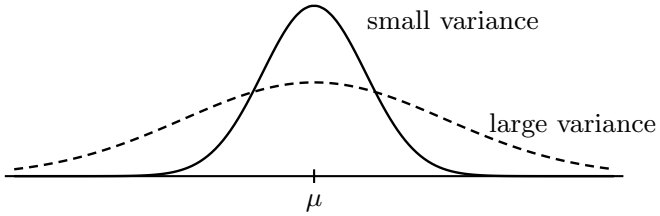
$$\text{Var}[X] = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \text{the expected squared distance of } X \text{ from its mean.}$$

Other notations include $\sigma^2\{X\} = \sigma_X^2 = \text{Var}[X]$.

$$\sigma_X = \sqrt{\text{Var}[X]} = \text{the standard deviation of } X,$$

which is interpreted as: "a typical distance of X to $\mathbb{E}[X]$."

Two densities with the same mean but different variance:



Example 4.4: Let X be the value of a fair die, so $f_X(x) = \frac{1}{6}$ for $x = 1, \ldots, 6$. The expectation and the variance are both sums over the support, weighted by the p.d.f.:

$$\mathbb{E}[X] = \sum_{\text{all } x} x\, f_X(x) = \sum_{x=1}^{6} x\, f_X(x) = 1\left(\tfrac{1}{6}\right) + 2\left(\tfrac{1}{6}\right) + \cdots + 6\left(\tfrac{1}{6}\right) = 3.5$$

$$\begin{aligned}\text{Var}[X] = \mathbb{E}\left[(X-\mu_X)^2\right] &= \sum_{\text{all } x} (x-\mu_X)^2\, f_X(x) = \sum_{x=1}^{6} (x-3.5)^2\, f_X(x) \\ &= (1-3.5)^2\left(\tfrac{1}{6}\right) + \cdots + (6-3.5)^2\left(\tfrac{1}{6}\right) = \tfrac{35}{12} = 2.91\overline{6}\end{aligned}$$

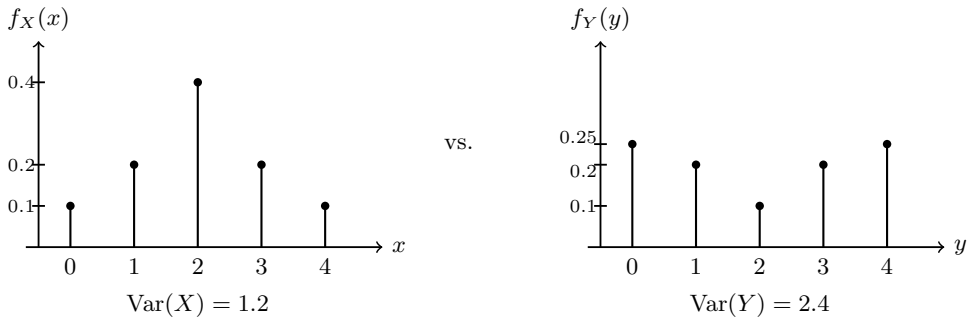
Big X is random; little x is the dummy variable running over the support S, and both sums collapse to a number. So $\mathbb{E}[X]$ and $\text{Var}[X]$ are deterministic constants attached to the distribution, not random themselves. The variance is the average squared distance of X from 3.5: it is $\mathbb{E}[g(X)]$ for $g(x) = (x-3.5)^2$, computed by LOTUS. Taking the square root,

$$\text{SD}[X] = \sqrt{\text{Var}[X]} = \sqrt{2.91\overline{6}} = 1.708,$$

so a die roll sits, on average, about 1.7 away from its mean.

Note: Why square the distance? Averaging the signed distance $X - \mathbb{E}[X]$ tells us nothing, because linearity makes it vanish for every RV: $\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[\mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[X] = 0$, since $\mathbb{E}[X]$ is a constant and constants are their own expectation. Squaring stops the positive and negative deviations from cancelling.

Example 4.5: Two RVs on $\{0, 1, 2, 3, 4\}$, both with mean 2:



$\mathrm{Var}(X) = 1.2$ $\mathrm{Var}(Y) = 2.4$

Y is visibly more spread out around the mean, and the variance measures that. Since $\mathbb{E}[X] = 2$, $\mathrm{Var}[X] = \mathbb{E}[(X-2)^2]$ is the expectation of the new RV $W = (X-2)^2$. LOTUS computes it directly, or we can first tabulate the p.d.f. of W by pooling the values of x that give the same $(x-2)^2$:

x	0	1	2	3	4
$f_X(x)$	0.1	0.2	0.4	0.2	0.1
$(x-2)^2$	4	1	0	1	4

w	0	1	4
$P(W=w)$	0.4	0.4	0.2

$$\mathrm{Var}[X] = \mathbb{E}[W] = 0(0.4) + 1(0.4) + 4(0.2) = 1.2.$$

The same steps for Y, with $Z = (Y-2)^2$:

y	0	1	2	3	4
$f_Y(y)$	0.25	0.2	0.1	0.2	0.25
$(y-2)^2$	4	1	0	1	4

z	0	1	4
$P(Z=z)$	0.1	0.4	0.5

$$\mathrm{Var}[Y] = \mathbb{E}[Z] = 0(0.1) + 1(0.4) + 4(0.5) = 2.4.$$

So $\mathrm{Var}[Y] = 2.4 > 1.2 = \mathrm{Var}[X]$, matching the picture. Computing from the definition means carrying the deviations around; the shortcut below avoids that.

Proposition 4.2 (Properties of variance): For a RV X and constants c, d:

i) $\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

ii) $\mathrm{Var}[c + dX] = \mathrm{Var}[dX] = d^2\mathrm{Var}[X]$; in particular $\mathrm{Var}[cX] = c^2\mathrm{Var}[X]$

iii) $\mathrm{Var}[g(X)] = \mathbb{E}[g^2(X)] - (\mathbb{E}[g(X)])^2$

Why i)? Write $\mu_X = \mathbb{E}[X]$, expand the square, and use linearity twice:

$$\begin{aligned}
\mathrm{Var}[X] = \mathbb{E}\big[(X-\mu_X)^2\big] &= \mathbb{E}\big[X^2 - 2\mu_X X + \mu_X^2\big] \\
&= \mathbb{E}[X^2] - \mathbb{E}[2\mu_X X] + \mathbb{E}[\mu_X^2] && \text{(linear)} \\
&= \mathbb{E}[X^2] - 2\mu_X\,\mathbb{E}[X] + \mu_X^2 && (\mu_X \text{ is a constant}) \\
&= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2
\end{aligned}$$

This is the working formula: two expectations and a subtraction, with no table of deviations to carry around.

Why ii)? The shift c drops out first, because $(c+dX)-\mathbb{E}[c+dX] = c+dX-c-d\mathbb{E}[X] = dX-d\mathbb{E}[X]$, so $\mathrm{Var}[c+dX] = \mathrm{Var}[dX]$. Then

$$\begin{aligned}\mathrm{Var}[dX] &= \mathbb{E}\big[(dX-\mathbb{E}[dX])^2\big] = \mathbb{E}\big[(dX-d\mathbb{E}[X])^2\big] = \mathbb{E}\big[(d\,(X-\mathbb{E}[X]))^2\big] \\ &= \mathbb{E}\big[d^2(X-\mathbb{E}[X])^2\big] && ((ab)^2=a^2b^2) \\ &= d^2\underbrace{\mathbb{E}\big[(X-\mathbb{E}[X])^2\big]}_{\mathrm{Var}[X]} = d^2\,\mathrm{Var}[X] && \text{(linear)}\end{aligned}$$

Shifting a RV moves every outcome but not the spread; rescaling by d multiplies the variance by d^2 and the standard deviation by $|d|$.

Why iii)? It is property i) applied to the RV $g(X)$, with both expectations computed by LOTUS.

Example 4.6: Let Y be the number of heads which come up on one flip of a fair coin:

y	0	1
$f_Y(y)$	$\frac{1}{2}$	$\frac{1}{2}$

i) Calculate $\mathrm{Var}[Y]$ using property i).

$$\mathbb{E}[Y] = \sum_{y=0}^{1} y\,f_Y(y) = 0\left(\tfrac{1}{2}\right)+1\left(\tfrac{1}{2}\right) = \tfrac{1}{2}$$

$$\mathbb{E}[Y^2] = \sum_{y=0}^{1} y^2 f_Y(y) = 0^2\left(\tfrac{1}{2}\right)+1^2\left(\tfrac{1}{2}\right) = \tfrac{1}{2}$$

$$\mathrm{Var}[Y] = \mathbb{E}[Y^2]-(\mathbb{E}[Y])^2 = \tfrac{1}{2}-\left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}, \qquad \mathrm{SD}[Y] = \sqrt{\tfrac{1}{4}} = \tfrac{1}{2}$$

Y^2 has the same p.d.f. as Y because $0^2=0$ and $1^2=1$; that is special to $\{0,1\}$-valued RVs, and in general $\mathbb{E}[Y^2]$ and $\mathbb{E}[Y]$ differ.

ii) Now let $W = 2Y+4$. By property ii),

$$\mathrm{Var}[W] = \mathrm{Var}[2Y+4] = \mathrm{Var}[2Y] = 2^2\,\mathrm{Var}[Y] = 4\left(\tfrac{1}{4}\right) = 1, \qquad \mathrm{SD}[W] = \sqrt{\mathrm{Var}[W]} = \sqrt{1} = 1.$$

The +4 moves the outcomes to $\{4,6\}$ without changing how far apart they are, and the factor 2 doubles the standard deviation.

Example 4.1, cont'd: Find and interpret σ_X.

$$\mathrm{Var}[X] = \mathbb{E}[X^2]-(\mathbb{E}[X])^2 = 14.55-(-0.23)^2 = 14.4971$$
$$\sigma_X = \sqrt{\mathrm{Var}[X]} = \sqrt{14.4971} = 3.8075$$

The typical distance of winnings from the average winnings is \$3.81.

Example 4.2, cont'd: Find σ_X^2.

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 10.677 - (3.222)^2 = 0.294$$

Example 4.7: If $\mu_X = 10$, $\sigma_X^2 = 2$, find μ_Y, σ_Y^2 if

i) $Y = 3 + 4X$:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[3 + 4X] = 3 + \mathbb{E}[4X] = 3 + 4\mathbb{E}[X] = 3 + 4\mu_X = 3 + 4(10) = 43\\ \text{Var}[Y] &= \text{Var}[3 + 4X] = \text{Var}[4X] = 4^2\text{Var}[X] = 16(\sigma_X^2) = 16(2) = 32\end{aligned}$$

ii) $Y = -4 - 2X$:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[c + dX] = c + d\mathbb{E}[X], \quad c = -4,\ d = -2\\ &= -4 + (-2)(10) = -24\\ \text{Var}[Y] &= \text{Var}[c + dX] = d^2\text{Var}[X], \quad d = -2\\ &= (-2)^2(2) = 8\end{aligned}$$

Moments and central moments. The quantities $\mathbb{E}[X^k]$ are the *moments* of X: $\mathbb{E}[X]$ is the first moment, $\mathbb{E}[X^2]$ the second, and so on. Centering first gives the *central moments* $\mathbb{E}\left[(X - \mathbb{E}[X])^k\right]$:

k	kth central moment $\mathbb{E}\left[(X - \mathbb{E}[X])^k\right]$
1	0, always (shown above)
2	$\text{Var}[X]$, the spread about the mean
3	skewness, the asymmetry of the distribution
4	kurtosis, the flatness of the distribution

A symmetric density has skewness 0. A long tail to the right pushes the cubed deviations positive on balance (right skewed); a long tail to the left makes them negative (left skewed):



symmetric: skew $= 0$

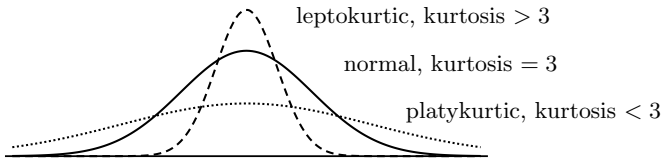


left skewed: skew < 0



right skewed: skew > 0

Kurtosis is read against the normal distribution: if $X \sim N(0,1)$ then $\text{kurt}(X) = 3$. A density that is more peaked, with heavier tails, has kurtosis above 3 and is called *leptokurtic*; a flatter one with lighter tails has kurtosis below 3 and is *platykurtic*:



Note: In practice the third and fourth central moments are divided by σ^3 and σ^4 before being called skewness and kurtosis, so that they do not change when X is rescaled. That standardization is what makes 3 the reference value for every normal, not just $N(0,1)$.

Exercises

Exercise 4.1. Suppose that 23 people in a room are comparing their birthdays. Assume that different people's birthdays are independent and that each person is equally likely to be born on any of 365 days.

(a) What is the probability that two particular people in the room, Adrian and Beatrix, have the same birthday?

(b) On average, how many *pairs* of people will share a birthday? Write $X = X_1 + \cdots + X_n$ where each X_i is a Bernoulli random variable, choosing the definition of X_i (and the value of n, which is not 23 or 365) so that the sum counts shared-birthday pairs. Then use linearity of expectation to compute $\mathbb{E}[X]$.

Exercise 4.2. You have a well-shuffled 52-card deck. On average, how many pairs of *adjacent* cards are there such that both cards are red? *Hint:* Use indicators.



Figure 4.1: *Oia, Santorini, at sunset.*[7]

Exercise 4.3 (Waiting for a sunset table in Santorini)**.** The wait X, in hours, for a cliffside table at a popular taverna on Santorini (the Greek island famous for its cliff-top sunsets; Fig. 4.1) has p.d.f. $f(x) = 2(1-x)$ for $0 \le x \le 1$ (and 0 otherwise).

(a) Verify that f integrates to 1, and find the c.d.f.

(b) What is the chance of waiting more than half an hour for a table? That is, find $P(X > 1/2)$.

(c) Find the mean wait.

(d) Find the median wait, and explain why it is below the mean.

Exercise 4.4 (An açaí stand in Salvador)**.** A beach vendor in Salvador, on Brazil's northeast coast, sells X bowls of açaí (a frozen purple-berry purée) a day, with mean 80 and standard deviation 12. Each bowl earns R$9 and the stand costs R$150 a day to run, so the daily profit is $Y = 9X - 150$.

(a) What is the stand's average daily profit, and how far does a day's profit typically fall from it? That is, find the mean and standard deviation of Y.

(b) Find the variance of Y, and say in one sentence why the 150 does not appear in it.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 4.1. (a) Whatever Adrian's birthday is, Beatrix matches it with probability $\frac{1}{365}$. **(b)** Let $X_{ij} = 1$ if persons i and j share a birthday, one indicator per pair, so $n = \binom{23}{2} = 253$ and $X = \sum X_{ij}$ counts shared-birthday pairs. By linearity, $\mathbb{E}[X] = 253 \cdot \frac{1}{365} = 0.693$: on average a bit under one matching pair, which is why a match in a room of 23 is a coin flip.

Exercise 4.2. There are 51 adjacent positions. Let $I_k = 1$ if cards k and $k+1$ are both red; $P(I_k = 1) = \frac{26}{52} \cdot \frac{25}{51} = \frac{25}{102}$. By linearity, $\mathbb{E}[\sum I_k] = 51 \cdot \frac{25}{102} = 12.5$ red-red adjacent pairs, even though the indicators are dependent.

Exercise 4.3. (a) $\int_0^1 2(1-x)\,dx = [2x - x^2]_0^1 = 1$, and $F(x) = 2x - x^2$ on $[0, 1]$. **(b)** $P(X > 1/2) = 1 - F(1/2) = 1 - \frac{3}{4} = \frac{1}{4}$. **(c)** $\mathbb{E}(X) = \int_0^1 2x(1-x)\,dx = 2(\frac{1}{2} - \frac{1}{3}) = \frac{1}{3}$ hour, twenty minutes. **(d)** Solve $2m - m^2 = \frac{1}{2}$: $m = 1 - 1/\sqrt{2} = 0.293$ hour, about 17.6 minutes. The density is right-skewed (most waits short, a tail of long ones), and the tail pulls the mean above the median.

Exercise 4.4. (a) $\mathbb{E}(Y) = 9(80) - 150 = 570$ and $\mathrm{SD}(Y) = 9(12) = 108$ reais. **(b)** $\mathrm{Var}(Y) = 81(144) = 11{,}664$; the 150 shifts every value by the same amount and so cannot change the spread.

4.3 Moment Generating Functions and Probability Inequalities

Moment Generating Function

By LOTUS the kth moment is $\mathbb{E}[X^k] = \int x^k f_X(x)\,dx$ (or the corresponding sum): a fresh integral for every k. The object below is a single function of one variable from which *every* moment can be read off by differentiation, a shortcut to any moment.

Definition 4.2: The **moment generating function (m.g.f.)** of a RV X is denoted $M_X(t)$ and defined as

$$M_X(t) = \mathbb{E}[e^{tX}], \quad \forall\, t \text{ s.t. } M_X(t) \text{ is finite.}$$

Properties of the m.g.f.

i) $M_X(0) = 1$

Proof: $M_X(0) = \mathbb{E}[e^{0\cdot X}] = \mathbb{E}[1] = 1$. ✓

ii) $M_X'(0) = \mathbb{E}[X]$

Proof: $\dfrac{d}{dt}\mathbb{E}[e^{tX}] = \mathbb{E}\left[\dfrac{d}{dt}e^{tX}\right] = \mathbb{E}[Xe^{tX}]$.

At $t = 0$: $M_X'(0) = \mathbb{E}[Xe^0] = \mathbb{E}[X]$. ✓

iii) $M_X^{(k)}(0) = \mathbb{E}[X^k]$ (the k-th derivative at 0 gives the k-th moment)

Proof: Iterate the argument of ii): each differentiation w.r.t. t pulls one more factor of X down from the exponent,

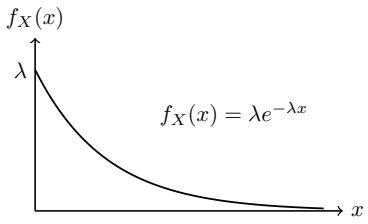
$$M_X^{(k)}(t) = \frac{d^k}{dt^k}\,\mathbb{E}[e^{tX}] = \mathbb{E}\left[\frac{d^k}{dt^k}\,e^{tX}\right] = \mathbb{E}[X^k e^{tX}]$$

At $t = 0$: $M_X^{(k)}(0) = \mathbb{E}[X^k e^0] = \mathbb{E}[X^k]$. ✓

Remark (a hidden assumption): The proofs of ii) and iii) each slip a derivative inside an expectation, $\frac{d}{dt}\mathbb{E}[\cdots] = \mathbb{E}[\frac{d}{dt}\cdots]$. Since an expectation is a sum or an integral, this swaps a limit (the derivative) with an infinite sum or integral, an exchange that is *not* automatic and can genuinely fail for badly behaved functions. It is legitimate here provided $M_X(t)$ is finite in an open interval around 0, which is exactly the hypothesis of the uniqueness theorem below; we will take that for granted in this course. The tools that justify the exchange are covered in real analysis: look up *differentiation under the integral sign* (the Leibniz rule) and the *dominated convergence theorem* if you would like to see when it works and how it can break.

Note: This is where the name "moment *generating* function" comes from. Expanding $e^{tX} = \sum_{k\geq 0}(tX)^k/k!$ inside the expectation shows that the coefficient of t^k in $M_X(t)$ is $\mathbb{E}[X^k]/k!$ (this is the Taylor series in the second theorem below), so the m.g.f. stores every moment of X at once. Differentiating k times and setting $t = 0$ isolates exactly one coefficient: the terms below t^k are differentiated away to 0, the terms above t^k still carry a factor of t and vanish at $t = 0$, and the t^k term alone survives, contributing $k! \cdot \mathbb{E}[X^k]/k! = \mathbb{E}[X^k]$.

Example 4.8: Let X have p.d.f. $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, with $\lambda > 0$ (the exponential distribution of Section 6.2).



a) Find the m.g.f. of X.

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx}\,\lambda e^{-\lambda x}\,dx = \int_0^\infty \lambda\, e^{(t-\lambda)x}\,dx = \frac{\lambda}{t-\lambda}\, e^{(t-\lambda)x}\Big|_0^\infty$$

If $t - \lambda < 0$, i.e. $t < \lambda$, then $e^{(t-\lambda)x} \to 0$ as $x \to \infty$ and the bracket is $[\,0 - 1\,]$:

$$M_X(t) = \frac{\lambda}{t-\lambda}\,[\,0 - 1\,] = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

The restriction is real: for $t \geq \lambda$ the integrand does not decay and $\mathbb{E}[e^{tX}] = \infty$, so the m.g.f. does not exist there. The moments only need derivatives at $t = 0$, which sits inside $(-\infty, \lambda)$.

b) Use the m.g.f. to find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

$$\mathbb{E}[X] = M_X'(0) = \frac{d}{dt}\,\lambda(\lambda - t)^{-1}\Big|_{t=0} = \lambda(-1)(\lambda - t)^{-2}(-1)\Big|_{t=0} = \frac{\lambda}{(\lambda - t)^2}\Big|_{t=0} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

$$\mathbb{E}[X^2] = M_X''(0) = \frac{d}{dt}\,\lambda(\lambda - t)^{-2}\Big|_{t=0} = \lambda(-2)(\lambda - t)^{-3}(-1)\Big|_{t=0} = \frac{2\lambda}{(\lambda - t)^3}\Big|_{t=0} = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}$$

c) Use part b) to calculate $\mathrm{Var}[X]$ and $\mathrm{SD}[X]$.

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}, \qquad \mathrm{SD}[X] = \frac{1}{\lambda}.$$

Both the mean and the standard deviation of an exponential are $1/\lambda$. Done directly, $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ would each be an integration by parts; the m.g.f. replaces them with two derivatives of $\lambda/(\lambda - t)$.

For $Y = -2 + 3X$:

$$M_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(-2+3X)}] = e^{-2t}\,\mathbb{E}[e^{3tX}] = e^{-2t} M_X(3t) = \frac{\lambda\, e^{-2t}}{\lambda - 3t}, \quad t < \tfrac{\lambda}{3}$$

Theorem 4.2: If $M_X(t)$ is finite in an open interval containing 0, then the m.g.f. uniquely determines the distribution of X.

Theorem 4.3: If all moments of X are finite and the series below converges in an open interval containing 0, then the moments uniquely determine the m.g.f. (via Taylor series) and hence the distribution:

$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!}\, t^k$$

Some Probability Inequalities

Sometimes we have information about a RV such as μ or σ^2 but do not know the p.d.f.

Theorem 4.4 (Markov / Chebyshev Inequalities):

i) **(Markov)** If $X \geq 0$ and $c > 0$, then $P(X \geq c) \leq \dfrac{\mathbb{E}[X]}{c}$.

ii) If $g(X) \geq 0$ and $c > 0$, then $P(g(X) \geq c) \leq \dfrac{\mathbb{E}[g(X)]}{c}$.

iii) $P(|X - \mu_X| \geq c) \leq \dfrac{\mathbb{E}[|X - \mu_X|]}{c}$ (Markov, applied to $|X - \mu_X|$)

iv) **(Chebyshev)** $P(|X - \mu_X| \geq c) \leq \dfrac{\mathrm{Var}[X]}{c^2}$

In the proof we use the *indicator* $\mathbb{1}_E$, which equals 1 when the event E occurs and 0 otherwise. The subscript is the event being checked, so a condition is written as the event it defines: $\mathbb{1}_{\{g(X)\geq c\}}$. Note that $\mathbb{E}[\mathbb{1}_E] = P(E)$.

Proof of (i), by the integral. Let $Y = \mathbb{1}_{\{X\geq c\}}$, so that $\mathbb{E}[Y] = (1)P(X \geq c) + (0)P(X < c) = P(X \geq c)$. Since $X \geq 0$, dropping the part of the integral on $[0, c)$ can only make it smaller, and on $[c, \infty)$ the factor x is at least c:

$$\mathbb{E}[X] = \int_0^\infty x f_X(x)\,dx \geq \int_c^\infty x f_X(x)\,dx \geq \int_c^\infty c\, f_X(x)\,dx = c\,P(X \geq c),$$

so $P(X \geq c) \leq \mathbb{E}[X]/c$.

Proof of (i), without the integral. Pointwise, $X = X(1 - Y) + XY \geq XY \geq cY$: on $\{X \geq c\}$ the right side is $c \leq X$, and on $\{X < c\}$ it is $0 \leq X$. Taking expectations preserves the inequality, so $\mathbb{E}[X] \geq c\,\mathbb{E}[Y] = c\,P(X \geq c)$. The picture is the line $y = x$ sitting above the step $c\,\mathbb{1}_{\{x\geq c\}}$.

Proof of (ii): the same argument with $g(X)$ in place of X:

$$P(g(X) \geq c) = \mathbb{E}\big[\mathbb{1}_{\{g(X)\geq c\}}\big] \leq \mathbb{E}\left[\frac{g(X)}{c}\,\mathbb{1}_{\{g(X)\geq c\}}\right] \leq \mathbb{E}\left[\frac{g(X)}{c}\right] = \frac{\mathbb{E}[g(X)]}{c}$$

(iii) and (iv) from (ii). Take $g(X) = |X - \mu_X|$ for (iii). For Chebyshev take $g(X) = (X - \mu_X)^2 \geq 0$ and write the threshold as $c = k^2$:

$$P\big((X - \mu_X)^2 \geq k^2\big) \leq \frac{\mathbb{E}[(X - \mu_X)^2]}{k^2} = \frac{\mathrm{Var}[X]}{k^2},$$

and $(X - \mu_X)^2 \geq k^2$ is the same event as $|X - \mu_X| \geq k$. So Chebyshev is Markov applied to the squared deviation, which is why it needs only a mean and a variance and nothing about the shape of the distribution. The price is that the bound is loose, as the examples show.

Example 4.9: Let X be equally likely to take any value between 0 and 10, i.e. X has p.d.f. $f_X(x) = \frac{1}{10}$ for $0 < x < 10$ (and 0 o.w.). (This is called the *uniform distribution*, written $X \sim U(0, 10)$; we will study it formally in Section 6.2.) Then

$$\mu_X = \int_0^{10} \frac{x}{10}\,dx = 5, \qquad \sigma_X^2 = \mathbb{E}[X^2] - \mu_X^2 = \int_0^{10} \frac{x^2}{10}\,dx - 5^2 = \frac{100}{3} - 25 = \frac{25}{3}$$

(a) **Exact:** $P(|X-5|>4) = P(X<1) + P(X>9) = \frac{1}{10} + \frac{1}{10} = 0.20$

(b) **Chebyshev:** $P(|X-5| \geq 4) \leq \dfrac{25/3}{4^2} = \dfrac{25}{48} \approx 0.52$

(c) **Difference:** $0.52 - 0.20 = 0.32$

Example 4.10: If $\mu_X = 3$, $\sigma_X = 10$, find a lower bound for $P(|X-\mu_X| < 4\sigma_X)$.

By Chebyshev with $c = 4\sigma_X = 40$:

$$P(|X-\mu_X| \geq 40) \leq \frac{\sigma_X^2}{40^2} = \frac{100}{1600} = \frac{1}{16}$$

$$\therefore \quad P(|X-\mu_X| < 4\sigma_X) \geq 1 - \frac{1}{16} = \frac{15}{16}$$

Exercises

Exercise 4.5. Let $X \sim \text{Exp}(3)$, so that $M_X(t) = \dfrac{3}{3-t}$ for $t < 3$.

(a) Use the $M_X(t)$ formula to compute $\mathbb{E}[e^X]$ and $\mathbb{E}[e^{2X}]$.

(b) Compute $\mathbb{E}[2^X]$ and $\mathbb{E}[3^X]$. *Hint:* $2 = e^{\ln 2}$ and $3 = e^{\ln 3}$.

(c) For which real numbers $a > 0$ is $\mathbb{E}[a^X]$ finite?

Exercise 4.6. Let X be a random variable and let $a > 0$. Define the indicator $I = \mathbb{1}_{\{|X| \geq a\}}$.

(a) Explain why $X^2 \geq a^2 I$ always holds. (Consider the two cases $|X| \geq a$ and $|X| < a$ separately.)

(b) Take expectations of both sides and use $\mathbb{E}[I] = P(|X| \geq a)$ to conclude that $P(|X| \geq a) \leq \mathbb{E}[X^2]/a^2$.

(c) Now apply part (b) to the random variable $X - \mu$, where $\mu = \mathbb{E}[X]$, to derive Chebyshev's inequality in its usual form.

(d) Where exactly in this argument did you use that $a > 0$?

Exercise 4.7 (The açaí stand, bounded)**.** Daily profit at a beach açaí stand has mean R\$570 and standard deviation R\$108, with the shape of its distribution unknown. Use Chebyshev's inequality to bound the probability that a day's profit is more than R\$216 away from the mean.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 4.5. (a) $\mathbb{E}[e^X] = M_X(1) = \frac{3}{2}$ and $\mathbb{E}[e^{2X}] = M_X(2) = 3$. **(b)** $\mathbb{E}[2^X] = \mathbb{E}[e^{(\ln 2)X}] = M_X(\ln 2) = \dfrac{3}{3 - \ln 2} = 1.30$ and $\mathbb{E}[3^X] = \dfrac{3}{3 - \ln 3} = 1.58$. **(c)** $M_X(t)$ is finite for $t < 3$, so $\mathbb{E}[a^X] = M_X(\ln a)$ is finite exactly when $\ln a < 3$, i.e. $0 < a < e^3 \approx 20.1$.

Exercise 4.6. (a) If $|X| \geq a$ then $I = 1$ and $X^2 \geq a^2 = a^2 I$; if $|X| < a$ then $I = 0$ and $X^2 \geq 0 = a^2 I$. **(b)** Expectation preserves inequalities: $\mathbb{E}[X^2] \geq a^2 \mathbb{E}[I] = a^2 P(|X| \geq a)$, so $P(|X| \geq a) \leq \mathbb{E}[X^2]/a^2$. **(c)** Apply (b) to $X - \mu$: $P(|X - \mu| \geq a) \leq \mathbb{E}[(X - \mu)^2]/a^2 = \sigma^2/a^2$, Chebyshev's inequality. **(d)** In (a): $|X| \geq a$ implies $X^2 \geq a^2$ only because $a > 0$ (for $a \leq 0$ the event $|X| \geq a$ is everything and the bound would be vacuous or false).

Exercise 4.7. $216 = 2(108)$ is two standard deviations, so Chebyshev gives $P(|Y - 570| \geq 216) \leq 1/2^2 = 1/4$, whatever the shape of the profit distribution.

4.4 Median, Quantiles, and the Mode of a Random Variable

Median and Quantiles of a RV

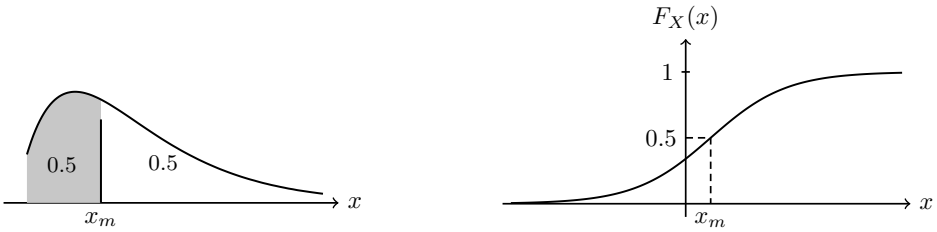
The median of a dataset (or RV) is defined to be: "the value for which 50% of the data is above, and 50% below."

The way we find this value for RV varies with the type of RV

Median for continuous X: The value x_m = median is the point x_m s.t.

$$P(X \le x_m) = P(X > x_m) = 0.50 \qquad \text{i.e.} \quad F_X(x_m) = 0.50$$

(x_m is unique whenever F_X is strictly increasing, which holds for every continuous distribution in these notes.) Equivalently $x_m = F_X^{-1}(0.50)$: the median is where the c.d.f. crosses one half. In pictures, it splits the area under the p.d.f. into two equal halves, and it is read off the c.d.f. by drawing a horizontal line at height 0.50 and dropping to the axis:



Median for discrete RV: Here we define the median as x_m s.t.

$$P(X \le x_m) \ge 0.50, \quad P(X \ge x_m) \ge 0.50$$

In practice, we find x_m by accumulating the c.d.f.:

1. Move through the support points in order until F_X first reaches or passes 0.50.
2. If F_X lands *exactly* on 0.50 at a support point x_k, average with the next support point: $x_m = \frac{x_k + x_{k+1}}{2}$. If F_X *jumps past* 0.50 at x_k, then $x_m = x_k$.

Example 4.11: Let X be a discrete RV with distribution:

x	1	2	3	4	5	6	7	8	9	10
$f(x)$	$2c$	c	$2c$	$6c$	$4c$	$2c$	c	$7c$	c	$6c$

a) Find c s.t. $f_X(x)$ is a valid p.d.f.

$$\sum_{i=1}^{10} f(x_i) = 1 \implies (2c + c + \cdots + 6c) = 1 \implies 32c = 1 \implies c = \tfrac{1}{32}$$

b) Find the median of X.

If we write out $F_X(x)$:

x	1	2	3	4	5	6	7	8	9	10
$F_X(x)$	$2c$	$3c$	$5c$	$11c$	$15c$	$17c$	$18c$	$25c$	$26c$	$32c$

Thus, $P(X < 6) = P(X \le 5) = \frac{15}{32} < 0.50, \quad P(X \le 6) = \frac{17}{32} > 0.50 \implies x_m = 6.$

Example 4.12: Let $f_X(x) = \begin{cases} 2x & 0 \le x \le 1 \\ 0 & \text{o.w.} \end{cases}$ be a continuous RV

a) Find $F_X(x)$.

$$\int_0^x f(t)\,dt = \int_0^x 2t\,dt = t^2\Big|_0^x = x^2$$

b) Find x_m, the median of X.

$$F_X(x_m) = 0.50 \implies x_m^2 = 0.50 \implies x_m = \sqrt{0.50} \implies x_m = 0.7071$$

Definition 4.3: For any $0 < p < 1$, the pth quantile (or percentile) of X is the value x_p s.t.

i) **Continuous** X**:** $P(X \le x_p) = p$, $P(X \ge x_p) = 1 - p$

ii) **Discrete** X**:** Use the same algorithm as for the median, with 0.50 replaced by p: accumulate F_X until it first reaches or passes p. If it lands exactly on p at a support point x_k, then $x_p = \dfrac{x_k + x_{k+1}}{2}$; if it jumps past p at x_k, then $x_p = x_k$.

$Q_1 = x_{.25}$ = First quartile, $\quad Q_3 = x_{.75}$ = Third quartile, $\quad Q_2 = x_m = x_{.50}$ = median = Second quartile.

The pth quantile is also written π_p; x_p and π_p mean the same thing. For continuous X it is $x_p = F_X^{-1}(p)$, and reading it off the c.d.f. is the median picture above with the horizontal line drawn at height p instead of 0.50.

The Mode

Definition 4.4: The mode of a RV X is any point which maximizes the p.d.f. (if a point exists that does). In symbols, $c = \text{argmax}_x\, f_X(x)$: the mode is the "most likely" value of X, the location of the highest peak of the p.d.f.

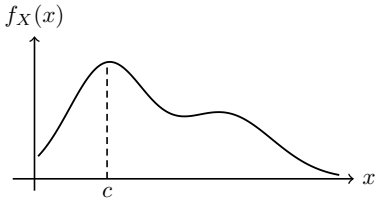




Figure 4.2: *A goldfish-scooping stall at a summer festival.*[8]

Example 4.13: At a goldfish-scooping stall at a Japanese summer festival (a matsuri; Fig. 4.2), the paper scoop tears after a few tries. Let X be the number of goldfish you take home, with distribution:

x	0	1	2	3	4
$f(x)$	0.35	0.25	0.10	0.15	0.15

a) What number of goldfish are you most likely to take home? That is, find the mode of X.

$x = 0$, since this gives the largest $f(x_i)$.

b) At most how many goldfish do half of players take home? That is, find the median of X.

Since $P(X < 1) = 0.35 < 0.5$, and $P(X \leq 1) = 0.60 > 0.50$, $x_m = 1$.

c) What number of goldfish do three-quarters of players take home at most? That is, find the 75th percentile of X.

Since $P(X < 3) = 0.70 < 0.75$, $P(X \leq 3) = 0.85 > 0.75$, $x_{.75} = 3$.

Example 4.14: Let X be a continuous RV with p.d.f.

$$f_X(x) = \begin{cases} \frac{3}{4}x^2(2-x) & 0 \leq x \leq 2 \\ 0 & \text{o.w.} \end{cases}$$

a) Find $F_X(x)$.

$$F_X(x) = \int_0^x f(t)\,dt = \int_0^x \left(\tfrac{6}{4}t^2 - \tfrac{3}{4}t^3\right) dt = \frac{6}{12}t^3 - \frac{3}{16}t^4\Big|_0^x$$
$$= \tfrac{1}{2}x^3 - \tfrac{3}{16}x^4 = \tfrac{1}{16}\left(8x^3 - 3x^4\right)$$

$F_X(2) = \frac{1}{16}\left(8(8) - 3(16)\right) = \frac{1}{16}\left(4(16) - 3(16)\right) = 1$ ✓

b) Find the mode of X.

$$\frac{d}{dx}f_X(x) = \frac{12}{4}x - \frac{9}{4}x^2 = 0 \;\Rightarrow\; 12 - 9x = 0$$
$$\Rightarrow\; 9x = 12 \;\Rightarrow\; x = \tfrac{4}{3}, \quad x = 0 \text{ are roots}$$

$$\frac{d^2}{dx^2} = \frac{12}{4} - \frac{18}{4}x \qquad @\ x = \tfrac{4}{3}: \ \frac{12}{4} - \frac{18}{4}\cdot\tfrac{4}{3} = \frac{36}{12} - \frac{72}{12} < 0 \implies x = \tfrac{4}{3} \text{ is a maximum.}$$

$$@\ x = 0: \ \frac{12}{4} - 0 = \frac{12}{4} > 0 \quad (x = 0 \text{ min})$$

c) Find x^* s.t. $P(X \geq x^*) = 0.20$.

$P(X \geq x^*) = 0.20 \implies P(X \leq x^*) = 0.80 \implies x^* = x_{.80}$

$F(x_{.80}) = 0.80 \implies \frac{1}{16}(8x^3 - 3x^4) = 0.80$, solve for x (not by hand)

$x_{.80} = 1.575$ or $x_{.80} = 2.33$. Since $0 \leq x \leq 2$, $x_{.80} = 1.575$.

Example 4.15: Let X have p.d.f. $f_X(x) = \lambda e^{-\lambda x}$ for $x > 0$, the exponential of Example 4.8, whose mean is $1/\lambda$.

a) Find the median x_m.

$$P(X \leq x_m) = \int_0^{x_m} \lambda e^{-\lambda x}\, dx = -e^{-\lambda x}\Big|_0^{x_m} = 1 - e^{-\lambda x_m} = 0.50 \implies e^{-\lambda x_m} = 0.50 \implies x_m = \frac{-\ln(0.50)}{\lambda} = \frac{\ln 2}{\lambda}$$

b) Find $x_{.90}$, the 90th percentile.

The same computation with 0.90 in place of 0.50: $1 - e^{-\lambda x_{.90}} = 0.90 \Rightarrow x_{.90} = \dfrac{-\ln(0.10)}{\lambda} = \dfrac{\ln 10}{\lambda}$. In general $x_p = -\ln(1-p)/\lambda$.

c) Find the mode.

f_X is decreasing on its whole support, so the p.d.f. is largest at the left endpoint: the mode is 0.

So for the exponential, mode $= 0 <$ median $= \dfrac{\ln 2}{\lambda} \approx \dfrac{0.69}{\lambda} <$ mean $= \dfrac{1}{\lambda}$. The long right tail pulls the mean past the median; for a symmetric density all three would coincide.

Some notes:

i) If the support or domain of X is $a < x < b$, then $\mathbb{E}[X]$ is bounded between a, b.

ii) $F_X(x) = P(X \leq x)$, so $P(X \geq x) = 1 - F_X(x)$.

iii) To find what percentile a given value x sits at, compute $F_X(x) = P(X \leq x)$: if $F_X(x) = 0.62$, then x is the 62nd percentile of X, i.e. $x = x_{0.62}$.

Exercises

Exercise 4.8 (Leafcutter ants on a rainforest trail)**.** Leafcutter ants march in long columns through the Costa Rican rainforest, each carrying a cut leaf fragment back to the underground fungus garden the colony feeds on. On a trail, the time X in seconds between successive ants passing a fixed point on the column is Exp(λ), with p.d.f. $f(x) = \lambda e^{-\lambda x}$ for $x > 0$.

(a) Find the mode of X by inspecting the p.d.f. (Be careful: the maximum occurs at an endpoint of the support.)

(b) Find the median of X by solving $F(x) = 1/2$.

(c) Write down $\mathbb{E}[X]$, and put the mode, median, and mean in increasing order.

(d) Sketch the p.d.f. and mark all three. Explain in one sentence how the shape of the distribution produces the ordering you found.

Exercise 4.9. Let X be uniform on $\{1, 2, 3, 4\}$.

(a) Write down the c.d.f. of X.

(b) Recall that m is a median if $P(X \le m) \ge 1/2$ and $P(X \ge m) \ge 1/2$. Show that *every* value in the interval $[2, 3]$ satisfies this definition.

(c) The convention in these notes resolves the tie by averaging. What value does it give?

(d) Now let Y be uniform on $\{1, 2, 3, 4, 5\}$. Is the median unique here? Explain what makes the two cases different.

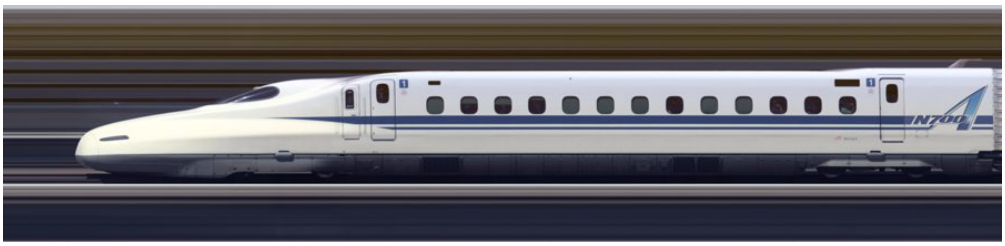


Figure 4.3: *A Shinkansen N700A.* [9]

Exercise 4.10 (Shinkansen delays)**.** Japan's Shinkansen bullet trains (Fig. 4.3) are famously punctual: model the delay X, in minutes, of a randomly chosen train by the c.d.f. $F(x) = 1 - e^{-x/0.4}$ for $x \ge 0$.

(a) Find the median delay.

(b) Find the 90th percentile of the delay.

(c) Find the probability a train is more than a minute late.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 4.8. (a) f is decreasing on $(0, \infty)$, so the maximum is at the left endpoint: the mode is 0. **(b)** $1 - e^{-\lambda m} = \frac{1}{2}$ gives $m = \ln 2/\lambda \approx 0.693/\lambda$. **(c)** $\mathbb{E}[X] = 1/\lambda$, so mode $0 <$ median $0.693/\lambda <$ mean $1/\lambda$. **(d)** The density starts at its peak and decays with a long right tail; the tail pulls the mean to the right of the median, and the peak sits at the far left.

Exercise 4.9. (a) $F(x) = \frac{1}{4}\lfloor x \rfloor$ for $1 \le x < 4$, with $F = 0$ below 1 and 1 from 4 on. **(b)** For any $m \in [2, 3]$: $P(X \le m) \ge P(X \le 2) = \frac{1}{2}$ and $P(X \ge m) \ge P(X \ge 3) = \frac{1}{2}$, so every such m qualifies. **(c)** The tied candidates 2 and 3 average to 2.5. **(d)** For Y uniform on five values, $P(Y \le 3) = \frac{3}{5} \ge \frac{1}{2}$ and $P(Y \ge 3) = \frac{3}{5} \ge \frac{1}{2}$ while 2 and 4 fail, so the median 3 is unique: with an odd number of equally likely values the c.d.f. jumps past $\frac{1}{2}$ at the middle value instead of landing exactly on it.

Exercise 4.10. (a) $1 - e^{-m/0.4} = \frac{1}{2}$ gives $m = 0.4 \ln 2 = 0.277$ minutes. **(b)** $e^{-x/0.4} = 0.1$ gives $x_{0.9} = 0.4 \ln 10 = 0.921$ minutes. **(c)** $P(X > 1) = e^{-1/0.4} = e^{-2.5} = 0.082$.

5 Named Discrete Distributions

5.1 Bernoulli and Binomial Distributions

Named Discrete Distributions

Some random variables occur often enough in practice that their behavior has been thoroughly studied, and $M_X(t)$, μ_X, σ_X^2, $F_X(x)$ can be defined.

Bernoulli Distribution

The simplest discrete RV records a single trial: let $X = 1$ if the trial is a "success" (with probability p) and $X = 0$ if it is a "failure" (with probability $q = 1 - p$). We write $X \sim \text{Bernoulli}(p)$ ("$\sim$" reads "is distributed as"). Directly from the definitions,

$$\mathbb{E}[X] = \sum_{x=0}^{1} x\, f_X(x) = 0(1-p) + 1(p) = p.$$

For the variance, X takes only the values 0 and 1, so $X^2 = X$ and $\mathbb{E}[X^2] = \mathbb{E}[X] = p$:

$$\sigma_X^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p) = pq.$$

The m.g.f. is a two-term sum:

$$M_X(t) = \mathbb{E}[e^{tX}] = e^{t(0)}(1-p) + e^{t(1)}p = pe^t + q.$$

A binomial RV is a sum of n independent Bernoulli trials, which is where the formulas $\mathbb{E}[X] = np$, $\sigma_X^2 = npq$, and $M_X(t) = (pe^t + q)^n$ below come from.

Binomial Distribution

This distribution assumes:

1. n independent experiments/trials are conducted
2. The outcome of each trial is a "success" or "failure" (arbitrary)
3. The probability of success is denoted p, and is the same for all trials
4. $X =$ # of successes out of n trials

Equivalently, let $X_i \sim \text{Bernoulli}(p)$ record trial i, with the X_i independent and p the same for all of them; then $X = X_1 + \cdots + X_n$. For $n = 2$ the four branches of the tree give the p.d.f. directly:

$$P(X=0) = (1-p)^2, \qquad P(X=1) = (1-p)p + p(1-p) = 2p(1-p), \qquad P(X=2) = p^2.$$

For $n = 3$ the eight leaves group by their total:

$$P(X=0) = \binom{3}{0}(1-p)^3, \quad P(X=1) = \binom{3}{1}p(1-p)^2, \quad P(X=2) = \binom{3}{2}p^2(1-p), \quad P(X=3) = \binom{3}{3}p^3.$$

Each leaf with x successes has probability $p^x(1-p)^{n-x}$, and the binomial coefficient counts how many leaves land on the same total: three paths lead to $X = 1$ because the success can come first, second, or third.

In general, $P(X = x) = \binom{n}{x}p^x(1-p)^{n-x}, \quad x = 0, 1, \ldots, n$

where $\binom{n}{x}$ = # of ways to get x successes out of n, $\quad p^x$ = probability of x successes, $\quad (1-p)^{n-x}$ = probability of $(n-x)$ ("the rest") failures.

X is said to be distributed binomial, with parameters n, p. I.e. $X \sim \text{Bin}(n, p)$. ($\sim$ = "is distributed as")

Notice:

$$\sum_{x=0}^{n}\binom{n}{x}p^x(1-p)^{n-x} = (p + (1-p))^n = 1^n = 1$$

One can also show:

$$\mu_X = np, \quad \sigma_X^2 = np(1-p), \quad M_X(t) = (pe^t + q)^n, \quad t \in \mathbb{R}$$

Writing $X = X_1 + \cdots + X_n$ does the work three times. Linearity gives the mean,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] = p + p + \cdots + p = np,$$

independence adds the variances (Section 9.5) to give $np(1-p) = npq$, and independence again factors the m.g.f.:

$$\begin{aligned} M_X(t) = \mathbb{E}\left[e^{t(X_1+\cdots+X_n)}\right] &= \mathbb{E}\left[e^{tX_1}e^{tX_2}\cdots e^{tX_n}\right] = \mathbb{E}\left[e^{tX_1}\right]\mathbb{E}\left[e^{tX_2}\right]\cdots\mathbb{E}\left[e^{tX_n}\right] \\ &= M_{X_1}(t)\cdots M_{X_n}(t) = (pe^t + q)^n. \end{aligned}$$

Differentiating at $t = 0$ returns np as a check. The direct sum for the mean is the algebra below.

Hint:

$$\begin{aligned} \mathbb{E}[X] = \mu_X &= \sum_{x=0}^{n} x\, f_X(x) = \sum_{x=0}^{n}\frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}\cdot x \\ &= \sum_{x=1}^{n}\frac{n!}{(x-1)!(n-x)!}p^x(1-p)^{n-x} \end{aligned}$$

Let $k = x - 1 \Rightarrow x = k + 1$:

$$= \sum_{k=0}^{n-1}\frac{n!}{k!(n-1-k)!}p^{k+1}(1-p)^{n-1-k} = np\Big(\sum \text{p.d.f. of Bin}(n-1, p)\Big) = np$$

Python note: In `scipy.stats`: `binom.pmf(8, 10, 0.8)` gives $P(X = 8)$, `binom.cdf(8, 10, 0.8)` gives $F_X(8)$, and `binom.rvs(10, 0.8, size=1000)` simulates 1000 draws. Every named distribution in these notes follows this same `pmf`/`pdf`, `cdf`, `ppf` (quantile), `rvs` pattern.

Example 5.1: Link, the hero of Nintendo's Zelda games, hits 80% of his bow shots at a target. He fires 10 arrows, and we assume the shots are independent. Let X be the number of hits, so $X \sim \text{Bin}(10, 0.8)$.

a) What is the chance Link hits exactly 8 of his 10 shots? That is, find $P(X = 8)$.

$$P(X = 8) = \binom{10}{8}(0.8)^8(0.2)^2 = 45(0.16777)(0.04) = 0.302$$

b) What is the chance he hits at least 9? That is, find $P(X \geq 9)$.

$$P(X \geq 9) = \binom{10}{9}(0.8)^9(0.2) + \binom{10}{10}(0.8)^{10} = 0.2684 + 0.1074 = 0.376$$

c) How many hits should he expect, and how much does that number typically vary? That is, find $\mathbb{E}[X]$, $\text{Var}[X]$, and $\text{SD}[X]$.

$$\mathbb{E}[X] = np = 10(0.8) = 8, \qquad \text{Var}[X] = npq = 10(0.8)(0.2) = 1.6, \qquad \text{SD}[X] = 1.265$$

Note: The most likely single outcome is $X = 8$, matching $\mathbb{E}[X]$, but its probability is only about 30%. A mean of 8 does *not* mean 8 is likely: it means 8 is the long-run average.

Example 5.2: Suppose a salsa class has 8 regulars, and each shows up to tonight's session independently with prob 0.60.

a) Find the prob at least two regulars show up.

$X = \#$ of regulars who show up out of 8, $X \sim \text{Bin}(8, 0.60)$

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - P(X \leq 1) = 1 - P(X = 0) - P(X = 1) \\ &= 1 - \tbinom{8}{0}(0.60)^0(0.40)^8 - \tbinom{8}{1}(0.60)^1(0.40)^7 \\ &= 1 - 0.00065536 - 0.00786432 \\ &= 0.9914803 \end{aligned}$$

b) Find the expected # of regulars who will show up, and the std dev.

$$\mu_X = np = 8(0.6) = 4.8, \qquad \sigma_X = \sqrt{np(1-p)} = 1.386$$

c) Find the prob that more regulars show up than two std dev above the mean.

$$\begin{aligned} P(X > \mu_X + 2\sigma_X) &= P(X > 4.8 + 2(1.386)) = P(X > 7.572) \\ &= P(X = 8) = \tbinom{8}{8}(0.6)^8(0.40)^0 = 0.01679 \end{aligned}$$

Theorem 5.1 (Mode of a Binomial): Let $X \sim \text{Bin}(n, p)$.

Let $m = \lfloor p(n+1) \rfloor$, i.e. $p(n+1)$ rounded down to the nearest integer. If $p(n+1)$ is an integer, then there are two modes of X, at m and $m-1$. If $p(n+1)$ is not an integer, the unique mode of X is m.

We omit the proof.

Exercises

Exercise 5.1. A book has n typos. Two proofreaders, Prue and Frida, independently read the book. Prue catches each typo with probability p_1, independently across typos, and likewise Frida catches each typo with probability p_2. Let X be the number of typos caught by at least one of the two proofreaders. Find the distribution of X.

Exercise 5.2. A hatchery moves sea turtle eggs from the beach in trays of 12.

(a) Suppose that in any particular tray, each individual egg has an independent 10% chance of being broken. What is the probability that out of the 12 eggs in the tray, 2 or fewer are broken? Write your answer as an exact formula and then as a number rounded to 3 decimal places.

(b) The hatchery flags a tray when 3 or more out of the 12 are broken. Thus trays with 2 or fewer broken eggs are "good" and trays with 3 or more are "bad." Tonight the hatchery will receive 100 trays. Let X be the number of good trays out of the 100. Write a formula for the p.d.f. of X.



Figure 5.1: *A folkloric dance and percussion company performing in Salvador, Brazil.* [10]

Exercise 5.3 (A folkloric show in Salvador)**.** Salvador, Bahia, is the heart of Afro-Brazilian culture, and its folkloric companies perform the dances of the orixás and capoeira to live drumming. Each of the 12 dancers and drummers in one such company (Fig. 5.1) is available for a given night's show independently with probability 0.9. Let X be the number who perform.

(a) Probability the full company performs.

(b) Probability at least 10 perform.

(c) Mean and standard deviation of X.

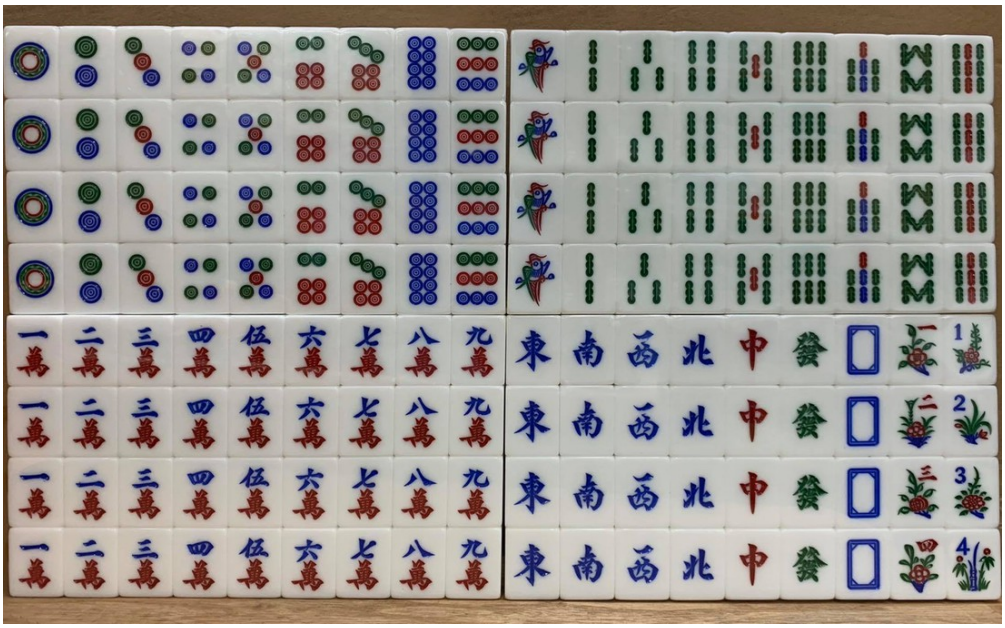


Figure 5.2: *A full mahjong set.* [11]

Exercise 5.4 (Dragons in the wall)**.** A mahjong set (Fig. 5.2) at a Chinatown parlor has 136 tiles, 12 of which are dragons (the red, green, and white honor tiles). Each hand begins by building and shuffling the wall (the stacked rows of face-down tiles) afresh, so the first tile you draw in one hand is independent of the first tile you drew in the last.

(a) Probability the first draw of a hand is a dragon.

(b) Expected number of hands, out of 13, whose first draw is a dragon.

(c) Probability that none of the 13 first draws is a dragon.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 5.1. A given typo is missed by both proofreaders with probability $(1-p_1)(1-p_2)$ (independence), so it is caught by at least one with probability $p = 1-(1-p_1)(1-p_2)$. Typos are independent trials, so $X \sim \text{Bin}\big(n,\ 1-(1-p_1)(1-p_2)\big)$.

Exercise 5.2. (a) The number broken is $\text{Bin}(12, 0.1)$: $P(\le 2) = \sum_{k=0}^{2}\binom{12}{k}(0.1)^k(0.9)^{12-k} = 0.2824 + 0.3766 + 0.2301 = 0.889$. **(b)** Each tray is good independently with probability 0.889, so $X \sim \text{Bin}(100, 0.889)$: $P(X=x) = \binom{100}{x}(0.889)^x(0.111)^{100-x}$ for $x = 0, \dots, 100$.

Exercise 5.3. $X \sim \text{Bin}(12, 0.9)$. **(a)** $0.9^{12} = 0.282$. **(b)** $P(X \ge 10) = \binom{12}{10}0.9^{10}0.1^2 + 12(0.9)^{11}(0.1) + 0.9^{12} = 0.230 + 0.377 + 0.282 = 0.889$. **(c)** $\mu = 10.8$, $\sigma = \sqrt{1.08} = 1.039$.

Exercise 5.4. (a) $12/136 = 3/34 = 0.088$. **(b)** The count over 13 hands is $\text{Bin}(13, 3/34)$ with mean $13(3/34) = 1.15$. **(c)** $(31/34)^{13} = 0.301$.

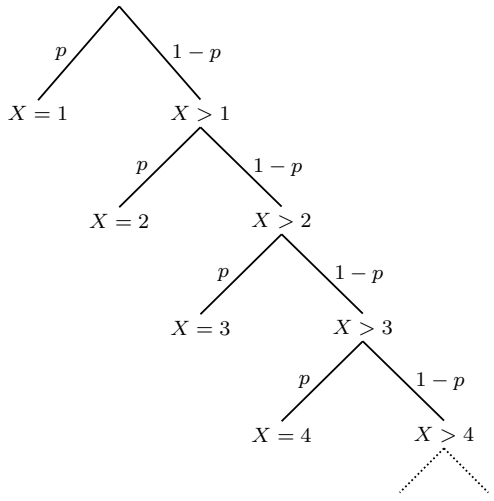
5.2 The Geometric Distribution

Geometric Distribution

A geometric distribution has the same assumptions as a binomial, except X = # of trials until first success. Then,

$$P(X=x) = (1-p)^{x-1}p, \quad x = 1, 2, \dots, \quad 0 < p < 1, \quad X \sim \text{Geo}(p)$$

The distribution is easiest to picture as a tree: at each trial the first success either arrives (probability p, and we stop) or it does not (probability $1-p$, and we keep going).



$P(X=1) = p, \qquad P(X>1) = 1-p$

$P(X=2) = (1-p)p, \qquad P(X>2) = (1-p)^2$

$P(X=3) = (1-p)^2p, \qquad P(X>3) = (1-p)^3$

$P(X=4) = (1-p)^3p, \qquad P(X>4) = (1-p)^4$

$\vdots$

Reading down the right-hand edge, "still going after k trials" is k failures in a row, so $P(X > k) = (1-p)^k$ with no sum needed.

This is a valid p.d.f.: each $f_X(x) = (1-p)^{x-1}p \geq 0$ since $0 < p < 1$ and $0 < 1-p < 1$, and the values sum to one because they form a geometric series with ratio $1-p$,

$$\sum_{x=1}^{\infty}(1-p)^{x-1}p = p\sum_{x=1}^{\infty}(1-p)^{x-1} = p\big(1+(1-p)+(1-p)^2+\cdots\big) = p\cdot\frac{1}{1-(1-p)} = 1. \qquad \square$$

Derivation of $\mathbb{E}[X]$*:*

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=1}^{\infty}x(1-p)^{x-1}p = \sum_{x=1}^{\infty}(x+1-1)(1-p)^{x-1}p \\ &= \sum_{x=1}^{\infty}(x-1)(1-p)^{x-1}p + \underbrace{\sum_{x=1}^{\infty}(1-p)^{x-1}p}_{=1\text{ (p.d.f. summed over } x=1)}\end{aligned}$$

Let $j = x-1 \Rightarrow x = j+1$:

$$\begin{aligned}&= \sum_{j=0}^{\infty}j(1-p)^j p + 1 = 0 + \sum_{j=1}^{\infty}j(1-p)^j p + 1 \\ &= \sum_{j=1}^{\infty}j(1-p)^{j-1}(1-p)p + 1 = (1-p)\sum_{j=1}^{\infty}j(1-p)^{j-1}p + 1\end{aligned}$$

$$\mathbb{E}[X] = (1-p)\mathbb{E}[X] + 1 \;\Rightarrow\; \mathbb{E}[X] = \mathbb{E}[X] - p\mathbb{E}[X] + 1 \;\Rightarrow\; p\mathbb{E}[X] = 1 \;\Rightarrow\; \mathbb{E}[X] = \tfrac{1}{p}$$

$\square$

Thus, if $X \sim \text{Geo}(p)$, $\mathbb{E}[X] = \frac{1}{p}$.

One may also show $\sigma_X^2 = \dfrac{1-p}{p^2}$ (so $\text{SD}[X] = \sqrt{1-p}/p$), $\quad M_X(t) = \dfrac{pe^t}{1-(1-p)e^t}, \quad t < -\ln(1-p)$.

Note: The derivation of $\mathbb{E}[X]$ turns the unknown into an equation in itself, $\mathbb{E}[X] = (1-p)\mathbb{E}[X] + 1$. That is the tree again: either the first trial succeeds, or one trial has been spent and the process starts over.

Example 5.3: A kingfisher makes independent dives and catches a fish on each with probability $p = 0.1$; let X be the number of dives up to and including its first catch, $X \sim \text{Geo}(0.1)$. Then $\mathbb{E}[X] = 1/p = 10$ dives on average, $\text{SD}[X] = \sqrt{0.9}/0.1 \approx 9.5$ dives, and the probability it needs more than five dives is the probability the first five all come up empty, $P(X > 5) = (0.9)^5 \approx 0.59$, read straight off the tree.

Python note: `scipy.stats.geom` uses the same convention as these notes (X counts trials up to and including the first success): `geom.pmf(4, 1/6)` is the $P(X = 4)$ of the die example below, and `geom.sf(6, 1/6)` (the *survival function* $P(X > 6)$) matches part b).

Example 5.4: Roll a fair die repeatedly until the first 6 appears. Let X be the number of rolls needed, so $X \sim \text{Geo}(1/6)$ (remember X counts the roll on which the 6 occurs, so $X \geq 1$).

a) Find $P(X = 4)$, the probability the first 6 appears on the fourth roll.

$$P(X = 4) = \left(\tfrac{5}{6}\right)^3 \left(\tfrac{1}{6}\right) = \tfrac{125}{1296} = 0.0965$$

The three failures come first, then the success.

b) What is the chance it takes more than six rolls to see the first 6? That is, find $P(X > 6)$.

Rather than summing, note that $X > 6$ means the first six rolls were all failures:

$$P(X > 6) = \left(\tfrac{5}{6}\right)^6 = 0.335$$

c) How many rolls should you expect to need, and how much does that number typically vary? That is, find $\mathbb{E}[X]$, σ_X^2, and $\text{SD}[X]$.

$$\mathbb{E}[X] = \tfrac{1}{p} = 6, \qquad \sigma_X^2 = \tfrac{1-p}{p^2} = \tfrac{5/6}{1/36} = 30, \qquad \text{SD}[X] = 5.48$$

d) Given that the first six rolls were not 6's, what is the probability we need more than four additional rolls?

Using the formula from b) twice,

$$P(X > 10 \mid X > 6) = \frac{P(X > 10)}{P(X > 6)} = \frac{(5/6)^{10}}{(5/6)^6} = \left(\tfrac{5}{6}\right)^4 = P(X > 4)$$

Note: The die does not remember the six failures: the count starts over. This is the *memoryless property*, and the geometric is the only discrete distribution that has it. The exponential (Section 6.2) is its continuous counterpart.

Example 5.5: A sleeping Bulborb (a big, slow predator from the Pikmin games) lies beside the path back to the Onion. Each time the Pikmin squad walks past, it wakes with probability 0.01, independently of every other pass.

a) What is the expected # of passes until the Bulborb first wakes?

$X \sim \text{Geo}(0.01)$, $\quad \mu_X = \tfrac{1}{p} = \tfrac{1}{0.01} = 100$ passes.

b) What is the prob the Bulborb first wakes after the 2nd pass?

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) = 1 - P(X = 1) - P(X = 2) \\ &= 1 - (0.01)(0.99)^0 - (0.01)(0.99)^1 \\ &= 1 - 0.01 - 0.0099 = 0.9801 \end{aligned}$$

Note: In general $P(X > k) = (1-p)^k$, the probability that the first k trials all fail; here $(0.99)^2 = 0.9801$, matching the computation above. The geometric distribution is also memoryless: given no wake-up in the first k passes, the number of *additional* passes needed is again $\text{Geo}(p)$. It is the discrete analogue of the exponential's memoryless property (Section 6.2).

Exercises

Exercise 5.5. Let $X \sim \text{Geo}(p)$, so $P(X = k) = q^{k-1}p$ for $k \geq 1$ with $q = 1 - p$. Write a summation formula for $\mathbb{E}[2^X]$. By identifying the sum as a geometric series, find the range of values of p that make the expected value finite, and compute $\mathbb{E}[2^X]$ for p in this range. Then do the same for $\mathbb{E}[2^{-X}]$ and explain why that answer is finite for every p.



Figure 5.4: *Ochre sea stars in a tide pool at Half Moon Bay.* [12]

Exercise 5.6 (Sea stars in the tide pools)**.** At a low tide at the Fitzgerald Marine Reserve tide pools in Half Moon Bay, each pool you check holds an ochre sea star (Fig. 5.4) with probability $1/8$, independently of the others. Let X be the number of pools you check until you find one.

(a) Mean and standard deviation of X.

(b) Probability you need to check more than 10 pools.

(c) After 10 empty pools, the probability you need more than 5 more. Which property did you use?



Figure 5.5: *Osiris, from the tomb of Nefertari.* [13]

Exercise 5.7 (Isis searches for Osiris)**.** In Egyptian mythology Osiris (Fig. 5.5) was the first king of Egypt and later the god of the dead, the green-skinned figure wrapped like a mummy in tomb paintings. His brother Set murdered him and scattered the pieces of his body along the Nile, and his wife Isis set out to search the river for them. Suppose that on each day of searching she finds a piece with probability 0.2, independently of other days, and let X be the number of days until she finds the first piece.

(a) Find $P(X = 3)$, the probability the first piece turns up on the third day.

(b) How many days should Isis expect to search before finding the first piece, and how variable is that? That is, find $\mathbb{E}(X)$ and $\text{Var}(X)$.

(c) After ten days of searching she has found nothing. Find the probability she needs more than five more days, and name the property you used.

(d) The myth counts 14 pieces. If each piece takes its own independent $\text{Geo}(0.2)$ number of days, find the mean and standard deviation of the total number of days to recover all 14.

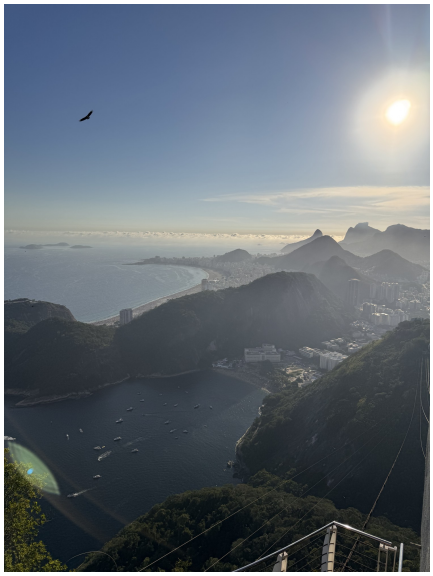


Figure 5.6: *Copacabana and Botafogo Bay from the summit of Sugarloaf, Rio de Janeiro.* [14]

Exercise 5.8 (Vultures over Botafogo Bay)**.** From the summit of Sugarloaf (Fig. 5.6) a visitor watches black vultures, the urubus that ride the thermals over Rio, cross the view above Botafogo Bay. They pass at an average rate of 2 per minute, and the count in any interval is Poisson.

(a) Probability that no vulture crosses during a given minute.

(b) Probability that at least 3 cross during a given minute.

(c) Expected number that cross during the 5-minute cable-car ride down, and the probability that exactly 10 do.

Exercise 5.9 (Learning to surf at Waikiki)**.** A beginner at Waikiki, Honolulu's gentle beginner surf beach, catches each wave they paddle for with probability 0.2, independently of every other wave.

(a) Let X be the number of waves paddled for up to and including the first ride. Find the mean of X and $P(X > 8)$.

(b) In a session of 10 waves, find the probability of at least 3 rides.

(c) Mean and standard deviation of the number of rides in 10 waves.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 5.5. $\mathbb{E}[2^X] = \sum_{k\geq 1} 2^k q^{k-1} p = 2p \sum_{k\geq 1} (2q)^{k-1}$, a geometric series with ratio $2q$, which converges exactly when $2q < 1$, i.e. $p > \frac{1}{2}$; then $\mathbb{E}[2^X] = \dfrac{2p}{1-2q} = \dfrac{2p}{2p-1}$. For $\mathbb{E}[2^{-X}] = \frac{p}{2} \sum_{k\geq 1} (q/2)^{k-1} = \dfrac{p/2}{1-q/2} = \dfrac{p}{1+p}$, the ratio $q/2 < 1$ for every p, so it is always finite: halving at each step cannot outrun the geometric tail.

Exercise 5.6. $X \sim \text{Geo}(1/8)$. **(a)** Mean 8, SD $\sqrt{7/8} \cdot 8 = \sqrt{56} = 7.48$. **(b)** $(7/8)^{10} = 0.263$. **(c)** By memorylessness, $(7/8)^5 = 0.513$.

Exercise 5.7. **(a)** $P(X = 3) = (0.8)^2(0.2) = 0.128$. **(b)** $\mathbb{E}(X) = 1/p = 5$ days and $\text{Var}(X) = (1-p)/p^2 = 0.8/0.04 = 20$. **(c)** By memorylessness, $P(X > 15 \mid X > 10) = P(X > 5) = 0.8^5 = 0.328$: ten empty days do not make the eleventh any more promising. **(d)** The total is a sum of 14 independent $\text{Geo}(0.2)$ variables, so its mean is $14(5) = 70$ days and its variance $14(20) = 280$, giving a standard deviation of $\sqrt{280} = 16.7$ days.

Exercise 5.8. $X \sim \text{Pois}(2)$ per minute. **(a)** $P(X = 0) = e^{-2} = 0.135$. **(b)** $P(X \geq 3) = 1 - e^{-2}(1+2+2) = 1 - 5e^{-2} = 0.323$. **(c)** Over 5 minutes the count is $\text{Pois}(10)$, so the expected number is 10 and $P(Y = 10) = e^{-10}10^{10}/10! = 0.125$: rates scale with the length of the window.

Exercise 5.9. **(a)** $X \sim \text{Geo}(0.2)$: mean 5, $P(X > 8) = 0.8^8 = 0.168$. **(b)** $Y \sim \text{Bin}(10, 0.2)$: $1 - (0.107 + 0.268 + 0.302) = 0.322$. **(c)** Mean 2, SD $\sqrt{1.6} = 1.26$.

5.3 The Poisson Distribution

Poisson Distribution

The Poisson distribution is typically used to model the # of times something occurred in a specific time period, or in a specific physical construct.

Example 5.6: # of typos on a page; # of bacteria in a vial; # of spam emails in a day; etc.

A Poisson RV X has parameter λ, denoted $X \sim \text{Pois}(\lambda)$, where λ = rate of event. If $X \sim \text{Pois}(\lambda)$, then the p.d.f. is

$$f_X(x) = e^{-\lambda}\frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots, \quad \lambda > 0$$

This is a valid p.d.f. because the sum is the Taylor series of e^λ:

$$\sum_{x=0}^{\infty} e^{-\lambda}\frac{\lambda^x}{x!} = e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^x}{x!} = e^{-\lambda}e^{\lambda} = 1.$$

Further, $\mathbb{E}[X] = \lambda$, $\sigma_X^2 = \lambda$ (so $\text{SD}[X] = \sqrt{\lambda}$), $M_X(t) = e^{\lambda(e^t-1)}$, $t \in \mathbb{R}$.

Python note: `1 - poisson.pmf(0, 4)` and `poisson.cdf(4, 4) - poisson.cdf(1, 4)` handle the karaoke example below.

Note: The Poisson arises as a limit of the binomial: if n is large, p is small, and $\lambda = np$ is moderate, then $\text{Bin}(n, p) \approx \text{Pois}(\lambda)$. This is why it models counts of rare events among many opportunities, such as typos among thousands of characters. Rates also rescale: if typos occur at rate λ per page, then the count over k pages is $\text{Pois}(k\lambda)$.

Proof of $\mathbb{E}[X]$*:*

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=0}^{\infty}\frac{x\,e^{-\lambda}\lambda^x}{x!} \qquad \left(\text{since at } x = 0,\ \tfrac{xe^{-\lambda}\lambda^x}{x!} = 0\right)\\ &= \sum_{x=1}^{\infty}\frac{e^{-\lambda}\lambda^x}{(x-1)!}\end{aligned}$$

Let $j = x - 1 \Rightarrow x = j + 1$:

$$= \sum_{j=0}^{\infty}\frac{e^{-\lambda}\lambda^{j+1}}{j!} = \lambda e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda e^{-\lambda}\left[e^{\lambda}\right] = \lambda \qquad \square$$

Derivation of the m.g.f. Fold e^{tx} into the power of λ; the sum is then the Taylor series of e^u with $u = \lambda e^t$:

$$\begin{aligned}M_X(t) = \mathbb{E}[e^{tX}] &= \sum_{x=0}^{\infty} e^{tx}\,e^{-\lambda}\frac{\lambda^x}{x!} = e^{-\lambda}\sum_{x=0}^{\infty}\frac{(e^t)^x\lambda^x}{x!}\\ &= e^{-\lambda}\underbrace{\sum_{x=0}^{\infty}\frac{(\lambda e^t)^x}{x!}}_{=e^{\lambda e^t}} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}, \qquad t \in \mathbb{R}. \qquad \square\end{aligned}$$

An alternative is to expand $e^{tX} = 1 + tX + t^2X^2/2! + \cdots$ inside the expectation, giving $M_X(t) = 1 + t\mathbb{E}[X] + t^2\mathbb{E}[X^2]/2! + \cdots$, and to sum the moments; it lands in the same place but needs every moment first, so the direct route is the one to remember.

Derivation of the variance. $\text{Var}[X] = \mathbb{E}[X^2] - \lambda^2$, and the cleanest route to $\mathbb{E}[X^2]$ is the factorial moment $\mathbb{E}[X(X-1)]$, because $x(x-1)$ cancels the top two factors of $x!$ exactly as x cancelled one of them in the proof of $\mathbb{E}[X]$. The $x = 0$ and $x = 1$ terms vanish, so

$$\mathbb{E}[X(X-1)] = \sum_{x=0}^{\infty} x(x-1)\, e^{-\lambda}\frac{\lambda^x}{x!} = e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^x}{(x-2)!} = \lambda^2 e^{-\lambda}\underbrace{\sum_{x=2}^{\infty}\frac{\lambda^{x-2}}{(x-2)!}}_{=e^{\lambda}} = \lambda^2.$$

Since $X(X-1) = X^2 - X$, linearity gives $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = \lambda^2 + \lambda$, and therefore

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda. \qquad \square$$

The same trick gives every factorial moment: $\mathbb{E}[X(X-1)(X-2)] = \lambda^3$, and in general $\mathbb{E}[X(X-1)\cdots(X-k+1)] = \lambda^k$.

Theorem 5.2 (Mode of a Poisson): If $X \sim \text{Pois}(\lambda)$ and λ is an integer, then X has two modes at λ and $\lambda - 1$. If λ is not an integer, then the mode is λ rounded down to nearest int.

Example 5.7: Let X be the number of earthquakes of magnitude 5 or greater within 100 km of San Francisco in one year, with rate $\lambda = 1/2$ (one earthquake every two years on average), so $X \sim \text{Pois}(1/2)$. The probability of between 2 and 4 earthquakes next year: the three events are disjoint, so

$$P(2 \le X \le 4) = P(X=2) + P(X=3) + P(X=4) = \frac{e^{-1/2}(1/2)^2}{2!} + \frac{e^{-1/2}(1/2)^3}{3!} + \frac{e^{-1/2}(1/2)^4}{4!} \approx 0.09.$$

Example 5.8: At karaoke night, the # of flubbed lyrics per song is Poisson distributed, with $\lambda = 4$ (four flubs per song, which anyone who has heard a late-night "Dancing Queen" will accept).

a) Calculate the probability there is at least one flub in a given song.

$$P(X \ge 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - \frac{e^{-\lambda}\lambda^0}{0!} = 1 - e^{-\lambda} = 1 - e^{-4} = 0.982$$

b) Find the probability that a song has between 2 and 4 flubs.

$$\begin{aligned} P(2 \le X \le 4) &= P(X=2) + P(X=3) + P(X=4) \\ &= e^{-\lambda}\left(\frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \frac{\lambda^4}{4!}\right) = e^{-\lambda}\frac{\lambda^2}{2!}\left[1 + \frac{\lambda}{3} + \frac{\lambda^2}{12}\right] \\ &= e^{-4}\cdot 8\left[1 + \frac{4}{3} + \frac{16}{12}\right] = e^{-4}\cdot 8\cdot\frac{11}{3} \approx 0.537 \end{aligned}$$

c) What is the most frequently occurring value of X?

Since $\lambda = 4$ is an integer, there are two modes, at $\lambda - 1 = 3$ and $\lambda = 4$ flubs.

d) Interpret the value σ_X.

$\sigma_X = \sqrt{4} = 2$: the number of flubs in a song typically differs from its mean of 4 by about 2 flubs, so most songs land somewhere between two and six.

Recap of Discrete RV

- **Bernoulli:** $X = 1$ if a single trial succeeds, 0 if it fails
- **Binomial:** $X = \#$ of "successes" out of n trials
- **Geometric:** $X = \#$ of trials up to and including first success
- **Poisson:** $X = \#$ of events in time frame or physical set

The p.d.f., $\mathbb{E}[X]$, σ_X^2, and the m.g.f. are recorded above for all four.

Exercises

Exercise 5.10. Suppose you have a 100-sided die and roll it 300 times. Let X be the total number of times you roll a 1, so $X \sim \text{Bin}(300, 0.01)$ exactly, with $\mathbb{E}[X] = 3$. The Poisson-limit note above says that approximately $X \sim \text{Pois}(3)$. Compute the exact values of $P(X = k)$ for $k = 0, 1, 2, 3, 4$ to four decimal places (software encouraged), and compare them to the Poisson approximations.

Exercise 5.11 (Cats on an Athens terrace)**.** Stray cats wander across a taverna terrace in Plaka, the old quarter below the Acropolis in Athens, at an average rate of 3 per hour, following a Poisson model.

(a) Probability no cat appears during a 20-minute lunch.

(b) Probability at least two appear during an hour.

(c) Mean and standard deviation of the count over a two-hour dinner.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 5.10. Exact $\text{Bin}(300, 0.01)$ versus $\text{Pois}(3)$:

k	0	1	2	3	4
binomial	0.0490	0.1486	0.2244	0.2252	0.1689
Poisson	0.0498	0.1494	0.2240	0.2240	0.1680

The two agree to about 0.001: n is large, p is small, and $np = 3$ is moderate, exactly the Poisson-limit regime.

Exercise 5.11. (a) Rate 1 in 20 minutes: $e^{-1} = 0.368$. **(b)** $\text{Pois}(3)$: $1 - e^{-3}(1 + 3) = 0.801$. **(c)** $\lambda = 6$: mean 6, SD 2.45.

6 Named Continuous Distributions

6.1 The Normal Distribution

Named Continuous RVs

There are also specific shapes of continuous distributions that are thought to be very common.

Normal Distribution

This is one of the most important distributions in statistics, for two main reasons.

I) It is believed that many distributions of RVs in the natural world follow a normal distribution.

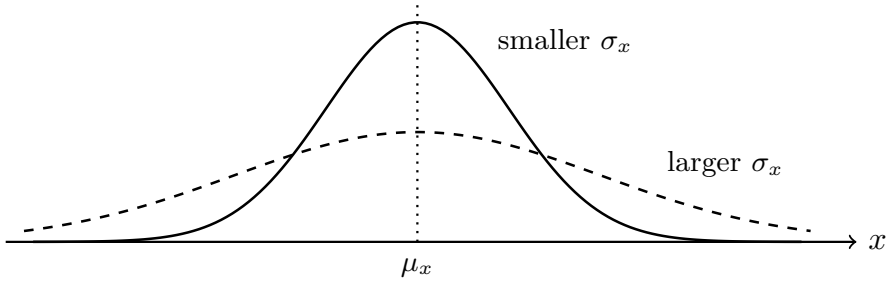
II) Certain combinations of RVs, regardless of their distribution, *approximately* follow a normal distribution (the CLT, Section 10.1).

If X is distributed normal, the p.d.f. is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left\{\frac{-(x-\mu_X)^2}{2\sigma_X^2}\right\}, \quad x \in \mathbb{R},\ \mu_X \in \mathbb{R},\ \sigma_X > 0$$

We then denote X as $X \sim N(\mu_X, \sigma_X^2)$.

The graph of $X \sim N(\mu_X, \sigma_X^2)$ is symmetric and unimodal, i.e. the bell curve, where μ_X gives the center and σ_X controls the spread.



A special case of a normal distribution is when $\mu_X = 0$, $\sigma_X = 1$. Then, if $X \sim N(0,1)$ it is called **standard normal**, which we typically denote by Z (i.e., $Z \sim N(0,1)$).

Facts about the normal distribution:

(i) The mode is μ_X

(ii) The median is μ_X

(iii) $f(\mu_X - x) = f(\mu_X + x)$ (The distribution is symmetric about μ)

(iv) The c.d.f. of $Z \sim N(0,1)$ is often denoted by Φ; if $X \sim N(\mu_X, \sigma_X^2)$, then $P(X \le x) = F_X(x) = \Phi\left(\dfrac{x - \mu_X}{\sigma_X}\right)$

The 68–95–99.7 rule: for any normal distribution,

$$P(|X - \mu_X| \le \sigma_X) \approx 0.683, \qquad P(|X - \mu_X| \le 2\sigma_X) \approx 0.954, \qquad P(|X - \mu_X| \le 3\sigma_X) \approx 0.997$$

About 68% of the probability lies within one standard deviation of the mean, 95% within two, and 99.7% within three. Compare these to Chebyshev's worst-case guarantees of ≥ 0, $\ge 3/4$, and $\ge 8/9$: the normal distribution concentrates much more tightly than Chebyshev requires.

Proposition 6.1: If $X \sim N(\mu_X, \sigma_X^2)$, then $Z = \dfrac{X - \mu_X}{\sigma_X} \sim N(0,1)$.

Proof:

$$F_Z(y) = P(Z \le y) = P\left(\frac{X - \mu_X}{\sigma_X} \le y\right) = P(X \le \mu_X + \sigma_X y)$$
$$= \int_{-\infty}^{\mu_X + \sigma_X y} \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-(t-\mu_X)^2/(2\sigma_X^2)}\, dt$$

Let $\dfrac{t - \mu_X}{\sigma_X} = z \;\Rightarrow\; t = \mu_X + \sigma_X z, \quad -\infty < t \le y, \quad dt = \sigma_X\, dz$

$$\Rightarrow \quad F_Z(y) = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-z^2/2}\, \sigma_X\, dz = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz$$

So finally,

$$f_Z(y) = \frac{d}{dy} F_Z(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \;\sim\; N(0,1).$$

Finally, if $X \sim N(\mu_X, \sigma_X^2)$, $\mathbb{E}[X] = \mu_X$, $\mathrm{Var}[X] = \sigma_X^2$. (Exercise: verify it for Z, then extrapolate to general X.)

The m.g.f. of the normal is

$$M_X(t) = \exp\left(\mu_X t + \frac{\sigma_X^2 t^2}{2}\right), \quad t \in \mathbb{R},$$

which for $Z \sim N(0,1)$ reduces to $M_Z(t) = e^{t^2/2}$. Note that the exponent is quadratic in t, so a sum of independent normals has an m.g.f. of exactly the same form; this is what makes the normal family closed under addition (Section 9.3), and it is also the target the m.g.f. converges to in the proof of the CLT (Section 10.3).

Standard normal tables give values of $\Phi(x) = P(Z \le x)$ for $Z \sim N(0,1)$, tabulated for $x \ge 0$ in steps of 0.01.

Python note: In practice `scipy.stats.norm` replaces the table: $\Phi(1.5)$ is `norm.cdf(1.5)`, the 75th percentile $z_{.75}$ is `norm.ppf(0.75)` $= 0.6745$, and $P(X \leq 4)$ for $X \sim N(10, 2^2)$ is `norm.cdf(4, loc=10, scale=2)`; note `scale` is the standard deviation σ, not the variance.

Note: For any $x < 0$, $\Phi(-x) = 1 - \Phi(x)$ (By symmetry)

Note: Integration of $X \sim N(\mu_X, \sigma_X^2)$ is generally not calculable by hand. We must use the normal tables.

Example 6.1: Let $X \sim N(\mu_X, \sigma_X^2)$. Denote the following in terms of Φ, the c.d.f. of $Z \sim N(0, 1)$.

a) $P(X > a)$, a a constant

$$\begin{aligned} P(X > a) &= 1 - P(X \leq a) = 1 - P\left(\frac{X - \mu_X}{\sigma_X} < \frac{a - \mu_X}{\sigma_X}\right) \\ &= 1 - P\left(Z < \left(\frac{a - \mu_X}{\sigma_X}\right)\right) = 1 - \Phi\left(\frac{a - \mu_X}{\sigma_X}\right) \end{aligned}$$

b) $P(a < X < b)$, a, b constants, $a < b$

$$P(a < X < b) = P(X < b) - P(X < a) = \Phi\left(\frac{b - \mu_X}{\sigma_X}\right) - \Phi\left(\frac{a - \mu_X}{\sigma_X}\right)$$

Example 6.2: The time X, in minutes, that a humpback whale stays submerged on one dive is normally distributed with mean 10, std dev 2.

a) What is the chance a dive lasts more than 4 minutes? That is, find $P(X > 4)$

$$\begin{aligned} P(X > 4) &= 1 - P(X \leq 4) = 1 - P\left(Z \leq \frac{4 - 10}{2}\right) = 1 - P(Z \leq -3) \\ &= \Phi(3) \approx 0.9987 \end{aligned}$$

b) Among dives shorter than 10 minutes, what fraction last more than 8 minutes? That is, find $P(X > 8 \mid X < 10)$

$$\begin{aligned} P(X > 8 \mid X < 10) &= \frac{P(X > 8 \cap X < 10)}{P(X < 10)} \\ &= \frac{P(8 < X < 10)}{P(X < 10)} \\ &= \frac{P(-1 < Z < 0)}{P(Z < 0)} \\ &= \frac{P(Z < 0) - P(Z < -1)}{P(Z < 0)} \\ &= \frac{0.50 - P(Z > +1)}{0.50} \\ &= \frac{0.50 - (1 - \Phi(1))}{0.50} \\ &= \frac{0.50 - (1 - \Phi(1))}{0.50} = \frac{0.50 - 0.1587}{0.50} = \frac{0.3413}{0.50} \approx 0.683 \end{aligned}$$

c) Three-quarters of dives end within how many minutes? That is, find the 75th percentile of X

For Z: $\Phi(z_p) \approx 0.75 \implies z_p = 0.67$

$$\implies \quad \frac{x_p - \mu_X}{\sigma_X} = .67 \implies x_p = \mu_X + .67\sigma_X \implies x_p = 10 + 0.67(2) = 11.34$$

d) Find $P\left(\frac{X+2}{13} > 1\right)$

$$P\left(\frac{X+2}{13} > 1\right) = P(X > 13(1) - 2) = P(X > 11)$$
$$= P\left(Z > \frac{1}{2}\right) = 1 - \Phi(.5) \approx 1 - 0.6915 = 0.3085$$

Facts: For any RV X, using the transformation $Z = \frac{X - \mu_X}{\sigma_X}$ gives $\mathbb{E}[Z] = 0$, $\text{Var}[Z] = 1$, but it is not generally true that the distribution remains what it was before.

Example 6.3: If we standardize $X \sim \Gamma(\alpha, \beta)$, we may not see $Z \sim \Gamma$.

Exercises

Exercise 6.1. The length X, in minutes, of an aurora display over Abisko, in Swedish Lapland, has a normal distribution with mean 30. If $P(25 < X < 35) = 0.8$, what is the standard deviation of X? Round to two decimal places. *Hint:* First find $P(X < 25)$ and $P(X > 35)$. How many standard deviations below the mean is 25?



Figure 6.1: *The weighing of the heart: Anubis checks the scale, Thoth records the verdict, and Ammit waits.* [15]

Exercise 6.2 (The weighing of the heart)**.** In the Egyptian afterlife a soul's heart is weighed in the Hall of Two Truths (Fig. 6.1) against the feather of Ma'at, the goddess of truth and order, while Thoth, the ibis-headed god of writing and reckoning, records the verdict. A heart in balance with the feather is justified and passes on; a heavier one is devoured by Ammit. Suppose a pure heart has a true excess weight of 0 feather-units and a guilty heart a true excess of 1 feather-unit, and that the scale is imperfect: its reading is the true excess plus $N(0, \sigma^2)$ noise, independent of the heart. Thoth records the heart as justified when the reading is below $1/2$ and as guilty otherwise.

(a) Find the probability that Thoth records the correct verdict, as a formula in terms of σ.

(b) What happens to this formula when σ is very small? When σ is very large? Explain intuitively why the results in these extreme cases make sense.

Exercise 6.3 (Soaking in an onsen)**.** The time X a guest spends in a hot-spring bath (onsen) in Hakone, a mountain resort town near Tokyo, is normally distributed with mean 25 minutes and standard deviation 6 minutes.

(a) Probability a guest soaks for more than 30 minutes.

(b) Probability a soak lasts between 20 and 35 minutes.

(c) The 90th percentile of soak times.

(d) Of 8 independent guests, the expected number who soak for more than 30 minutes.

Exercise 6.4 (The cable car line at Powell and Market)**.** The time X a visitor spends in line at the Powell Street cable car turnaround is normally distributed with mean 20 minutes and standard deviation 5 minutes.

(a) Probability a visitor waits more than 30 minutes.

(b) Probability a wait lasts between 15 and 25 minutes.

(c) The 10th percentile of wait times.

(d) Probability that at least one of 4 independent visitors waits more than 30 minutes.

Exercise 6.5 (An April afternoon in Athens)**.** Afternoon temperatures in Athens in April are normally distributed with mean 68°F and standard deviation 5°F.

(a) Probability an afternoon tops 75°F.

(b) Probability an afternoon lands between 60°F and 70°F.

(c) The 95th percentile of afternoon temperature.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 6.1. By symmetry $P(X < 25) = P(X > 35) = 0.1$, so 25 is the 10th percentile: $(25 - 30)/\sigma = -1.2816$, giving $\sigma = 5/1.2816 = 3.90$ minutes.

Exercise 6.2. (a) For a guilty heart the reading is $1 + N$ and the verdict is wrong when $1 + N < \frac{1}{2}$, i.e. $N < -\frac{1}{2}$; for a pure heart the reading is N and the verdict is wrong when $N \geq \frac{1}{2}$. By symmetry of the normal both error probabilities equal $P(N > \frac{1}{2}) = 1 - \Phi(\frac{1}{2\sigma})$, so whichever kind of heart is on the scale,

$$P(\text{correct verdict}) = \Phi\left(\frac{1}{2\sigma}\right).$$

(b) As $\sigma \to 0$, $\Phi(1/(2\sigma)) \to 1$: a precise scale separates the two kinds of heart perfectly. As $\sigma \to \infty$, it tends to $\Phi(0) = \frac{1}{2}$: the noise swamps the one-unit difference and the judgment is no better than a coin toss, so Ammit eats pure and guilty hearts alike.

Exercise 6.3. $X \sim N(25, 36)$. **(a)** $1-\Phi(0.83) = 0.203$. **(b)** $\Phi(1.67)-\Phi(-0.83) = 0.749$. **(c)** $25+1.2816(6) = 32.7$ min. **(d)** $8(0.203) = 1.63$.

Exercise 6.4. $X \sim N(20, 25)$. **(a)** $1 - \Phi(2) = 0.023$. **(b)** $\Phi(1) - \Phi(-1) = 0.683$. **(c)** $20 - 1.2816(5) = 13.6$ min. **(d)** $1 - (1 - 0.023)^4 = 0.089$.

Exercise 6.5. $X \sim N(68, 25)$. **(a)** $1 - \Phi(1.4) = 0.081$. **(b)** $\Phi(0.4) - \Phi(-1.6) = 0.601$. **(c)** $68 + 1.645(5) = 76.2°$F.

6.2 Uniform and Exponential Distributions

Uniform Distribution

This is the simplest distribution, and is used in mostly theoretical settings. Naturally occurring variables are not typically uniform. Its most important practical role is in simulation: computers generate $U(0,1)$ random numbers, and transformations of them (Section 11.2) produce draws from any other distribution.

If X is uniform with $-\infty < \alpha < x < \beta < \infty$, then we denote it as $X \sim U(\alpha, \beta)$, and

$$f_X(x) = \begin{cases} \dfrac{1}{\beta - \alpha} & \alpha \le x \le \beta \\ 0 & \text{o.w.} \end{cases}$$

$$\mathbb{E}[X] = \frac{\alpha + \beta}{2}, \qquad \text{Var}[X] = \frac{(\beta - \alpha)^2}{12}, \qquad M_X(t) = \frac{e^{\beta t} - e^{\alpha t}}{(\beta - \alpha)\, t} \ \ (t \neq 0), \quad M_X(0) = 1$$

Notice the p.d.f. is a rectangle.

Python note: `scipy.stats.uniform` takes `loc` $= \alpha$ and `scale` $= \beta - \alpha$ (not β): the stage-changeover example below is `uniform.cdf(x, loc=5, scale=20)`.

Example 6.4: At a music festival the crew needs between 5 and 25 minutes to reset the main stage between acts, and within that range no changeover length is more likely than another. Let $X =$ the changeover time in minutes, so $X \sim U(5, 25)$.

a) What is the chance a changeover takes at least x minutes? That is, find $P(X \ge x)$

$$P(X \ge x) = \int_x^{25} \frac{1}{25 - 5}\, dt = \frac{t}{20}\bigg|_x^{25} = \frac{25 - x}{20}$$

b) What is the chance a changeover is done within x minutes? That is, find $P(X \le x)$

Since continuous,

$$P(X \le x) = 1 - P(X \ge x) = 1 - \frac{25 - x}{20} = \frac{20 - (25 - x)}{20} = \frac{x - 5}{20}$$

c) The quickest 10% of changeovers are done within how many minutes? That is, find the 10th percentile of X

$$P(X \le x_p) = 0.10 \ \Rightarrow \ \frac{x_p - 5}{20} = 0.10 \ \Rightarrow \ x_p - 5 = 2 \ \Rightarrow \ x_p = 7$$

d) The quickest p% of changeovers are done within how many minutes? That is, find the general equation for the pth percentile of X, $p \in (0, 100)$. (Note we are on the percent scale here: the pth percentile is the quantile $x_{p/100}$ of Section 4.4.)

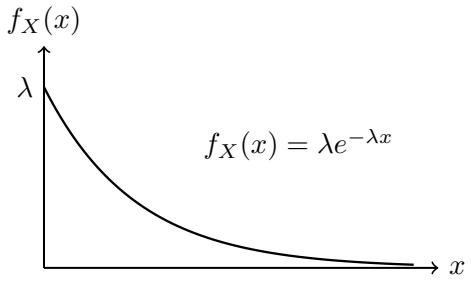
$$P(X \le x_p) = \frac{p}{100} \ \Rightarrow \ \frac{x_p - 5}{20} = \frac{p}{100} \ \Rightarrow \ x_p - 5 = \frac{20p}{100} \ \Rightarrow \ x_p - 5 = \frac{p}{5}$$
$$\Rightarrow \ x_p = \frac{p}{5} + 5 = \frac{25 + p}{5}$$

Exponential Distribution

If events occur at rate λ per unit time, the waiting time until the first event is exponential, with p.d.f.

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0,\ \lambda > 0$$

Denoted $X \sim \text{Exp}(\lambda)$. This graph typically looks like a decreasing exponential, where λ = rate of event per unit time (typically).



Connection to the Poisson: if events occur at rate λ per unit time, the *count* of events in a time window is Poisson, while the *waiting time* until the next event is $\text{Exp}(\lambda)$. The count is discrete and the waiting time is continuous, and the same λ governs both.

Further, if $X \sim \text{Exp}(\lambda)$,

$$\mathbb{E}[X] = \tfrac{1}{\lambda}, \qquad \sigma_X^2 = \tfrac{1}{\lambda^2}, \qquad M_X(t) = \frac{1}{1 - t/\lambda}, \quad t < \lambda$$

$$F_X(x) = 1 - e^{-\lambda x}, \qquad P(X > x) = e^{-\lambda x}$$

Python note: `scipy.stats.expon` takes `scale` $= 1/\lambda$ (scale, not rate): $P(X \le x)$ for $\text{Exp}(\lambda)$ is `expon.cdf(x, scale=1/lam)`.

Example 6.5: The time X, in days, until a cat's next burst of zoomies (a sudden frantic sprint around the house) is exponential with $\lambda = 2$ (two bursts per day, on average).

a) What is the chance the cat goes more than 3 days without zoomies? That is, find $P(X > 3)$

$$P(X > 3) = 1 - P(X \le 3) = 1 - F_X(3) = 1 - (1 - e^{-\lambda \cdot 3}) = e^{-2(3)} = e^{-6}$$

b) If the cat has gone 2 days without zoomies, what is the chance the wait lasts more than 5 days in all? That is, find the probability X is larger than 5, if X is known to be larger than 2.

$$\begin{aligned} P(X > 5 \mid X > 2) &= \frac{P(X > 5 \cap X > 2)}{P(X > 2)} \\ &= \frac{P(X > 5)}{P(X > 2)} = \frac{1 - P(X \le 5)}{1 - P(X \le 2)} \\ &= \frac{1 - (1 - e^{-2(5)})}{1 - (1 - e^{-2(2)})} = \frac{e^{-10}}{e^{-4}} = e^{-10+4} = e^{-6} \end{aligned}$$

Notice (a), (b) are the same. This gives the following result:

Theorem 6.1 (Memoryless Property): If $X \sim \text{Exp}(\lambda)$, then for $s > 0$, $t > 0$,

$$P(X > s + t \mid X > s) = P(X > t)$$

This is equivalent to saying "the fact that $X > s$ had no effect on whether X was larger by t more".

Suppose $X =$ lifetime of battery. The memoryless property would state that: "How much time has already passed has no bearing on the additional time the battery will last".

Example 6.6: The lifetime of the motor that spins a club's disco ball is 6 years on average. Assume this lifetime is distributed Exponential.

a) Identify λ and $\mathbb{E}[X]$.

$\mathbb{E}[X] = \text{given} = 6 = \dfrac{1}{\lambda} \implies \lambda = \frac{1}{6}$ (rate of failure per year)

b) Find the prob a motor lasts over 7 years.

$$P(X > 7) = 1 - F_X(7) = e^{-7(1/6)} = 0.311$$

c) If a motor has lasted 8 years, what is the prob it will last an additional 3 or longer?

$$P(X > 8 + 3 \mid X > 8) = P(X > 3) = e^{-3/6} = 0.606$$

d) Estimate the prob X is within two std dev from the mean, and calculate it exactly.

Chebyshev's Inequality: $P(|X - \mathbb{E}[X]| < k\sigma_X) \geq 1 - \frac{1}{k^2}$

$k = 2$, so $P(|X - 6| < k\sigma_X) \geq 1 - \frac{1}{4} = \frac{3}{4}$

$\sigma_X = \sqrt{\frac{1}{\lambda^2}} = \sqrt{6^2} = 6$, so

$$\begin{aligned} P(|X - 6| < 2(6)) &= P\big((X - 6 < 12) \cap (X - 6 > -12)\big) \\ &= P(X < 18 \text{ and } X > -6) = P(X < 18) \quad (\text{since } x > 0) \\ &= F_X(18) = 1 - e^{-18/6} = 0.9502 \end{aligned}$$

Exercises

Exercise 6.6 (A sloth on a branch)**.** A three-toed sloth spends its days hanging in cecropia trees, the fast-growing tropical trees whose leaves it feeds on. One settles at a point X meters along a 10-meter horizontal cecropia branch, with $X \sim U(0, 10)$. What is $P(2 < X < 8)$, the chance it is on the middle six meters? If we are given that $2 < X < 8$, what is the conditional probability that $6 < X < 7$? Answer these questions by looking at areas on the graph of the p.d.f., without writing down any integrals.

Exercise 6.7. Let $X \sim U(\alpha, \beta)$.

(a) Write down the c.d.f. of X and solve $F(m) = 1/2$ to find the median.

(b) Using $\mathbb{E}[X] = \frac{\alpha+\beta}{2}$, show that the mean equals the median.

(c) For which of the other distributions we have studied does the mean equal the median? For which is the median strictly smaller? Explain the connection to skewness.

Exercise 6.8 (The Copacabana bus)**.** A bus runs along Copacabana, Rio's famous beachfront, every 15 minutes. Arriving at the stop at a random moment, your wait X is $U(0, 15)$.

(a) Probability you wait less than 4 minutes.

(b) Mean, variance, and standard deviation of the wait.

(c) The 80th percentile of the wait.

(d) Probability you wait more than 10 minutes, given you have already waited 5. Does the uniform have the memoryless property?

Exercise 6.9. The wait X, in hours after dark, for the aurora to appear over Kiruna, in Swedish Lapland, is $\text{Exp}(1/4)$. Use the memoryless property to compute $P(X < 7 \mid X > 5)$.

Exercise 6.10. A barber takes 30 minutes on average per walk-in haircut. Assume the length of a haircut is exponentially distributed, and that this is the only barber.

(a) What is the probability a specific haircut runs over 1 hour?

(b) Given that the barber will finish in under 60 minutes, what is the probability they finish in under 40 minutes?

(c) Find the probability a specific haircut ends within one standard deviation of the average time. *Careful:* what is the standard deviation of an exponential, and does the lower endpoint fall inside the support?

(d) What is the probability they finish more than two haircuts in one hour? *Hint:* if the gaps between completions are i.i.d. exponential with rate λ, what is the distribution of the *number* of completions in a fixed window?

(e) Parts (a) and (d) describe the same process from two directions. State the general relationship between the waiting-time distribution and the count distribution.

Exercise 6.11 (Rainbows over Mānoa)**.** The time between rainbow sightings in Honolulu's Mānoa Valley (a rainy valley known for them) is modeled as exponential with mean 2 days.

(a) Probability of going more than 5 days without a rainbow.

(b) The median gap between rainbows.

(c) It has been 3 days since the last one. Probability of at least 2 more rainbow-free days.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 6.6. The density is a flat $\frac{1}{10}$, so probabilities are lengths over 10: $P(2 < X < 8) = \frac{6}{10} = 0.6$. Conditioning on $2 < X < 8$ rescales to that interval of length 6, so $P(6 < X < 7 \mid 2 < X < 8) = \frac{1}{6}$.

Exercise 6.7. (a) $F(x) = \dfrac{x - \alpha}{\beta - \alpha}$ on $[\alpha, \beta]$; $F(m) = \frac{1}{2}$ gives $m = \dfrac{\alpha + \beta}{2}$. **(b)** That is exactly $\mathbb{E}[X]$. **(c)** Mean equals median for every symmetric distribution we have met (uniform, normal). For right-skewed ones (exponential, Poisson for small λ) the median is strictly smaller: the long right tail pulls the mean past the median while the median only counts where the mass is.

Exercise 6.8. $X \sim U(0, 15)$. **(a)** $4/15 = 0.267$. **(b)** 7.5, $15^2/12 = 18.75$, 4.33. **(c)** $0.8(15) = 12$ min. **(d)** $P(X > 10 \mid X > 5) = \frac{5/15}{10/15} = \frac{1}{2} \neq P(X > 5) = \frac{2}{3}$, so no: the uniform is not memoryless.

Exercise 6.9. By memorylessness, $P(X < 7 \mid X > 5) = P(X < 2) = 1 - e^{-2/4} = 1 - e^{-0.5} = 0.393$.

Exercise 6.10. $X \sim \text{Exp}(\lambda = 1/30)$ per minute. **(a)** $P(X > 60) = e^{-2} = 0.135$. **(b)** $P(X < 40 \mid X < 60) = \dfrac{1 - e^{-4/3}}{1 - e^{-2}} = \dfrac{0.736}{0.865} = 0.851$. **(c)** SD $= 1/\lambda = 30$, so the interval is $(0, 60)$ (the lower endpoint 0 is the edge of the support): $P(X < 60) = 1 - e^{-2} = 0.865$. **(d)** The number of completions in an hour is $\text{Pois}(60/30) = \text{Pois}(2)$: $P(N > 2) = 1 - e^{-2}(1 + 2 + 2) = 1 - 5e^{-2} = 0.323$. **(e)** Exponential gaps with rate $\lambda \iff$ Poisson counts with mean λt in a window of length t; "the kth event is after time t" is the same event as "fewer than k events by time t."

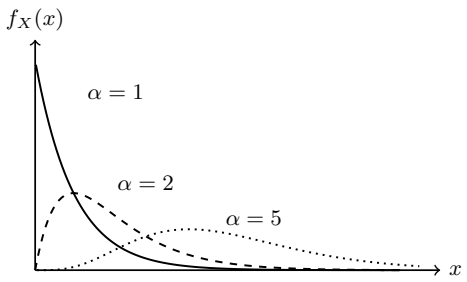
Exercise 6.11. Rate $1/2$ per day. **(a)** $e^{-2.5} = 0.082$. **(b)** $2 \ln 2 = 1.39$ days. **(c)** By memorylessness $P(X > 2) = e^{-1} = 0.368$.

6.3 Gamma and Chi-Squared Distributions

Gamma Distribution

An extremely flexible distribution that is used in many areas of statistics is the gamma distribution with parameters α, β, denoted $X \sim \Gamma(\alpha, \beta)$.

If events occur at rate λ over time, the waiting time until the α-th event follows $\Gamma(\alpha,\ \beta = \lambda)$. The exponential is the special case $\alpha = 1$, the waiting time for the first event; we verify in Section 9.4 that a sum of α independent exponential waiting times is exactly $\Gamma(\alpha,\ \beta = \lambda)$. The two parameters are the source of the flexibility: α controls the shape and β controls the rate.



The figure shows $\Gamma(\alpha,\ \beta = 1)$ densities for $\alpha = 1, 2, 5$: sharply decreasing at $\alpha = 1$ (the exponential), then increasingly bell-shaped as α grows.

The p.d.f. of X is then

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\beta x}, \quad x > 0,\ \alpha > 0,\ \beta > 0$$

Where $\Gamma(\alpha)$ is the gamma function, and

(i) $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y}\, dy, \quad \alpha > 0$

(ii) $\Gamma(\alpha) = (\alpha - 1)\,\Gamma(\alpha - 1)$

(iii) When α is an integer, $\Gamma(\alpha) = (\alpha - 1)!$

(iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

For instance $\Gamma(7) = 6! = 720$, which (ii) writes as $6\Gamma(6) = 6 \cdot 5!$. Property (ii) also reaches the half-integers from (iv):

$$\Gamma\left(\tfrac{5}{2}\right) = \tfrac{3}{2}\Gamma\left(\tfrac{3}{2}\right) = \tfrac{3}{2} \cdot \tfrac{1}{2}\Gamma\left(\tfrac{1}{2}\right) = \tfrac{3}{4}\sqrt{\pi}.$$

With $\alpha = 1$ the constant is $\Gamma(1) = 0! = 1$ and the p.d.f. collapses to $\beta e^{-\beta x}$, the exponential density; this is the sense in which the gamma generalizes the exponential.

If $X \sim \Gamma(\alpha, \beta)$, then

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \qquad \sigma_X^2 = \frac{\alpha}{\beta^2}, \qquad M_X(t) = \left(\frac{\beta}{\beta - t}\right)^\alpha, \quad t < \beta$$

Python note: `scipy.stats.gamma` is parameterized by shape `a` $= \alpha$ and `scale` $= 1/\beta$ (scale, not rate), so $\Gamma(\alpha, \beta)$ probabilities are `gamma.cdf(x, a=alpha, scale=1/beta)`. This is the same convention as `expon`, which is the `a=1` case. Forgetting this rate-vs-scale conversion is the most common scipy error with these families.

Many times we may find a function that is proportional to a Gamma distribution, and if we recognize that we can do some useful manipulations. The strategy is to multiply and divide by the normalizing constant of the recognized p.d.f., since a complete p.d.f. integrates to 1.

Example 6.7: Integrate the following:

$$\int_0^\infty x^5 e^{-x/2}\, dx$$

$$\begin{aligned}
&= \int_0^\infty x^{6-1} e^{-x/2}\, dx \;\propto\; \Gamma\left(\alpha = 6,\, \beta = \tfrac{1}{2}\right) \\
&= \frac{\Gamma(6)}{(1/2)^6} \int_0^\infty \frac{(1/2)^6}{\Gamma(6)}\, x^{6-1} e^{-x/2}\, dx = 2^6 \Gamma(6) \cdot 1, \quad X \sim \Gamma\left(\alpha = 6,\, \beta = \tfrac{1}{2}\right) \\
&= 2^6 \Gamma(6)(1) = 64(5!) = 64(120) = 7680
\end{aligned}$$

The same trick derives the moments. For the mean, $x \cdot x^{\alpha-1} = x^{(\alpha+1)-1}$ is the kernel of a $\Gamma(\alpha+1, \beta)$ density:

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^\infty x\, \frac{\beta^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\beta x}\, dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}} \underbrace{\int_0^\infty \frac{\beta^{\alpha+1}}{\Gamma(\alpha+1)}\, x^{(\alpha+1)-1} e^{-\beta x}\, dx}_{=1} \\
&= \frac{1}{\beta} \cdot \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \frac{1}{\beta} \cdot \frac{\alpha\Gamma(\alpha)}{\Gamma(\alpha)} = \frac{\alpha}{\beta},
\end{aligned}$$

using (ii) in the last line. For the m.g.f., $e^{tx} e^{-\beta x} = e^{-(\beta - t)x}$ is the kernel of a $\Gamma(\alpha, \beta - t)$ density, which is a valid density only when $\beta - t > 0$:

$$M_X(t) = \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-(\beta-t)x}\, dx = \frac{\beta^\alpha}{(\beta-t)^\alpha} \underbrace{\int_0^\infty \frac{(\beta-t)^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-(\beta-t)x}\, dx}_{=1} = \left(\frac{\beta}{\beta - t}\right)^\alpha, \quad t < \beta.$$

The variance then follows from the m.g.f. Writing $M_X(t) = \beta^\alpha (\beta - t)^{-\alpha}$,

$$M_X'(t) = \alpha\beta^\alpha (\beta - t)^{-\alpha-1}, \qquad M_X''(t) = \alpha(\alpha+1)\beta^\alpha (\beta - t)^{-\alpha-2},$$

so $\mathbb{E}[X^2] = M_X''(0) = \alpha(\alpha+1)/\beta^2$ and $\sigma_X^2 = \dfrac{\alpha^2 + \alpha}{\beta^2} - \dfrac{\alpha^2}{\beta^2} = \dfrac{\alpha}{\beta^2}$. (Recognizing the $\Gamma(\alpha+2, \beta)$ kernel in $\mathbb{E}[X^2]$ gives the same answer directly.)

Example 6.8: A club's disco ball motor fails at a rate of $\lambda = 1/6$ per year, and a failed motor is repaired at once. The time X until the ball has broken twice is the waiting time for the second event at rate $1/6$, so $X \sim \Gamma(\alpha = 2,\, \beta = 1/6)$. Then

$$\mathbb{E}[X] = \frac{2}{1/6} = 12 \text{ years}, \qquad \sigma_X^2 = \frac{2}{(1/6)^2} = 72 \text{ years}^2, \qquad \sigma_X = \sqrt{72} = 6\sqrt{2} \approx 8.49 \text{ years}.$$

Each of the two waits is $\text{Exp}(1/6)$, with mean 6 and variance 36; the gamma mean and variance are just twice those, as they must be for a sum of two independent waits. The mean grows with α (more events to wait for) and shrinks with β (events arrive faster); so does the spread.

Chi-Squared (χ^2) Distribution

This special case of the gamma is the distribution of squared standard normals: if $Z \sim N(0,1)$, then $Z^2 \sim \chi^2_1$ (Section 11.1, with a second proof by the transformation method in Section 11.2), and a sum of r independent squared standard normals is χ^2_r. Since sample variances are built from sums of squared deviations, the χ^2 distribution is central to variance estimation.

If $X \sim \Gamma\left(\alpha = \frac{r}{2},\ \beta = \frac{1}{2}\right)$, then $X \sim \chi^2_r$ and

$$f_X(x) = \frac{1}{\Gamma(r/2)\,2^{r/2}}\, x^{(r/2)-1} e^{-x/2}, \quad x > 0,\ r > 0 \text{ integer.}$$

$$\mathbb{E}[X] = r, \qquad \sigma^2_X = 2r, \qquad M_X(t) = \frac{1}{(1-2t)^{r/2}}, \quad t < \tfrac{1}{2}$$

Example 6.9: Take $r = 2$. Then $\chi^2_2 = \Gamma(\alpha = 1,\ \beta = 1/2)$, which is the exponential distribution with mean 2, so

$$P(X \le 3) = 1 - e^{-3/2} \approx 0.777$$

For general r, probabilities come from χ^2 tables or software.

Python note: `chi2.cdf(3, df=2)` $= 0.777$ reproduces the example above.

Review of Continuous Random Variables

Normal Distribution

$X \sim N(\mu_X, \sigma^2_X)$. X is very general. Most commonly used distribution in statistics. Convert to $Z \sim N(0,1)$ to find probabilities.

Uniform Distribution

$X \sim U(\alpha, \beta)$. X can be general, but this distribution is mostly used in theoretical applications. Very easy to integrate.

Exponential

$X \sim \text{Exp}(\lambda)$. X = time to event (usually). Special case of Gamma. Has memory-less property. Somewhat easy to integrate.

Gamma

$X \sim \Gamma(\alpha, \beta)$. X is very general. Specific cases of α, β give other distributions. Typically do not integrate by hand.

Chi-Squared Distribution

$X \sim \chi^2_r$. Returns in Section 11.1 (squared normals, sums, and the sample variance). Special case of Gamma. Will never integrate by hand.

Exercises

Exercise 6.12 (Whale blows off the headland)**.** Watching from a headland, the gap between consecutive blows of a feeding humpback whale is exponential with mean 4 minutes.

(a) Probability a gap exceeds 6 minutes.

(b) Name the distribution of the time until the third blow, with its parameters, mean, and standard deviation.

(c) Probability the third blow comes within 12 minutes. (Count blows instead of integrating.)

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 6.12. Rate $1/4$ per minute. **(a)** $e^{-1.5} = 0.223$. **(b)** $\Gamma(3, 1/4)$: mean 12 min, SD $4\sqrt{3} = 6.93$ min. **(c)** At least 3 blows in 12 minutes, $\text{Pois}(3)$: $1 - 8.5e^{-3} = 0.577$.

7 Joint Distributions

7.1 Joint Densities: Discrete and Continuous

Joint Distributions

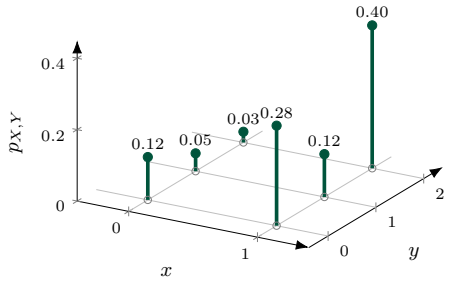
We may also be interested in how two outcomes of two separate experiments (or two outcomes of the same experiment) are associated with each other.



Figure 7.1: *Everest at sunset from Kala Patthar.* [16]

Example 7.1: The Himalayan Database, the archive begun by the journalist Elizabeth Hawley that logs every expedition in the Nepal Himalaya, records for every climber on an Everest expedition (Fig. 7.1) whether they used supplemental oxygen and how high they got. For a randomly chosen climber let X indicate oxygen use (1 = used, 0 = not) and let Y record the outcome: 0 = turned back below 8000 m, 1 = turned back above 8000 m, 2 = reached the summit. Suppose the probabilities are those in the table below; the right column and bottom row are the row and column totals.

	$y=0$	$y=1$	$y=2$	
$x=0$	0.12	0.05	0.03	0.20
$x=1$	0.28	0.12	0.40	0.80
	0.40	0.17	0.43	1



Each variable has its own distribution: the row totals say 80% of climbers use oxygen, and the column totals say $P(Y = 2) = 0.43$ of climbers summit. But the interesting questions concern the pair. Does oxygen go with summiting? Reading the table, $P(Y = 2 \mid X = 1) = 0.40/0.80 = 0.50$ while $P(Y = 2 \mid X = 0) = 0.03/0.20 = 0.15$: a climber on oxygen is more than three times as likely to reach the top, so the two variables are strongly associated. If we know X, can we sharpen our prediction of Y? Yes: without knowing about oxygen we would give a climber a 0.43 chance of summiting; told they climb without it, that drops to 0.15. And the probability that a climber both uses oxygen and summits is $P(X = 1, Y = 2) = 0.40$, a number no one-variable distribution can give. Every one of these questions needs the distribution of (X, Y) jointly. These two questions, association and prediction, drive this section: marginals, conditionals, covariance, and independence.

Intuition (Dependence, informally). Roll a red die and a blue die, and bet that the red die shows a 1. Before bets close, the croupier announces only the *sum* X of the two dice. If the croupier says $X = 2$, the red die must be a 1 and your bet is a sure thing; if the croupier says $X = 7$, every face of the red die is still equally likely, and the announcement tells you nothing. Because what we learn about X can change the distribution of the red die Y, the two are *dependent*. In general, two RVs are dependent when learning the value of one changes the distribution of the other.

We will consider outcomes of RV X, Y as pairs (X, Y), and consider the cases where

(i) Both X, Y are discrete

(ii) Both X, Y are continuous

We next extend c.d.f.'s, p.d.f.'s, etc. to distributions of (X, Y).

Joint Distribution Function (Joint c.d.f.)

Definition 7.1: The joint probability distribution of (X, Y), denoted $P_{X,Y}$, is defined as

$$\begin{aligned} P_{X,Y}(B) &= P[(X, Y) \in B], \quad B \subseteq \mathbb{R}^2 = \mathbb{R} \times \mathbb{R} \quad \text{(The 2 dimensional Euclidean space)} \\ &= P[\exists\, s \in S : (X(s),\, Y(s)) \in B] \end{aligned}$$

Definition 7.2: The joint distribution function (c.d.f.) is

$$F_{X,Y}(x, y) = P(X \le x,\ Y \le y), \quad x, y \in \mathbb{R}$$

The c.d.f. $F_{X,Y}$ has similar properties as before:

(i) $0 \le F_{X,Y}(x, y) \le 1 \quad \forall\, x, y \in \mathbb{R}$

(ii) The variation of $F_{X,Y}$ over rectangles is ≥ 0.
That is, if $x_1 \le x_2$ and $y_1 \le y_2$, then

$$F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1) \ge 0.$$

Simply states that the probability $F_{X,Y}$ assigns to any rectangle, $P(x_1 < X \le x_2,\ y_1 < Y \le y_2)$, is ≥ 0.

(iii) $F_{X,Y}(x, y)$ is right continuous

(iv) $\lim_{n\to\infty} F_{X,Y}(x_n, y_n) = 1, \quad x_n \to \infty, y_n \to \infty$

$\lim_{n\to\infty} F_{X,Y}(x_n, y_n) = 0, \quad x_n \to -\infty, y_n \to -\infty$

Joint p.d.f. of Discrete RV

If X, Y are discrete RVs, then we have the following:

(i) The joint p.d.f. is:

$$f_{X,Y}(x, y) = \begin{cases} P(X = x_i,\ Y = y_j) & \text{if } x = x_i,\ y = y_j,\ i, j \geq 1 \\ 0 & \text{o.w.} \end{cases}$$

Note: $\sum_{x\in\mathbb{R}} \sum_{y\in\mathbb{R}} f_{X,Y}(x, y)$ must $= 1$ as well.

(ii) $f_{X,Y}(x, y) \geq 0 \quad \forall\, x, y \in \mathbb{R}$

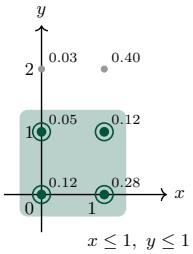
(iii) For any $B \subseteq \mathbb{R}^2$, $P((X, Y) \in B) = \sum_{(x_i,y_j)\in B} f_{X,Y}(x_i, y_j)$

(iv) $F_{X,Y}(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} f_{X,Y}(x_i, y_j)$

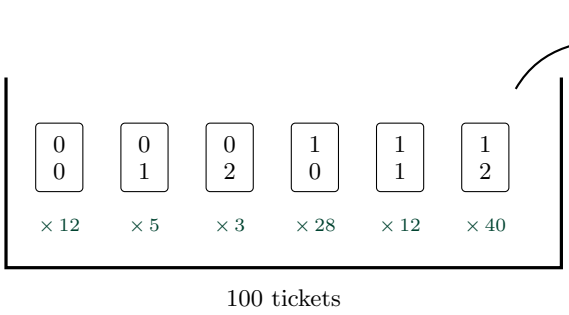
In the Everest table above, the body of the table is the joint p.d.f. and the margins are the distributions of X and Y alone. So $f_{X,Y}(1, 2) = P(X = 1,\ Y = 2) = 0.40$, and the c.d.f. at $(1, 1)$ adds the four cells with $x \leq 1$ and $y \leq 1$:

$$F_{X,Y}(1, 1) = f_{X,Y}(0, 0) + f_{X,Y}(0, 1) + f_{X,Y}(1, 0) + f_{X,Y}(1, 1) = 0.12 + 0.05 + 0.28 + 0.12 = 0.57,$$

the probability that a climber does not summit.



A box of tickets. A joint p.m.f. is easiest to picture as a box of tickets that each carry *two* numbers, x on top and y underneath. For Everest, fill a box with 100 tickets, 100 $f_{X,Y}(x, y)$ of each kind, and draw one at random:



$\Longleftrightarrow$

	$y=0$	$y=1$	$y=2$
$x=0$	0.12	0.05	0.03
$x=1$	0.28	0.12	0.40

Any table can serve as a joint p.m.f., provided each pair (x, y) appears once, every entry is between 0 and 1, and the entries add up to 1; then it describes some box of tickets.

Same marginals, different joint. Here is a second table for oxygen use X and outcome Y, with exactly the same row and column totals as the Everest table:

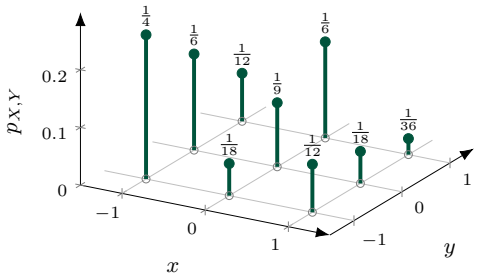
	$y=0$	$y=1$	$y=2$	
$x=0$	0.080	0.034	0.086	0.20
$x=1$	0.320	0.136	0.344	0.80
	0.40	0.17	0.43	1

Every cell here is its row total times its column total, so oxygen tells us nothing about the outcome: $P(Y=2 \mid X=1) = 0.344/0.80 = 0.43$, the same as $P(Y=2)$. In the real Everest table, $P(Y=2 \mid X=1) = 0.50$ while $P(Y=2 \mid X=0) = 0.15$. The two marginal distributions cannot tell these two situations apart; only the joint distribution carries the dependence.

Example 7.2: A club player at the Mechanics' Institute Chess Club in San Francisco, the oldest chess club in the United States, founded in 1854, plays a two-game match against a stronger opponent. Let X be the result of game 1 and Y the result of game 2, each coded -1 for a loss, 0 for a draw, and 1 for a win. From the club's records of such matches, the joint p.d.f. of X, Y is given by the table:

$x \backslash y$	-1	0	1
-1	1/4	1/6	1/12
0	1/18	1/9	1/6
1	1/12	1/18	1/36

$f_{X,Y}(x_i, y_j)$

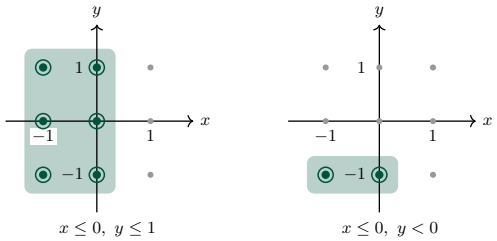


a) Find $F_{X,Y}(0,1)$, the probability that the player does not win game 1

$$F_{X,Y}(0,1) = P(X \le 0,\ Y \le 1) = \sum_{x \le 0} \sum_{y \le 1} f_{X,Y}(x_i, y_j)$$
$$= \tfrac{1}{4} + \tfrac{1}{6} + \tfrac{1}{12} + \tfrac{1}{18} + \tfrac{1}{9} + \tfrac{1}{6} = \frac{9+6+3+2+4+6}{36} = \frac{30}{36}$$

b) Find the prob $X \le 0$ and $Y < 0$: no win in game 1 and a loss in game 2

$$P(X \le 0,\ Y < 0) = P(X \le 0,\ Y = -1) = \tfrac{1}{18} + \tfrac{1}{4} = \tfrac{2}{36} + \tfrac{9}{36} = \tfrac{11}{36}$$



Joint p.d.f. of Continuous X, Y

If X, Y are continuous RVs, then we have:

(i) The joint p.d.f. is any $f_{X,Y}$ s.t.:

1. $f_{X,Y}(x,y) \ge 0 \quad \forall\, x, y \in \mathbb{R}$
2. $P[(X,Y) \in B] = \iint_B f_{X,Y}(x,y)\, dx\, dy, \quad B \subseteq \mathbb{R}^2$
3. $\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\, dx\, dy = 1$

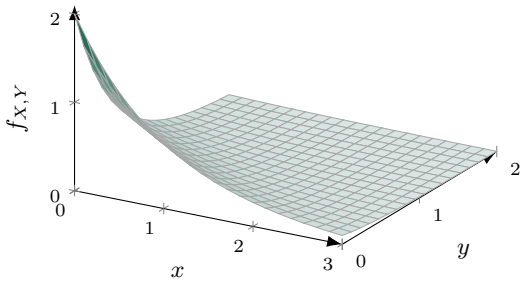
(ii) $F_{X,Y}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u,v)\, du\, dv \quad \forall\, x, y \in \mathbb{R}$

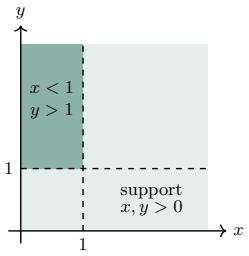
(iii) $\dfrac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x,y) = f_{X,Y}(x,y) \quad \forall\, x, y \in \mathbb{R}$ which are continuity points of $F_{X,Y}$

Example 7.3: A birder on the headlands at Point Reyes watches for peregrine falcons and gray whales. Let X be the time in hours until the first peregrine sighting and Y the time in hours until the first whale sighting. Peregrines appear at a rate of 1 per hour and whales at a rate of 2 per hour, and the two waits are independent, so the joint p.d.f. of X, Y is

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty,\ 0 < y < \infty \\ 0 & \text{o.w.} \end{cases}$$

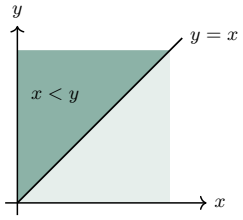
a) Find $P(X < 1,\ Y > 1)$: a peregrine within the first hour but no whale in that hour





$$\begin{aligned}\int_0^1 \int_1^\infty 2e^{-x}e^{-2y}\,dy\,dx &= \int_0^1 2e^{-x}\left(-\tfrac{1}{2}e^{-2y}\right)\Big|_1^\infty dx \\ &= \int_0^1 -e^{-x}(0-e^{-2})\,dx \\ &= \int_0^1 e^{-(x+2)}\,dx \\ &= -e^{-(x+2)}\Big|_0^1 = e^{-2}-e^{-3}\end{aligned}$$

b) Find $P(X < Y)$, the probability that the peregrine shows up before the whale



$$\begin{aligned}\int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy &= \int_0^\infty 2e^{-2y}(-e^{-x})\Big|_0^y\,dy \\ &= \int_0^\infty 2e^{-2y}(-e^{-y}+1)\,dy \\ &= \int_0^\infty -2e^{-3y}+2e^{-2y}\,dy \\ &= \tfrac{2}{3}e^{-3y}-e^{-2y}\Big|_0^\infty \\ &= (0+0)-\left(\tfrac{2}{3}-1\right) = \tfrac{1}{3}\end{aligned}$$

Notice we must be very careful about our bounds.

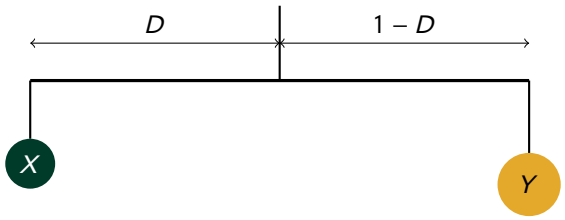
Example 7.4: A mobile hangs from a string tied to a light rod of length 1 m, with an ornament of mass X at the left end and one of mass Y at the right. The rod hangs level only when the string sits at the center of mass, which torque balance ($X \cdot D = Y \cdot (1-D)$) places at distance

$$D = \frac{Y}{X+Y}$$

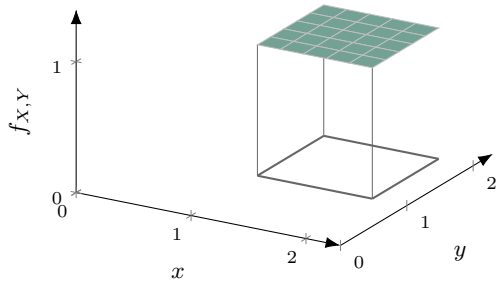
from the left end. The artist casts ornaments whose masses, in grams, are independent and uniform on $(1, 2)$. Each mass on its own has the $U(1, 2)$ density $f_X(x) = \frac{1}{2-1} = 1$ for $1 < x < 2$ (and $f_Y(y) = 1$ for $1 < y < 2$), and 0 outside that interval. Because the two masses are independent, the joint p.d.f. is the product of the two marginals (as in the Point Reyes example; made precise Section 9.1):

$$f_{X,Y}(x, y) = f_X(x)\, f_Y(y) = \begin{cases} 1 \cdot 1 = 1 & 1 < x < 2,\ 1 < y < 2 \\ 0 & \text{o.w.} \end{cases}$$

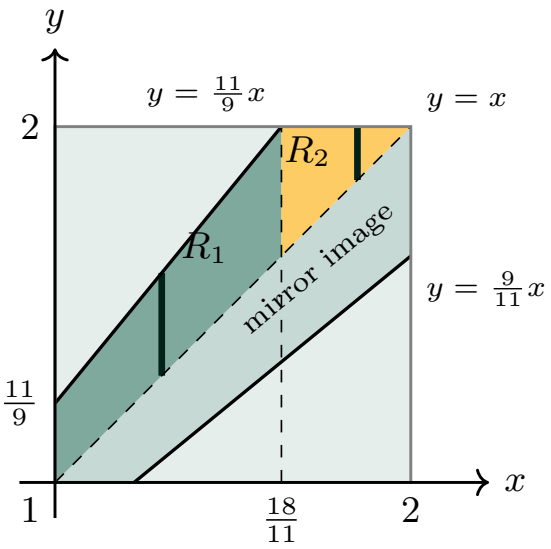
The product is 0 as soon as either factor is, so the density lives on the square $(1, 2) \times (1, 2)$ and is constant there: (X, Y) is uniform on the square. Since the square has area 1, the constant height 1 is exactly what makes the total volume $\iint f_{X,Y}\, dx\, dy = 1$, and the probability of any event inside the square is just its area.



a) The string is tied at the center of the rod. Find the probability that the center of mass lies within 5 cm of it, so the mobile hangs nearly level. Now $|D - \frac{1}{2}| \le 0.05$ means $|Y - X| \le 0.1(X + Y)$, i.e. $\frac{9}{11}X \le Y \le \frac{11}{9}X$: a narrow wedge around the diagonal of the square. Since the density is 1, the probability is the wedge's area, and by symmetry that is twice the part above the diagonal. The line $y = \frac{11}{9}x$ meets the top edge $y = 2$ at $x = \frac{18}{11}$, so



As always, the probability is the double integral of the joint p.d.f. over the event. Integrate y first: at a fixed x in the upper half of the wedge, y runs from the diagonal $y = x$ up to the line $y = \frac{11}{9}x$ while that line is inside the square ($x < \frac{18}{11}$), and up to the top edge $y = 2$ after it has left ($x > \frac{18}{11}$). The upper limit for y changes formula at $x = \frac{18}{11}$, so a single double integral cannot describe the region: we cut the upper half there into the two pieces R_1 and R_2 in the picture and integrate over each. The density is $f_{X,Y}(x, y) = 1$ on the whole square, so each inner integral is $\int 1\, dy$, which is just the length of the y-interval: $\frac{11}{9}x - x$ in the first piece and $2 - x$ in the second.



R_1: $1 < x < \frac{18}{11}$, $x < y < \frac{11}{9}x$

R_2: $\frac{18}{11} < x < 2$, $x < y < 2$

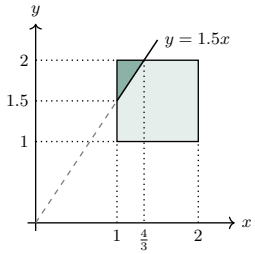
lower half: same area by symmetry

$$P\left(|D - \tfrac{1}{2}| \le 0.05\right) = \iint_{\text{wedge}} f_{X,Y}(x,y)\,dy\,dx = 2\left[\iint_{R_1} 1\,dy\,dx + \iint_{R_2} 1\,dy\,dx\right]$$
$$= 2\left[\int_1^{18/11}\int_x^{11x/9} 1\,dy\,dx + \int_{18/11}^{2}\int_x^{2} 1\,dy\,dx\right]$$
$$= 2\left[\int_1^{18/11}\left(\tfrac{11}{9}x - x\right)dx + \int_{18/11}^{2}(2-x)\,dx\right] = 2\left[\tfrac{203}{1089} + \tfrac{8}{121}\right] = \tfrac{50}{99} \approx 0.505.$$

Roughly half of the mobiles hang within tolerance; the other half need the string moved.

b) On average, how far from the left end does the balance point sit? That is, find $\mathbb{E}(D)$. No integral is needed. X and Y are independent with the same distribution, so the pairs (X, Y) and (Y, X) have the same joint distribution, and therefore $D = \frac{Y}{X+Y}$ and $1 - D = \frac{X}{X+Y}$ have the same distribution. Then $\mathbb{E}(D) = \mathbb{E}(1 - D) = 1 - \mathbb{E}(D)$, so $\mathbb{E}(D) = \frac{1}{2}$: on average the balance point is at the center, even though it almost never is exactly.

c) Find the probability that the balance point is more than 60 cm from the left end. $D > 0.6$ means $Y > 1.5X$, a triangle in the corner of the square with x from 1 to $\frac{4}{3}$:

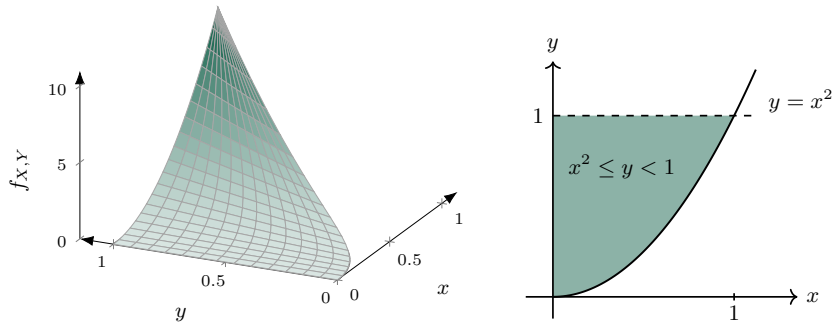


At a fixed x, y runs from the line $y = 1.5x$ up to the top edge $y = 2$; the line leaves the square through the top at $x = \frac{4}{3}$, which is why x stops there. Again $f_{X,Y} = 1$, so the inner integral is the length $2 - 1.5x$:

$$P(D > 0.6) = \int_1^{4/3} \int_{1.5x}^{2} f_{X,Y}(x,y)\, dy\, dx = \int_1^{4/3} \int_{1.5x}^{2} 1\, dy\, dx = \int_1^{4/3} (2 - 1.5x)\, dx$$
$$= \left[2x - \tfrac{3}{4}x^2\right]_1^{4/3} = \tfrac{4}{3} - \tfrac{5}{4} = \tfrac{1}{12} \approx 0.083.$$

Example 7.5: Cave isopods (*Asellus*) are small crustaceans, relatives of the pill bug, that live in pitch-dark cave pools and have lost their eyes and pigment over generations. For a juvenile sampled from a cave pool let X be its body mass as a fraction of adult mass and Y its body length as a fraction of adult length; length matures ahead of mass, and the scatter of (X, Y) across the population is modeled by the joint p.d.f.

$$f_{X,Y}(x,y) = \begin{cases} cx^2 y & 0 < x^2 \le y < 1,\ x > 0 \quad \text{(This bound important!)} \\ 0 & \text{o.w.} \end{cases}$$



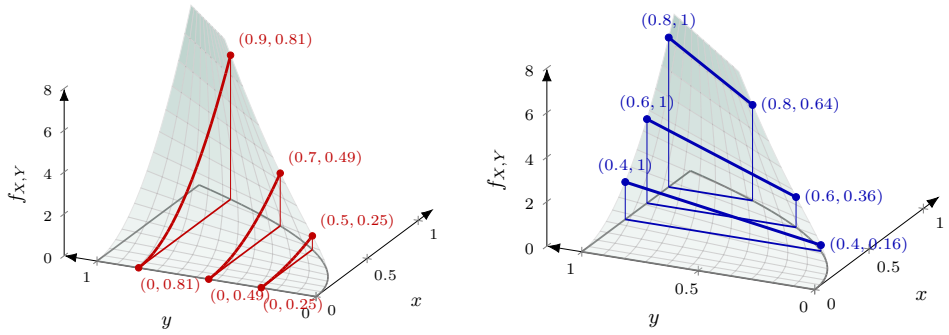
a) Find c s.t. $f_{X,Y}(x,y)$ is a valid p.d.f.

We have $0 < y < 1$, $x^2 \le y$, and $0 < x < 1$.

Thus, to find c s.t. $\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\, dx\, dy = 1$,

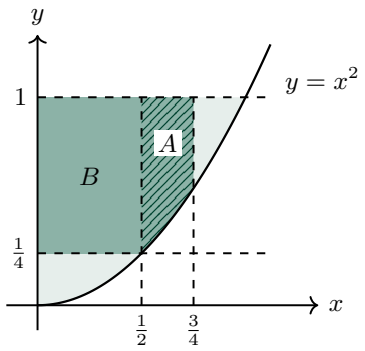
$$\int_0^1 \int_0^{\sqrt{y}} cx^2 y\, dx\, dy = \int_0^1 \int_{x^2}^1 cx^2 y\, dy\, dx$$
$$= \int_0^1 cx^2 \frac{y^2}{2}\Big|_{x^2}^1 dx = \int_0^1 cx^2\left(\tfrac{1}{2} - \tfrac{x^4}{2}\right) dx$$
$$= c\int_0^1 \tfrac{x^2}{2} - \tfrac{x^6}{2}\, dx = c\left(\frac{x^3}{6} - \frac{x^7}{14}\right)\Big|_0^1$$
$$= c\left(\tfrac{1}{6} - \tfrac{1}{14}\right) = c\left(\tfrac{7}{42} - \tfrac{3}{42}\right) = c\,\frac{4}{42} = 1$$
$$\Rightarrow\ c = \frac{42}{4} = \frac{21}{2}$$

The two orders of integration slice the surface $f_{X,Y} = \frac{21}{2}x^2 y$ in two ways. With dx inside (left), each cross-section is at a fixed y: a parabola in x running from $(0, y)$, at height 0, out to the curve at $(\sqrt{y}, y)$, at height $\frac{21}{2}y^2$. With dy inside (right), each cross-section is at a fixed x: a line in y running from the curve at (x, x^2), at height $\frac{21}{2}x^4$, up to $(x, 1)$, at height $\frac{21}{2}x^2$. Adding up either family of slices gives the same total volume.



b) What is the chance a sampled juvenile is under three-quarters of adult mass but over a quarter of adult length? That is, find $P\left(0 < X < \frac{3}{4},\ \frac{1}{4} < Y \leq 1\right)$

At $y = \frac{1}{4}$, $x^2 = \frac{1}{4}$, so $x = \frac{1}{2}$. We split the integral in two:



$$\underbrace{\int_{1/2}^{3/4}\int_{x^2}^{1} cx^2 y\,dy\,dx}_{\text{Area A}} + \underbrace{\int_{0}^{1/2}\int_{1/4}^{1} cx^2 y\,dy\,dx}_{\text{Area B}} \approx 0.63$$

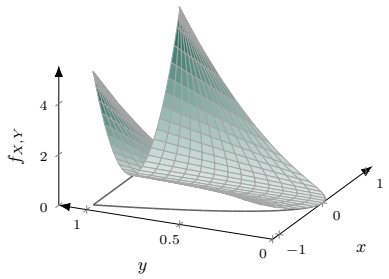
7.2 Joint Density Examples and Marginal Densities

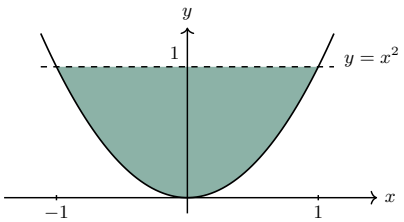
Example 7.6 (follow-up): Same density, but now *without* the restriction $x > 0$:

$$f_{X,Y}(x, y) = \begin{cases} Cx^2 y & 0 < x^2 < y < 1 \\ 0 & \text{o.w.} \end{cases}$$

Find C.

The region is now $x \in (-1, 1)$, $y \in (x^2, 1)$:





Setting $\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, dx\, dy = 1$:

$$\int_{-1}^{1}\int_{x^2}^{1} Cx^2 y\, dy\, dx = \int_{-1}^{1} Cx^2 \frac{y^2}{2}\Big|_{x^2}^{1} dx = \int_{-1}^{1} Cx^2\left(\tfrac{1}{2} - \tfrac{x^4}{2}\right) dx$$
$$= C\int_{-1}^{1} \tfrac{x^2}{2} - \tfrac{x^6}{2}\, dx = C\left(\tfrac{2}{6} - \tfrac{2}{14}\right) = C \cdot \frac{4}{21} = 1 \;\Rightarrow\; C = \frac{21}{4}$$

Notice $C = \frac{21}{4}$ is exactly half of the $c = \frac{21}{2}$ found above: dropping $x > 0$ doubles the support (the integrand $x^2 y$ is symmetric in x), so the normalizing constant is cut in half. This is why that bound was important!

Marginal p.d.f.'s of X, Y

Consider two discrete RVs X, Y with joint distribution $f_{X,Y}(x, y)$.

(i) The marginal p.d.f. of X only, $f_X(x)$, can be found as:

$$f_X(x_i) = \sum_{y_j \in \mathbb{R}} f_{X,Y}(x_i, y_j)$$

(ii) The marginal p.d.f. of Y only, $f_Y(y)$, can be found as:

$$f_Y(y_j) = \sum_{x_i \in \mathbb{R}} f_{X,Y}(x_i, y_j)$$

(iii) The conditional distribution of Y given $X = x$ can be found as:

$$f_{Y|X=x}(y) = f_{X,Y}(x, y) \,/\, f_X(x) = \frac{P(X = x,\, Y = y)}{P(X = x)}$$

(iv) The conditional distribution of X given $Y = y$ can be found as:

$$f_{X|Y=y}(x) = f_{X,Y}(x, y) \,/\, f_Y(y) = \frac{P(X = x,\, Y = y)}{P(Y = y)}$$

(v) $F_X(x) = \lim_{y \to \infty} F_{X,Y}(x, y)$

(vi) $F_Y(y) = \lim_{x\to\infty} F_{X,Y}(x,y)$

Notice the marginal p.d.f.'s are found by using the rule of total probability,

$$\begin{aligned} P(Y=y) &= P(X=x_1, Y=y) + P(X=x_2, Y=y) + \cdots \\ &= P(X=x_1 \cap Y=y) + \cdots \end{aligned}$$

And conditional p.d.f.'s are found by using the definition of conditional probabilities:

$$P(A \mid B) = \frac{P(A\cap B)}{P(B)}, \quad \text{Let } A=(Y=y),\ B=(X=x)$$

$$= P(Y=y \mid X=x) = \frac{P(X=x \cap Y=y)}{P(X=x)}$$

Example 7.7 (Everest, continued): Return to the Everest table (Fig. 7.1).

a) What share of climbers use oxygen, and how far does a climber get, if we ignore the other variable? That is, find the marginal p.d.f.s $f_X(x)$ and $f_Y(y)$.

For f_X we hold x fixed and add across its row, over every value of y:

$$f_X(x) = \begin{cases} 0.12+0.05+0.03 = 0.2, & x=0 \\ 0.28+0.12+0.40 = 0.8, & x=1 \end{cases} \qquad f_Y(y) = \begin{cases} 0.12+0.28 = 0.40, & y=0 \\ 0.05+0.12 = 0.17, & y=1 \\ 0.03+0.40 = 0.43, & y=2 \end{cases}$$

For f_Y we add down each column instead. These are exactly the row and column totals printed in the margins of the table, which is where the name *marginal* comes from.

b) Among climbers who use oxygen, how far do they get? That is, find the conditional p.d.f. $f_{Y|X=1}(y)$.

Keep only the $x=1$ row and divide it by its total $f_X(1) = P(X=1) = 0.8$, so that it sums to 1:

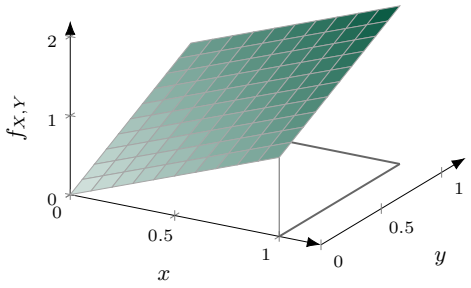
$$f_{Y|X=1}(y) = \frac{f_{X,Y}(1,y)}{f_X(1)} = \begin{cases} 0.28/0.8 = 0.35, & y=0 \\ 0.12/0.8 = 0.15, & y=1 \\ 0.40/0.8 = 0.50, & y=2 \end{cases}$$

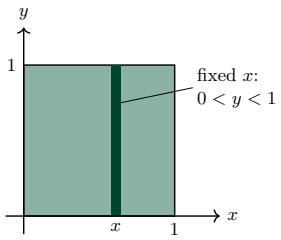
Half of the climbers on oxygen reach the summit, against 43% of all climbers.

Note: *Conditioning in a table: zero out and renormalize.* Every conditional calculation in a table follows the same two steps: set to 0 the cells that are inconsistent with what we have learned, then divide the surviving cells by their total so that they add to 1 again. The information need not be a single value. If we learn only that a climber did *not* summit ($Y \le 1$), the surviving cells are the $y=0$ and $y=1$ columns, with total $0.40+0.17 = 0.57$, so $P(X=1 \mid Y\le 1) = (0.28+0.12)/0.57 \approx 0.70$.

Example 7.8: Let X, Y be continuous with joint p.d.f.

$$f_{X,Y}(x,y) = x+y, \qquad 0<x<1,\ 0<y<1.$$





a) Verify this is a valid joint p.d.f.

$$\int_0^1 \int_0^1 (x+y)\,dy\,dx = \int_0^1 \left(x+\tfrac{1}{2}\right) dx = \tfrac{1}{2}+\tfrac{1}{2} = 1\checkmark$$

b) Find the marginal p.d.f. of X. Integrate the *other* variable out:

$$f_X(x) = \int_0^1 (x+y)\,dy = \left[xy + \tfrac{y^2}{2}\right]_{y=0}^1 = x+\tfrac{1}{2}, \qquad 0 < x < 1$$

By symmetry, $f_Y(y) = y + \frac{1}{2}$ on $0 < y < 1$.

c) Are X and Y independent?

$$f_X(x)f_Y(y) = \left(x+\tfrac{1}{2}\right)\left(y+\tfrac{1}{2}\right) = xy + \tfrac{x}{2} + \tfrac{y}{2} + \tfrac{1}{4} \;\neq\; x+y$$

so they are **not** independent.

d) Once we have f_X, questions about X alone need only the marginal:

$$P\left(X < \tfrac{1}{2}\right) = \int_0^{1/2} \left(x+\tfrac{1}{2}\right) dx = \tfrac{1}{8}+\tfrac{1}{4} = \tfrac{3}{8}, \qquad \mathbb{E}[X] = \int_0^1 x\left(x+\tfrac{1}{2}\right) dx = \tfrac{1}{3}+\tfrac{1}{4} = \tfrac{7}{12}$$

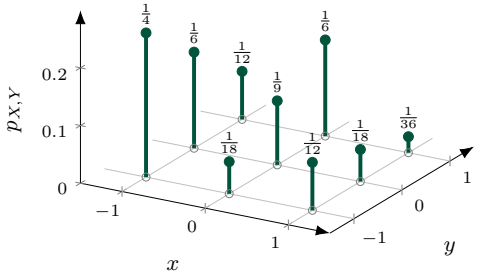
Note: The marginal is what is left after "forgetting" the other variable. Note that $\mathbb{E}[X] = \frac{7}{12} > \frac{1}{2}$: the density puts more weight at larger x, so X is pulled above the midpoint of its range.

7.3 Marginal and Conditional Densities

Example 7.9: The joint p.d.f. of X, Y is:

$x \backslash y$	-1	0	1
-1	$1/4 = 9/36$	$1/6 = 6/36$	$1/12 = 3/36$
0	$1/18 = 2/36$	$1/9 = 4/36$	$1/6 = 6/36$
1	$1/12 = 3/36$	$1/18 = 2/36$	$1/36$

$f_{X,Y}(x_i, y_j)$



a) Find $f_Y(y)$, $f_X(x)$

We simply add the rows in each column:

$$\begin{aligned} f_Y(y=-1) &= f_{X,Y}(y=-1, x=-1) + f_{X,Y}(y=-1, x=0) + f_{X,Y}(y=-1, x=1) \\ &= \tfrac{1}{4} + \tfrac{1}{18} + \tfrac{1}{12} = \frac{9+2+3}{36} = \frac{14}{36} \\ f_Y(y=0) &= \tfrac{1}{6} + \tfrac{1}{9} + \tfrac{1}{18} = \frac{6+4+2}{36} = \frac{12}{36} \\ f_Y(y=1) &= \tfrac{1}{12} + \tfrac{1}{6} + \tfrac{1}{36} = \frac{3+6+1}{36} = \frac{10}{36} \end{aligned}$$

Y	-1	0	1
$f_Y(y)$	$14/36$	$12/36$	$10/36$

Similarly we add the columns in each row for f_X:

X	-1	0	1
$f_X(x)$	$18/36$	$12/36$	$6/36$

b) Find $f_{X|Y=0}(x)$

$$\begin{aligned} f_{X|Y=0}(x=-1) &= f_{X,Y}(-1,0) \,/\, f_Y(0) = \tfrac{1/6}{12/36} = \tfrac{6/36}{12/36} = \tfrac{6}{12} \\ f_{X|Y=0}(x=0) &= f_{X,Y}(0,0) \,/\, f_Y(0) = \tfrac{1/9}{12/36} = \tfrac{4/36}{12/36} = \tfrac{4}{12} \\ f_{X|Y=0}(x=1) &= f_{X,Y}(1,0) \,/\, f_Y(0) = \tfrac{1/18}{12/36} = \tfrac{2/36}{12/36} = \tfrac{2}{12} \end{aligned}$$

X	-1	0	1
$f_{X\|Y=0}(x)$	$6/12$	$4/12$	$2/12$

c) Calculate $F_{X,Y}(x,y)$

$F_{X,Y}(x,y)$	$y=-1$	$y=0$	$y=1$
$x=-1$	$1/4$	$\frac{6}{36}+\frac{9}{36}=\frac{15}{36}$	$\frac{6}{36}+\frac{9}{36}+\frac{3}{36}=\frac{18}{36}$
$x=0$	$\frac{9}{36}+\frac{2}{36}=\frac{11}{36}$	$\frac{9}{36}+\frac{6}{36}+\frac{2}{36}+\frac{4}{36}=\frac{21}{36}$	$\frac{9}{36}+\frac{6}{36}+\frac{3}{36}+\frac{2}{36}+\frac{4}{36}+\frac{6}{36}=\frac{30}{36}$
$x=1$	$\frac{9}{36}+\frac{2}{36}+\frac{3}{36}=\frac{14}{36}$	$\frac{9}{36}+\frac{6}{36}+\frac{2}{36}+\frac{4}{36}+\frac{3}{36}+\frac{2}{36}=\frac{26}{36}$	1

Notice we are adding every row, column up to and including (x,y).

Marginal/Conditional p.d.f. of X, Y

Consider two continuous RVs X, Y with joint distribution $f_{X,Y}(x,y)$.

(i) The marginal p.d.f. of X only, $f_X(x)$, can be found as:

$$f_X(x)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

(ii) The marginal p.d.f. of Y only, $f_Y(y)$, can be found as:

$$f_Y(y)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

(iii) The conditional distribution of Y given $X=x$ can be found as:

$$f_{Y|X=x}(y)=f_{X,Y}(x,y)\,/\,f_X(x),\quad x \text{ fixed and } f_X(x)>0$$

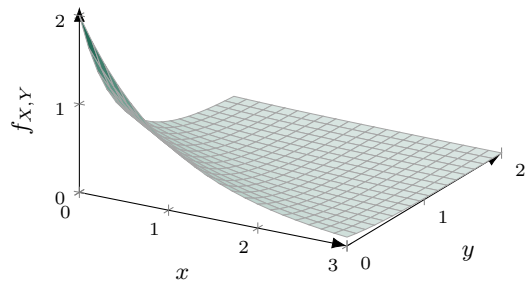
(iv) The conditional distribution of X given $Y=y$ can be found as:

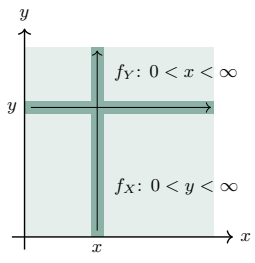
$$f_{X|Y=y}(x)=f_{X,Y}(x,y)\,/\,f_Y(y),\quad y \text{ fixed and } f_Y(y)>0$$

Example 7.10: The joint p.d.f. of X, Y is:

$$f_{X,Y}(x,y)=\begin{cases}2e^{-x}e^{-2y} & 0<x<\infty,\ 0<y<\infty\\ 0 & \text{o.w.}\end{cases}$$

a) Find $f_Y(y)$, $f_X(x)$.





$$f_Y(y) = \int_0^\infty 2e^{-x}e^{-2y}\,dx = -2e^{-x}e^{-2y}\Big|_0^\infty$$
$$= 0 - (-2e^{-2y}) = 2e^{-2y}, \quad Y \sim \text{Exp}(\lambda = 2)$$

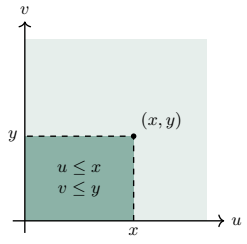
$$f_X(x) = \int_0^\infty 2e^{-x}e^{-2y}\,dy = -e^{-x}e^{-2y}\Big|_0^\infty$$
$$= 0 - (-e^{-x}) = e^{-x}, \quad X \sim \text{Exp}(\lambda = 1)$$

b) Find $f_{Y|X}(y)$

$$f_{Y|X} = f_{X,Y}(x,y) \,/\, f_X(x) = \frac{2e^{-x}e^{-2y}}{e^{-x}} = 2e^{-2y}$$

c) Find $F_{X,Y}(x,y)$

$$F_{X,Y}(x,y) = P(X \le x,\ Y \le y)$$



$$= \int_0^y \int_0^x 2e^{-u}e^{-2v}\,du\,dv = \int_0^y -2e^{-u}e^{-2v}\Big|_0^x\,dv$$
$$= \int_0^y 2e^{-2v}(1-e^{-x})\,dv = (1-e^{-x})\left(-e^{-2v}\Big|_0^y\right)$$
$$= (1-e^{-x})(1-e^{-2y}), \quad 0 < x < \infty,\ 0 < y < \infty$$

Example 7.11 (cave isopods, continued): Return to the cave isopods, with joint p.d.f. $f_{X,Y}(x,y) = \frac{21}{2}x^2 y$ on $0 < x^2 \le y < 1$, $x > 0$ (we found $c = \frac{21}{2}$).

a) How is body mass distributed, ignoring length? That is, find the marginal p.d.f. $f_X(x)$.

For a fixed x, y runs from the curve $y = x^2$ up to 1 (a vertical strip in the region picture), so

$$f_X(x) = \int_{x^2}^1 \tfrac{21}{2}x^2 y\,dy = \frac{21x^2}{2}\cdot\frac{y^2}{2}\Big|_{x^2}^1 = \frac{21x^2}{2}\cdot\frac{1-x^4}{2} = \frac{21}{4}x^2\left(1-x^4\right), \qquad 0 < x < 1.$$

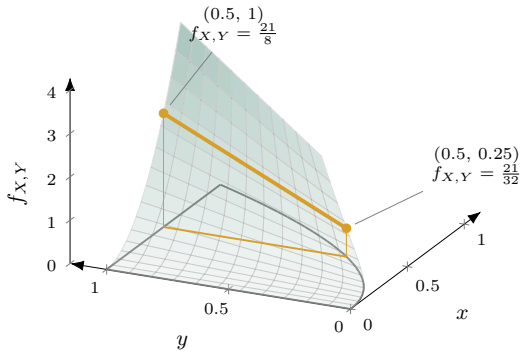
b) Among juveniles of a given mass, how is length distributed? That is, find the conditional p.d.f. $f_{Y|X=x}(y)$.

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{21}{2}x^2 y}{\frac{21}{4}x^2(1-x^4)} = \frac{2y}{1-x^4}, \qquad x^2 \le y \le 1.$$

The support of the conditional p.d.f. is the strip above x, and it depends on x.

c) A juvenile is at half of adult mass. How is its length distributed? That is, find $f_{Y|X=0.5}(y)$.

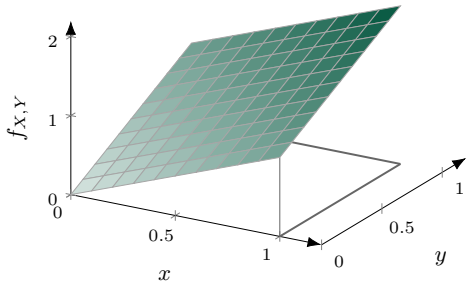
$$f_{Y|X=0.5}(y) = \frac{2y}{1-(0.5)^4} = \frac{2y}{15/16} = \frac{32}{15}y, \qquad 0.25 \le y \le 1.$$



Geometrically, we slice the surface above $x = 0.5$. The slice is a line in y, from height $\frac{21}{32}$ at $(0.5, 0.25)$ up to $\frac{21}{8}$ at $(0.5, 1)$. Dividing by $f_X(0.5) = \frac{21}{4}(0.25)(1 - \frac{1}{16}) = \frac{315}{256}$ rescales the slice to area 1, so the conditional p.d.f. runs from $\frac{8}{15}$ to $\frac{32}{15}$. Such a juvenile is at least a quarter of adult length, and longer lengths are more likely.

Exercises

Exercise 7.1. The random variables X, Y have joint p.d.f. $f(x, y) = c(x + y)$ for $0 < x < 1, 0 < y < 1$.



(a) Are X and Y independent?

(b) Find the value of c that makes this a valid joint p.d.f.

(c) Find the marginal p.d.f. of X, specifying the support.

(d) Find the conditional p.d.f. of Y given that $X = x$.

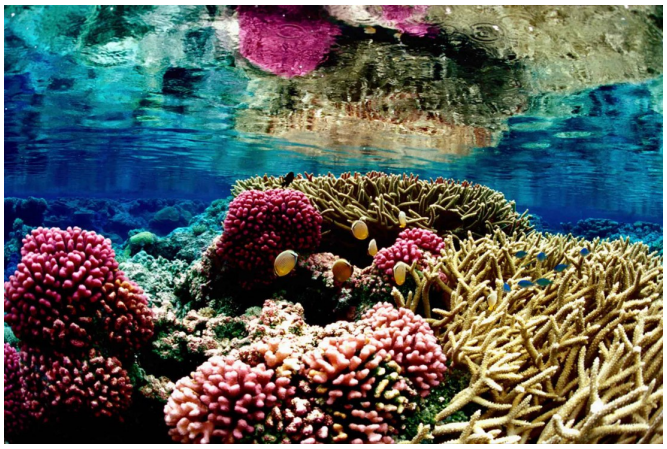


Figure 7.2: *A coral reef.* [17]

Exercise 7.2. An underwater camera at a reef monitoring station (Fig. 7.2) breaks down every so often. The dive team tries to fix it when it breaks; if they can't, they send it back to the manufacturer. In a particular month, let X be the number of times the camera breaks, and let Y be the number of times it is sent back. Assume

$$P(X = 0) = 0.3, \quad P(X = 1) = 0.4, \quad P(X = 2) = 0.2, \quad P(X = 3) = 0.1,$$

and that when the camera breaks, the team can fix it 60% of the time and must send it back 40% of the time (independently across breakdowns).

(a) Draw a table for the joint distribution of X and Y.

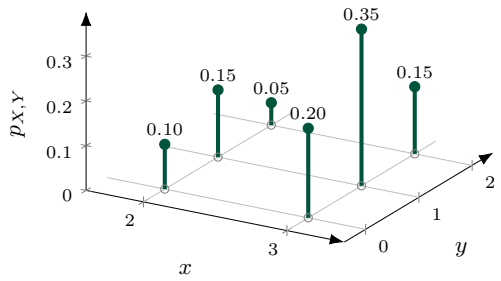
(b) Find the marginal p.d.f. of Y.

(c) Use the table to demonstrate that X and Y are dependent. (You need only one table entry that fails $P(X = a, Y = b) = P(X = a)P(Y = b)$.)

(d) Find the conditional distribution of X given that $Y = 2$.

Exercise 7.3 (A week on a Greek island)**.** A visitor's number of beach days X and ferry hops Y in a week have joint p.m.f.

$x \backslash y$	0	1	2
2	0.10	0.15	0.05
3	0.20	0.35	0.15



(a) What is the distribution of a visitor's beach days on their own, and of ferry hops on their own? That is, find both marginal p.m.f.s.

(b) Among visitors with 3 beach days, what is the chance of at least one ferry hop? That is, find $P(Y \geq 1 \mid X = 3)$.

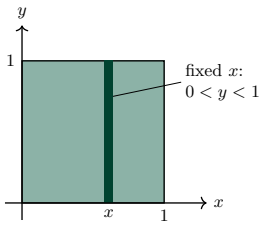
(c) Are X and Y independent? Check one cell.

(d) Do visitors with more beach days tend to take more or fewer ferry hops? That is, find the covariance of X and Y.

Solutions to the exercises

Work each exercise on the previous page before comparing.

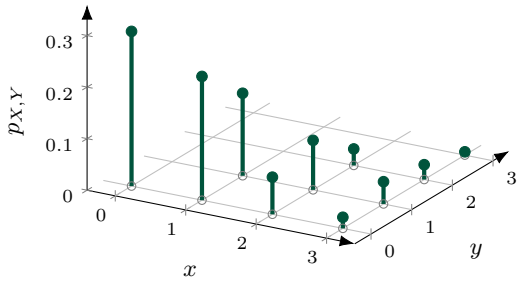
Exercise 7.1. (a) No: $x + y$ does not factor as $g(x)h(y)$.



(b) $\int_0^1\int_0^1 c(x+y)\,dx\,dy = c(\frac{1}{2}+\frac{1}{2}) = c$, so $c = 1$. **(c)** $f_X(x) = \int_0^1 (x+y)\,dy = x + \frac{1}{2}$ for $0 < x < 1$. **(d)** $f_{Y|X}(y \mid x) = \dfrac{x+y}{x+1/2}$ for $0 < y < 1$ (and fixed $0 < x < 1$).

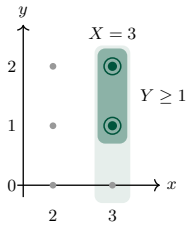
Exercise 7.2. (a) Given $X = x$ breakdowns, $Y \sim \text{Bin}(x, 0.4)$. Multiplying by $P(X = x)$:

$x \backslash y$	0	1	2	3
0	0.300	0	0	0
1	0.240	0.160	0	0
2	0.072	0.096	0.032	0
3	0.0216	0.0432	0.0288	0.0064



(b) Column sums: $f_Y = (0.6336,\ 0.2992,\ 0.0608,\ 0.0064)$. **(c)** $P(X = 0, Y = 1) = 0 \neq 0.3(0.2992)$. **(d)** $P(X = 2 \mid Y = 2) = 0.032/0.0608 = 0.526$ and $P(X = 3 \mid Y = 2) = 0.0288/0.0608 = 0.474$ (and 0 for $x < 2$).

Exercise 7.3. (a) $f_X = (0.30, 0.70)$ on $x = 2, 3$; $f_Y = (0.30, 0.50, 0.20)$. **(b)** $(0.35 + 0.15)/0.70 = 5/7 = 0.714$.



(c) No: $f(2,0) = 0.10 \neq 0.30(0.30) = 0.09$. **(d)** $\mathbb{E}(X) = 2.7$, $\mathbb{E}(Y) = 0.9$, $\mathbb{E}(XY) = 0.30 + 0.20 + 1.05 + 0.90 = 2.45$, so Cov $= 2.45 - 2.43 = 0.02$.

7.4 Conditional Expectation and Conditional Variance

Conditional Expectation

We can also find the expected value of X, given that we know $Y = y$. We simply replace $f_X(x)$ with $f_{X|Y=y}(x)$:

$$\mathbb{E}[X \mid Y = y_i] = \sum_{x \in \mathbb{R}} x_i \, f_{X|Y=y_i}(x_i) \qquad \text{(Discrete)}$$

or

$$\mathbb{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x \, f_{X|Y=y}(x) \, dx \qquad \text{(Continuous)}$$

We can also find $\mathbb{E}[Y \mid X = x]$ in a similar way.

Example 7.12 (Everest, continued): On the 0/1/2 scale, how far does a climber on oxygen get on average? That is, find $\mathbb{E}[Y \mid X = 1]$.

Weight each outcome by the conditional p.d.f. $f_{Y|X=1}$ found above:

$$\mathbb{E}[Y \mid X = 1] = \sum_{y} y \, f_{Y|X=1}(y) = 0(0.35) + 1(0.15) + 2(0.50) = 1.15.$$

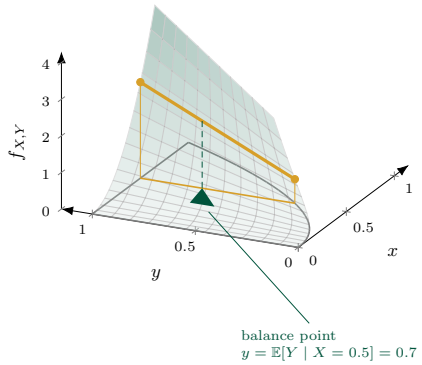
For comparison, $\mathbb{E}[Y] = 0(0.40) + 1(0.17) + 2(0.43) = 1.03$ over all climbers, and $\mathbb{E}[Y \mid X = 0] = \big(0(0.12) + 1(0.05) + 2(0.03)\big)/0.2 = 0.55$ for climbers without oxygen.

Example 7.13 (cave isopods, continued): For the cave isopods, a juvenile's mass is a fraction x of adult mass. What fraction of adult length do we expect it to have reached? That is, find $\mathbb{E}[Y \mid X = x]$, and evaluate it at $x = 0.5$.

Since this is a conditional expectation, we integrate against the conditional p.d.f. $f_{Y|X=x}(y) = \frac{2y}{1-x^4}$ on $x^2 \le y \le 1$:

$$\mathbb{E}[Y \mid X = x] = \int_{x^2}^{1} y \cdot \frac{2y}{1-x^4} \, dy = \frac{1}{1-x^4} \cdot \frac{2}{3} y^3 \Big|_{x^2}^{1} = \frac{1}{1-x^4} \cdot \frac{2}{3}\big(1 - (x^2)^3\big) = \frac{2}{3} \cdot \frac{1-x^6}{1-x^4}, \qquad 0 < x < 1.$$

At half of adult mass, $\mathbb{E}[Y \mid X = 0.5] = \frac{2}{3} \cdot \frac{1-1/64}{1-1/16} = \frac{2}{3} \cdot \frac{63/64}{15/16} = \frac{7}{10}$: a juvenile at 50% of adult mass is expected to be at 70% of adult length.



The slice of the surface above $X = x$, rescaled to area 1, is the conditional p.d.f. of Y, and $\mathbb{E}[Y \mid X = x]$ is the point where that slice balances.

Conditional Variance

Similarly, we have

$$\begin{aligned}\text{Var}[X \mid Y = y_i] &= \mathbb{E}[X^2 \mid Y = y_i] - (\mathbb{E}[X \mid Y = y_i])^2 \\ &= \sum_{x_i} x_i^2 \, f_{X|Y=y_i}(x_i) - \left(\sum_{x_i} x_i \, f_{X|Y=y_i}(x_i) \right)^2\end{aligned}$$

For the discrete case.

$$\begin{aligned}\text{Var}[X \mid Y = y] &= \mathbb{E}[X^2 \mid Y = y] - (\mathbb{E}[X \mid Y = y])^2 \\ &= \int x^2 \, f_{X|Y=y}(x) \, dx - \left(\int x \, f_{X|Y=y}(x) \, dx \right)^2\end{aligned}$$

For the continuous case.

The Tower Property

First, a remark on the object involved. For each fixed y, $\mathbb{E}[X \mid Y = y]$ is a number. If we let the conditioning value vary, we obtain a *function* of Y, written $\mathbb{E}[X \mid Y]$. Since it is a function of the RV Y, $\mathbb{E}[X \mid Y]$ is itself a random variable, and we may take its expectation w.r.t. the distribution of Y. Doing so recovers the unconditional mean:

Proposition 7.1 (Tower Property):

$$\mathbb{E}_Y\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X], \quad \text{and} \quad \mathbb{E}_X\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}[Y]$$

Proof for the continuous case:

$$\begin{aligned}\mathbb{E}[g(Y)] = \int_{-\infty}^{\infty} g(y) \, f_Y(y) \, dy, \quad \text{Let } g(y) = \mathbb{E}[X \mid Y = y] \\ &= \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] \, f_Y(y) \, dy \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x \, f_{X|Y=y}(x \mid y) \, dx \right] f_Y(y) \, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \left[\frac{f_{X,Y}(x,y)}{f_Y(y)} \right] f_Y(y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \, f_{X,Y}(x,y) \, dy \, dx \\ &= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X,Y}(x,y) \, dy \right] dx \\ &= \int_{-\infty}^{\infty} x \, f_X(x) \, dx = \mathbb{E}[X] \qquad \blacksquare\end{aligned}$$

In words: averaging the group means $\mathbb{E}[X \mid Y = y]$, weighted by how likely each group is, recovers the overall mean of X.

The tower property also goes by the *law of total expectation* and *double expectation*. Writing $g(y) = \mathbb{E}[X \mid Y = y]$, it reads $\mathbb{E}[X] = \mathbb{E}[g(Y)]$, so the outer expectation is computed with LOTUS against f_Y; likewise $\mathbb{E}[Y] = \mathbb{E}[h(X)]$ with $h(x) = \mathbb{E}[Y \mid X = x]$.

Example 7.14 (cave isopods, continued): Averaged over all juveniles, what fraction of adult length have they reached? That is, use the tower property to find $\mathbb{E}[Y]$.

From the isopod example above, $h(x) = \mathbb{E}[Y \mid X = x] = \frac{2}{3} \cdot \frac{1-x^6}{1-x^4}$, and the marginal is $f_X(x) = \frac{21}{4}x^2(1-x^4)$ on $0 < x < 1$. By LOTUS,

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \int_0^1 \frac{2}{3} \cdot \frac{1-x^6}{1-x^4} \cdot \frac{21}{4}x^2(1-x^4)\,dx = \frac{2 \cdot 21}{3 \cdot 4}\int_0^1 \left(x^2 - x^8\right) dx$$
$$= \frac{7}{2}\left(\frac{x^3}{3} - \frac{x^9}{9}\right)\Big|_0^1 = \frac{7}{2} \cdot \frac{2}{9} = \frac{7}{9}.$$

The factor $1 - x^4$ cancels, which is what makes this route short. An average juvenile is $\frac{7}{9}$ of the way to adult length. As a check, $f_Y(y) = \int_0^{\sqrt{y}} \frac{21}{2}x^2 y\,dx = \frac{7}{2}y^{5/2}$ on $0 < y < 1$, and $\mathbb{E}[Y] = \int_0^1 \frac{7}{2}y^{7/2}\,dy = \frac{7}{9}$.

Law of Total Variance

The variance of X decomposes along the same conditioning:

Theorem 7.1 (Law of Total Variance)**.** *For RVs X, Y with finite variances,*

$$\mathrm{Var}[X] = \mathbb{E}_Y\big[\mathrm{Var}[X \mid Y]\big] + \mathrm{Var}_Y\big(\mathbb{E}[X \mid Y]\big)$$

(and likewise with the roles of X and Y exchanged).

Proof. Both terms on the RHS are expectations of functions of Y. First, by the tower property applied to X^2,

$$\mathbb{E}\big[\mathrm{Var}[X \mid Y]\big] = \mathbb{E}\big[\mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2\big] = \mathbb{E}[X^2] - \mathbb{E}\big[(\mathbb{E}[X \mid Y])^2\big]$$

Second, by the tower property applied to X,

$$\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \mathbb{E}\big[(\mathbb{E}[X \mid Y])^2\big] - \big(\mathbb{E}\big[\mathbb{E}[X \mid Y]\big]\big)^2 = \mathbb{E}\big[(\mathbb{E}[X \mid Y])^2\big] - (\mathbb{E}[X])^2$$

Adding the two lines, the $\mathbb{E}[(\mathbb{E}[X \mid Y])^2]$ terms cancel, leaving $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathrm{Var}[X]$. ∎

In words: total variance = (average spread *within* the groups defined by Y) + (spread *between* the group means). A worked instance appears in Section 8.4. This decomposition returns constantly in later courses: in regression it splits variation into "explained" and "unexplained" parts, and in machine learning it underlies the bias–variance decomposition.

Conditional Expectation as the Best Predictor

Why is $\mathbb{E}[Y \mid X]$ the right notion of "predicting Y from X"? Because among *all* functions of X, it has the smallest mean squared error:

Proposition 7.2. *For any function g with $\mathbb{E}[g(X)^2] < \infty$,*

$$\mathbb{E}\big[(Y - g(X))^2\big] \;\geq\; \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2\big],$$

with equality iff $g(X) = \mathbb{E}[Y \mid X]$ with probability 1.

Proof sketch. Write $Y - g(X) = \big(Y - \mathbb{E}[Y \mid X]\big) + \big(\mathbb{E}[Y \mid X] - g(X)\big)$ and expand the square. The cross term vanishes: conditionally on X, the factor $\mathbb{E}[Y \mid X] - g(X)$ is a constant while $\mathbb{E}\big[Y - \mathbb{E}[Y \mid X] \mid X\big] = 0$, so the tower property sends the cross term to 0. What remains is

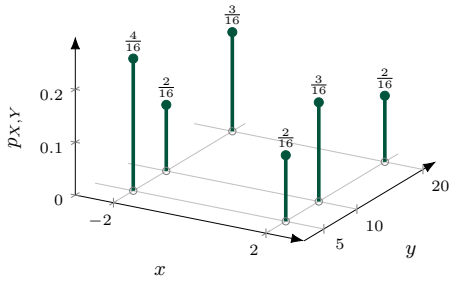
$$\mathbb{E}\big[(Y - g(X))^2\big] = \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2\big] + \underbrace{\mathbb{E}\big[(\mathbb{E}[Y \mid X] - g(X))^2\big]}_{\geq 0} \qquad \blacksquare$$

The function $x \mapsto \mathbb{E}[Y \mid X = x]$ is called the *regression function* of Y on X. This proposition is the reason your regression course estimates conditional means, and why machine learning methods trained under squared-error loss are all, at bottom, estimating $\mathbb{E}[Y \mid X]$.

7.5 Conditional Expectation: Worked Examples

Example 7.15: Consider a discrete joint p.d.f. as follows $P(X = x, Y = y)$:

$x \backslash y$	5	10	20	$f_X(x)$
-2	4/16	2/16	3/16	9/16
2	2/16	3/16	2/16	7/16
$f_Y(y)$	6/16	5/16	5/16	total: 1



Note the marginal p.d.f.'s are given.

a) Find $f_{Y|X=-2}(y)$, $f_{Y|X=+2}(y)$:

Y	5	10	20
$P(Y = y \mid X = -2)$	4/9	2/9	3/9

Y	5	10	20
$P(Y = y \mid X = 2)$	2/7	3/7	2/7

b) Find $\mathbb{E}[Y \mid X = -2]$, $\mathbb{E}[Y \mid X = +2]$, $\text{Var}[Y \mid X = +2]$

$$\mathbb{E}[Y \mid X = -2] = \sum_{y_i} y_i \, f_{Y|X}(y_i) = 5\left(\tfrac{4}{9}\right) + 10\left(\tfrac{2}{9}\right) + 20\left(\tfrac{3}{9}\right) = \frac{100}{9}$$

$$\mathbb{E}[Y \mid X = 2] = \sum_{y_i} y_i \, f_{Y|X}(y_i) = 5\left(\tfrac{2}{7}\right) + 10\left(\tfrac{3}{7}\right) + 20\left(\tfrac{2}{7}\right) = \frac{80}{7}$$

$$\mathbb{E}[Y^2 \mid X = 2] = \sum_{y_i} y_i^2 \, f_{Y|X}(y_i) = 5^2\left(\tfrac{2}{7}\right) + 10^2\left(\tfrac{3}{7}\right) + 20^2\left(\tfrac{2}{7}\right) = \frac{1150}{7}$$

$$\begin{aligned}\text{Var}[Y \mid X = 2] &= \mathbb{E}[Y^2 \mid X = 2] - (\mathbb{E}[Y \mid X = 2])^2 \\ &= \frac{1150}{7} - \left(\frac{80}{7}\right)^2 = \frac{1150}{7} - \frac{6400}{49} = \frac{8050}{49} - \frac{6400}{49} = \frac{1650}{49}\end{aligned}$$

c) Find $\mathbb{E}[Y]$, and verify it is equal to $\mathbb{E}[\mathbb{E}[Y \mid X]]$

$$\mathbb{E}[Y] = \sum_{y_i} y_i \, f_Y(y_i) = 5\left(\tfrac{6}{16}\right) + 10\left(\tfrac{5}{16}\right) + 20\left(\tfrac{5}{16}\right) = 11.25 \quad \checkmark$$

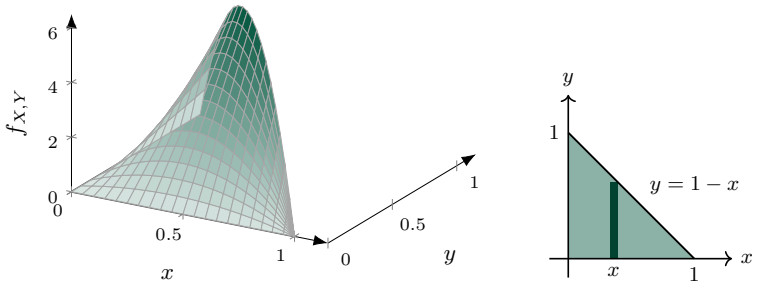
$$\begin{aligned}\mathbb{E}_X[\mathbb{E}[Y \mid X]] &= \sum_x \mathbb{E}[Y \mid X = x_i] \, f_X(x_i) = \left(\frac{100}{9}\right)\frac{9}{16} + \left(\frac{80}{7}\right)\frac{7}{16} \\ &= 11.25 \quad \checkmark\end{aligned}$$

Example 7.16: In a kelp-forest survey, divers record what covers each square-meter quadrat of reef. Let X be the fraction of the quadrat covered by kelp and Y the fraction covered by purple urchins. Kelp and urchins take up separate parts of the same square meter, so together they cover at most all of it: $X + Y \le 1$, with whatever is left being bare rock or other cover. Across the surveyed reef the pair is modeled by

$$f_{X,Y}(x,y) = \begin{cases} 24xy & 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le y + x \le 1 \\ 0 & \text{o.w.} \end{cases}$$

$$f_X(x) = \begin{cases} 12x(1-x)^2 & 0 < x < 1 \\ 0 & \text{o.w.} \end{cases} \qquad f_Y(y) = \begin{cases} 12y(1-y)^2 & 0 < y < 1 \\ 0 & \text{o.w.} \end{cases}$$

a) Among quadrats where kelp covers a fraction x, how is urchin cover distributed? That is, find $f_{Y|X=x}(y)$



$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{24xy}{12x(1-x)^2} = \frac{2y}{(1-x)^2}, \quad 0 \le y \le 1-x,\ 0 \le y + x \le 1$$

b) On average, how much of a quadrat do urchins cover when kelp covers a fraction x? When kelp covers half? That is, find $\mathbb{E}[Y \mid X = x]$, and $\mathbb{E}[Y \mid X = \frac{1}{2}]$

$$\mathbb{E}[Y \mid X = x] = \int_0^{1-x} y \cdot \frac{2y}{(1-x)^2}\, dy = \frac{2}{(1-x)^2} \cdot \frac{y^3}{3}\Big|_0^{1-x} = \frac{2(1-x)^3}{3(1-x)^2} = \frac{2}{3}(1-x)$$

$$\mathbb{E}[Y \mid X = \tfrac{1}{2}] = \frac{2}{3}(1-x) = \frac{2}{3}\left(\tfrac{1}{2}\right) = \frac{2}{6} = \frac{1}{3}$$

Given that kelp covers a fraction x of the quadrat, urchins cover two-thirds of what is left on average; at half kelp cover, a third of the quadrat is urchins.

c) Averaged over the whole reef, what fraction of a quadrat do urchins cover? That is, find $\mathbb{E}_X[\mathbb{E}[Y \mid X = x]]$

$$\begin{aligned}
\mathbb{E}_X[\mathbb{E}[Y \mid X = x]] &= \int_0^1 \tfrac{2}{3}(1-x)\, f_X(x)\, dx \\
&= \int_0^1 \tfrac{2}{3}(1-x)\big(12x(1-x)^2\big)\, dx \\
&= \int_0^1 \tfrac{2}{3}\cdot 12x(1-x)^3 \\
&= \int_0^1 8x(1-x)^3\, dx \\
&= 8\int_0^1 x\big(-x^3+3x^2-3x+1\big)\, dx = 8\int_0^1 -x^4+3x^3-3x^2+x\, dx \\
&= 8\Big[-\tfrac{x^5}{5}+\tfrac{3x^4}{4}-x^3+\tfrac{x^2}{2}\Big]_0^1 = 8\Big(-\tfrac{4}{20}+\tfrac{15}{20}-\tfrac{20}{20}+\tfrac{10}{20}\Big) \\
&= 8\Big(\tfrac{1}{20}\Big) = \tfrac{2}{5}
\end{aligned}$$

Averaged over all quadrats, urchins cover 40% of the reef.

Exercises

Exercise 7.4. The random variables X, Y have joint p.d.f. $f(x,y) = 6(x-y)$ for $0 < y < x < 1$.

(a) Sketch the region where f is positive. Verify that f integrates to 1 over that region.

(b) Show that the marginal density of X is $f_X(x) = 3x^2$ on $0 < x < 1$.

(c) Find the conditional density $f_{Y|X=x}(y)$, specifying for which y it is valid.

(d) Show that $\mathbb{E}[Y \mid X = x] = x/3$, and hence that the *random variable* $\mathbb{E}[Y \mid X]$ equals $X/3$.

(e) Compute $\mathbb{E}[X]$, then use the tower property $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$ to find $\mathbb{E}[Y]$.

(f) Check your answer by finding $f_Y(y)$ directly and computing $\mathbb{E}[Y]$ from it. Which route was less work?

Exercise 7.5 (Poison dart frogs in two quadrats)**.** Strawberry poison dart frogs are thumbnail-sized, bright red frogs of the Central American rainforest floor, whose colors warn predators of the toxins in their skin. A field team counts them in two equal leaf-litter quadrats on a rainforest floor. The counts X and Y are independent $\text{Pois}(\lambda)$ random variables; let $T = X + Y$ be the total. (Section 9.4 will show that $T \sim \text{Pois}(2\lambda)$; assume that here.)

(a) For $0 \le j \le k$, write $P(X = j \mid T = k)$ as a ratio of probabilities and substitute the Poisson p.d.f.s.

(b) Simplify. Every λ and every $e^{-\lambda}$ should cancel, leaving $P(X = j \mid T = k) = \binom{k}{j}\left(\tfrac{1}{2}\right)^k$.

(c) Identify the conditional distribution of X given $T = k$ by name.

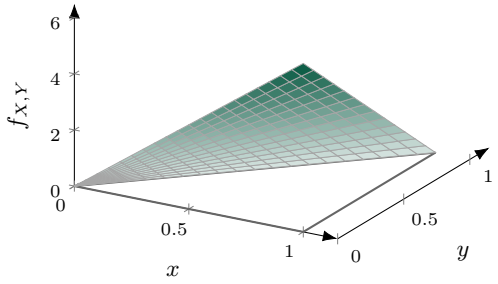
(d) Use part (c) to write down $\mathbb{E}[X \mid T]$ without further computation. Then give a one-sentence symmetry argument for the same answer that uses no algebra at all.

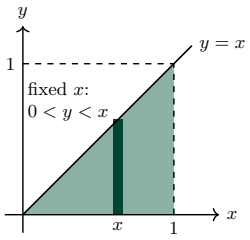
(e) Verify the tower property: check that $\mathbb{E}[\mathbb{E}[X \mid T]] = \mathbb{E}[X]$.

Solutions to the exercises

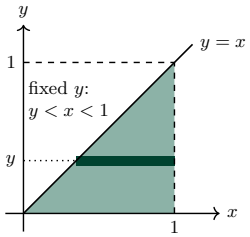
Work each exercise on the previous page before comparing.

Exercise 7.4. (a) The triangle below the diagonal: $0 < y < x < 1$. $\int_0^1 \int_0^x 6(x-y)\,dy\,dx = \int_0^1 3x^2\,dx = 1$.



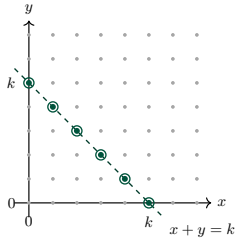


(b) $f_X(x) = \int_0^x 6(x-y)\,dy = 6x^2 - 3x^2 = 3x^2$. **(c)** $f_{Y|X=x}(y) = \dfrac{6(x-y)}{3x^2} = \dfrac{2(x-y)}{x^2}$ for $0 < y < x$. **(d)** $\mathbb{E}[Y \mid X = x] = \int_0^x y\,\dfrac{2(x-y)}{x^2}\,dy = \dfrac{2}{x^2}\left(\frac{x^3}{2} - \frac{x^3}{3}\right) = \dfrac{x}{3}$, so $\mathbb{E}[Y \mid X] = X/3$. **(e)** $\mathbb{E}[X] = \int_0^1 3x^3\,dx = \frac{3}{4}$, so $\mathbb{E}[Y] = \mathbb{E}[X/3] = \frac{1}{4}$.



(f) $f_Y(y) = \int_y^1 6(x-y)\,dx = 3(1-y)^2$, and $\mathbb{E}[Y] = \int_0^1 3y(1-y)^2\,dy = \frac{1}{4}$. The tower route needed one easy integral instead of two.

Exercise 7.5. (a) $P(X = j \mid T = k) = \dfrac{P(X=j)P(Y=k-j)}{P(T=k)} = \dfrac{e^{-\lambda}\lambda^j/j! \cdot e^{-\lambda}\lambda^{k-j}/(k-j)!}{e^{-2\lambda}(2\lambda)^k/k!}$.



(b) The exponentials cancel and $\lambda^k/(2\lambda)^k = 2^{-k}$, leaving $\dfrac{k!}{j!(k-j)!}2^{-k} = \binom{k}{j}\left(\frac{1}{2}\right)^k$. **(c)** $X \mid T = k \sim \text{Bin}(k, \frac{1}{2})$. **(d)** $\mathbb{E}[X \mid T] = T/2$; by symmetry X and Y split the total evenly in expectation. **(e)** $\mathbb{E}[T/2] = 2\lambda/2 = \lambda = \mathbb{E}[X]$.

8 Covariance and Correlation

8.1 Joint Expectations and the Joint m.g.f.

Joint Expectations

The law of the unconscious statistician (LOTUS) works the same way with two RVs: weight $g(x, y)$ by the joint p.d.f. and add up over the support S. For a real-valued function $g(x, y)$ on $\mathbb{R}^2$, we have the following results:

(i) $\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x_i, y_i)\, f_{X,Y}(x_i, y_i)$ (discrete)

$\mathbb{E}[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)\, f_{X,Y}(x,y)\, dx\, dy$ (continuous)

(ii) $\text{Var}[g(X,Y)] = \mathbb{E}[(g(X,Y))^2] - (\mathbb{E}[g(X,Y)])^2$ (think: $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$)

Assuming all expectations finite, and c, d constants,

(iii) $\mathbb{E}[cg(X,Y)] = c\,\mathbb{E}[g(X,Y)]$
$\mathbb{E}[c + dg(X,Y)] = c + d\,\mathbb{E}[g(X,Y)]$

(iv) $\mathbb{E}[cX + dY] = c\,\mathbb{E}[X] + d\,\mathbb{E}[Y]$

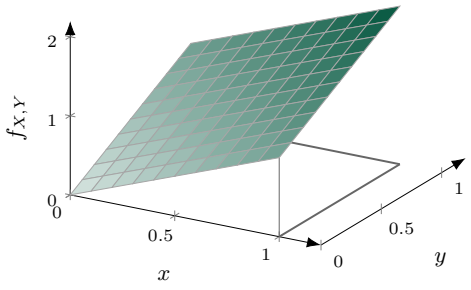
(v) $g(x,y) \le h(x,y) \implies \mathbb{E}[g(X,Y)] \le \mathbb{E}[h(X,Y)]$
$g(x) \le h(x) \implies \mathbb{E}[g(x)] \le \mathbb{E}[h(x)]$

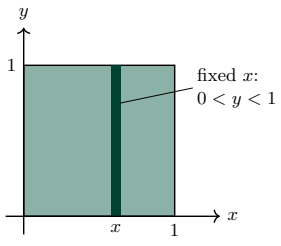
(vi) $\text{Var}[cg(X,Y)] = c^2\,\text{Var}[g(X,Y)]$
$\text{Var}[c + dg(X,Y)] = d^2\,\text{Var}[g(X,Y)]$

Just as $M_X(t)$ packages all the moments of one RV, a two-argument version packages all joint moments, in particular $\mathbb{E}[XY]$, which we will need when we define covariance in Section 8.2. It also yields a clean criterion for independence in Section 9.1.

Example 8.1: Take the joint p.d.f. $f_{X,Y}(x,y) = x + y$ on the unit square from Section 7.2, where we found $f_X(x) = x + \frac{1}{2}$ and $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{7}{12}$.

a) Compute $\mathbb{E}[XY]$ using the two-dimensional version of LOTUS:





$$\mathbb{E}[XY] = \int_0^1 \int_0^1 xy\,(x+y)\,dx\,dy = \int_0^1 \int_0^1 \left(x^2y + xy^2\right) dx\,dy$$
$$= \int_0^1 \left(\tfrac{y}{3} + \tfrac{y^2}{2}\right) dy = \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{3}$$

b) Compare $\mathbb{E}[XY]$ with $\mathbb{E}[X]\,\mathbb{E}[Y]$:

$$\mathbb{E}[X]\mathbb{E}[Y] = \left(\tfrac{7}{12}\right)^2 = \tfrac{49}{144} \approx 0.3403, \qquad \mathbb{E}[XY] = \tfrac{1}{3} = \tfrac{48}{144} \approx 0.3333$$

These are *not* equal, which confirms what we found in Section 7.2: X and Y are not independent.

Note: The gap $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = -\frac{1}{144}$ is the covariance of X and Y, which we define formally in Section 8.2. It is negative and small: knowing X is large makes Y slightly *less* likely to be large, because the density must integrate to 1 over a fixed square.

Definition 8.1: The joint moment generating function of X, Y is:

$$M_{X,Y}(t_1, t_2) = \mathbb{E}[e^{t_1 X} e^{t_2 Y}] = \mathbb{E}[e^{t_1 X + t_2 Y}]$$

Thus for the discrete case:

$$M_{X,Y}(t_1, t_2) = \sum_x \sum_y e^{t_1 x + t_2 y}\, f_{X,Y}(x_i, y_i)$$

And for the continuous case:

$$M_{X,Y}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y}\, f_{X,Y}(x, y)\, dy\, dx$$

Proposition 8.1:

(i) $M_{X,Y}(0,0) = 1$ for any RVs X, Y

(ii) $M_{X,Y}(t_1, 0) = M_X(t_1), \quad M_{X,Y}(0, t_2) = M_Y(t_2)$

(iii) For c_1, c_2, d_1, d_2 constants:
$M_{c_1 X + d_1,\, c_2 Y + d_2}(t_1, t_2) = e^{d_1 t_1} e^{d_2 t_2}\, M_{X,Y}(c_1 t_1,\, c_2 t_2)$

(iv) $\dfrac{\partial}{\partial t_1} M_{X,Y}(t_1, t_2)\Big|_{t_1 = t_2 = 0} = \mathbb{E}[X]$

$\dfrac{\partial}{\partial t_2} M_{X,Y}(t_1, t_2)\Big|_{t_1 = t_2 = 0} = \mathbb{E}[Y]$

$\dfrac{\partial^2}{\partial t_1\, \partial t_2} M_{X,Y}(t_1, t_2)\Big|_{t_1 = t_2 = 0} = \mathbb{E}[XY]$

(so long as we may exchange differentiation and taking expectations)

Example 8.2: Suppose the joint m.g.f. of X, Y is given by

$$M_{X,Y}(t_1, t_2) = \left(\frac{2}{2 - t_1}\right)\left(\frac{3}{3 - t_2}\right), \quad t_1 < 2,\ t_2 < 3$$

Find

a) $\mathbb{E}[X], \mathbb{E}[Y]$

$$\left.\frac{\partial}{\partial t_1} M_{X,Y}(t_1, t_2)\right|_{t_1=0=t_2} = \left(\frac{2}{(2-t_1)^2}\right)\left(\frac{3}{3-t_2}\right)\Bigg|_{t_1=t_2=0} = \frac{2}{4} = \frac{1}{2}$$

$$\left.\frac{\partial}{\partial t_2} M_{X,Y}(t_1, t_2)\right|_{t_1=0=t_2} = \left.\frac{3}{(3-t_2)^2} \cdot \frac{2}{2-t_1}\right|_{t_1=t_2=0} = \frac{3}{9} = \frac{1}{3}$$

b) $\mathbb{E}[XY]$

$$\frac{\partial^2}{\partial t_1 \, \partial t_2} = \left.\frac{3}{(3-t_2)^2} \cdot \frac{2}{(2-t_1)^2}\right|_{t_1=t_2=0} = \frac{3}{9} \cdot \frac{2}{4} = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{6}$$

Covariance/Correlation

Next, we discuss how/if X and Y vary together, specifically if they have a linear relationship.

If as $X \uparrow$, Y tends to $\uparrow$, X and Y have a positive linear relationship.

If as $X \uparrow$, Y tends to $\downarrow$, X and Y have a negative linear relationship.

8.2 Covariance: Definition and Bounds

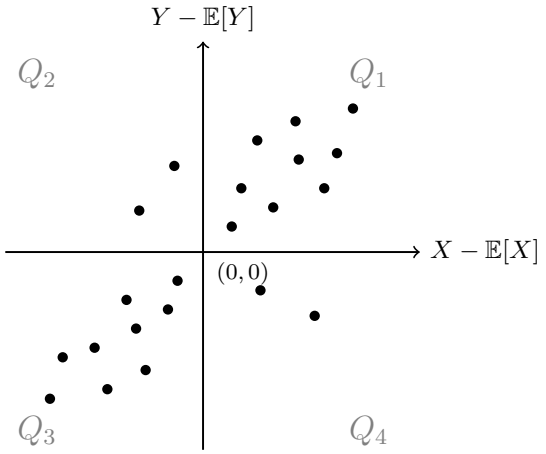
Covariance is one of the ways in which the strength of a linear relationship is measured.

Definition 8.2: The covariance between two random variables X and Y is:

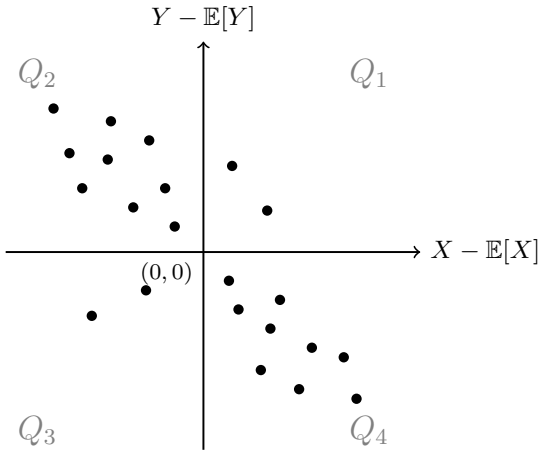
$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

In words, covariance measures how much X and Y *co-vary*: whether they tend to land on the same side of their means at the same time.

Consider $\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[Y - \mathbb{E}[Y]] = 0$. Subtracting the mean from a RV "centers" it about 0. Consider X, Y to be discrete RVs. Then, a plot of $(X - \mathbb{E}[X])$ vs $(Y - \mathbb{E}[Y])$:



Positive relationship: $\text{Cov}(X, Y) > 0$



Negative relationship: $\text{Cov}(X, Y) < 0$

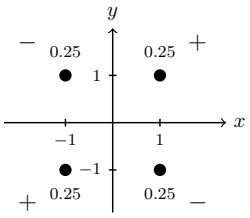
Notice that with a positive relationship (left), the majority of points are in Q_1, Q_3, where $(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) > 0$; with a negative relationship (right), the majority are in Q_2, Q_4, where $(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) < 0$.

This suggests that if

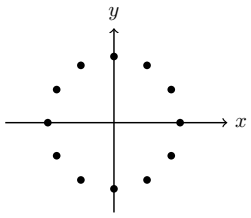
$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \sum_{x,y} f_{X,Y}(x_i, y_i)(x_i - \mathbb{E}[X])(y_i - \mathbb{E}[Y]) > 0,$$

then X, Y have a positive linear relationship, while $\text{Cov}(X, Y) < 0$ suggests a negative linear relationship. One way to read the picture is as a see-saw. Each point carries the weight $(x - \mathbb{E}[X])(y - \mathbb{E}[Y])$: put the Q_1 and Q_3 points on one side and the Q_2 and Q_4 points on the other. The covariance is positive when the Q_1/Q_3 side is heavier and negative when the Q_2/Q_4 side is.

The third case: as $X \uparrow$, Y tends to stay the same, and we call X, Y *uncorrelated*. Now the points balance across the quadrants, the positive products in Q_1, Q_3 cancel the negative ones in Q_2, Q_4, and we want $\text{Cov}(X, Y) = 0$:



Four corners: $\text{Cov}(X, Y) = 0$



A ring: $\text{Cov}(X, Y) = 0$

Example 8.3 (four corners): Let X, Y have joint p.d.f. $f_{X,Y}(x, y) = 0.25$ at each of $(-1, -1)$, $(-1, 1)$, $(1, -1)$, $(1, 1)$ (left picture above). In table form, with the marginals in the margins:

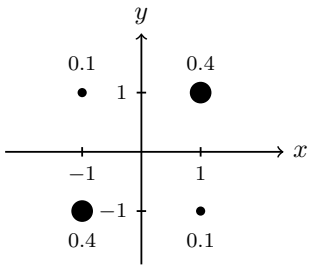
$y \backslash x$	-1	1	$f_Y(y)$
1	0.25	0.25	0.5
-1	0.25	0.25	0.5
$f_X(x)$	0.5	0.5	1

Using the marginal, $\mathbb{E}[X] = \sum_x x\, f_X(x) = (1)(0.5) + (-1)(0.5) = 0$, and likewise $\mathbb{E}[Y] = 0$. By LOTUS with $g(x, y) = xy$,

$$\mathbb{E}[XY] = (-1)(-1)(0.25) + (1)(-1)(0.25) + (-1)(1)(0.25) + (1)(1)(0.25) = 0,$$

so $\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] = 0$: the four corners are uncorrelated.

What if we move weight onto the diagonal: $f_{X,Y}(1, 1) = f_{X,Y}(-1, -1) = 0.4$ and $f_{X,Y}(-1, 1) = f_{X,Y}(1, -1) = 0.1$?



$y \backslash x$	-1	1	$f_Y(y)$
1	0.1	0.4	0.5
-1	0.4	0.1	0.5
$f_X(x)$	0.5	0.5	1

The marginals have not changed, so still $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, but now

$$\mathbb{E}[XY] = (-1)(-1)(0.4) + (1)(-1)(0.1) + (-1)(1)(0.1) + (1)(1)(0.4) = 0.8 - 0.2 = 0.6,$$

so $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.6 - (0)(0) = 0.6 > 0$. Most of the probability sits in Q_1 and Q_3: as $X \uparrow$, Y tends to $\uparrow$.

Theorem 8.1. *Consider RVs X, Y with finite expectation, finite and positive variances σ_X^2, σ_Y^2. Then,*

$$-\sigma_X \sigma_Y \le \text{Cov}(X, Y) \le \sigma_X \sigma_Y$$

and

$$\text{Cov}(X,Y) = \sigma_X\sigma_Y \quad \textit{iff} \quad P\left[Y = \mathbb{E}[Y] + \tfrac{\sigma_Y}{\sigma_X}(X - \mathbb{E}[X])\right] = 1$$
$$\text{Cov}(X,Y) = -\sigma_X\sigma_Y \quad \textit{iff} \quad P\left[Y = \mathbb{E}[Y] - \tfrac{\sigma_Y}{\sigma_X}(X - \mathbb{E}[X])\right] = 1$$

Proposition 8.2: $\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$

Proof:

$$\begin{aligned}
\text{Cov}(X,Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\
&= \mathbb{E}[XY - Y\mathbb{E}[X] - X\mathbb{E}[Y] + \mathbb{E}[X]\mathbb{E}[Y]] \\
\text{E[X] constant,} \quad &= \mathbb{E}[XY] - \mathbb{E}[Y\mathbb{E}[X]] - \mathbb{E}[X\mathbb{E}[Y]] + \mathbb{E}[X]\mathbb{E}[Y] \\
\text{E[Y] constant,} \quad &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\
&= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \qquad \blacksquare
\end{aligned}$$

How does this relate to variance? Taking $Y = X$,

$$\text{Var}[X] = \mathbb{E}[(X - \mu_X)(X - \mu_X)] = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])] = \text{Cov}(X,X),$$

so variance is the covariance of a RV with itself, and the computational formula becomes $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Proof of Theorem 8.1:

WLOG (justified by the standardization below), let $\sigma_X^2 = \sigma_Y^2 = 1$, $\mathbb{E}[X] = \mathbb{E}[Y] = 0$. We need to show $-1 \le \text{Cov}(X,Y) \le 1$, i.e. (since here $\text{Cov}(X,Y) = \mathbb{E}[XY]$) that $-1 \le \mathbb{E}[XY] \le 1$.

Clearly, $0 \le \mathbb{E}[(X - Y)^2]$

$$\begin{aligned}
&= \mathbb{E}[X^2] + \mathbb{E}[Y^2] - 2\mathbb{E}[XY] \\
\text{Since } \mathbb{E}[X] = \mathbb{E}[Y] = 0, \quad &= 2 - 2\mathbb{E}[XY] \\
&\Rightarrow \; -2 \le 2\mathbb{E}[XY] \\
&\Rightarrow \; 1 \ge \mathbb{E}[XY] \;\Rightarrow\; \mathbb{E}[XY] \le 1
\end{aligned}$$

Similarly $0 \le \mathbb{E}[(X + Y)^2]$

$$\begin{aligned}
&= \mathbb{E}[X^2] + \mathbb{E}[Y^2] + 2\mathbb{E}[XY] = 2 + 2\mathbb{E}[XY] \\
&-2 \le 2\mathbb{E}[XY] \;\Rightarrow\; -1 \le \mathbb{E}[XY]
\end{aligned}$$

Combining the two bounds, $-1 \le \mathbb{E}[XY] \le 1$. $\blacksquare$

Now, notice for any RVs X, Y with mean μ_X, μ_Y, variance σ_X^2, σ_Y^2.

Let $X^* = \dfrac{X - \mu_X}{\sigma_X}$, $Y^* = \dfrac{Y - \mu_Y}{\sigma_Y}$, so that $\sigma_{X^*}^2 = \sigma_{Y^*}^2 = 1$, $\mu_{X^*} = \mu_{Y^*} = 0$.

Then, we have:

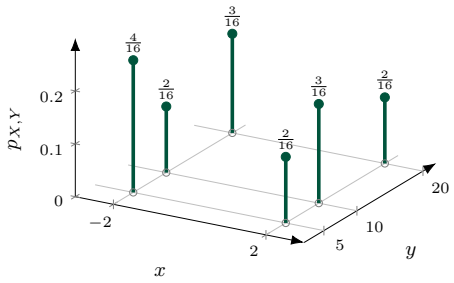
$$\begin{aligned}
-1 &\le \mathbb{E}\left[\left(\frac{X - \mu_X}{\sigma_X}\right)\left(\frac{Y - \mu_Y}{\sigma_Y}\right)\right] \le 1 \\
-1 &\le \frac{1}{\sigma_X\sigma_Y}\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \le 1 \\
&\Rightarrow \; -\sigma_X\sigma_Y \le \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \le \sigma_X\sigma_Y \\
&\Rightarrow \; -\sigma_X\sigma_Y \le \text{Cov}(X,Y) \le \sigma_X\sigma_Y \qquad \blacksquare
\end{aligned}$$

8.3 Covariance Computations and Correlation

Example 8.4: Recall X, Y with joint p.d.f.:

$x \backslash y$	5	10	20	$f_X(x)$
-2	4/16	2/16	3/16	9/16
2	2/16	3/16	2/16	7/16
$f_Y(y)$	6/16	5/16	5/16	total: 1

$$\mathbb{E}[X] = -0.25, \quad \mathbb{E}[Y] = 11.25$$



a) Find $\mathbb{E}[XY]$

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{x_i, y_i} x_i y_i f_{X,Y}(x_i, y_i) \\ &= (5)(-2)\left(\tfrac{4}{16}\right) + (10)(-2)\left(\tfrac{2}{16}\right) + (20)(-2)\left(\tfrac{3}{16}\right) \\ &\quad + (5)(2)\left(\tfrac{2}{16}\right) + (10)(2)\left(\tfrac{3}{16}\right) + (20)(2)\left(\tfrac{2}{16}\right) \\ &= -\tfrac{40}{16} = -2.5\end{aligned}$$

b) Find $\mathrm{Cov}(X, Y)$

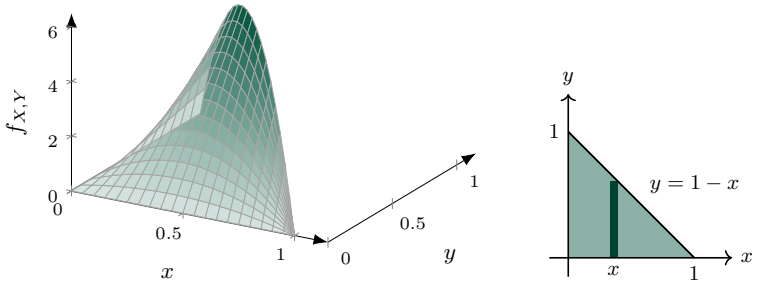
$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = -2.5 - (-0.25)(11.25) = 0.3125$$

Notice that $\mathrm{Cov}(X, Y)$ is dependent on the magnitude of X, Y, since it is bounded by $\pm\sigma_X\sigma_Y$.

Example 8.5: Recall X, Y continuous with joint p.d.f.

$$f_{X,Y}(x, y) = \begin{cases} 24xy & 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le y + x \le 1 \\ 0 & \text{o.w.} \end{cases} \qquad \mathbb{E}[X] = \tfrac{2}{5}, \quad \mathbb{E}[Y] = \tfrac{2}{5}$$

a) Find $\mathbb{E}[XY]$



$$\begin{aligned}
\mathbb{E}[XY] &= \int\int xy \cdot 24xy \, dy \, dx \\
&= 24 \int_0^1 \int_0^{1-x} x^2 y^2 \, dy \, dx \\
&= 24 \left[\int_0^1 \tfrac{1}{3} y^3 x^2 \Big|_0^{1-x} \, dx \right] \\
&= \tfrac{24}{3} \int_0^1 x^2 (1-x)^3 \, dx \\
&= 8 \int_0^1 -x^5 + 3x^4 - 3x^3 + x^2 \, dx \\
&= 8\left(-\tfrac{1}{6}x^6 + \tfrac{3}{5}x^5 - \tfrac{3}{4}x^4 + \tfrac{1}{3}x^3 \right) \Big|_0^1 \\
&= 8\left(-\tfrac{1}{6} + \tfrac{3}{5} - \tfrac{3}{4} + \tfrac{1}{3} \right) \\
&= 8\left(\tfrac{-10+36-45+20}{60} \right) = 8\left(\tfrac{1}{60} \right) = \tfrac{2}{15}
\end{aligned}$$

b) Find $\text{Cov}(X, Y)$

$$\text{Cov}(X, Y) = \tfrac{2}{15} - \left(\tfrac{2}{5}\right)\left(\tfrac{2}{5}\right) = \tfrac{2}{15} - \tfrac{4}{25} = \tfrac{20}{150} - \tfrac{24}{150} = -\tfrac{4}{150} = -\tfrac{2}{75}$$

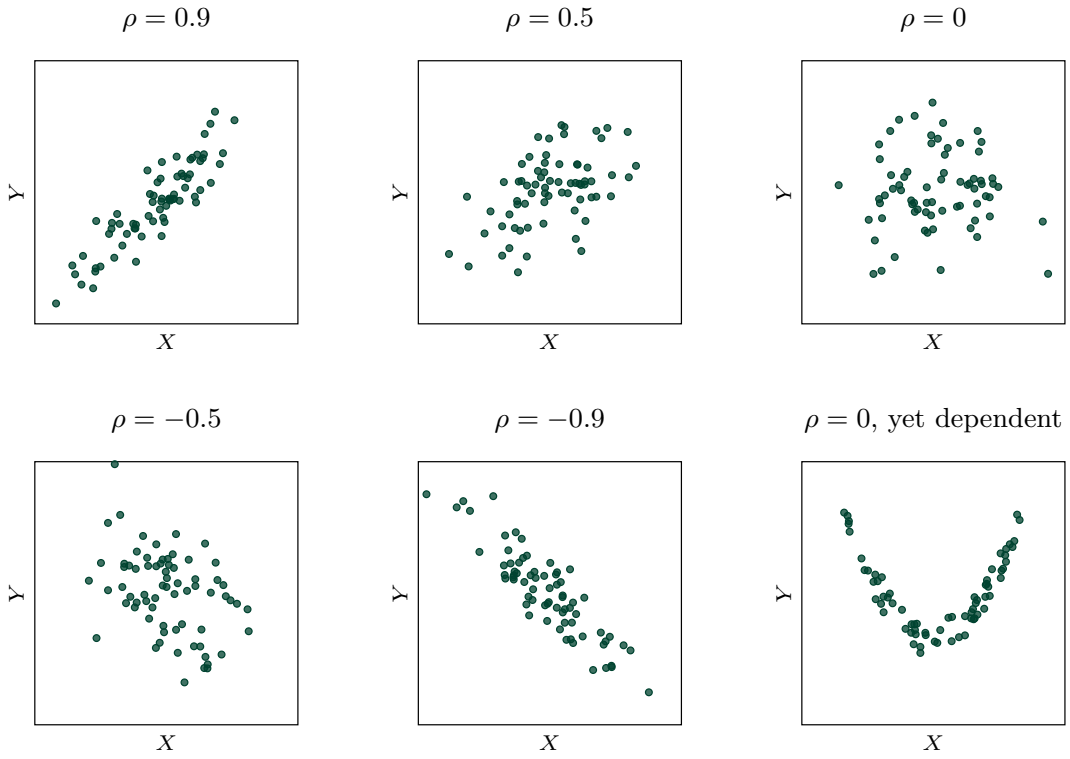
Because we want a measure of linear strength that is not dependent on the range of X, Y, we may standardize $\text{Cov}(X, Y)$.

Definition 8.3: The correlation between two RVs X, Y is:

$$\rho(X, Y) = \text{Cov}(X, Y) \,/\, (\sigma_X \, \sigma_Y)$$

Notice $-1 \leq \rho(X, Y) \leq 1$.

The pictures below each show 70 draws of (X, Y). The closer ρ is to ± 1, the more tightly the cloud hugs a line. A correlation of 0 means no *linear* trend, which is not the same as no relationship (last panel).



Proposition 8.3: For constants a, b, c, d

(i) $\text{Cov}(a + X,\ b + Y) = \text{Cov}(X, Y)$

(ii) $\text{Cov}(cX,\ dY) = cd\,\text{Cov}(X, Y)$

(iii) $\text{Cov}(a + cX,\ b + dY) = cd\,\text{Cov}(X, Y)$

(iv) $\rho(a + X,\ b + Y) = \rho(X, Y)$

(v) $\rho(cX,\ dY) = \text{sign}(cd)\,\rho(X, Y)$ $\qquad \text{sign}(cd) = \begin{cases} 1 & \text{if } cd > 0 \\ -1 & \text{if } cd < 0 \end{cases}$

(vi) $\rho(a + cX,\ b + dY) = \text{sign}(cd)\,\rho(X, Y)$

8.4 Correlation: Worked Examples

Example 8.6: Suppose X is a RV with $\sigma_X^2 = 0.5899$, $\mathbb{E}[X] = 1.01$

Y is a RV with $\sigma_Y^2 = 0.6099$, $\mathbb{E}[Y] = 0.99$, and $\mathbb{E}[XY] = 1.3$.

a) Find $\text{Cov}(X, Y)$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 1.3 - (1.01)(0.99) = 0.3001$$

b) Find $\rho(X, Y)$

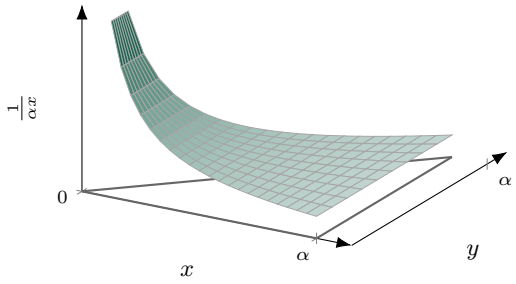
$$\begin{aligned}\rho(X, Y) &= \text{Cov}(X, Y) \,/\, \sqrt{\sigma_X^2\, \sigma_Y^2} \\ &= 0.3001 \,/\, \sqrt{0.5899(0.6099)} = 0.5\end{aligned}$$

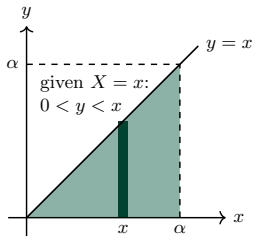
A correlation of 0.5 indicates a moderate positive linear relationship: larger values of X tend to come with larger values of Y, but far from deterministically.

c) Find $\rho(-5X,\ 10Y)$

$$\begin{aligned}\rho(bX,\ dY) &= \text{sign}(bd)\,\rho(X, Y) \\ &= -0.50\end{aligned}$$

Example 8.7: Let X be distributed $U(0, \alpha)$ for $\alpha > 0$, and $Y \mid X = x \sim U(0, x)$.





a) Find $\mathbb{E}[Y]$, $\text{Var}[Y]$

Recall $\mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}[Y \mid X = x]]$

$$= \mathbb{E}_X\left[\frac{x}{2}\right] = \int_0^\alpha \frac{x}{2} \cdot \frac{1}{\alpha}\, dx = \frac{x^2}{4\alpha}\bigg|_0^\alpha = \frac{\alpha^2}{4\alpha} = \frac{\alpha}{4}$$

$$\mathbb{E}[Y^2 \mid X = x] = \int_0^x y^2 \cdot \frac{1}{x}\, dy = \frac{y^3}{3} \cdot \frac{1}{x}\bigg|_0^x = \frac{x^2}{3}$$

Then,

$$\mathbb{E}[Y^2] = \mathbb{E}_X[\mathbb{E}_Y[Y^2 \mid X = x]] = \mathbb{E}_X\left[\frac{x^2}{3}\right]$$

$$= \int_0^\alpha \frac{x^2}{3} \cdot \frac{1}{\alpha}\, dx = \frac{x^3}{9\alpha}\bigg|_0^\alpha = \frac{\alpha^2}{9}$$

Thus,

$$\text{Var}[Y] = \frac{\alpha^2}{9} - \left(\frac{\alpha}{4}\right)^2 = \frac{\alpha^2}{9} - \frac{\alpha^2}{16} = \frac{7\alpha^2}{144}$$

Note (law of total variance, Section 7.4): the same answer comes from decomposing the variance:

$$\text{Var}[Y] = \mathbb{E}[\text{Var}[Y \mid X]] + \text{Var}[\mathbb{E}[Y \mid X]]$$

Here $\text{Var}[Y \mid X] = \frac{X^2}{12}$ and $\mathbb{E}[Y \mid X] = \frac{X}{2}$, so

$$\mathbb{E}\left[\tfrac{X^2}{12}\right] + \text{Var}\left(\tfrac{X}{2}\right) = \frac{\alpha^2/3}{12} + \frac{1}{4}\cdot\frac{\alpha^2}{12} = \frac{\alpha^2}{36} + \frac{\alpha^2}{48} = \frac{7\alpha^2}{144} \checkmark$$

The first term is the average spread of Y around its conditional mean; the second is the spread of the conditional means themselves.

b) Find $\text{Cov}(X, Y)$

$$\mathbb{E}[XY] = \mathbb{E}_X[\mathbb{E}_Y[XY \mid X = x]] = \mathbb{E}_X[X\,\mathbb{E}_Y[Y \mid X = x]] \quad (X \text{ is constant w.r.t. } \mathbb{E}_Y)$$
$$= \int_0^\alpha x\left(\frac{x}{2}\right)\frac{1}{\alpha}\,dx = \frac{x^3}{6\alpha}\Big|_0^\alpha = \frac{\alpha^2}{6}$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{\alpha^2}{6} - \left(\frac{\alpha}{4}\right)\left(\frac{\alpha}{2}\right) = \frac{\alpha^2}{24}$$

c) Find $\rho(X, Y)$

Note: $\sigma_X^2 = \frac{\alpha^2}{12}$, since $X \sim U(0, \alpha)$

$$\rho(X, Y) = \text{Cov}(X, Y) \,/\, \sqrt{\sigma_X^2\,\sigma_Y^2}$$
$$= \left(\frac{\alpha^2}{24}\right) \Big/ \sqrt{\left(\frac{\alpha^2}{12}\right)\left(\frac{7\alpha^2}{144}\right)}$$
$$= \frac{1/24}{\sqrt{7/1728}} = \frac{1/24}{\sqrt{7}/(24\sqrt{3})} = \sqrt{\frac{3}{7}} \approx 0.655$$

This moderately strong positive correlation makes sense: Y is drawn from $(0, X)$, so it tends to track X. Notice also that ρ is free of α, since correlation is scale-invariant.

Note: X and Y are said to be uncorrelated (have no linear relationship) if $\rho(X, Y) = 0$.

Theorem 8.2: For two RVs X, Y, with finite first and second moments and positive std dev's,

$$\text{Var}[X + Y] = \sigma_X^2 + \sigma_Y^2 + 2\text{Cov}(X, Y) = \sigma_X^2 + \sigma_Y^2 + 2\sigma_X\sigma_Y\,\rho(X, Y)$$

Proof:

$$\begin{aligned}
\text{Var}[X + Y] &= \mathbb{E}[(X + Y)^2] - (\mathbb{E}[X + Y])^2 \\
&= \mathbb{E}[X^2 + 2XY + Y^2] - \left((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2\right) \\
&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 + \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 + 2\mathbb{E}[XY] - 2\mathbb{E}[X]\mathbb{E}[Y] \\
&= \sigma_X^2 + \sigma_Y^2 + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]) \\
&= \sigma_X^2 + \sigma_Y^2 + 2\text{Cov}(X, Y) \qquad \blacksquare
\end{aligned}$$

Proposition 8.4: For a, b, c, d constants

(i) $\text{Var}[X - Y] = \sigma_X^2 + \sigma_Y^2 - 2\text{Cov}(X, Y)$

(ii) $\text{Var}[(a + X) + (c + Y)] = \text{Var}[X + Y]$

(iii) $\text{Var}[cX + dY] = c^2\sigma_X^2 + d^2\sigma_Y^2 + cd \cdot 2\text{Cov}(X, Y)$

Exercises

Exercise 8.1. Let X and Y have the joint p.d.f.

$x \backslash y$	-1	0
0	$20/c$	$15/c$
2	$10/c$	$20/c$
4	$25/c$	$10/c$

(a) Find the value of c that makes this a valid joint p.d.f.

(b) Find the marginal p.d.f.s, and verify that $\mathbb{E}[X] = 2$ and $\mathbb{E}[Y] = -0.55$.

(c) Find $\text{Cov}(X, Y)$. What kind of relationship does this indicate between X and Y?

(d) Find the conditional distribution of Y given $X = 4$, and compute $\mathbb{E}[Y \mid X = 4]$.

(e) Suppose we sample 5 values of X independently. What is the probability that exactly two of them equal 4? Name the distribution you are using.

Exercise 8.2. Let $Z \sim N(0, 1)$ and let S be independent of Z with $P(S = 1) = P(S = -1) = 1/2$. Define $W = SZ$.

(a) Show that $W \sim N(0, 1)$. *Hint:* condition on the value of S and use the symmetry of the standard normal.

(b) Show that $\text{Cov}(Z, W) = 0$, so Z and W are uncorrelated.

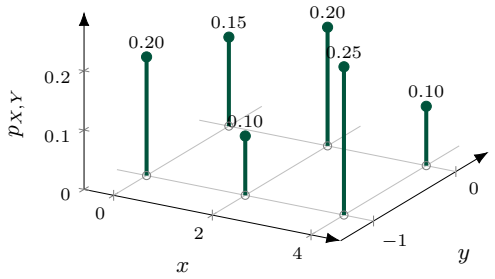
(c) Show that Z and W are nevertheless *not* independent. *Hint:* what is $|W|$ in terms of $|Z|$? Find an event involving W whose conditional probability given $\{|Z| < 1\}$ differs from its unconditional probability.

(d) Both Z and W are standard normal and uncorrelated, yet dependent. Explain why this does not contradict the fact (Section 11.3) that *jointly* normal uncorrelated random variables are independent. What fails here?

Solutions to the exercises

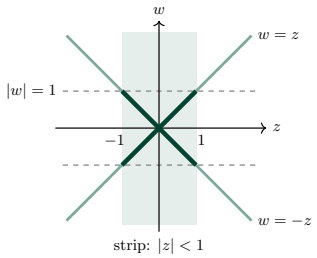
Work each exercise on the previous page before comparing.

Exercise 8.1. (a) The entries sum to $100/c$, so $c = 100$.



(b) $f_X = (0.35, 0.30, 0.35)$ on $x = 0, 2, 4$ and $f_Y = (0.55, 0.45)$ on $y = -1, 0$; $\mathbb{E}[X] = 0.6 + 1.4 = 2$ and $\mathbb{E}[Y] = -0.55$. **(c)** $\mathbb{E}[XY] = 2(-1)(0.10) + 4(-1)(0.25) = -1.2$, so $\text{Cov} = -1.2 - 2(-0.55) = -0.10$: a weak negative association (larger X goes with $y = -1$ more often). **(d)** $P(Y = -1 \mid X = 4) = 0.25/0.35 = 5/7$, $P(Y = 0 \mid X = 4) = 2/7$, so $\mathbb{E}[Y \mid X = 4] = -5/7 = -0.714$. **(e)** $\text{Bin}(5, 0.35)$: $\binom{5}{2}(0.35)^2(0.65)^3 = 10(0.1225)(0.2746) = 0.336$.

Exercise 8.2. (a) $P(W \le w) = \frac{1}{2}P(Z \le w) + \frac{1}{2}P(-Z \le w) = \frac{1}{2}\Phi(w) + \frac{1}{2}\Phi(w) = \Phi(w)$ by symmetry of Z. **(b)** $\mathbb{E}[ZW] = \mathbb{E}[SZ^2] = \mathbb{E}[S]\mathbb{E}[Z^2] = 0 \cdot 1 = 0$ and $\mathbb{E}[Z] = \mathbb{E}[W] = 0$, so $\text{Cov}(Z, W) = 0$. **(c)** $|W| = |Z|$, so $P(|W| < 1 \mid |Z| < 1) = 1$ while $P(|W| < 1) = 0.683$: dependent.



(d) (Z, W) is not jointly normal: $Z + W$ equals $2Z$ or 0, each with probability $\frac{1}{2}$, which is not normal. The uncorrelated-implies-independent fact needs joint normality, not just normal marginals.

9 Independence and Sums of Random Variables

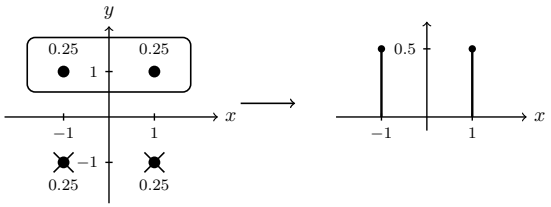
9.1 Independence of Random Variables

Independence of RVs

Just like we have conditional distributions, distributions may be independent of one another.

Example 9.1 (four corners, continued): For the four corners of Example 8.3, calculate the conditional p.d.f. of $X \mid Y = 1$.

Keep only the row $y = 1$ and divide by its total $f_Y(1) = 0.5$:



keep the row $y = 1$ $\qquad$ $f_{X|Y=1}(x) = f_X(x)$

$$f_{X|Y=1}(x \mid y = 1) = \frac{f_{X,Y}(x, 1)}{f_Y(1)} = \begin{cases} 0.25/0.5 = 0.5, & x = -1 \\ 0.25/0.5 = 0.5, & x = 1 \end{cases}$$

Notice in this case $f_{X|Y=1}(x \mid y = 1) = f_X(x)$: learning $Y = 1$ does not change the distribution of X (and the row $y = -1$ gives the same). This idea motivates the definition of independence for RVs.

Definition 9.1: If X, Y are RVs such that

$$f_{X|Y=y}(x \mid y) = f_X(x) \quad \text{for all } y \text{ with } f_Y(y) > 0,$$

then we say X, Y are independent RVs. (Think: $P(A \mid B) = P(A)$.)

Corollary: if X, Y are independent, then $f_{X,Y}(x, y) = f_X(x)\, f_Y(y)$. (Think: $P(A \cap B) = P(A)\, P(B)$.) For the four corners, every cell is $0.25 = (0.5)(0.5)$. With weight 0.4 on the diagonal (Example 8.3) they are not independent: $f_{X|Y=1}(1 \mid y = 1) = 0.4/0.5 = 0.8 \neq 0.5 = f_X(1)$.

The same idea in terms of events:

Definition 9.2: Two RVs X, Y are said to be independent of one another iff for all $B_1, B_2 \subseteq \mathbb{R}$

(i) $P(X \in B_1 \cap Y \in B_2) = P(X \in B_1)\, P(Y \in B_2)$

(ii) $P(X \in B_1 \mid Y \in B_2) = P(X \in B_1)$ $\quad$ (whenever $P(Y \in B_2) > 0$)

This can be extended to k RVs:

Definition 9.3: RVs $X_1, X_2, \ldots, X_k$ are said to be jointly independent iff for all $B_1, B_2, \ldots, B_k \subseteq \mathbb{R}$,

$$P(X_i \in B_i,\ i = 1, \ldots, k) = \prod_{i=1}^{k} P(X_i \in B_i)$$

To check for independence, we may use the following results:

Theorem 9.1: RVs $X_1, \ldots, X_k$ are jointly independent if and only if

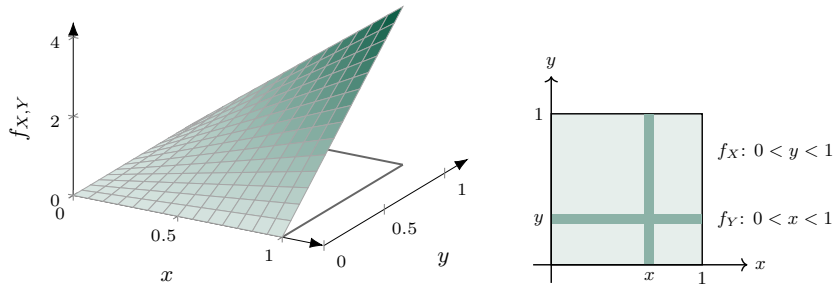
(i) $F_{X_1,\ldots,X_k}(x_1, \ldots, x_k) = F_{X_1}(x_1) \cdots F_{X_k}(x_k)$
(You can multiply their marginal c.d.f.'s together to get the joint c.d.f.)

(ii) $f_{X_1,\ldots,X_k}(x_1, \ldots, x_k) = f_{X_1}(x_1) \cdots f_{X_k}(x_k)$
(You can multiply their marginal p.d.f.'s together to get the joint p.d.f.)

(iii) $M_{X_1,X_2,\ldots,X_k}(t_1, \ldots, t_k) = M_{X_1}(t_1) \cdots M_{X_k}(t_k)$
(You can multiply their marginal m.g.f.'s together to get the joint m.g.f., provided the m.g.f.'s exist)

Example 9.2: Compare two joint densities that both "factor."

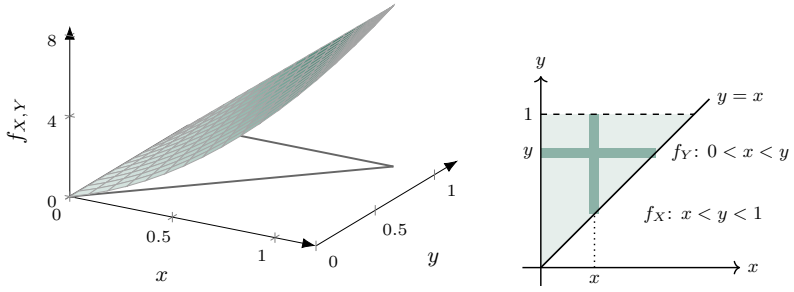
a) Let $f_{X,Y}(x, y) = 4xy$ on the square $0 < x < 1, 0 < y < 1$. Then



$$f_X(x) = \int_0^1 4xy\,dy = 2x, \qquad f_Y(y) = \int_0^1 4xy\,dx = 2y$$

and $f_X(x)f_Y(y) = 4xy = f_{X,Y}(x, y)$, so X and Y **are** independent.

b) Now let $f_{X,Y}(x, y) = 8xy$ on the *triangle* $0 < x < y < 1$. The formula $8xy$ still factors into an x-part times a y-part, but now



$$f_X(x) = \int_x^1 8xy\,dy = 4x(1 - x^2), \qquad f_Y(y) = \int_0^y 8xy\,dx = 4y^3$$

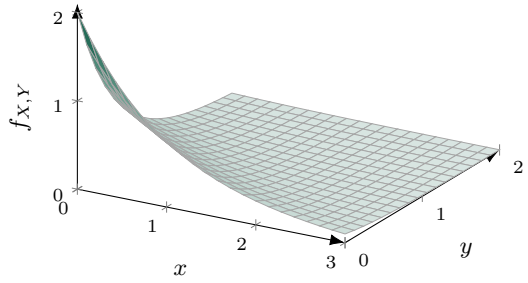
and $f_X(x)f_Y(y) = 16xy^3(1 - x^2) \neq 8xy$. So X and Y are **not** independent.

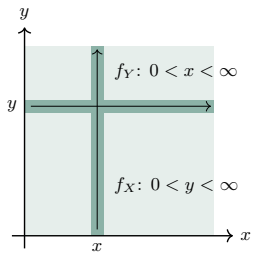
Note: The formula factoring is not enough: the *support* must not introduce dependence either, so it must be a rectangle. On the triangle, learning that $X > \frac{1}{2}$ forces $Y > \frac{1}{2}$, so X certainly carries information about Y. The cave isopods (Section 7.1) are the same: on $0 < x^2 \le y < 1$, learning $X = 0.5$ forces $Y \ge 0.25$. Always check the region before concluding independence.

Example 9.3: Suppose X, Y are continuous RVs with joint p.d.f.

$$f_{X,Y}(x,y) = \begin{cases} \lambda_1\lambda_2\, e^{-\lambda_1 x - \lambda_2 y} & x > 0,\ y > 0 \\ 0 & \text{o.w.} \end{cases}$$

where λ_1, λ_2 are positive constants. Show X, Y are independent.





$$f_{X,Y}(x,y) = \lambda_1 e^{-\lambda_1 x}\, \lambda_2 e^{-\lambda_2 y} \quad \forall\, x > 0,\ y > 0$$

$$f_X(x) = \int_0^\infty \lambda_1\lambda_2\, e^{-\lambda_1 x} e^{-\lambda_2 y}\, dy = \frac{\lambda_1\lambda_2}{(-\lambda_2)}\, e^{-\lambda_1 x} e^{-\lambda_2 y}\Big|_0^\infty = -\lambda_1\, e^{-\lambda_1 x}(0-1) = \lambda_1\, e^{-\lambda_1 x}$$

Similarly $\quad f_Y(y) = \lambda_2 e^{-\lambda_2 y}$

$\therefore \quad f_{X,Y}(x,y) = f_X(x)\, f_Y(y) \quad \checkmark$

Notice we can also show:

$$f_{X|Y=y} = f_{X,Y}(x,y) \,/\, f_Y(y) = \frac{\lambda_1\lambda_2\, e^{-\lambda_1 x} e^{-\lambda_2 y}}{\lambda_2\, e^{-\lambda_2 y}} = \lambda_1 e^{-\lambda_1 x} = f_X(x)$$

$$f_{Y|X=x} = f_{X,Y}(x,y) \,/\, f_X(x) = \frac{\lambda_1\lambda_2\, e^{-\lambda_1 x} e^{-\lambda_2 y}}{\lambda_1\, e^{-\lambda_1 x}} = \lambda_2 e^{-\lambda_2 y} = f_Y(y)$$

Either is acceptable in showing independence.

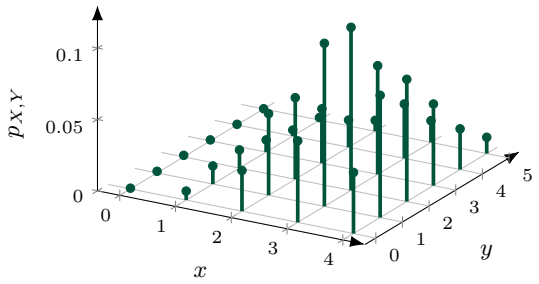
Warning (check the support!): The factorization must hold on a *rectangular* support. Recall the density $f_{X,Y}(x,y) = 24xy$ on the triangle $0 \le y \le 1 - x$, $x, y \ge 0$ (Section 7.5). The formula $24xy$ factors into an x-part times a y-part, but X, Y are *not* independent: the support is not a rectangle, so knowing $X > \frac{1}{2}$ forces $Y < \frac{1}{2}$. Indeed, the marginals give $f_X(x)\, f_Y(y) = 12x(1-x)^2 \cdot 12y(1-y)^2 \ne 24xy$.

9.2 Expectation and Variance of Sums

Example 9.4: Let $X \sim \text{Bin}(4, \frac{3}{4})$ and $Y \sim \text{Pois}(2)$, independent.

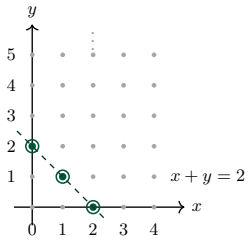
m.g.f.'s:

$$M_X(t) = \left(\tfrac{1}{4} + \tfrac{3}{4}e^t\right)^4, \qquad M_Y(t) = e^{2(e^t-1)}$$



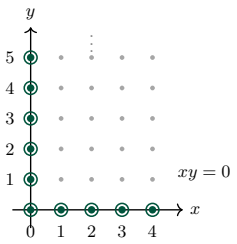
(a) $P(X + Y = 2)$

The event $X + Y = 2$ can happen as $(X, Y) = (0, 2)$, $(1, 1)$, or $(2, 0)$, so we sum over the ways to split the total, using independence to factor each term (this sum is called a *convolution*):



$$\begin{aligned} P(X + Y = 2) &= \sum_{k=0}^{2} P(X = k)\, P(Y = 2 - k) \\ &= P(X = 0)P(Y = 2) + P(X = 1)P(Y = 1) + P(X = 2)P(Y = 0) \\ &= \binom{4}{0}\left(\tfrac{3}{4}\right)^0\left(\tfrac{1}{4}\right)^4 \cdot \frac{e^{-2}2^2}{2!} + \binom{4}{1}\left(\tfrac{3}{4}\right)^1\left(\tfrac{1}{4}\right)^3 \cdot \frac{e^{-2}2^1}{1!} + \binom{4}{2}\left(\tfrac{3}{4}\right)^2\left(\tfrac{1}{4}\right)^2 \cdot e^{-2} \\ &= \tfrac{80}{256}\, e^{-2} \approx 0.042293 \end{aligned}$$

(b) $P(XY = 0) = P(X = 0 \text{ or } Y = 0)$



$$
\begin{aligned}
&= P(X = 0) + P(Y = 0) - P(X = 0)P(Y = 0) \\
&= \left(\tfrac{1}{4}\right)^4 + e^{-2} - \left(\tfrac{1}{4}\right)^4 e^{-2} \\
&\approx 0.1387
\end{aligned}
$$

(c) $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = 3 \cdot 2 = 6$ (by item 1 of the Proposition below)

Proposition 9.1: If X and Y are independent, then:

1. $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$
2. $\mathrm{Cov}(X, Y) = 0$ (i.e. X and Y are uncorrelated)
3. $g(X)$ is independent of Y for any function g

Proof of 1 and 2. Independence factors the joint p.d.f., and the double integral (a double sum in the discrete case) then splits into two single ones:

$$\mathbb{E}[XY] = \int_{S_X}\int_{S_Y} xy\, f_X(x)\, f_Y(y)\, dy\, dx = \int_{S_X} x f_X(x) \underbrace{\left[\int_{S_Y} y f_Y(y)\, dy\right]}_{\mathbb{E}[Y]} dx = \mathbb{E}[Y] \underbrace{\int_{S_X} x f_X(x)\, dx}_{\mathbb{E}[X]} = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Then $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0$. □

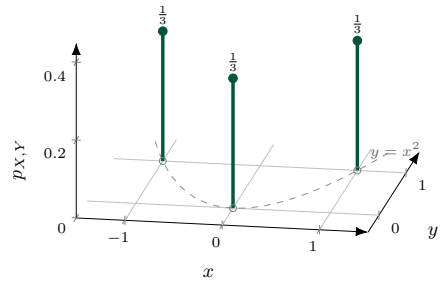
The implication runs one way only: independence $\Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \Rightarrow \mathrm{Cov}(X, Y) = 0$, but zero covariance does *not* give independence. Independence is the stronger condition.

Note: Uncorrelated does *not* imply independent in general.

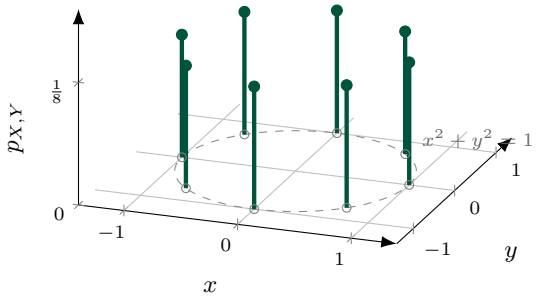
Example 9.5: Let X take values $-1, 0, 1$, each with probability $\frac{1}{3}$, and let $Y = X^2$. Then $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = \mathbb{E}[X] = 0$ (since $x^3 = x$ for $x = -1, 0, 1$), so

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0,$$

i.e. X, Y are uncorrelated. But Y is completely determined by X, and they are far from independent: e.g. $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = \frac{1}{3}$. Correlation only detects *linear* relationships; here the relationship is perfectly quadratic.



Example 9.6 (eight points on a circle): Let (X, Y) land on one of the eight equally spaced points of the unit circle, $(\cos\theta, \sin\theta)$ for $\theta = 0°, 45°, \ldots, 315°$, each with probability $\frac{1}{8}$. Every point satisfies $x^2 + y^2 = 1$.



The joint p.m.f. as a table, with the marginal of Y in the last column:

$y \backslash x$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1	$f_Y(y)$
1	0	0	$\frac{1}{8}$	0	0	$\frac{1}{8}$
$\frac{\sqrt{2}}{2}$	0	$\frac{1}{8}$	0	$\frac{1}{8}$	0	$\frac{2}{8}$
0	$\frac{1}{8}$	0	0	0	$\frac{1}{8}$	$\frac{2}{8}$
$-\frac{\sqrt{2}}{2}$	0	$\frac{1}{8}$	0	$\frac{1}{8}$	0	$\frac{2}{8}$
-1	0	0	$\frac{1}{8}$	0	0	$\frac{1}{8}$

The points are perfectly balanced between Q_1, Q_3 (where $xy > 0$) and Q_2, Q_4 (where $xy < 0$), and each marginal is symmetric about 0, so $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ and $\mathbb{E}[XY] = \frac{1}{8}\sum xy = 0$: $\mathrm{Cov}(X, Y) = 0$.

But X and Y are not independent. Knowing X pins Y down to $y = \pm\sqrt{1 - x^2}$, two values rather than five. Given $X = \frac{\sqrt{2}}{2}$, for instance,

$$f_{Y|X=\sqrt{2}/2}(y) = \frac{f_{X,Y}(\sqrt{2}/2,\ y)}{f_X(\sqrt{2}/2)} = \begin{cases} \frac{1/8}{2/8} = \frac{1}{2}, & y = \frac{\sqrt{2}}{2} \\ \frac{1/8}{2/8} = \frac{1}{2}, & y = -\frac{\sqrt{2}}{2} \end{cases} \neq f_Y(y),$$

so learning X changes the distribution of Y. The support is a circle rather than a rectangle, so the joint p.m.f. has no chance of factoring as $f_X(x)\, f_Y(y)$, even though the covariance is 0.

Proposition 9.2: If $X_1, \ldots, X_n$ are independent, then

$$\mathbb{E}\left[\prod_{i=1}^{n} g_i(X_i)\right] = \prod_{i=1}^{n} \mathbb{E}[g_i(X_i)]$$

Corollary 9.1: If $X_1, \dots, X_n$ are i.i.d., then

$$M_{X_1+\cdots+X_n}(t) = \big[M_X(t)\big]^n$$

Expectation and variance of sums

For RVs X, Y and constants a, b:

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y], \qquad \mathbb{E}[aX+b] = a\,\mathbb{E}[X] + b,$$
$$\operatorname{Var}[X+Y] = \operatorname{Var}[X] + \operatorname{Var}[Y] + 2\operatorname{Cov}(X,Y), \qquad \operatorname{Var}[aX+b] = a^2 \operatorname{Var}[X].$$

If $X_1, \dots, X_n$ are independent, every covariance term vanishes and $\operatorname{Var}[X_1 + \cdots + X_n] = \operatorname{Var}[X_1] + \cdots + \operatorname{Var}[X_n]$; if in addition each has mean μ and variance σ^2, then $\mathbb{E}[\bar{X}] = \mu$ and $\operatorname{Var}[\bar{X}] = \sigma^2/n$.

Example 9.7 (two investments: covariance as a hedge): Two investments each either gain \$100 or lose \$50 on the next trade, with probability $\frac{1}{2}$ each, so $\mathbb{E}[X] = \mathbb{E}[Y] = 25$ and $\operatorname{Var}[X] = \operatorname{Var}[Y] = 75^2 = 5625$. How they move *together* is described by a box of two tickets (X on top, Y underneath), one drawn at random.

Scenario 1: tickets $\binom{100}{100}$ and $\binom{-50}{-50}$. The two rise and fall together, $\operatorname{Cov}(X,Y) = 5625$, and $\operatorname{Var}[X+Y] = 5625 + 5625 + 2(5625) = 22{,}500$, so the total profit is \$50 on average with a standard deviation of \$150.

Scenario 2: tickets $\binom{100}{-50}$ and $\binom{-50}{100}$. Now $\operatorname{Cov}(X,Y) = -5625$ and $\operatorname{Var}[X+Y] = 0$: the total profit is \$50 every time.

Same expected profit, very different risk. A negative covariance is a hedge (gold is often held for this reason, since it tends to move against the stock market), while a positive covariance compounds risk: mortgage-backed securities in 2008 were priced as if the individual mortgages defaulted nearly independently, when in fact their defaults were positively correlated and arrived together.

Exercises

Exercise 9.1. Four quick drills on linear combinations.

(a) X_1, X_2, X_3 are independent with common variance 2. Find $\operatorname{Var}[X_1 + 2X_2 + 3X_3]$.

(b) $\operatorname{Var}[Z] = 10$, $\operatorname{Var}[W] = 5$, $\operatorname{Cov}(Z,W) = -4$. Find $\operatorname{Var}[Z - W]$.

(c) $\operatorname{Var}[X] = 2$, $\operatorname{Var}[Y] = 4$, $\operatorname{Cov}(X,Y) = -0.5$. Find $\operatorname{SD}[nX - mY]$ in terms of n and m.

(d) X, Y, Z are independent with $\operatorname{Var}[X] = 4$, $\operatorname{Var}[Y] = 4$, $\operatorname{Var}[Z] = 1/n$. Find $\operatorname{Var}\left[\dfrac{X/2 - Y + \sqrt{n}\,Z}{-2}\right]$.

Exercise 9.2 (Three rain gauges in a cloud forest)**.** Three rain gauges in the Monteverde cloud forest of Costa Rica, a mountain forest kept wet year round by the clouds that drift through it, record daily rainfall X, Y, Z in millimeters. Gauges Y and Z sit on the same slope and X is across the ridge, and the records satisfy

$$\operatorname{Var}[X] = \operatorname{Var}[Y] = \operatorname{Var}[Z] = 7, \qquad \operatorname{Cov}(Y,Z) = 6, \qquad \operatorname{Cov}(X, Y+Z) = 8.$$

The quantity of interest is the variance of the combined daily total $X + Y + Z$.

(a) Expand $\mathrm{Var}[X + Y + Z]$ in terms of the three variances and the three pairwise covariances.

(b) You are not told $\mathrm{Cov}(X, Y)$ or $\mathrm{Cov}(X, Z)$ individually. Explain why you do not need them.

(c) Find $\mathrm{Var}[X + Y + Z]$.

Exercise 9.3 (Daily spending on an island)**.** On a Greek island a traveler's daily ferry cost has mean 25 euros and standard deviation 5 euros, and daily food cost has mean 40 euros and standard deviation 12 euros; the two are independent, and so are the days.

(a) Mean and standard deviation of one day's total.

(b) Mean and standard deviation of a seven-day total.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 9.1. (a) $2(1+4+9) = 28$. **(b)** $10+5-2(-4) = 23$. **(c)** $\text{Var} = 2n^2+4m^2-2nm(-0.5) = 2n^2+4m^2+nm$, so $\text{SD} = \sqrt{2n^2+4m^2+nm}$. **(d)** $\frac{1}{4}\left(\frac{1}{4}\cdot 4 + 4 + n\cdot\frac{1}{n}\right) = \frac{1}{4}(1+4+1) = \frac{3}{2}$.

Exercise 9.2. (a) $\text{Var}[X+Y+Z] = \text{Var}X+\text{Var}Y+\text{Var}Z+2\text{Cov}(X,Y)+2\text{Cov}(X,Z)+2\text{Cov}(Y,Z)$. **(b)** Only the sum $\text{Cov}(X,Y)+\text{Cov}(X,Z) = \text{Cov}(X,Y+Z) = 8$ appears, by bilinearity. **(c)** $21+2(8)+2(6) = 49$.

Exercise 9.3. (a) $25+40 = 65$ euros; $\sqrt{25+144} = 13$ euros. **(b)** $7(65) = 455$ euros; $\sqrt{7\cdot 169} = 13\sqrt{7} = 34.4$ euros.

9.3 Sums of Normal Random Variables

Sums of Normal RVs

Example 9.8 (preview): Let $X \sim N(100, 144)$ and $Y \sim N(80, 25)$ be independent, so $\sigma_X = 12$ and $\sigma_Y = 5$. Using the result of this section,

$$X+Y \sim N(180,\ 169), \qquad X-Y \sim N(20,\ 169),$$

since *variances add in both cases*; the common standard deviation is $\sqrt{169} = 13$.

a) $P(X+Y > 200) = P\left(Z > \frac{200-180}{13}\right) = 1-\Phi(1.538) = 0.062$

b) $P(X < Y) = P(X-Y < 0) = P\left(Z < \frac{0-20}{13}\right) = \Phi(-1.538) = 0.062$

Note: The two answers agree, and not by accident: both ask how far 20 is from the relevant mean in units of 13. Note also that $\text{Var}[X-Y] = \text{Var}[X]+\text{Var}[Y]$, *not* the difference. Subtracting random variables makes the result more variable, not less.

Suppose $X_1, \ldots, X_k$ are independent, and let $X_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, \ldots, k$. Then

$$\sum X_i \sim N\left(\sum \mu_i,\ \sum \sigma_i^2\right)$$

Proof. Recall $M_{X_i}(t) = \exp\left(\mu_i t + \frac{\sigma_i^2}{2}t^2\right)$ and since all X_i are independent,

$$M_{\sum X_i}(t) = \prod_{i=1}^{k} M_{X_i}(t) = \prod_{i=1}^{k} \exp\left(\mu_i t + \tfrac{\sigma_i^2}{2}t^2\right) = \exp\left(\sum \mu_i\, t + \tfrac{\sum \sigma_i^2}{2}\, t^2\right)$$

which is the m.g.f. of $N\left(\sum \mu_i,\ \sum \sigma_i^2\right)$. □

Further, if $\mu_i = \mu_X$, $\sigma_i^2 = \sigma_X^2$ $\forall\, i = 1, \ldots, n$, then

$$\sum X_i \sim N(n\mu_X,\ n\sigma_X^2)$$

Further, if we have constants c_i, $i = 1, \ldots, k$,

$$\sum c_i X_i \sim N\Big(\sum c_i \mu_i,\ \sum c_i^2 \sigma_i^2\Big)$$

The above gives us that if $X_1, \ldots, X_n$ are independent and $X_i \sim N(\mu_X, \sigma_X^2)$:

$$\bar{X} \sim N\Big(\mu_X,\ \frac{\sigma_X^2}{n}\Big), \qquad \frac{\bar{X} - \mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1)$$

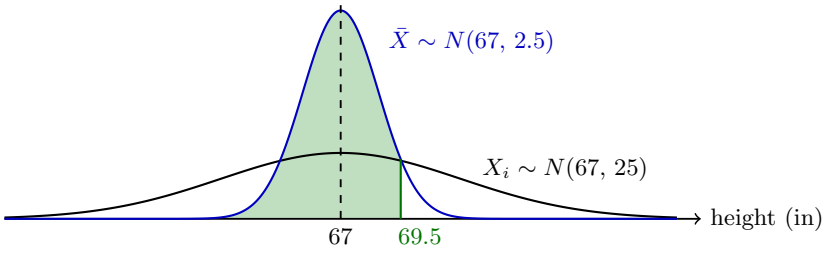
$$S = \sum_{i=1}^{n} X_i \sim N(n\mu_X,\ n\sigma_X^2), \qquad \frac{S - n\mu_X}{\sqrt{n\sigma_X^2}} \sim N(0,1)$$

Example 9.9: Suppose a random USF student's height, in inches, is $X_i \sim N(67,\ 25)$, and we take a random sample of $n = 10$ students, independent of one another. What is the distribution of their average height $\bar{X} = \frac{1}{10}\sum_{i=1}^{10} X_i$, and what is the probability that it is below 69.5 inches?

The mean stays put and the variance divides by n:

$$\bar{X} \sim N\Big(67,\ \frac{25}{10}\Big) = N(67,\ 2.5)$$

The average is far less spread out than a single student's height:



Standardize with $\sigma_{\bar{X}} = \sqrt{2.5}$:

$$P(\bar{X} < 69.5) = P\left(Z < \frac{69.5 - 67}{\sqrt{2.5}}\right) = P\left(Z < \frac{2.5}{\sqrt{2.5}}\right) = P\big(Z < \sqrt{2.5}\big) = \Phi(1.58) = 0.9429$$

Note: A single student is below 69.5 inches with probability $\Phi\big(\frac{2.5}{5}\big) = \Phi(0.5) = 0.6915$. The same 2.5 inches is half a standard deviation for one student but 1.58 standard deviations for the average of ten, which is why the average is much more likely to land below it.

Example 9.10: A music festival's permit requires a sound-level meter at the nearest houses, and the noise consultant logs one reading, in dB(A) (treated here as plain numbers), during each of the four headline sets. Each reading is normally distributed with mean 60, std dev 6, independent across sets. The organizer is fined if the average of the four readings is 70 or more.

a) What is the prob of a fine?

$X_i \sim N(\mu_X = 60,\ \sigma_X^2 = 36)$

$\bar{X} \sim N(\mu_{\bar{X}} = 60,\ \sigma_{\bar{X}}^2 = \sigma_X^2/n = 36/4 = 9)$, so $\sigma_{\bar{X}} = 3$

Thus,

$$\begin{aligned} P(\bar{X} > 70) &= P\left(\frac{\bar{X} - \mu_X}{\sigma_{\bar{X}}} > \frac{70 - 60}{3}\right) = P(Z > 3.33) \\ &= 1 - P(Z \leq 3.33) = 1 - \Phi(3.33) \\ &= 1 - 0.999566 = 0.000434 \end{aligned}$$

b) Two of the sets are on Saturday and two on Sunday. What is the prob that the Saturday average reading is higher than the Sunday one?

Let $\bar{X}_A$ = Saturday average, $\bar{X}_A \sim N(60,\ \sigma_{\bar{X}_A}^2 = 36/2 = 18)$

$\bar{X}_B$ = Sunday average, $\bar{X}_B \sim N(60,\ \sigma_{\bar{X}_B}^2 = 36/2 = 18)$

$$P(\bar{X}_A > \bar{X}_B) = P(\bar{X}_A - \bar{X}_B > 0)$$

$$\begin{aligned} X^* = (\bar{X}_A - \bar{X}_B) = \left(\bar{X}_A + (-1)\bar{X}_B\right) &\sim N\left(\sum c_i \mu_{\bar{X}_i},\ \sigma_{X^*}^2 = \sum c_i^2 \sigma_{\bar{X}_i}^2\right) \\ &\sim N\left(\mu_X + (-1)\mu_X,\ \sigma_{X^*}^2 = \tfrac{6^2}{2} + (-1)^2 \tfrac{6^2}{2}\right) \\ &\sim N\left(0,\ \tfrac{36}{2} + \tfrac{36}{2}\right) \sim N(0,\ 36) \end{aligned}$$

So,

$$P(\bar{X}_A - \bar{X}_B > 0) = P\left(Z > \frac{0 - 0}{\sqrt{36}}\right) = P(Z > 0) = 0.5$$

9.4 Sums of Independent Random Variables

Sums of Independent RVs

Let $S = X_1 + \cdots + X_n$ where X_i are i.i.d. with mean μ_X and variance σ_X^2. Let $\bar{X} = S/n$. Then:

$$\mathbb{E}[S] = n\mu_X, \qquad \text{Var}[S] = n\sigma_X^2$$

$$\mathbb{E}[\bar{X}] = \mu_X, \qquad \text{Var}[\bar{X}] = \frac{\sigma_X^2}{n}$$

Example 9.11: Run times, in minutes, of nature documentaries have mean $\mu_X = 70$ and standard deviation $\sigma_X = 10$. A random sample of $n = 25$ documentaries is drawn. Let S be the total of their run times and $\bar{X}$ their average.

$$\mathbb{E}[S] = 25(70) = 1750, \qquad \text{Var}[S] = 25(100) = 2500, \qquad \text{SD}[S] = 50$$

$$\mathbb{E}[\bar{X}] = 70, \qquad \text{Var}[\bar{X}] = \frac{100}{25} = 4, \qquad \text{SD}[\bar{X}] = 2$$

Note: The sum becomes *more* variable as n grows ($\text{SD}[S] = \sigma_X\sqrt{n}$), while the average becomes *less* variable ($\text{SD}[\bar{X}] = \sigma_X/\sqrt{n}$). Both scale with $\sqrt{n}$, in opposite directions. This is the single most important fact in the rest of the course.

The m.g.f. of a Sum

Let $X_1, \ldots, X_n$ be independent and let $S_n = X_1 + \cdots + X_n$. Then

$$\begin{aligned} M_{S_n}(t) = \mathbb{E}\left[e^{tS_n}\right] &= \mathbb{E}\left[e^{tX_1}e^{tX_2}\cdots e^{tX_n}\right] \\ &= \mathbb{E}\left[e^{tX_1}\right]\mathbb{E}\left[e^{tX_2}\right]\cdots\mathbb{E}\left[e^{tX_n}\right] = \prod_{i=1}^{n} M_{X_i}(t). \end{aligned}$$

The second line is where independence comes in. Each sum below is found the same way: multiply the m.g.f.s, then recognize the product as the m.g.f. of a named distribution.

Sums of Binomial RVs

If $X_i \sim \text{Bin}(n_i, p)$ are independent, then

$$\sum_{i=1}^{k} X_i \sim \text{Bin}\left(\sum_{i=1}^{k} n_i,\ p\right)$$

Proof. Using the m.g.f.:

$$M_{\sum X_i}(t) = \prod_{i=1}^{k} M_{X_i}(t) = \prod_{i=1}^{k} \left(q + pe^t\right)^{n_i} = \left(q + pe^t\right)^{\sum n_i}$$

This is the m.g.f. of $\text{Bin}(\sum n_i, p)$. □

Example 9.12: A rhythm game scores a set of two songs by how many of the scrolling notes the player hits on the beat. The first song's chart (its sequence of notes) has 20 notes and the second has 30, and the player hits each note with probability 0.9 independently. Let $X_1 \sim \text{Bin}(20, 0.9)$ and $X_2 \sim \text{Bin}(30, 0.9)$ be independent. Then

$$X_1 + X_2 \sim \text{Bin}(50, 0.9), \qquad \mathbb{E}[X_1 + X_2] = 45, \qquad \text{Var}[X_1 + X_2] = 50(0.9)(0.1) = 4.5$$

Note the success probability p must be the *same* on both charts, or the sum is not binomial at all.

Sums of Poisson RVs

If $X_i \sim \text{Pois}(\lambda_i)$ are independent, then

$$\sum_{i=1}^{k} X_i \sim \text{Pois}\left(\sum_{i=1}^{k} \lambda_i\right)$$

Proof.

$$M_{\sum X_i}(t) = \prod_{i=1}^{k} M_{X_i}(t) = \prod_{i=1}^{k} e^{\lambda_i(e^t - 1)} = e^{(\sum \lambda_i)(e^t - 1)}$$

This is the m.g.f. of $\text{Pois}(\sum \lambda_i)$. □

In particular, if $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Pois}(\lambda)$, then $S_n = X_1 + \cdots + X_n \sim \text{Pois}(n\lambda)$.

The same result without m.g.f.s. We can also reach $\text{Pois}(n\lambda)$ from the joint p.m.f. By independence,

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{x_1+\cdots+x_n}}{x_1!\cdots x_n!}.$$

Every outcome with total s shares the factor $e^{-n\lambda}\lambda^s$. To find $P(S_n = s)$ we add the joint p.m.f. over all outcomes with total s:

$$P(S_n = s) = e^{-n\lambda}\lambda^s \sum_{x_1+\cdots+x_n=s} \frac{1}{x_1!\cdots x_n!} = \frac{e^{-n\lambda}\lambda^s}{s!} \sum_{x_1+\cdots+x_n=s} \frac{s!}{x_1!\cdots x_n!} = \frac{e^{-n\lambda}(n\lambda)^s}{s!}.$$

The last step uses the multinomial theorem: the coefficients $s!/(x_1!\cdots x_n!)$ add up to $(1+1+\cdots+1)^s = n^s$. The result is the $\text{Pois}(n\lambda)$ p.m.f. Note that the joint p.m.f. at a single outcome is not yet the p.m.f. of S_n; the sum over outcomes is what produces the $s!$ in the denominator. The m.g.f. proof avoids this bookkeeping, which is why we prefer it.

Sums of Exponential RVs

If $X_i \sim \text{Exp}(\lambda_i)$ are independent, the m.g.f. of $\sum X_i$ is

$$M_{\sum X_i}(t) = \prod_{i=1}^{k} \frac{\lambda_i}{\lambda_i - t}$$

This does *not* simplify to a recognizable named distribution in general.

Special case: If all $\lambda_i = \lambda$ (i.e. i.i.d.), then

$$M_{\sum X_i}(t) = \left(\frac{\lambda}{\lambda - t}\right)^k$$

which is the m.g.f. of $\Gamma(\alpha = k,\ \beta = \lambda)$.

Thus, the sum of k i.i.d. $\text{Exp}(\lambda)$ random variables is $\Gamma(\alpha = k,\ \beta = \lambda)$.

Exercises

Exercise 9.4. Let X and Y be independent $\text{Geo}(p)$ random variables, so $P(X = k) = q^{k-1}p$ for $k \geq 1$ with $q = 1 - p$, and let $Z = X + Y$.

(a) Write down the convolution sum $P(Z = n) = \sum_k P(X = k)P(Y = n - k)$, being careful about the range of k.

(b) Show that every term in the sum is equal to p^2q^{n-2}: it does not depend on k at all.

(c) Conclude that $P(Z = n) = (n - 1)p^2q^{n-2}$ for $n \geq 2$.

(d) Give a direct probabilistic interpretation: if you flip a coin until the second Heads, why should the total number of flips be n with probability proportional to $n - 1$?

Exercise 9.5 (Flashes in Toyama Bay)**.** Each spring, firefly squid, finger-length squid studded with blue light organs, rise by the millions to the surface of Toyama Bay, Japan, to spawn. A camera moored in the bay during the season counts $X \sim \text{Pois}(\lambda)$ flashes per minute from firefly squid and $Y \sim \text{Pois}(\mu)$ flashes per minute from lanternfish, independently. Let $Z = X + Y$ be the total number of flashes in a minute.

(a) Write down the convolution sum for $P(Z = n)$.

(b) Pull the factor $e^{-(\lambda+\mu)}$ out of the sum. Then multiply and divide by $n!$ so that the summand becomes $\binom{n}{k}\lambda^k\mu^{n-k}$.

(c) Apply the binomial theorem to conclude that $Z \sim \text{Pois}(\lambda + \mu)$.

(d) This section proved the same result using m.g.f.s. Which proof do you find more convincing, and which more informative about *why* it is true?

Exercise 9.6 (Bike rentals at two Golden Gate kiosks)**.** The numbers of bikes rented in an hour at two rental kiosks near the Golden Gate Bridge are independent Poisson counts with means 4 and 6.

(a) Name the distribution of the combined hourly count.

(b) Find the probability the two kiosks together rent at most 7 bikes in an hour.

Exercise 9.7 (Sums of Bernoulli RVs)**.** Let $X_1, \ldots, X_n$ be i.i.d. $\text{Bern}(p)$ and let $S_n = X_1 + \cdots + X_n$.

(a) Find the m.g.f. of a single X_i.

(b) Find the m.g.f. of S_n.

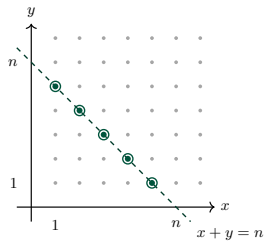
(c) Name the distribution of S_n.

(d) Explain the answer to (c) by saying what S_n counts.

Solutions to the exercises

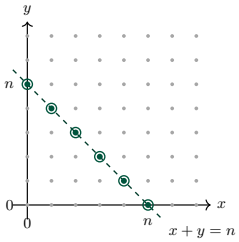
Work each exercise on the previous page before comparing.

Exercise 9.4. (a) $P(Z = n) = \sum_{k=1}^{n-1} P(X = k)P(Y = n - k)$, since each of X, Y is at least 1.



(b) Each term is $q^{k-1}p \cdot q^{n-k-1}p = p^2q^{n-2}$. **(c)** There are $n-1$ equal terms, so $P(Z = n) = (n-1)p^2q^{n-2}$ for $n \geq 2$. **(d)** The second Heads lands on flip n exactly when the nth flip is Heads and exactly one of the first $n-1$ is Heads: $n-1$ positions for that first Heads, each sequence having probability p^2q^{n-2}.

Exercise 9.5. (a) $P(Z = n) = \sum_{k=0}^{n} \dfrac{e^{-\lambda}\lambda^k}{k!} \cdot \dfrac{e^{-\mu}\mu^{n-k}}{(n-k)!}$.



(b) $= \dfrac{e^{-(\lambda+\mu)}}{n!} \sum_{k=0}^{n} \binom{n}{k} \lambda^k \mu^{n-k}$. **(c)** The sum is $(\lambda+\mu)^n$, so $P(Z = n) = e^{-(\lambda+\mu)}(\lambda+\mu)^n/n!$: $Z \sim \text{Pois}(\lambda+\mu)$. **(d)** The m.g.f. proof is shorter and more convincing as a proof; the convolution shows *why*: splitting n events between two independent sources is a binomial choice, which is what the binomial theorem is counting.

Exercise 9.6. (a) Pois(10). **(b)** $e^{-10} \sum_{k=0}^{7} 10^k/k! = 0.220$.

Exercise 9.7. (a) $M_{X_i}(t) = \mathbb{E}[e^{tX_i}] = q + pe^t$, where $q = 1 - p$. **(b)** $M_{S_n}(t) = \prod_{i=1}^{n}(q + pe^t) = (q + pe^t)^n$. **(c)** This is the m.g.f. of Bin(n, p), so $S_n \sim \text{Bin}(n, p)$. **(d)** Each X_i is 1 on a success and 0 otherwise, so S_n counts the successes in n independent trials, which is the definition of a binomial count.

9.5 Linear Combinations of Normals



Figure 9.1: *A sea otter.* [18]

Example 9.13: Suppose two sea otters (Fig. 9.1) forage independently, and their daily clam counts are

- Otter 1's daily clam count is normally distributed with mean 12, std dev 3
- Otter 2's daily clam count is normally distributed with mean 20, std dev 5

a) Find the prob that otter 2 eats more than twice as many clams as otter 1.

Let X_1 = otter 1's clam count, $X_1 \sim N(12,\ \sigma^2_{X_1} = 9)$

Let X_2 = otter 2's clam count, $X_2 \sim N(20,\ \sigma^2_{X_2} = 25)$

$$\begin{aligned} P(X_2 > 2X_1) &= P(X_2 - 2X_1 > 0) \\ \mathbb{E}[X_2 - 2X_1] &= \mathbb{E}[X_2] - 2\mathbb{E}[X_1] = 20 - 2(12) = -4 \\ \text{Var}[X_2 - 2X_1] &= \sigma^2_{X_2} + (-2)^2\sigma^2_{X_1} + 2(1)(-2)\text{Cov}(X_1, X_2) \\ &= 5^2 + 4(3^2) + 0 = 25 + 36 = 61 \end{aligned}$$

Thus, $X_2 - 2X_1 \sim N(-4,\ 61)$

$$\begin{aligned} P(X_2 - 2X_1 > 0) &= P\left(Z > \frac{0 - (-4)}{\sqrt{61}}\right) = P(Z > 0.5121) \\ &= 1 - P(Z \le 0.5121) = 1 - \Phi(0.5121) = 1 - 0.6957 = 0.3043 \end{aligned}$$

b) Determine the distribution of the difference in clam count between otter 1 vs 2, and the total clam count (per day).

$$D = X_1 - X_2 \sim N\big(12 - 20,\ \sigma_D^2 = 3^2 + (-1)^2 5^2\big) \sim N(-8,\ 34)$$
$$S = X_1 + X_2 \sim N\big(12 + 20,\ \sigma_S^2 = 3^2 + 5^2\big) \sim N(32,\ 34)$$

c) What is the distribution of the total amount that otter 2 eats per week?

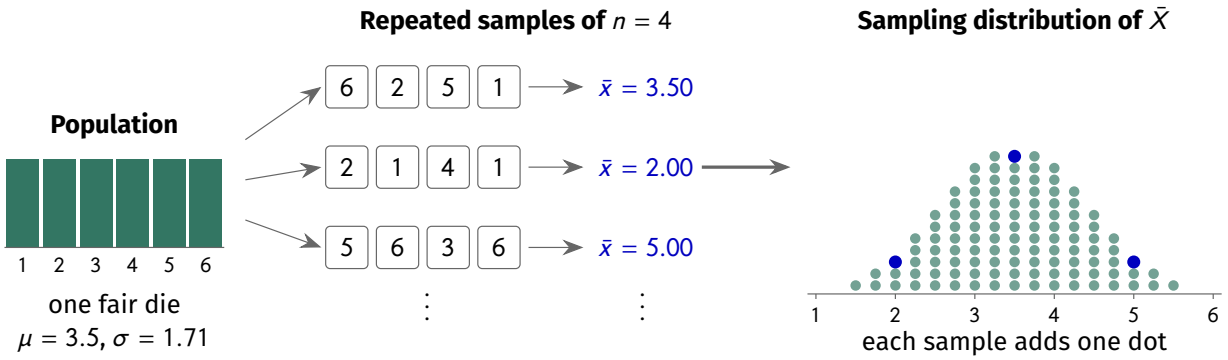
Since X_2 is clam count per day,

$$\begin{aligned} S = \sum_{i=1}^{7} X_{2,i} &\sim N\big(7\mu_{X_2},\ \sigma_S^2 = n\sigma_{X_2}^2\big) \\ &\sim N\big(7(20),\ \sigma_S^2 = 7(5^2)\big) \\ &\sim N(140,\ \sigma_S^2 = 175) \end{aligned}$$

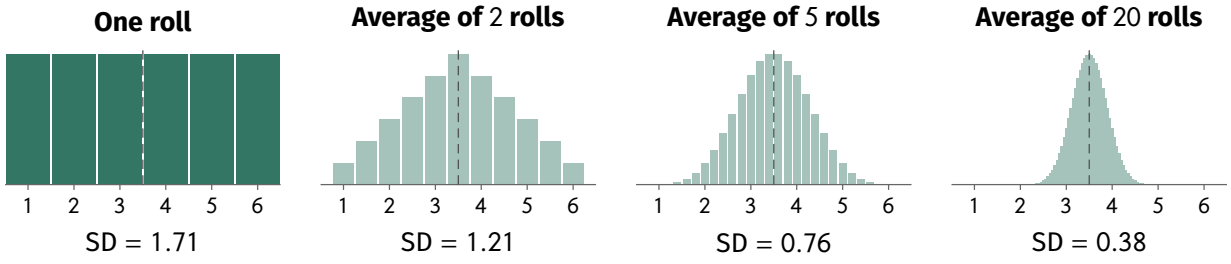
10 Convergence and Limit Theorems

10.1 The Central Limit Theorem

What is a sampling distribution? A statistic changes from sample to sample, so it has a distribution of its own, called its *sampling distribution*. Take four rolls of a die. Each sample gives one value of $\bar{X}$, and each value adds one dot to the picture on the right.



For a die we can compute the sampling distribution exactly. As n grows, the center stays at 3.5, the SD shrinks like $1/\sqrt{n}$, and the shape turns into a bell, even though the population is flat.



Central Limit Theorem (CLT)

Recall if $X_1, \ldots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu_X$, $\text{Var}[X_i] = \sigma_X^2$. Then

$$S_n = \sum_{i=1}^{n} X_i \quad \text{has } \mathbb{E}[S] = n\mu_X,\ \text{Var}[S] = n\sigma_X^2$$

$$\bar{X}_n = \frac{1}{n} S_n = \frac{1}{n} \sum X_i \quad \text{has } \mathbb{E}[\bar{X}] = \mu_X,\ \text{Var}[\bar{X}] = \frac{\sigma_X^2}{n}$$

Then, we also have

$$\frac{S_n - \mathbb{E}[S_n]}{\sqrt{\text{Var}[S_n]}} = \frac{S_n - n\mu_X}{\sqrt{n\sigma_X^2}} = \frac{\bar{X}_n - \mu_X}{\sqrt{\sigma_X^2/n}} = \frac{\sqrt{n}\,(\bar{X}_n - \mu_X)}{\sigma_X}$$

Then, the CLT states:

Theorem 10.1 (Central Limit Theorem): Let $X_1, \ldots, X_n$ be i.i.d. RVs with $\mathbb{E}[X_i] = \mu_X$, $\text{Var}[X_i] = \sigma_X^2$, μ_X and σ_X^2 finite. Let $S_n = \sum X_i$, $\bar{X}_n = \frac{1}{n} S_n = \frac{1}{n} \sum X_i$. Then,

$$\frac{S_n - n\mu_X}{\sqrt{n\sigma_X^2}} = \frac{\sqrt{n}\,(\bar{X}_n - \mu_X)}{\sigma_X} \xrightarrow{d} Z \sim N(0,1)$$

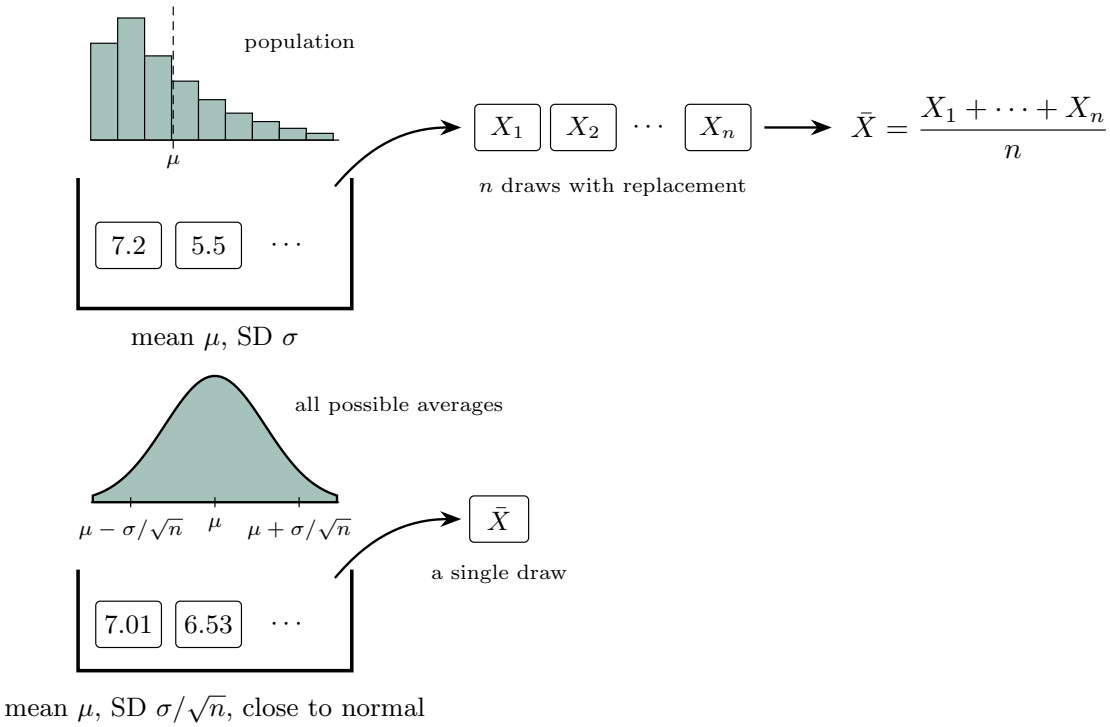
Here $\xrightarrow{d}$ denotes convergence in distribution, which we define in Section 10.2: the c.d.f. of the left side approaches the c.d.f. of Z as n grows.

In words. If $X_1, \ldots, X_n$ are a simple random sample from some unknown distribution F_X, with $\mu_X = \mathbb{E}[X_i]$ and $\sigma_X^2 = \text{Var}[X_i]$, then for n large enough

$$\bar{X}_n \dot{\sim} N\left(\mu_X, \frac{\sigma_X^2}{n}\right),$$

where the dot over the tilde reads "approximately distributed as." When the population is normal this holds exactly (Section 9.3). The CLT says that for large n it holds approximately for nearly any population. To see this happen one sample at a time, work through the notebook `sampling_distributions_clt.ipynb` in the Simulation Demos folder on Canvas.

A box-model picture. Think of the population as a box of tickets with mean μ_X and standard deviation σ_X, whatever the shape of its histogram. A sample is n draws with replacement, and $\bar{X}$ is their average. Now imagine a second box holding one ticket for every possible sample average. The CLT describes that second box: its tickets have mean μ_X and standard deviation $\sigma_X/\sqrt{n}$, and their histogram is close to a normal curve even when the first box's histogram is not. Our sample mean is a single draw from the second box.



A Galton board (Fig. 10.1) is the theorem built in wood: each ball falls through rows of pegs, bouncing left or right at random at every row, and the balls pile up in the bins below in the bell-shaped heap that the theorem predicts.

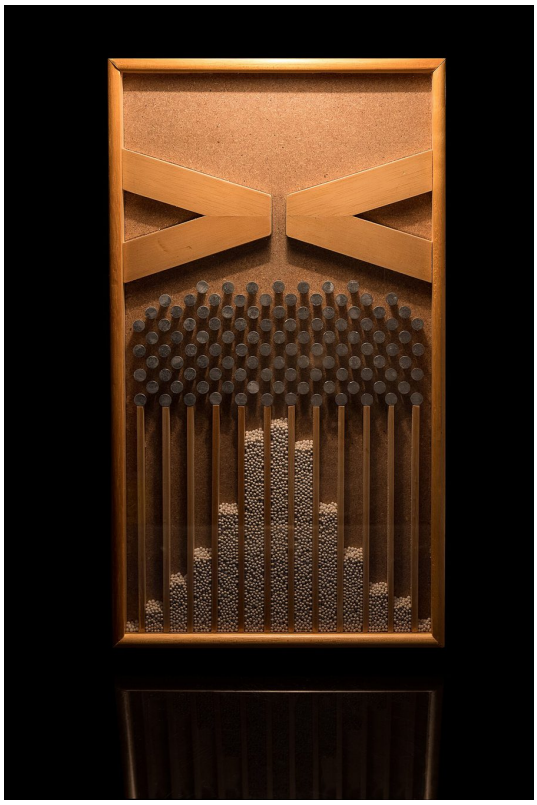


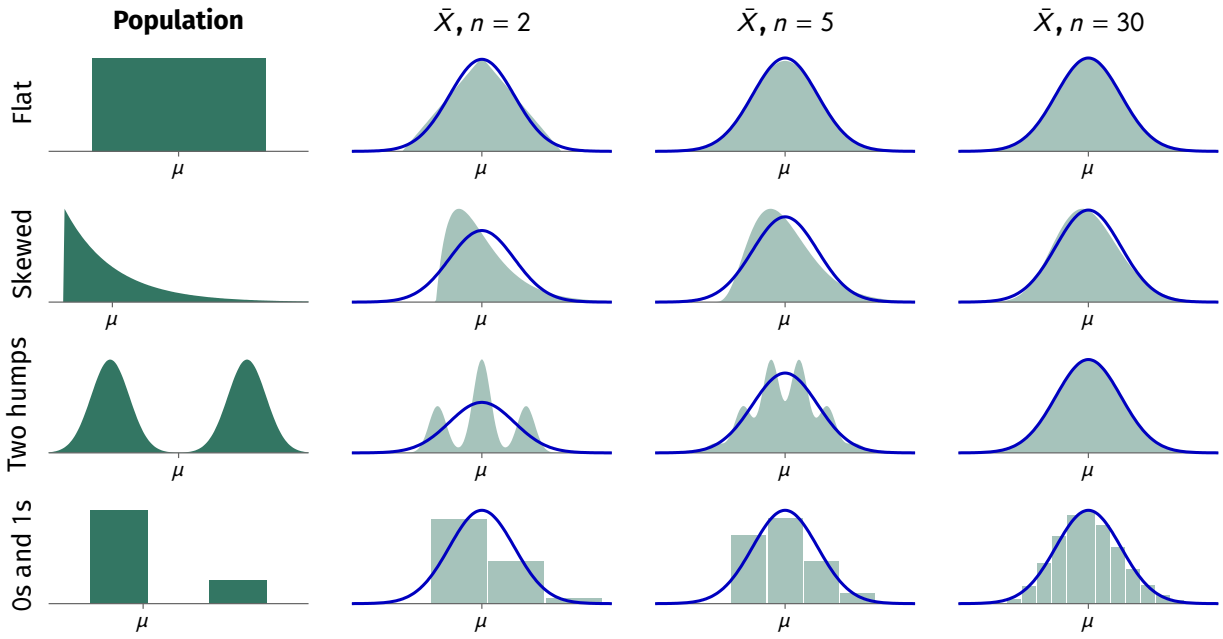
Figure 10.1: *A Galton board.* [19]

I.e., for a sufficiently large sample of size n from the same population, we may use the normal distribution to estimate probabilities of the form

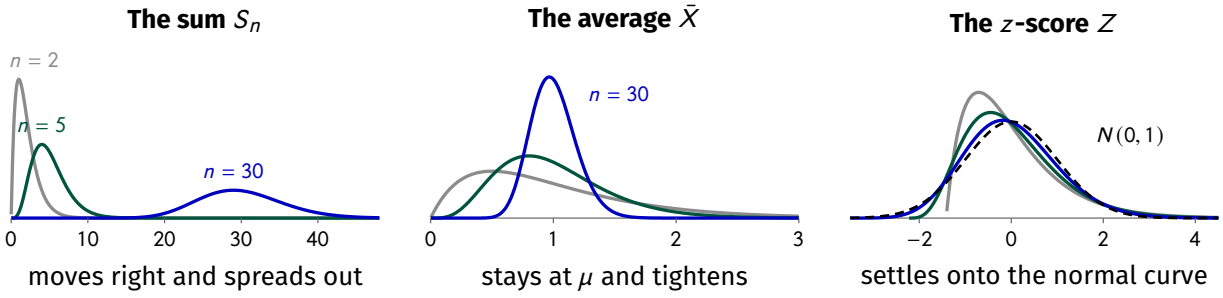
$$P\left(\frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} < a\right) \approx \Phi(a), \qquad P\left(\frac{S_n - n\mu_X}{\sigma_X\sqrt{n}} < b\right) \approx \Phi(b)$$

"Sufficiently large" in practice is $n \geq 30$. This is a rule of thumb, not a theorem: heavily skewed populations need larger n, while nearly symmetric populations get away with less.

The shape of the population does not matter. Each row below starts from a different population. The shaded areas are the exact sampling distributions of $\bar{X}$, and the blue curve is the normal curve that the CLT predicts from μ and $\sigma/\sqrt{n}$ alone.



Three views of one sample. Here the population is skewed, and each panel uses the same three sample sizes. The sum and the average both keep changing with n, so the theorem is stated for the z-score, which settles onto a single curve.



Sums and averages of a random sample

For $X_1, \ldots, X_n$ i.i.d. with mean μ_X and standard deviation σ_X:

$$\mathbb{E}[S_n] = n\mu_X, \qquad \mathrm{SD}[S_n] = \sqrt{n}\,\sigma_X,$$
$$\mathbb{E}[\bar{X}_n] = \mu_X, \qquad \mathrm{SD}[\bar{X}_n] = \sigma_X/\sqrt{n}.$$

If the population is normal, S_n and $\bar{X}_n$ are exactly normal; otherwise they are approximately normal for large n (CLT). Either way, a probability comes from the z-score (value − mean)/SD and the normal table.

Note: The CLT also gives the distribution of a sample proportion: if $X \sim \text{Bin}(n, p)$, then $\hat{p} = X/n \approx N\left(p,\ \frac{p(1-p)}{n}\right)$, which is the basis of polling margins of error.

The same reasoning explains why the 68–95–99.7 rule fits so many measured quantities: a measurement that adds up many small, nearly independent effects is approximately normal. In z-score terms, a value within ± 2 standard deviations of its mean is typical; about 3 in 1000 land beyond ± 3; beyond ± 4 is rare (about 6 in 100,000); and beyond ± 5 is extremely rare (fewer than 1 in a million).

Example 10.1 (guessing at trivia night): A trivia night at a bar in the Mission has 22 multiple-choice questions, each with 3 options. A team that guesses every answer at random gets $X \sim \text{Bin}(22, \frac{1}{3})$ questions right. If 100 teams guess independently, how likely is their average score to be above 7? That is, find $P(\bar{X} > 7)$ for $\bar{X} = (X_1 + \cdots + X_{100})/100$.

Each team has $\mathbb{E}[X_i] = 22 \cdot \frac{1}{3} = 7.333$ and $\text{Var}[X_i] = 22 \cdot \frac{1}{3} \cdot \frac{2}{3} = 4.889$, so $\mathbb{E}[\bar{X}] = 7.333$ and $\text{SD}[\bar{X}] = \sqrt{4.889/100} = 0.2211$. By the CLT,

$$P(\bar{X} > 7) \approx P\left(Z > \frac{7 - 7.333}{0.2211}\right) = P(Z > -1.508) = \Phi(1.508) = 0.934.$$

Here we can check the answer: the total $100\bar{X}$ is exactly $\text{Bin}(2200, \frac{1}{3})$, and a computer gives $P(100\bar{X} > 700) = 0.9316$. The CLT is off by less than 0.003.

Example 10.2 (a full cable car): Suppose a cable car is rated for 4500 kg of passengers, and rider weights have mean 75 kg and standard deviation 16 kg (skewed right, so not normal). With 57 riders aboard, how likely is their total weight to exceed the rating? That is, find $P(S > 4500)$ for $S = X_1 + \cdots + X_{57}$.

Here $\mathbb{E}[S] = 57(75) = 4275$ kg and $\text{SD}[S] = \sqrt{57}\,(16) = 120.8$ kg, so by the CLT

$$P(S > 4500) \approx P\left(Z > \frac{4500 - 4275}{120.8}\right) = P(Z > 1.86) = 0.031,$$

about one full car in 32. We never needed the shape of the weight distribution, only its mean and standard deviation.

Example 10.3: A festival bar's tap pours pints. The pour volume has mean $\mu_X = 500$ mL and standard deviation $\sigma_X = 8$ mL. *The shape of the pour distribution is unknown.* A tray holds $n = 36$ pints.

a) How is the average pour across a tray of 36 pints distributed? That is, give the approximate distribution of the sample mean $\bar{X}_{36}$.

$$\bar{X}_{36} \dot{\sim} N\left(500, \tfrac{8^2}{36}\right), \qquad \text{SD}[\bar{X}_{36}] = \tfrac{8}{6} = 1.333$$

b) How likely is the tray's average pour to exceed 502 mL? That is, approximate $P(\bar{X}_{36} > 502)$.

$$P(\bar{X}_{36} > 502) \approx P\left(Z > \frac{502 - 500}{1.333}\right) = P(Z > 1.5) = 1 - \Phi(1.5) = 0.0668$$

c) How is the total volume poured for the tray distributed, and how likely is it to come up short of 17,900 mL? That is, give the approximate distribution of the tray total S_{36}, and approximate $P(S_{36} < 17{,}900)$.

$$S_{36} \dot{\sim} N(36(500),\ 36(64)) = N(18{,}000,\ 2304), \qquad \text{SD}[S_{36}] = 48$$

$$P(S_{36} < 17{,}900) \approx P\left(Z < \frac{17{,}900 - 18{,}000}{48}\right) = \Phi(-2.083) = 0.0186$$

Note: We never needed the shape of the fill distribution, only its mean and variance. That is the whole point of the CLT: it converts a question about an unknown distribution into a question about the normal, provided we are asking about a sum or an average and n is large enough.

I.e., if $n \geq 30$ and a random sample from the same population was taken,

$$P(\bar{X}_n < x) = P\left(\frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} < a\right) \approx \Phi(a), \qquad a = \frac{x - \mu_X}{\sigma_X/\sqrt{n}}$$

$$P(S_n < x) = P\left(\frac{S_n - n\mu_X}{\sigma_X\sqrt{n}} < b\right) \approx \Phi(b), \qquad b = \frac{x - n\mu_X}{\sigma_X\sqrt{n}}$$

We will prove this result in Section 10.3.

Example 10.4: Suppose that the time it takes to net, measure, and release one dragonfly at a pond survey has a mean of 4, std dev of 2 minutes. Suppose dragonflies are independent, and a random sample of 50 was taken.

a) How are the average and the total handling times for the 50 dragonflies distributed? That is, identify the approximate distribution of $\bar{X} = \frac{1}{n}\sum X_i$, $S = \sum X_i$.

$$\bar{X} \dot{\sim} N\left(\mu_X,\ \sigma^2_{\bar{X}} = \frac{\sigma^2}{n}\right) = N\left(\mu_X = 4,\ \sigma^2_{\bar{X}} = \frac{4}{50}\right)$$

$$S \dot{\sim} N(\mu_S = n\mu_X,\ \sigma^2_S = n\sigma^2_X) = N(200,\ 200)$$

b) Estimate the prob that the total handling time for the 50 dragonflies is under 180 minutes.

$$P(S < 180) \approx P\left(Z < \frac{180 - 200}{\sqrt{200}}\right) = P(Z < -1.41)$$
$$= 1 - \Phi(+1.41) = 1 - 0.9207 = 0.0793$$

c) Estimate the probability that the average handling time per dragonfly for the 50 dragonflies is over 5 minutes.

$$P(\bar{X} > 5) = 1 - P(\bar{X} < 5) = 1 - P\left(Z < \frac{5 - 4}{2/\sqrt{50}}\right) = 1 - \Phi(3.536)$$
$$= 1 - 0.9998 = 0.0002$$

Note: The CLT essentially states that the sum of i.i.d. RVs, or some linear combination of a sum of i.i.d. RVs, has an approximate normal distribution for large n (see the rule-of-thumb note above).

The above is why the normal distribution is the most frequently used distribution in statistics.

Exercises

Exercise 10.1. For a dive boat, the average weight of a diver who boards, before gear, is 165 pounds, with standard deviation 22 pounds. Let X_i be the weight of the ith diver.

(a) Find the probability that the average weight of 36 divers boarding the boat is above 170 pounds. Which theorem lets you do this without knowing the distribution of X_i?

(b) Now assume X_i is normally distributed. What is the probability that the *total* weight of 16 divers is over 2500 pounds?

(c) Same question for 3500 pounds.

(d) The placard on the boat will say "Maximum capacity X pounds or 16 divers." Based on (b) and (c), which value of X should be posted so that the stated limits are consistent with each other? Explain.

(e) In part (a) you did not need normality, but in (b) and (c) you assumed it. Explain why the assumption was needed there but not in (a).

Exercise 10.2. For (a) and (b), decide whether the statement is true or false, and *fully explain* your answer. Part (c) asks for a short derivation.

(a) If $Y = g(X)$ is a transformation of a continuous random variable X, where g is continuous, differentiable and invertible, then the support of Y is the same as the support of X.

(b) The Central Limit Theorem implies that for $X_1, X_2, \ldots$ i.i.d. with common mean μ_X and variance σ_X^2, the distribution of X_i approaches a normal distribution as $n \to \infty$.

(c) Let $X_1, X_2, \ldots$ be i.i.d. with finite mean and variance. Show that $\sum_{i=1}^{n} X_i$ does *not* converge in probability to $\mathbb{E}\left[\sum_{i=1}^{n} X_i\right]$. *Hint:* what happens to the variance of the sum as n grows? Contrast this with the behavior of the *average*.

Exercise 10.3 (A yak train to Everest Base Camp)**.** An expedition's gear goes up the Khumbu valley to Everest Base Camp by yak. Each duffel weighs 18 kg on average with standard deviation 5 kg; the shape of the distribution is unknown and duffels are independent. The outfitter loads 100 duffels and has booked enough yaks for 1900 kg.

(a) Mean and standard deviation of the total load S.

(b) Approximate $P(S > 1900)$, the probability the booked yaks are not enough.

(c) Approximate the probability the average duffel weighs under 17.5 kg.

Exercise 10.4 (Wave heights at Pipeline)**.** Wave heights at Pipeline, the famous big-wave surf break on Oahu's North Shore, on a winter day have mean 3 m and standard deviation 0.8 m, with an unknown, right-skewed shape. A surf forecaster averages 64 waves sampled across the day, far enough apart to treat as independent.

(a) Give the approximate distribution of the average, and justify it.

(b) Approximate the probability the average exceeds 3.2 m.

(c) Approximate the probability the average is below 2.8 m.

Exercise 10.5 (Aurora nights at Abisko)**.** The global Kp index (a 0 to 9 scale of geomagnetic activity, issued for the whole planet) on winter nights at Abisko, in Swedish Lapland, has mean 2.5 and standard deviation 1.2, with an unknown, skewed shape. A visitor stays $n = 30$ nights; treat the nights as independent.

(a) Give the approximate distribution of the 30-night average Kp.

(b) Approximate the probability the average exceeds 3.

(c) The aurora fills the sky overhead in a corona when Kp ≥ 4, which happens on 20% of nights independently. Probability of at least 10 such nights out of 30.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 10.1. (a) $\mathrm{SD}(\bar{X}) = 22/6 = 3.667$; by the CLT, $P(\bar{X} > 170) \approx P(Z > 1.364) = 0.086$. **(b)** $S \sim N(16 \cdot 165,\ 16 \cdot 22^2) = N(2640,\ 88^2)$ exactly: $P(S > 2500) = P(Z > -1.59) = 0.944$. **(c)** $P(S > 3500) = P(Z > 9.77) \approx 0$. **(d)** Post about 3500 pounds: 16 divers essentially never exceed it, so the two limits agree; 2500 would be exceeded 94% of the time by 16 divers and the limits would contradict each other. **(e)** In (a) $n = 36$ is large enough for the CLT to supply normality of the average; in (b) and (c) $n = 16$ is smallish and we needed an exact distribution for the sum, which only holds if each X_i is normal.

Exercise 10.2. (a) False: g maps the support of X onto its image, which is generally a different set (e.g. $Y = e^X$ turns $\mathbb{R}$ into $(0, \infty)$). **(b)** False: the CLT is about the *average* (or sum) of many X_i; the distribution of each X_i is fixed and never changes with n. **(c)** $\mathrm{Var}[\sum X_i] = n\sigma^2 \to \infty$, so the sum keeps spreading out around its mean rather than concentrating; by contrast $\mathrm{Var}[\bar{X}] = \sigma^2/n \to 0$, which is why the average converges.

Exercise 10.3. (a) 1800 kg and $\sqrt{100} \cdot 5 = 50$ kg. **(b)** $P(Z > 2) = 0.0228$. **(c)** $\mathrm{SD}(\bar{X}) = 0.5$: $P(Z < -1) = 0.1587$.

Exercise 10.4. (a) CLT: $\bar{X} \approx N(3, 0.01)$, SD 0.1 m. **(b)** $P(Z > 2) = 0.023$. **(c)** $P(\bar{X} < 2.8) \approx P(Z < -2) = 0.023$, the mirror image of (b).

Exercise 10.5. (a) $N(2.5, 0.048)$, SD 0.219. **(b)** $P(Z > 2.28) = 0.011$. **(c)** $\mathrm{Bin}(30, 0.2)$: CLT $P(Z > (10 - 6)/2.19 = 1.83) = 0.034$; exact 0.061.

10.2 Convergence in Distribution

The convergence definitions in this section apply to *any* sequence of RVs, discrete or continuous; no independence or identical-distribution assumptions are needed. When we study $\bar{X}_n$ and the Law of Large Numbers, we will additionally assume the RVs are "independent and identically distributed" (i.i.d.).

There are two natural senses in which a sequence of RVs Y_n may settle down as n grows: 1) the *values* of Y_n get close to the value of Y, which is convergence in probability, a statement about the RVs themselves; or 2) the *distribution* of Y_n gets close to the distribution of Y, which is convergence in distribution, a statement about c.d.f.'s only. The second is weaker: two RVs can have identical distributions while their values never agree.

Definition 10.1: Let $Y_1, \ldots, Y_n$ be RVs with respective c.d.f.'s $F_1, \ldots, F_n$, and let Y be a RV with c.d.f. G.

We say the sequence of RVs $\{Y_n\}$ **converges in distribution** to the RV Y as $n \to \infty$ if

$$F_n(x) \xrightarrow{n\to\infty} G(x) \qquad \text{for all continuity points } x \text{ of } G$$

This is typically denoted as $Y_n \xrightarrow{d} Y$.

Example 10.5: Suppose we have a discrete RV with the following c.d.f.'s F_n and G, $n \geq 1$:

$$F_n(x) = \begin{cases} 0 & \text{if } x < 1 - \frac{1}{n} \\ \frac{1}{2} & \text{if } 1 - \frac{1}{n} \leq x < 1 + \frac{1}{n} \\ 1 & \text{if } x \geq 1 + \frac{1}{n} \end{cases} \qquad G(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

G is continuous for all $x \neq 1$.

For $x < 1$, let $n_0 > \dfrac{1}{1-x}$, i.e. $x < 1 - \dfrac{1}{n_0}$. Then $x < 1 - \dfrac{1}{n}$ $\forall\, n \geq n_0$.

Thus, by $F_n(x)$ definition, we see $F_n(x) = 0 = G(x)$ for $x < 1 - \frac{1}{n}$, i.e. $x < 1$ ($n \to \infty$).

For $x > 1$, let $n_0 \geq \dfrac{1}{x-1}$, i.e. $x \geq 1 + \dfrac{1}{n_0}$. Then $x \geq 1 + \dfrac{1}{n}$ $\forall\, n \geq n_0$.

Thus, $F_n(x) = 1$ $\forall\, n \geq n_0$, i.e. as $n \to \infty$, $F_n(x) = 1 = G(x)$.

Thus $F_n(x) \xrightarrow{d} G(x)$, or $Y_n \xrightarrow{d} Y$.

Note: We need only consider all continuity points, so that $x = 1$ (which is not a continuity point) need not be considered.

Example 10.6: Suppose

$$F_n(x) = \begin{cases} 1 - \left(1 - \frac{1}{n}\right)^{nx} & x > 0 \\ 0 & \text{o.w.} \end{cases}$$

Show $F_n(x) \xrightarrow{d} F_X(x) = 1 - e^{-x}$.

$$\lim_{n\to\infty} 1-\left(1-\tfrac{1}{n}\right)^{nx} = 1-\lim_{n\to\infty}\left(1-\tfrac{1}{n}\right)^{nx} = 1-e^{-x} = \text{c.d.f. of } \mathrm{Exp}(\lambda=1)$$

10.3 Proof of the CLT

This proof requires the following lemma, which we state without proof:

Lemma 10.1: Let $Z_1, Z_2, \ldots$ be a sequence of RVs with c.d.f.s $F_{Z_n}(z)$ and m.g.f.s $M_{Z_n}(t)$, and let Z be a RV with c.d.f. $F_Z(z)$ and m.g.f. $M_Z(t)$, all finite for every t in an open interval around 0. If $M_{Z_n}(t) \to M_Z(t)$ for every t in that interval, then

$$F_{Z_n}(z) \to F_Z(z) \qquad \text{at every } z \text{ where } F_Z(z) \text{ is continuous,}$$

i.e. $Z_n \xrightarrow{d} Z$.

Proof of CLT. Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Assume also that the m.g.f. $M_X(t)$ is finite for every t in an open interval around 0. (This is a genuine restriction; the theorem holds more generally, but the m.g.f. argument requires it.)

Standardize. Since $\sigma > 0$, let $Y_i = (X_i - \mu)/\sigma$. The Y_i are i.i.d. with $\mathbb{E}[Y_i] = 0$ and $\mathbb{E}[Y_i^2] = \mathrm{Var}[Y_i] = 1$, and their common m.g.f. $M_Y(t) = e^{-\mu t/\sigma} M_X(t/\sigma)$ is finite near 0 because M_X is. Then

$$Z_n = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} = \frac{\sum X_i - n\mu}{\sigma\sqrt{n}} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{X_i-\mu}{\sigma} = \frac{\sum Y_i}{\sqrt{n}},$$

and we want to show that $M_{Z_n}(t) \to e^{t^2/2}$, the m.g.f. of $Z \sim N(0,1)$.

The m.g.f. of Z_n. Since $1/\sqrt{n}$ is a constant, $M_{\sum Y_i/\sqrt{n}}(t) = M_{\sum Y_i}\left(\frac{t}{\sqrt{n}}\right)$. And since the Y_i are independent with a common m.g.f. M_Y (the m.g.f. of a sum of independent RVs is the product of their m.g.f.s),

$$M_{Z_n}(t) = M_{\sum Y_i}\left(\tfrac{t}{\sqrt{n}}\right) = \left[M_Y\left(\tfrac{t}{\sqrt{n}}\right)\right]^n.$$

For a fixed t, the point $t/\sqrt{n}$ lies in the interval where M_Y is finite once n is large enough, so this is finite for all large n. We need to show

$$\left[M_Y\left(\tfrac{t}{\sqrt{n}}\right)\right]^n \to e^{t^2/2}, \quad \text{or equivalently} \quad n\ln\left[M_Y\left(\tfrac{t}{\sqrt{n}}\right)\right] \to \frac{t^2}{2}.$$

Take logs. Let $L(s) = \ln\big(M_Y(s)\big)$, which is defined because $M_Y(s) > 0$. An m.g.f. that is finite on an open interval around 0 can be differentiated as many times as we like there, so L, L' and L'' exist

and are continuous near 0. Then

$$\begin{aligned} L(0) &= \ln(M_Y(0)) = \ln(1) = 0 \\ L'(0) &= \frac{M_Y'(0)}{M_Y(0)} = \frac{\mathbb{E}[Y]}{1} = 0 \\ L''(0) &= \frac{M_Y(0)\,M_Y''(0) - \big[M_Y'(0)\big]^2}{[M_Y(0)]^2} \qquad \text{(by quotient rule)} \\ &= \frac{(1)\,\mathbb{E}[Y^2] - (\mathbb{E}[Y])^2}{(1)^2} = \mathbb{E}[Y^2] = 1 \end{aligned}$$

L'Hôpital. Write $n \ln\big[M_Y(t/\sqrt{n})\big] = \dfrac{L(t/\sqrt{n})}{1/n}$. As $n \to \infty$, the numerator $\to L(0) = 0$ and the denominator $\to 0$. Treat n as a continuous variable so that we may differentiate w.r.t. n; if the limit exists as $n \to \infty$ through all real numbers, it exists through the integers too. By the chain rule,

$$\frac{d}{dn} L\Big(\frac{t}{\sqrt{n}}\Big) = L'\Big(\frac{t}{\sqrt{n}}\Big) \cdot \Big(-\frac{t}{2}\, n^{-3/2}\Big), \qquad \frac{d}{dn}\,\frac{1}{n} = -n^{-2},$$

so L'Hôpital's rule gives

$$\lim_{n\to\infty} \frac{L(t/\sqrt{n})}{1/n} = \lim_{n\to\infty} \frac{-\frac{t}{2}\, L'(t/\sqrt{n})\, n^{-3/2}}{-n^{-2}} = \lim_{n\to\infty} \frac{t\, L'(t/\sqrt{n})}{2\, n^{-1/2}}.$$

Since L' is continuous, $L'(t/\sqrt{n}) \to L'(0) = 0$, and $2n^{-1/2} \to 0$, so this is again $0/0$. Differentiate the new numerator and denominator:

$$\frac{d}{dn}\Big[t\, L'\Big(\frac{t}{\sqrt{n}}\Big)\Big] = t\, L''\Big(\frac{t}{\sqrt{n}}\Big) \cdot \Big(-\frac{t}{2}\, n^{-3/2}\Big), \qquad \frac{d}{dn}\, 2n^{-1/2} = -n^{-3/2},$$

and apply L'Hôpital again:

$$\lim_{n\to\infty} \frac{t\, L'(t/\sqrt{n})}{2\, n^{-1/2}} = \lim_{n\to\infty} \frac{-\frac{t^2}{2}\, L''(t/\sqrt{n})\, n^{-3/2}}{-n^{-3/2}} = \lim_{n\to\infty} \frac{t^2}{2}\, L''\Big(\tfrac{t}{\sqrt{n}}\Big) = \frac{t^2}{2}\, L''(0) = \frac{t^2}{2} \qquad \checkmark$$

using the continuity of L'' at 0. (For $t = 0$ both sides are 0 directly.)

Conclude. So $M_{Z_n}(t) \to e^{t^2/2}$, the m.g.f. of $N(0,1)$, for every t, and the Lemma gives $Z_n \xrightarrow{d} Z \sim N(0,1)$. This proves the CLT for i.i.d. X_i with finite mean, positive finite variance, *and* an m.g.f. that is finite near 0. The CLT holds without the m.g.f. condition, but that needs a different proof. □

Exercises

Exercise 10.6. The random variables $X_1, \ldots, X_{50}$ are independent and identically distributed as χ^2_1. Let $S = \sum_{i=1}^{50} X_i$ and $\bar{X} = S/50$.

(a) Using moment generating functions, find the *exact* distribution of S.

(b) Recall that a χ^2_1 random variable has mean 1 and variance 2. Use this to find $\mathbb{E}[S]$ and $\mathrm{Var}[S]$ directly, and check them against the distribution you named in part (a).

(c) Using the Central Limit Theorem, give the approximate distribution of S, and of $\bar{X}$.

(d) Use the approximation from (c) to estimate $P(S > 50)$.

(e) The exact value is $P(\chi^2_{50} > 50) = 0.4734$, so the CLT estimate is close but systematically too large. Explain why, using the fact that the χ^2_{50} distribution is right-skewed while the normal is symmetric. Where does the mean of a right-skewed distribution sit relative to its median?

(f) What do parts (a) and (c) together tell you about the behavior of χ^2_n random variables as $n \to \infty$?

(g) Prove that $\bar{X} \xrightarrow{p} 1$.

Exercise 10.7 (A capsule hotel)**.** A capsule hotel (a Japanese budget hotel of stacked sleeping pods) in Shinjuku, Tokyo, has 200 pods, each booked on a given night independently with probability 0.8. Approximate the probability that at least 170 pods are booked.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 10.6. (a) $M_S(t) = \prod(1-2t)^{-1/2} = (1-2t)^{-25}$, the m.g.f. of χ^2_{50}. **(b)** $\mathbb{E}[S] = 50$, $\mathrm{Var}[S] = 100$, matching χ^2_{50} (mean 50, variance $2 \cdot 50$). **(c)** $S \approx N(50, 100)$ and $\bar{X} \approx N(1, 1/25)$. **(d)** $P(S > 50) \approx P(Z > 0) = 0.5$. **(e)** For a right-skewed distribution the mean sits above the median, so the true $P(S > \text{mean})$ is below $\frac{1}{2}$; the symmetric normal puts it at exactly $\frac{1}{2}$. **(f)** χ^2_n is approximately $N(n, 2n)$ for large n. **(g)** $\mathbb{E}[\bar{X}] = 1$ and $\mathrm{Var}[\bar{X}] = 2/50 \to 0$, so by Chebyshev $P(|\bar{X} - 1| > \varepsilon) \le \dfrac{2/n}{\varepsilon^2} \to 0$: the weak law of large numbers.

Exercise 10.7. $X \sim \mathrm{Bin}(200, 0.8)$, mean 160, SD 5.66: $P(X \ge 170) \approx P(Z > (170 - 160)/5.66 = 1.77) = 0.038$. (The exact binomial answer is 0.043.)

10.4 Convergence in Probability

Definition 10.2: The sequence of RVs $\{Y_n\}$, $n \geq 1$, **converges in probability** to the RV Y if $\forall\, \varepsilon > 0$,

$$P(|Y_n - Y| \geq \varepsilon) \xrightarrow{n\to\infty} 0 \qquad \text{or, equivalently,} \qquad P(|Y_n - Y| \leq \varepsilon) \xrightarrow{n\to\infty} 1$$

This is typically denoted $Y_n \xrightarrow{p} Y$.

Example 10.7: Let $X_n \sim \text{Exp}(n)$. Show that $X_n \xrightarrow{p} 0$.

$$\begin{aligned}
\lim_{n\to\infty} P(|X_n - 0| \geq \varepsilon) &= \lim_{n\to\infty} P(X_n \geq \varepsilon) \qquad \text{(since } X_n \geq 0\text{)} \\
&= \lim_{n\to\infty} e^{-n\varepsilon} \qquad \text{(since } X_n \sim \text{Exp}(n)\text{)} \\
&= 0 \qquad \forall\, \varepsilon > 0
\end{aligned}$$

Thus, $X_n \xrightarrow{p} 0$. ✓

Example 10.8: Let $X_n = X + Y_n$, where $\mathbb{E}[Y_n] = \dfrac{1}{n}$, $\text{Var}[Y_n] = \dfrac{\sigma^2}{n}$, where $\sigma > 0$ and finite. Show $X_n \xrightarrow{p} X$.

First, recall $|a + b| \leq |a| + |b|$ (triangle inequality). Let $a = Y_n - \mathbb{E}[Y_n]$, $b = \mathbb{E}[Y_n]$. Then,

$$|Y_n| \leq |Y_n - \mathbb{E}[Y_n]| + \mathbb{E}[Y_n]$$

$$\star \quad |Y_n| \leq |Y_n - \mathbb{E}[Y_n]| + \tfrac{1}{n}$$

And, for any $\varepsilon > 0$ we have

$$\begin{aligned}
P(|X - X_n| \geq \varepsilon) = P(|Y_n| \geq \varepsilon) &\leq P\left(|Y_n - \mathbb{E}[Y_n]| + \tfrac{1}{n} \geq \varepsilon\right) \quad \text{(by } \star\text{)} \\
&= P\left(|Y_n - \mathbb{E}[Y_n]| \geq \varepsilon - \tfrac{1}{n}\right) \\
&\leq \frac{\text{Var}[Y_n]}{\left[\varepsilon - \frac{1}{n}\right]^2} \qquad \text{(Chebyshev's inequality)} \\
&= \frac{\sigma^2/n}{\left(\varepsilon - \frac{1}{n}\right)^2} = \frac{\sigma^2}{n\left(\varepsilon - \frac{1}{n}\right)^2} \xrightarrow{n\to\infty} 0
\end{aligned}$$

Thus, $P(|X - X_n| \geq \varepsilon) \to 0$ as $n \to \infty$, and $X_n \xrightarrow{p} X$. ✓

Converging in probability or distribution are often used in many proofs.

We can relate the two types of convergence with the following theorem:

Theorem 10.2: Let $\{Y_n\}$, $n \geq 1$, be a sequence of RVs and let Y be a RV. Then,

$$Y_n \xrightarrow{p} Y \quad \text{always implies} \quad Y_n \xrightarrow{d} Y$$

However, convergence in distribution need not imply convergence in probability.

Example 10.9: Let $Z \sim N(0,1)$ and $Y_n = -Z$ for every n. Since $-Z \sim N(0,1)$ as well, $Y_n \xrightarrow{d} Z$ trivially; the c.d.f.'s are identical. But $|Y_n - Z| = 2|Z|$, which does not shrink as n grows, so Y_n does not converge in probability to Z. Convergence in distribution compares distributions, not values.

Proof of the theorem is left for more advanced classes.

10.5 The Weak Law of Large Numbers

Weak Law of Large Numbers (WLLN)

Theorem 10.3 (Weak Law of Large Numbers): Let $X_1, X_2, \ldots$ be i.i.d. RVs with $\mathbb{E}[X_i] = \mu_X \; \forall \, i = 1, 2, \ldots,$ μ_X finite, and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$ be the sample mean of $X_1, \ldots, X_n$. Then,

$$\bar{X}_n \xrightarrow{p} \mu_X$$

Proof. Suppose $\text{Var}[X_i] = \sigma_X^2 \; \forall \, i = 1, 2, \ldots$ Then, by Chebyshev's inequality we have:

$$P\big(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| > \varepsilon\big) \leq \frac{1}{\varepsilon^2} \text{Var}[\bar{X}_n]$$

$$\Rightarrow \quad P\big(|\bar{X}_n - \mu_X| > \varepsilon\big) \leq \frac{1}{\varepsilon^2} \cdot \frac{\sigma_X^2}{n}$$

And since $\lim_{n \to \infty} \frac{1}{\varepsilon^2} \cdot \frac{\sigma_X^2}{n} = 0,$

$$P\big(|\bar{X}_n - \mu_X| > \varepsilon\big) \xrightarrow{n \to \infty} 0$$

i.e. $\bar{X}_n \xrightarrow{p} \mu_X$. □

If we drop the assumption that $\text{Var}[X_i] = \sigma_X^2 \; \forall \, i = 1, \ldots, n$, the result still holds, but the proof is omitted.

Some applications of the WLLN follow:

1. If $X_i \sim \text{Bin}(1, p)$, then $\mathbb{E}[X_i] = p$, and $\bar{X}_n \xrightarrow{p} p$
2. If $X_i \sim \text{Pois}(\lambda)$, then $\mathbb{E}[X_i] = \lambda$, and $\bar{X}_n \xrightarrow{p} \lambda$
3. If $X_i \sim N(\mu_X, \sigma_X^2)$, then $\mathbb{E}[X_i] = \mu_X$, and $\bar{X}_n \xrightarrow{p} \mu_X$
4. If $X_i \sim \text{Exp}(\lambda)$, then $\mathbb{E}[X_i] = \frac{1}{\lambda}$, and $\bar{X}_n \xrightarrow{p} \frac{1}{\lambda}$

Example 10.10: Roll a fair die repeatedly and let $\bar{X}_n$ be the average of the first n rolls, so $\mu_X = 3.5$ and $\sigma_X^2 = \frac{35}{12}$. How large must n be to guarantee $P(|\bar{X}_n - 3.5| \geq 0.1) \leq 0.05$?

From the Chebyshev bound in the WLLN proof:

$$P(|\bar{X}_n - 3.5| \geq 0.1) \leq \frac{\sigma_X^2}{n\,\varepsilon^2} = \frac{35/12}{n\,(0.1)^2} \leq 0.05 \quad \Rightarrow \quad n \geq \frac{35/12}{(0.05)(0.01)} = 5833.3$$

so $n = 5834$ rolls suffice. Chebyshev is conservative; the CLT (Section 10.1) certifies the same guarantee with a much smaller n.

Example 10.11: A casino game returns, on average, −\$0.05 per \$1 played, with SD = \$2 per play. Let $\bar{X}_n$ be the player's average winnings per play after n plays.

a) What does the WLLN say?

Since $\mu_X = -0.05$, we have $\bar{X}_n \xrightarrow{p} -0.05$. The player's long-run average converges to a loss of five cents per play, regardless of any short-run luck.

b) Bound the probability that a player is *ahead* after n plays.

Being ahead means $\bar{X}_n > 0$, which requires $\bar{X}_n$ to miss its mean by at least 0.05:

$$P(\bar{X}_n > 0) \le P\big(|\bar{X}_n - (-0.05)| \ge 0.05\big) \le \frac{\sigma_X^2}{n\varepsilon^2} = \frac{4}{n(0.05)^2} = \frac{1600}{n}$$

c) Evaluate for $n = 10{,}000$ and $n = 1{,}000{,}000$.

$$n = 10^4 : \ \le 0.16 \qquad\qquad n = 10^6 : \ \le 0.0016$$

Note: This is why casinos are profitable and why gamblers are not. The house does not need to win any individual bet; it only needs n to be large. Note also that the bound says nothing useful for small n: for $n = 100$ it gives 16, which is vacuous. In the short run, anything can happen.

Empirical c.d.f. and WLLN

When the true c.d.f. of a RV is unknown, we typically estimate it with the following:

$$F_n(x, s) = \frac{1}{n}\big(\#\text{ of } X_1(s), X_2(s), \ldots, X_n(s) \le x\big)$$

Recall $X_i(s) =$ value of RV at sample point s.

In practice, we have a sample $x_1, x_2, \ldots, x_n$, and

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{x_i \le x\}}, \qquad \mathbb{1}_{\{x_i \le x\}} = \begin{cases} 1 & \text{if } x_i \le x \\ 0 & \text{if } x_i > x \end{cases}$$

But, notice $\mathbb{1}_{\{X_i \le x\}} \sim \text{Bin}(n = 1,\ p = F_X(x))$ (where $F_X(x)$ is the "true" c.d.f.).

Thus,

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \le x\}} \xrightarrow{p} F_X(x) \qquad \forall\, x \in \mathbb{R}$$

I.e., the empirical c.d.f. converges in probability to the "true" c.d.f.

Exercises

Exercise 10.8 (Steps around Athens)**.** A traveler's daily step count in Athens has mean 14,000 and standard deviation 4,000, with days independent. Let $\bar{X}$ be the average over 25 days.

(a) Use Chebyshev's inequality to bound the probability that $\bar{X}$ is more than 2,000 steps from 14,000.

(b) Approximate the same probability with the central limit theorem, and compare.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 10.8. (a) $\mathrm{SD}(\bar{X}) = 800$, so 2000 steps is 2.5 SDs: Chebyshev gives at most $1/2.5^2 = 0.16$. **(b)** CLT: $2(1 - \Phi(2.5)) = 0.012$, so the bound is loose by a factor of about 13.

10.6 The Sample Variance

Just as $\bar{X}_n$ estimates μ_X, we want an estimate of σ_X^2 from the same sample. If μ_X is known, a natural estimate is the **known-mean sample variance**

$$S_0^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu_X)^2,$$

the average squared deviation about μ_X. When μ_X is unknown, we replace it with $\bar{X}$ and divide by $n - 1$, which gives the usual **sample variance**

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Two properties of S_0^2 follow from what we already have.

1) It is right on average. Each term has $\mathbb{E}[(X_i - \mu_X)^2] = \mathrm{Var}[X_i] = \sigma_X^2$, so by linearity $\mathbb{E}[S_0^2] = \sigma_X^2$.
2) It settles down at σ_X^2. Since S_0^2 is a sample mean of the i.i.d. RVs $(X_i - \mu_X)^2$, the WLLN gives $S_0^2 \xrightarrow{p} \sigma_X^2$.

The usual sample variance shares both properties: $\mathbb{E}[S^2] = \sigma_X^2$ and $S^2 \xrightarrow{p} \sigma_X^2$. Its exact distribution for a normal population involves the chi-squared distribution, which we take up in Section 11.1.

Many of the properties we discussed are used in proofs of hypothesis tests results, and confidence interval results. Notice that these tests typically begin with the assumptions that:

a) A random sample was taken
b) Subjects are independent
c) Sample size is large

The primary reasons for the above assumptions stem from the theoretical results of this section.

11 Transformations of Random Variables

11.1 Squared Normals and the Chi-Squared Distribution

Distribution of Z^2

Suppose $Z \sim N(0,1)$. Then $Y = Z^2 \sim \chi^2_1$, the $\Gamma(\alpha = \frac{1}{2}, \beta = \frac{1}{2})$ distribution of Section 6.3. One proof shows that the m.g.f. of Z^2 is $(1-2t)^{-1/2}$, the m.g.f. of χ^2_1. Section 11.2 gives a second proof, which finds the density of Y directly.

Sums of χ^2 RVs

If $X_i \sim \chi^2_{r_i}$ are independent, then

$$\sum_{i=1}^{k} X_i \sim \chi^2_{\sum r_i}$$

Proof.

$$M_{\sum X_i}(t) = \prod_{i=1}^{k} M_{X_i}(t) = \prod_{i=1}^{k} (1-2t)^{-r_i/2} = (1-2t)^{-(\sum r_i)/2}$$

This is the m.g.f. of $\chi^2_{\sum r_i}$. □

Together, these two results give the following:

Sum of Squared $N(0,1)$

Let $X_1, \ldots, X_k \sim N(\mu_i, \sigma_i^2)$, $i = 1, \ldots, k$, and all X_i independent. Then,

$$\sum_{i=1}^{k} \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2 \sim \chi^2_k$$

and if $\mu_i = \mu_X$, $\sigma_i^2 = \sigma_X^2$ $\forall\, i = 1, \ldots, k$:

$$\begin{aligned} \sum_{i=1}^{k} \left(\frac{X_i - \mu_X}{\sigma_X} \right)^2 &= \frac{1}{\sigma_X^2} \sum (X_i - \mu_X)^2 \\ &= \frac{1}{\sigma_X^2} \cdot k \cdot \frac{1}{k} \sum (X_i - \mu_X)^2 \\ &= \frac{k}{\sigma_X^2} S_0^2 \sim \chi^2_k \end{aligned}$$

where $S_0^2 = \dfrac{1}{k}\sum (X_i - \mu_X)^2$ is the known-mean sample variance of Section 10.6. (This differs from the usual sample variance of the glossary, $S^2 = \frac{1}{k-1}\sum (X_i - \bar{X})^2$, which replaces μ_X with $\bar{X}$ and satisfies $(k-1)S^2/\sigma_X^2 \sim \chi^2_{k-1}$, a result you will prove in mathematical statistics.)

Mean, Variance of S_0^2 (known-mean sample variance)

Since we have shown that $\dfrac{k}{\sigma_X^2} S_0^2 \sim \chi_k^2$, this gives:

$$\mathbb{E}\left[\frac{k}{\sigma_X^2} S_0^2\right] = k \qquad (\text{Using } \mathbb{E}[\chi_r^2] = r)$$

$$\Rightarrow \quad \frac{k}{\sigma_X^2}\mathbb{E}[S_0^2] = k$$

$$\Rightarrow \quad \mathbb{E}[S_0^2] = \frac{k\,\sigma_X^2}{k} = \sigma_X^2$$

$$\mathrm{Var}\left[\frac{k}{\sigma_X^2} S_0^2\right] = 2k \qquad (\text{Using } \mathrm{Var}[\chi_r^2] = 2r)$$

$$\Rightarrow \quad \frac{k^2}{\sigma_X^4}\mathrm{Var}[S_0^2] = 2k$$

$$\Rightarrow \quad \mathrm{Var}[S_0^2] = \frac{\sigma_X^4}{k^2}(2k) = \frac{2\sigma_X^4}{k}$$

Example 11.1: Let Z_1, Z_2, Z_3 be i.i.d. $N(0,1)$.

a) Find the distribution of $W = 2Z_1 - Z_2 + 3$, and compute $P(W > 5)$.

This is a linear combination of independent normals, so it is normal. Its mean and variance are

$$\mathbb{E}[W] = 2(0) - 0 + 3 = 3, \qquad \mathrm{Var}[W] = 2^2(1) + (-1)^2(1) = 5$$

so $W \sim N(3,5)$ and

$$P(W > 5) = P\left(Z > \frac{5-3}{\sqrt{5}}\right) = 1 - \Phi(0.894) = 0.186$$

b) Find the distribution of $V = Z_1^2 + Z_2^2 + Z_3^2$, together with its mean and variance.

By the result above with $k = 3$, $V \sim \chi_3^2$. Using $\mathbb{E}[\chi_k^2] = k$ and $\mathrm{Var}[\chi_k^2] = 2k$,

$$\mathbb{E}[V] = 3, \qquad \mathrm{Var}[V] = 6$$

c) Is V normal? Is it ever negative?

No and no. V is a sum of *squares*, so $P(V \geq 0) = 1$ and its distribution is right-skewed, not symmetric.

Note: Parts a) and b) show two operations that behave completely differently. Adding normals with constants keeps you inside the normal family. *Squaring* them leaves it entirely: you land in the chi-squared family, on the positive half-line. The chi-squared is the distribution of *squared error*, which is why it governs sample variances.

11.2 Transformations of a Random Variable

We often want the distribution of $Y = g(X)$, when the distribution of X is known. Throughout this section, assume X is a continuous RV with p.d.f. f_X and c.d.f. F_X.

The main tool is the **c.d.f. method**: express $F_Y(y) = P(g(X) \leq y)$ as a statement about X, then differentiate to get f_Y.

Linear case: If $Y = a + bX$ with $b \neq 0$, then for $b > 0$:

$$F_Y(y) = P(a + bX \leq y) = P\left(X \leq \frac{y-a}{b}\right) = F_X\left(\frac{y-a}{b}\right)$$

and for $b < 0$ (the inequality flips when dividing by b):

$$F_Y(y) = P\left(X \geq \frac{y-a}{b}\right) = 1 - F_X\left(\frac{y-a}{b}\right)$$

Differentiating either case w.r.t. y (chain rule) gives one formula covering both:

$$f_Y(y) = f_X\left(\frac{y-a}{b}\right)\frac{1}{|b|}$$

Example 11.2: Let $X \sim U(0,1)$, so $f_X(x) = 1$ for $0 < x < 1$, and let $Y = 3 + 2X$. Then for $3 \leq y \leq 5$:

$$f_Y(y) = f_X\left(\frac{y-3}{2}\right) \cdot \frac{1}{2} = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

so $Y \sim U(3,5)$: a linear map sends a uniform RV to another uniform RV.

General monotone case: If $Y = g(X)$, where g is strictly monotone and differentiable on the support of X, then

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

for y in the range of g, and $f_Y(y) = 0$ otherwise.

Derivation (increasing case): $F_Y(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate w.r.t. y. For decreasing g the inequality flips, and the derivative of g^{-1} is negative, which produces the absolute value.

Note: If g is *not* monotone (e.g. $Y = X^2$), this formula does not apply directly; go back to the c.d.f. method, as we do for Z^2 below.

Example 11.3: Let $X \sim U(0,1)$ and $Y = -\frac{1}{\lambda}\ln(X)$, where $\lambda > 0$. Here $g(x) = -\frac{1}{\lambda}\ln(x)$ is strictly decreasing on $(0,1)$, with $g^{-1}(y) = e^{-\lambda y}$. For $y > 0$:

$$f_Y(y) = f_X\left(e^{-\lambda y}\right)\left|-\lambda e^{-\lambda y}\right| = 1 \cdot \lambda e^{-\lambda y} = \lambda e^{-\lambda y}$$

i.e. $Y \sim \text{Exp}(\lambda)$, the exponential distribution of Section 6.2. This is how exponential RVs are simulated from uniform random numbers.

Theorem 11.1 (Probability Integral Transform): If F_X is continuous, and $Y = F_X(X)$, then

$$Y \sim U(0,1)$$

Proof (for F_X also strictly increasing): For $0 < y < 1$,

$$F_Y(y) = P(F_X(X) \le y) = P(X \le F_X^{-1}(y)) = F_X(F_X^{-1}(y)) = y$$

which is the c.d.f. of $U(0,1)$. □

Note: Continuity of F_X is essential: if X is discrete, $F_X(X)$ takes only countably many values and cannot be $U(0,1)$. Read in the other direction, the result says $X = F_X^{-1}(U)$ has c.d.f. F_X when $U \sim U(0,1)$ ("inverse transform sampling"), which is exactly what the previous example did for the exponential.

Python note: Inverse transform sampling in NumPy: with `rng = numpy.random.default_rng()`, `u = rng.random(10000)` draws $U(0,1)$ values, and `x = -numpy.log(u)/lam` turns them into $\text{Exp}(\lambda)$ draws, exactly as above.

Squared Normals Revisited

The map $g(z) = z^2$ is not monotone on $\mathbb{R}$, so the formula above does not apply, but the c.d.f. method still works. Let $Z \sim N(0,1)$ and $Y = Z^2$. We show again that $Y \sim \chi^2_1$ (Section 11.1), this time from the density.

Proof.

$$\begin{aligned} F_Y(y) = P(Y \le y) = P(Z^2 \le y) &= P\left(|Z| \le \sqrt{y}\right) \\ &= P\left(-\sqrt{y} < Z < \sqrt{y}\right) \\ &= \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) \\ &= \Phi(\sqrt{y}) - \left(1 - \Phi(\sqrt{y})\right) \\ &= 2\Phi(\sqrt{y}) - 1 \end{aligned}$$

Now,

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy}\left(2\Phi(\sqrt{y}) - 1\right) = 2\,\frac{d}{dy}\Phi(\sqrt{y}) \\ &= 2\, f_Z(\sqrt{y})\,\frac{d}{dy}\sqrt{y} \qquad \text{(chain rule)} \\ &= 2 \cdot \frac{1}{\sqrt{2\pi}}\, e^{-y/2} \cdot \frac{1}{2\sqrt{y}} \\ &= \frac{1}{\sqrt{\pi}\,(2)^{1/2}}\, y^{-1/2}\, e^{-y/2} \\ &= \frac{1}{\Gamma(1/2)\,2^{1/2}}\, y^{1/2-1}\, e^{-y/2} \qquad \text{since } \Gamma(1/2) = \sqrt{\pi} \end{aligned}$$

Let $\alpha = 1/2$, $\beta = 1/2$:

$$= \frac{\beta^\alpha}{\Gamma(\alpha)}\, y^{\alpha-1}\, e^{-\beta y} \sim \Gamma\left(\alpha = \tfrac{1}{2},\ \beta = \tfrac{1}{2}\right) \sim \chi^2_1 \qquad \square$$

Exercises

Exercise 11.1. Let X be a continuous random variable with p.d.f. $f_X(x) = 4x^3$ for $0 < x \leq 1$ (and 0 o.w.).

(a) Let $Y = 1/X$. Find the p.d.f. of Y, specifying its support. *Hint:* use the c.d.f. method; note that $X \leq 1$ forces $Y \geq 1$.

(b) Find $\mathbb{E}[Y]$. Then use Markov's inequality to bound $P(Y > 6)$.

(c) Find $\mathbb{E}\left[\dfrac{4}{X} - 10\right]$.

(d) If we sample n values from X and average them, what does that average converge to in probability? Be as specific as you can, and name the theorem you are using.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 11.1. (a) For $y \geq 1$, $F_Y(y) = P(1/X \leq y) = P(X \geq 1/y) = 1 - (1/y)^4$, so $f_Y(y) = 4/y^5$ on $y > 1$. **(b)** $\mathbb{E}[Y] = \int_1^\infty 4y^{-4}\,dy = \frac{4}{3}$; Markov: $P(Y > 6) \leq \dfrac{4/3}{6} = \frac{2}{9}$. **(c)** $4\mathbb{E}[1/X] - 10 = 4(\frac{4}{3}) - 10 = -\frac{14}{3}$. **(d)** By the weak law of large numbers, $\bar{X}_n \xrightarrow{p} \mathbb{E}[X] = \int_0^1 4x^4\,dx = \frac{4}{5}$.

11.3 Min and Max Distributions; Transformations of Two Variables

Min and Max Distributions

Let $X_1, \ldots, X_n$ be i.i.d. continuous RVs, each with c.d.f. F_X and p.d.f. f_X, and let

$$Y_{(1)} = \min(X_1, \ldots, X_n), \qquad Y_{(n)} = \max(X_1, \ldots, X_n)$$

Max: The max is $\leq y$ iff *every* $X_i \leq y$, so by independence:

$$F_{Y_{(n)}}(y) = P(X_1 \leq y, \ldots, X_n \leq y) = [F_X(y)]^n \quad \Rightarrow \quad f_{Y_{(n)}}(y) = n[F_X(y)]^{n-1} f_X(y)$$

Min: The min is $> y$ iff *every* $X_i > y$, so:

$$\begin{aligned} & F_{Y_{(1)}}(y) = 1 - P(X_1 > y, \ldots, X_n > y) = 1 - [1 - F_X(y)]^n \\ \Rightarrow \quad & f_{Y_{(1)}}(y) = n[1 - F_X(y)]^{n-1} f_X(y) \end{aligned}$$

Example 11.4: A string of n disco-ball spotlights is wired in series (it goes dark as soon as any one bulb burns out), with i.i.d. bulb lifetimes $X_i \sim \text{Exp}(\lambda)$. The string's lifetime is $Y_{(1)} = \min(X_i)$. Since $1 - F_X(y) = e^{-\lambda y}$:

$$f_{Y_{(1)}}(y) = n\left(e^{-\lambda y}\right)^{n-1} \lambda e^{-\lambda y} = n\lambda\, e^{-n\lambda y}, \quad y > 0$$

so $Y_{(1)} \sim \text{Exp}(n\lambda)$: the min of n i.i.d. exponentials is exponential with n times the rate, and the expected lifetime is $\mathbb{E}[Y_{(1)}] = \frac{1}{n\lambda}$.

Example 11.5: Let $X_1, \ldots, X_n$ be i.i.d. $U(0, \theta)$, and let $Y_{(n)} = \max(X_i)$. For $0 \leq y \leq \theta$, $F_X(y) = y/\theta$, so

$$f_{Y_{(n)}}(y) = n\left(\frac{y}{\theta}\right)^{n-1} \frac{1}{\theta} = \frac{n\, y^{n-1}}{\theta^n}, \qquad \mathbb{E}[Y_{(n)}] = \int_0^\theta y\, \frac{n\, y^{n-1}}{\theta^n}\, dy = \frac{n}{n+1}\,\theta$$

As n grows, the sample max approaches θ, as we would expect.

Transformation of Two Variables

In Section 11.2 we found the distribution of a function of one RV. The same question arises for pairs: given the joint p.d.f. of (X, Y), we want the joint p.d.f. of a transformed pair, such as $(X + Y,\ X)$.

Let (X, Y) have joint p.d.f. $f_{X,Y}(x, y)$, and define

$$X^* = g_1(X, Y), \quad Y^* = g_2(X, Y)$$

If the map $(x, y) \mapsto (x^*, y^*)$ is one-to-one, we can solve for the inverse transformation $x = h_1(x^*, y^*)$, $y = h_2(x^*, y^*)$, and then

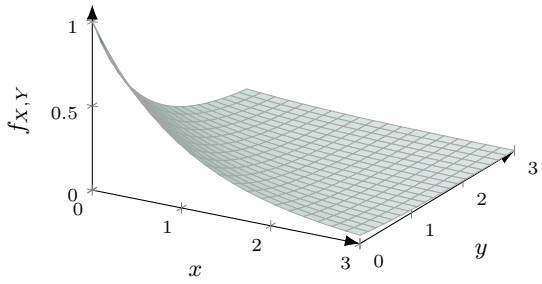
$$f_{X^*,Y^*}(x^*, y^*) = f_{X,Y}\big(h_1(x^*, y^*),\, h_2(x^*, y^*)\big) \cdot |J|$$

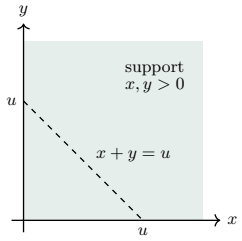
Note that $f_{X,Y}$ is evaluated at the *inverse* functions, and J is the Jacobian determinant of the inverse map:

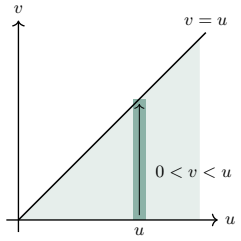
$$J = \begin{vmatrix} \frac{\partial x}{\partial x^*} & \frac{\partial x}{\partial y^*} \\ \frac{\partial y}{\partial x^*} & \frac{\partial y}{\partial y^*} \end{vmatrix}$$

Example 11.6: Let X, Y be independent with $X \sim \text{Exp}(1)$, $Y \sim \text{Exp}(1)$, so $f_{X,Y}(x, y) = e^{-x}e^{-y}$ for $x, y > 0$. Define $U = X + Y$, $V = X$.

The inverse transformation is $x = v$, $y = u - v$. The support carries over from the original variables: we need $x = v > 0$ and $y = u - v > 0$, which together give $0 < v < u$.







$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1 \quad \Rightarrow \quad |J| = 1$$

$$f_{UV}(u, v) = e^{-v}\, e^{-(u-v)} \cdot 1 = e^{-u}, \quad 0 < v < u$$

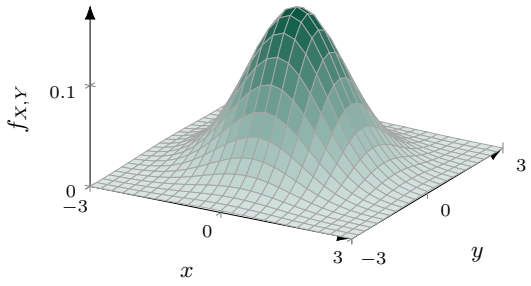
Marginal of U:

$$f_U(u) = \int_0^u e^{-u}\, dv = u\, e^{-u}, \quad u > 0$$

So $U \sim \Gamma(\alpha = 2, \beta = 1)$, i.e., the sum of two independent Exp(1) RVs has a $\Gamma(\alpha = 2, \beta = 1)$ distribution.

Example 11.7: Let X, Y be i.i.d. $N(0, 1)$, so $f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$. Define $U = X + Y$, $V = X - Y$.

The inverse is $x = \frac{u+v}{2}$, $y = \frac{u-v}{2}$, and $|J| = \frac{1}{2}$.



$$\begin{aligned} f_{UV}(u,v) &= \frac{1}{2\pi}\exp\!\left(-\frac{1}{2}\left[\left(\frac{u+v}{2}\right)^2+\left(\frac{u-v}{2}\right)^2\right]\right)\cdot\frac{1}{2} \\ &= \frac{1}{4\pi}\exp\!\left(-\frac{u^2+v^2}{4}\right) \\ &= \underbrace{\frac{1}{\sqrt{4\pi}}e^{-u^2/4}}_{U\sim N(0,2)}\cdot\underbrace{\frac{1}{\sqrt{4\pi}}e^{-v^2/4}}_{V\sim N(0,2)} \end{aligned}$$

So $U \sim N(0,2)$, $V \sim N(0,2)$, and U, V are independent. Linear combinations of independent normals remain normal.

Looking Ahead: the Bivariate Normal

The pair (X, Y) is said to have a **bivariate normal** distribution if every linear combination $aX + bY$ is normally distributed. The joint distribution is then completely determined by five numbers: μ_X, μ_Y, σ_X^2, σ_Y^2, and the correlation $\rho = \rho(X, Y)$. (The (U, V) example above is the special case with zero means and $\rho = 0$.) Two facts about this family are worth carrying into your next courses; we state them without proof.

Fact 1 (the regression line). If (X, Y) is bivariate normal, the conditional distribution of Y given $X = x$ is again normal:

$$Y \mid X = x \;\sim\; N\!\left(\mu_Y + \rho\,\frac{\sigma_Y}{\sigma_X}(x-\mu_X),\;\; \sigma_Y^2(1-\rho^2)\right)$$

The conditional mean is *linear* in x: this line is exactly the population regression line of your regression course, and by the best-predictor result of Section 7.4 it is the best prediction of Y from X. The conditional variance $\sigma_Y^2(1-\rho^2)$ does not depend on x and shrinks as $|\rho| \to 1$: the fraction ρ^2 of the variance of Y is "explained" by X.

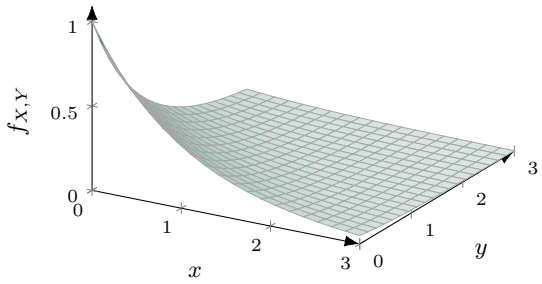
Fact 2 (uncorrelated = independent, here only). Within the bivariate normal family, $\rho = 0$ implies that X and Y are independent. Recall from Section 9.2 that this implication *fails* for general RVs; the bivariate normal is the important special case where uncorrelated and independent coincide.

Exercises

Exercise 11.2 (An anglerfish's lure)**.** A deep-sea anglerfish lights the lure on its forehead in brief bursts to draw prey. The waits X and Y, in minutes, between the first three bursts are independent $\text{Exp}(1)$ random variables. Define

$$U = X + Y, \qquad V = \frac{X}{X+Y},$$

the time from the first burst to the third and the fraction of it spent waiting for the second.



(a) Solve for x and y in terms of u and v, and determine the range of (U, V).

(b) Compute the Jacobian and show that its absolute value is u.

(c) Show that the joint p.d.f. of (U, V) is $f_{U,V}(u, v) = ue^{-u}$ for $u > 0$, $0 < v < 1$.

(d) Factor this and conclude that $U \sim \Gamma(\alpha = 2, \beta = 1)$, $V \sim U(0, 1)$, and U and V are *independent*. Comment on why this independence is surprising, given that both U and V are built from the same two variables.



Figure 11.1: *A green darner dragonfly.* [20]

Exercise 11.3. Each of n adult dragonflies (Fig. 11.1) patrols a pond independently, and the time T_i until dragonfly i first crosses an observer's transect line is $\text{Exp}(\lambda)$, the T_i being i.i.d.

(a) Let $L = \min(T_1, \ldots, T_n)$. Observe that if $L > x$, then none of the n dragonflies has crossed by time x. Use this to compute $P(L > x)$, and identify the distribution of L as another exponential whose rate can be written in terms of λ and n. What are its mean and variance?

(b) Let $M = \max(T_1, \ldots, T_n)$. Compute the mean and variance of M *without* using calculus. *Hint:* Start by waiting for the first crossing, whose distribution you found in part (a). Then apply the memoryless property, repeatedly.

Exercise 11.4 (Meeting up at Carnival)**.** Five friends set out independently for a meeting point in Salvador, and each one's lateness, in minutes, is exponential with mean 10.

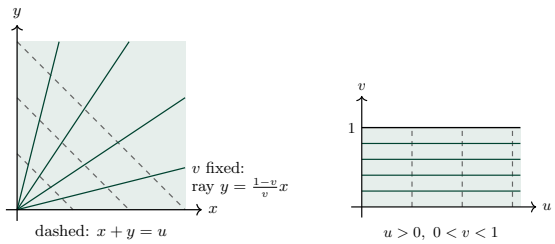
(a) Name the distribution of the time until the first friend arrives, and give its mean.

(b) Write the c.d.f. of the time until all five have arrived, and find the probability everyone is there within 20 minutes.

Solutions to the exercises

Work each exercise on the previous page before comparing.

Exercise 11.2. (a) $x = uv$, $y = u(1-v)$; since $x, y > 0$ we need $u > 0$ and $0 < v < 1$.



(b) $\dfrac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} v & u \\ 1-v & -u \end{vmatrix} = -uv - u(1-v) = -u$, absolute value u. **(c)** $f_{U,V}(u,v) = f_{X,Y}(uv, u(1-v)) \cdot u = e^{-uv}e^{-u(1-v)}u = ue^{-u}$ on $u > 0, 0 < v < 1$. **(d)** ue^{-u} is the $\Gamma(2,1)$ density in u times the $U(0,1)$ density 1 in v, so U and V are independent with those marginals. The total U carries no information about the fraction V that came from X: once you know the two waits add to u, the split is uniformly random regardless of u.

Exercise 11.3. (a) $P(L > x) = \prod P(T_i > x) = e^{-n\lambda x}$, so $L \sim \text{Exp}(n\lambda)$ with mean $1/(n\lambda)$ and variance $1/(n\lambda)^2$. **(b)** After the first crossing, the remaining $n-1$ dragonflies are, by memorylessness, fresh $\text{Exp}(\lambda)$ clocks, so the next gap is $\text{Exp}((n-1)\lambda)$, and so on. M is a sum of independent gaps $\text{Exp}(n\lambda), \text{Exp}((n-1)\lambda), \ldots, \text{Exp}(\lambda)$:

$$\mathbb{E}[M] = \frac{1}{\lambda}\sum_{k=1}^{n}\frac{1}{k}, \qquad \text{Var}[M] = \frac{1}{\lambda^2}\sum_{k=1}^{n}\frac{1}{k^2}.$$

Exercise 11.4. (a) The minimum of five $\text{Exp}(1/10)$ times is $\text{Exp}(5/10)$: rate $1/2$ per minute, mean 2 minutes. **(b)** $F_{\max}(y) = (1 - e^{-y/10})^5$; at $y = 20$, $(1 - e^{-2})^5 = 0.483$.

Review Topics

- Set theory
- Combinatorics
- Probability theory
- Discrete/continuous RV's
- Expectation, variance, m.g.f.
- Inequalities
- Named distributions
- Joint distributions

- Conditional distributions, conditional expectation
- Covariance, correlation
- Independence
- Sampling distributions (χ^2, sample variance)
- Convergence (WLLN, CLT)
- Transformations
- Min/max distributions

Image credits

Photographs are from Wikimedia Commons under the licenses listed.

1 Kinkaku-ji (left) and Ginkaku-ji (right), Kyoto. Photos: Basile Morin, CC BY-SA 4.0, and Laitr Keiows, CC BY-SA 3.0, via Wikimedia Commons.

2 Sandhill cranes over the Sangre de Cristo Mountains. Photo: Great Sand Dunes National Park and Preserve, public domain, via Wikimedia Commons.

3 The Shibuya scramble crossing. Photo: Jakub Hałun, CC BY 4.0, via Wikimedia Commons.

4 Egyptian scarab amulets. Photo: the Met, CC0, via Wikimedia Commons.

5 Ozzy and Olive in boxes. Photo: Cody Carroll.

6 Torii gates on the path at Fushimi Inari, Kyoto. Photo: Cody Carroll.

7 Oia, Santorini, at sunset. Photo: Pedro Szekely, CC BY-SA 2.0, via Wikimedia Commons.

8 A goldfish-scooping stall at a summer festival. Photo: Takayoshi Nishida, CC BY-SA 2.0, via Wikimedia Commons.

9 A Shinkansen N700A. Photo: Dllu, CC BY-SA 4.0, via Wikimedia Commons.

10 A folkloric dance and percussion company performing in Salvador, Brazil. Photo: Cody Carroll.

11 A full mahjong set. Photo: Cangjie6, CC BY-SA 4.0, via Wikimedia Commons.

12 Ochre sea stars in a tide pool at Half Moon Bay. Photo: Brocken Inaglory, CC BY-SA 4.0, via Wikimedia Commons.

13 Osiris, from the tomb of Nefertari. Wall painting from the tomb of Nefertari; public domain, via Wikimedia Commons.

14 Copacabana and Botafogo Bay from the summit of Sugarloaf, Rio de Janeiro. Photo: Cody Carroll.

15 The weighing of the heart: Anubis checks the scale, Thoth records the verdict, and Ammit waits. Papyrus of Hunefer, British Museum; public domain, via Wikimedia Commons.

16 Everest at sunset from Kala Patthar. Photo: Nirajha, CC BY-SA 4.0, via Wikimedia Commons.

17 A coral reef. Photo: Jim E Maragos, U.S. Fish and Wildlife Service, public domain, via Wikimedia Commons.

18 A sea otter. Photo: Mrkoww, CC BY-SA 3.0, via Wikimedia Commons.

19 A Galton board. Photo: Exhibit made by Estes Objethos Atelier, photo, CC BY-SA 4.0, via Wikimedia Commons.

20 A green darner dragonfly. Photo: Chuck Evans Mcevan, CC BY-SA 3.0, via Wikimedia Commons.